



HOL-2770-01-ANS-L

# Table of Contents

|                                                                      |    |
|----------------------------------------------------------------------|----|
| VMware vDefend Firewall Security Journey - (HOL-2770-01-ANS-L) ..... | 5  |
| Lab Guidance .....                                                   | 6  |
| First Time Using Hands-on-Labs? .....                                | 8  |
| Module 1 – vDefend & Security Services Platform Introduction .....   | 10 |
| vDefend Security Services Platform Overview .....                    | 11 |
| vDefend Security Services Platform Architecture .....                | 13 |
| vDefend Security Services Platform Installer Hands On .....          | 15 |
| vDefend Security Services Platform Instance Hands On .....           | 20 |
| vDefend Security Services Platform Summary .....                     | 28 |
| Module 2 – Security Journey Overview .....                           | 29 |
| Module 2 Overview: Securing VCF with DFW 1-2-3-4 & ATP 1-2-3 .....   | 30 |
| Distributed Firewall (DFW) 1-2-3-4 .....                             | 31 |
| Advanced Threat Protection (ATP) 1-2-3 .....                         | 35 |
| Module 3 - Importing Datacenter Assets .....                         | 36 |
| Datacenter Topology: Acme .....                                      | 37 |
| CSV Import Introduction .....                                        | 38 |
| Login to Security Services Platform (SSP) .....                      | 43 |
| Import the CSV .....                                                 | 45 |
| Publish Assets .....                                                 | 50 |
| Validate Published Assets in SSP .....                               | 55 |
| Login to NSX .....                                                   | 56 |
| Validate Published Assets in NSX .....                               | 59 |
| Conclusion .....                                                     | 66 |
| Module 4 – Stage 1: Security Segmentation Assessment & Report .....  | 67 |
| Security Segmentation Assessment & Report Introduction .....         | 68 |
| Login to Security Services Platform (SSP) .....                      | 73 |

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| Calculate Security Segmentation Score.....                                    | 75  |
| View Security Assessment Report .....                                         | 83  |
| Conclusion .....                                                              | 90  |
| Module 5 – Stage 2: Infrastructure (Shared) Services Segmentation.....        | 91  |
| Infrastructure (Shared) Services Segmentation Introduction .....              | 92  |
| Login to Security Services Platform (SSP).....                                | 94  |
| Publish Infrastructure Services Rules to Distributed Firewall & IDS/IPS ..... | 96  |
| Login to NSX .....                                                            | 110 |
| Validate Published Infrastructure Services Rules in NSX.....                  | 113 |
| Re-calculate Security Segmentation Score.....                                 | 115 |
| Conclusion .....                                                              | 120 |
| Module 6 – Stage 3: Environment (Zone) Segmentation .....                     | 121 |
| Environment (Zone) Segmentation Introduction .....                            | 122 |
| Login to Security Services Platform (SSP).....                                | 124 |
| Generate and Publish Default Environment Policies .....                       | 126 |
| Login to NSX .....                                                            | 129 |
| Validate Environment Rule Publish in NSX .....                                | 132 |
| Monitor Inter-Environment Traffic.....                                        | 134 |
| Create Exceptions.....                                                        | 139 |
| Lockdown Environment Pairs .....                                              | 148 |
| View Environment Pair Protection Progress with SSP.....                       | 153 |
| Re-calculate Security Segmentation Score.....                                 | 157 |
| Conclusion .....                                                              | 162 |
| Module 7 – Stage 4: Application Microsegmentation.....                        | 163 |
| Application Micro-segmentation Introduction .....                             | 164 |
| Login to Security Services Platform (SSP).....                                | 167 |
| Define Application Criticality.....                                           | 169 |

|                                                |     |
|------------------------------------------------|-----|
| Monitor Applications.....                      | 174 |
| Protect Applications .....                     | 177 |
| Login to NSX .....                             | 187 |
| Validate Application Rule Publish in NSX ..... | 190 |
| Lockdown Applications .....                    | 192 |
| Re-calculate Security Segmentation Score.....  | 197 |
| Conclusion .....                               | 201 |



# **VMware vDefend Firewall Security Journey - (HOL-2770-01-ANS-L)**

# Lab Guidance

Want to know the fastest way to get value from vDefend Distributed Firewall? Take this lab to learn how vDefend DFW 1-2-3-4 can help you deploy Zero-Trust Micro-segmentation in a few weeks to rapidly secure VCF workloads.

This lab is available for you to repeat as many times as you want. Use the Table of Contents in the upper right-hand corner of the Lab Manual to jump ahead to any module.

| Module | Title                                                          | Time       | Level    |
|--------|----------------------------------------------------------------|------------|----------|
| 1      | vDefend & Security Services Platform Architecture Introduction | 10 minutes | Beginner |
| 2      | Security Journey Overview                                      | 10 minutes | Beginner |
| 3      | Importing Datacenter Assets                                    | 25 minutes | Beginner |
| 4      | Stage 1: Security Segmentation Assessment & Report             | 10 minutes | Beginner |
| 5      | Stage 2: Infrastructure (Shared) Services Segmentation         | 15 minutes | Beginner |
| 6      | Stage 3: Environment (Zone) Segmentation                       | 30 minutes | Beginner |
| 7      | Stage 4: Application Micro-segmentation                        | 30 minutes | Beginner |

A module can be either 15, 30, 45, or 60 minutes in length. The module level can be either Beginner, Intermediate, or Advanced.

Lab Authors:

- Quinton Coelho, Solution Architect, Application Network & Security, EMEA

- 
- Lynne Dark, Senior Application Network Security Technical Adoption Manager, Americas
  - Joshua Newton, Staff Application Network Security Technical Adoption Manager, Americas
  - Apoorv Malmane, Principal Engineer, Application Network & Security, Americas
  - Bhavik Mehta, Application Network Security Technical Adoption Manager, Americas

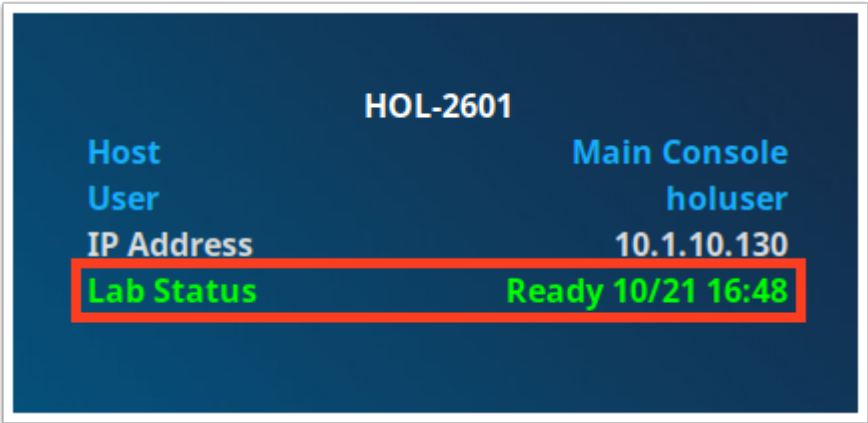
Special Thanks:

- Frank Snyder, Solution Architect, Application Network & Security, Americas
- Antoine Deleporte, Senior Technical Product Manager, Application Network & Security, EMEA
- Kausum Kumar, Senior Technical Product Manager, Application Network & Security, Americas

# First Time Using Hands-on-Labs?

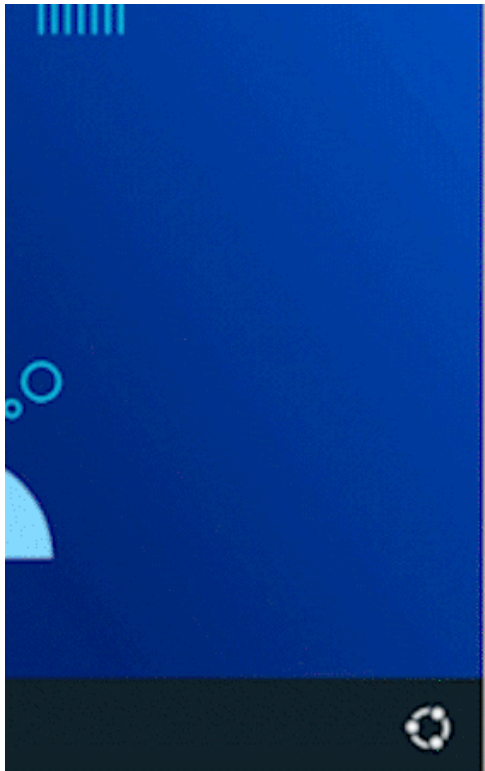
💡 If this is your first time taking a lab you can review the [VMware Lab Platform interface](#) before proceeding.

This includes help with copy/paste from the manual into the console and other user-interface questions.



The lab console will indicate when your lab has finished all the startup routines and is ready for you to start. If you see anything other than "Ready", please wait for the status to update. If after 5 minutes your lab has not changed to "Ready", please ask for assistance.

## Drag and Drop Text



Click to enlarge

At the VCF Operations login prompt, select the login method and type in the following user and password information:

1. At the Login Method dropdown, select **Local Account**.
2. At the username field, type **admin**.
3. At the password field, type **VMware123!VMware123!**.
4. Click **LOG IN**.

The user automatically credentials.

 **Hi! How may I help you?**

Save time by dragging and dropping required text from the manual into the lab console.



# **Module 1 – vDefend & Security Services Platform Introduction**

# vDefend Security Services Platform Overview

The Security Services Platform (SSP) for VMware® vDefend™ accelerates and simplifies the journey to Zero Trust security within the VCF environment. It is a high-performance and scalable platform that runs numerous vDefend security services to protect workloads in the VMware vDefend(VCF) private cloud.

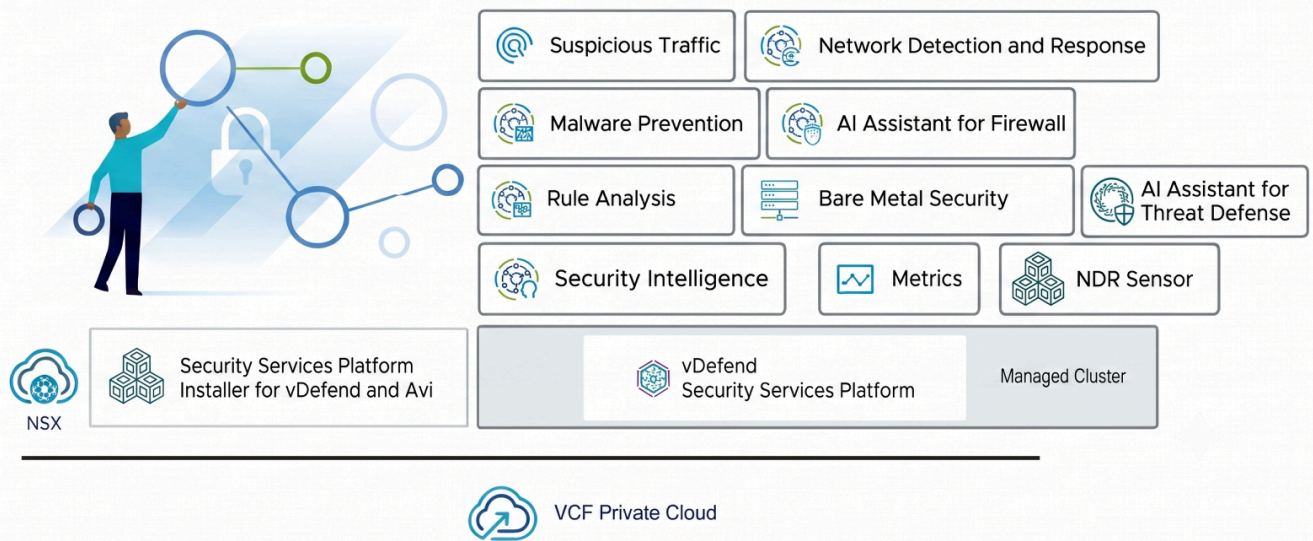
## Key Features

Security Services Platform centralizes vDefend advanced features. The primary services that are hosted on the platform include:

- **Security Intelligence:** Provides real-time application flow discovery, deep analytics, and machine learning-based firewall rule recommendations to accelerate micro-segmentation.
- **Security Assessment Report:** A comprehensive report, critical to both the vDefend segmentation and ATP journeys. It assesses the current Zero Trust security posture of VCF workloads across segmentation and Advanced Threat Prevention. The report provides a score for each and offers actionable recommendations for improving the security posture.
- **Rapid Segmentation Journey:** A prescriptive, automated workflow and phased methodology to easily and quickly implement macro/micro-segmentation for lateral security within the VCF environment.
- **Rapid ATP Journey:** A prescriptive deployment approach for Advanced Threat Prevention, allowing organizations to progressively enable IDS/IPS, Malware Prevention Service, Network Detection and Response, and Network Traffic Analysis to systematically improve their threat defense posture.
- **Bare Metal Security:** A horizontally scalable architecture that supports large-scale micro-segmentation for Bare Metal Servers.
- **Rule Analysis:** Rule Analysis is a policy-optimization feature, which analyzes active Distributed Firewall (DFW) rules to identify inefficiencies, misconfigurations, and potential security risks without requiring manual audits or third-party tools.
- **AI Assistant for Firewall:** Uses integrated AI technologies to drastically simplify firewall operations, policy lifecycle management, and troubleshooting for security teams.
- **AI Assistant for Threat Defense:** Uses AI-powered threat analytics to assist security operations in rapidly investigating alerts, understanding complex attack chains, and accelerating incident response times.
- **Malware Prevention Service:** Detects and prevents zero-day exploits and malicious activities by analyzing anomalous network behavior.
- **Network Detection and Response:** Correlates multi-context threat signals into high-fidelity campaigns that are mapped to the MITRE ATT&CK framework.
- **NDR Sensor:** NDR Sensors continuously capture network traffic data and anomalous behaviors, feeding multi-context threat signals into the NDR engine to identify threat campaigns that are mapped to the MITRE ATT&CK framework.

- **Network Traffic Analysis(NTA):** NTA uses machine learning (ML) and behavioral analytics to detect unknown, evasive, and zero-day threats moving within your environment.

## Security Services Platform



# vDefend Security Services Platform Architecture

vDefend Security Services Platform Architecture review.

## vDefend Security Services Platform Concepts

**SSP Installer** - SSP Installer is a VM which deploys and manages the Security Services Platform Instance and Avi Operations Instance. The SSP Installer hosts packages needed for the Security Services Platform instance lifecycle management, which includes deployment and upgrade. The installer features a browser-based user interface for managing the environment.

**SSP Instance** - Security Services Platform is a managed cluster that hosts various core security features like Security Intelligence, Network Detection and Response, Malware Prevention Service, and so on. Multiple instances of the platform can be deployed and managed by the SSP Installer.

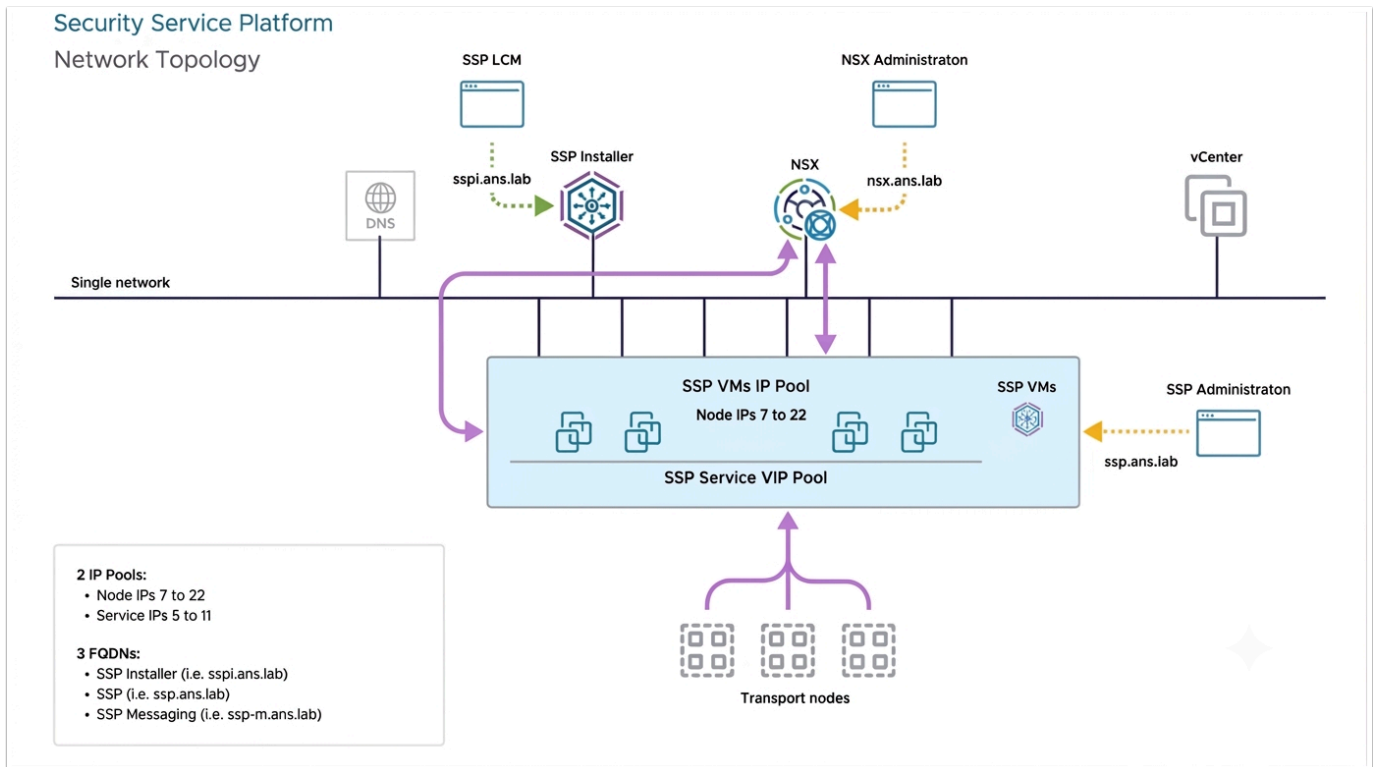
**Security Feature** - A security feature is a network security tool or technology, such as Security Intelligence, Network Detection and Response, Network Traffic Analysis, and so on.

**Avi Operations** - Avi Operations instance integrates with the VMware Avi Load Balancer Controller and enables AI Assistant capabilities for Avi, facilitating enhanced operational capabilities using generative AI.

## Network Topologies

Security Services Platform can work in a single-network environment (for example, a single Virtual Private Cloud (VPC), Local Area Network (LAN), dedicated VLAN, or air-gapped environment) or in a routed-network environment.

# Single-network Topology

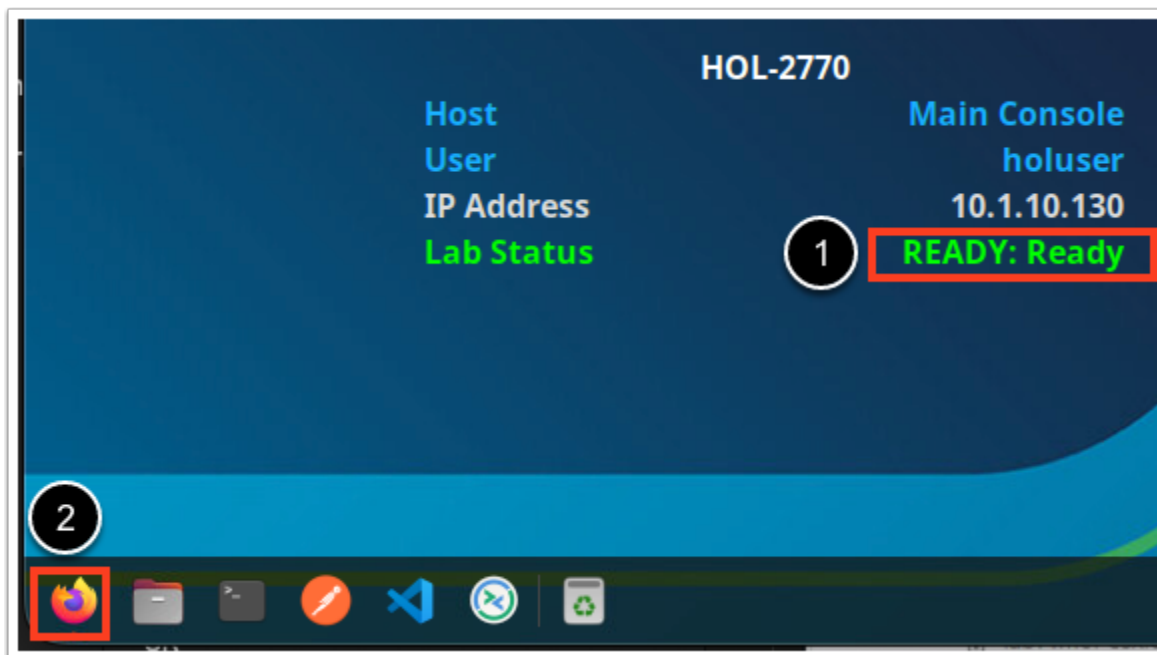


- In a single-network environment, all components reside on the same network. There is direct connectivity across all components, such as vCenter, NSX Manager, transport nodes, SSP Installer, and the SSP instance. The IP addresses of all the components are on the same network.
- Security Services Platform can also be deployed where SSP Installer is on a management network and the SSP instance is on a different network (for example, the network where the transport nodes are on). Both networks are routable.

# vDefend Security Services Platform Installer Hands On

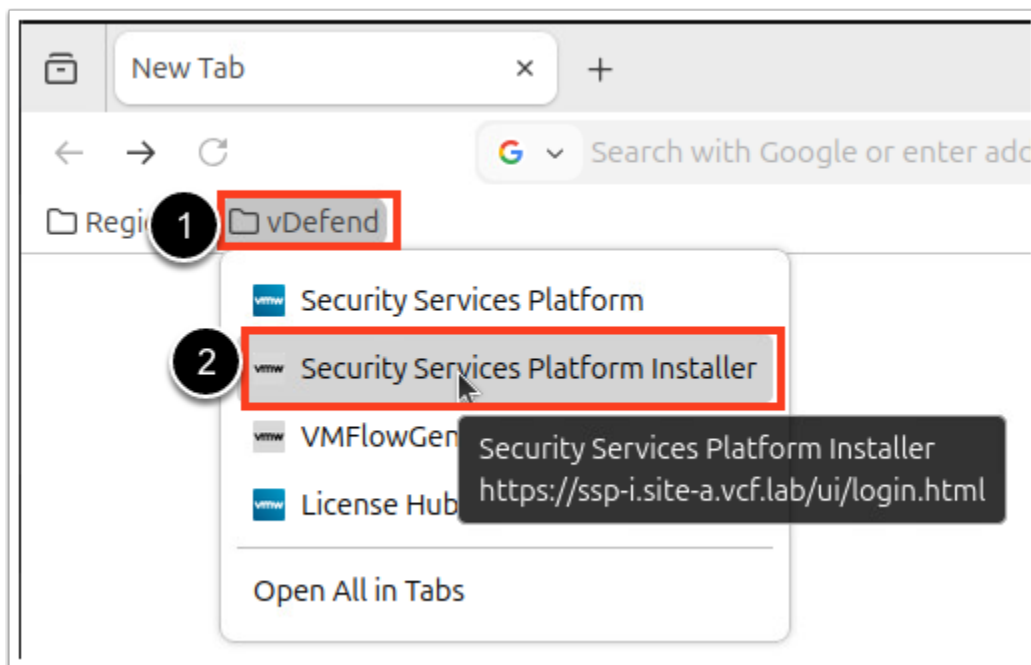
Let's get some hands-on exposure and experience with the vDefend Security Services Platform Installer.

## Accessing Firefox



1. Confirm the Lab Status shows **Ready**
2. Click on the Firefox icon to launch the browser

## Accessing Security Services Platform Installer



1. Click the **vDefend** Folder on the browser
2. Click on the **Security Services Platform Installer** bookmark.

## Accessing Security Services Platform, Log In

VMware vDefend  
Security Services Platform Installer

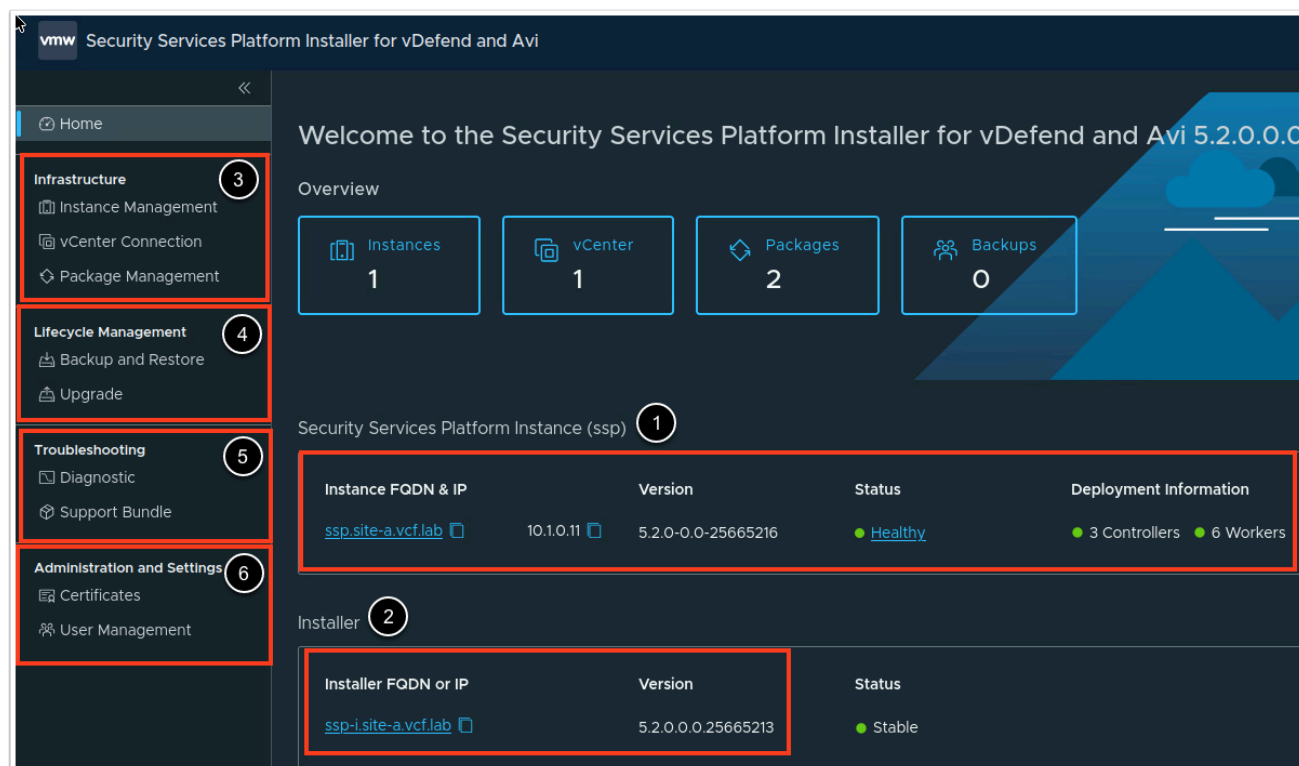
1 Username

2 Password  [Forgot Password](#)

3

1. Username **admin** should be pre-populated
2. Password **VMware123!VMware123!** should be pre-populated
3. Click **LOGIN**

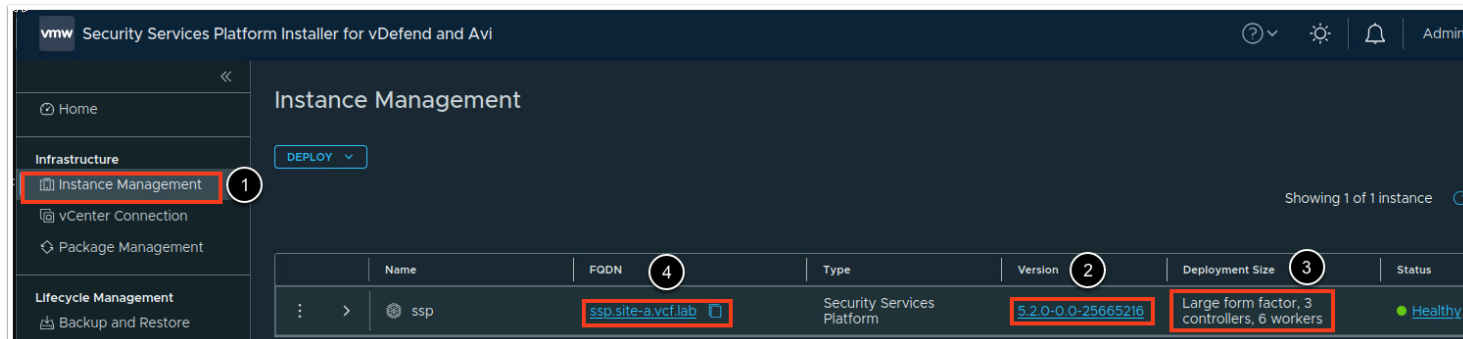
# vDefend Security Services Platform Installer UI Overview



**i** In this lab we have pre-deployed the Security Services Platform (SSP) Installer and created a SSP instance.

1. **Security Services Platform Instance (ssp):** Displays the primary operational status, network details, version, and Deployment Information ( controllers & workers) of the platform instance.
2. **Installer:** Shows the network address, running software version, and current status of the platform installer utility.
3. **Infrastructure:** Configures core integrations and connections, including cluster instances, vCenter environments, and software packages.
4. **Lifecycle Management:** Handles routine platform maintenance tasks, including system backups, restores, and version upgrades.
5. **Troubleshooting:** Provides system diagnostics (Health chart & Components Grid) and support bundle generation capability.
6. **Administration and Settings:** Manages platform security parameters, including SSL/TLS certificates and user access controls.

# vDefend Security Services Platform - Instance Management

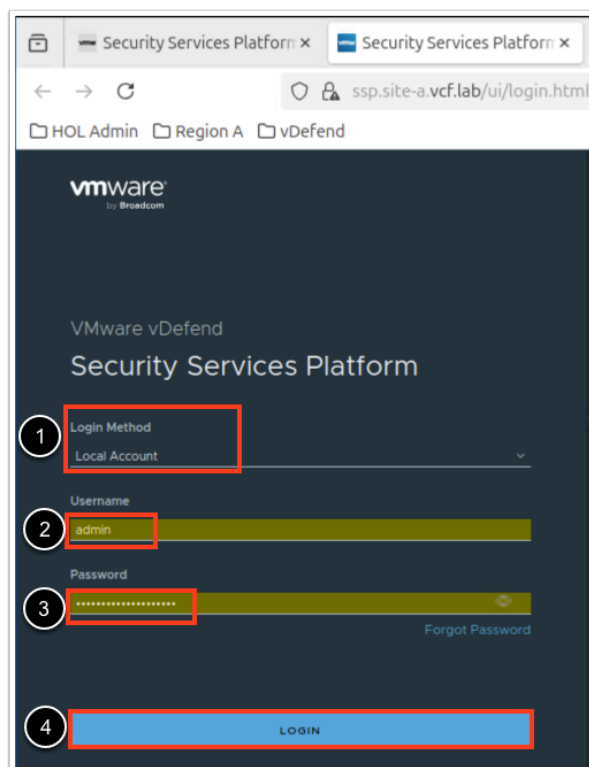


1. Click **Instance Management**
2. **Version:** This indicates the version of the SSP instance deployed.
3. **Deployment Size:** Details the form factor and Deployment Information (number of controllers & workers)
4. **FQDN:** Click on the FQDN **ssp.site-a.vcf.lab** to access the pre-deployed SSP instance.

# vDefend Security Services Platform Instance Hands On

Let's get some hands-on exposure and experience with vDefend Security Services Platform features.

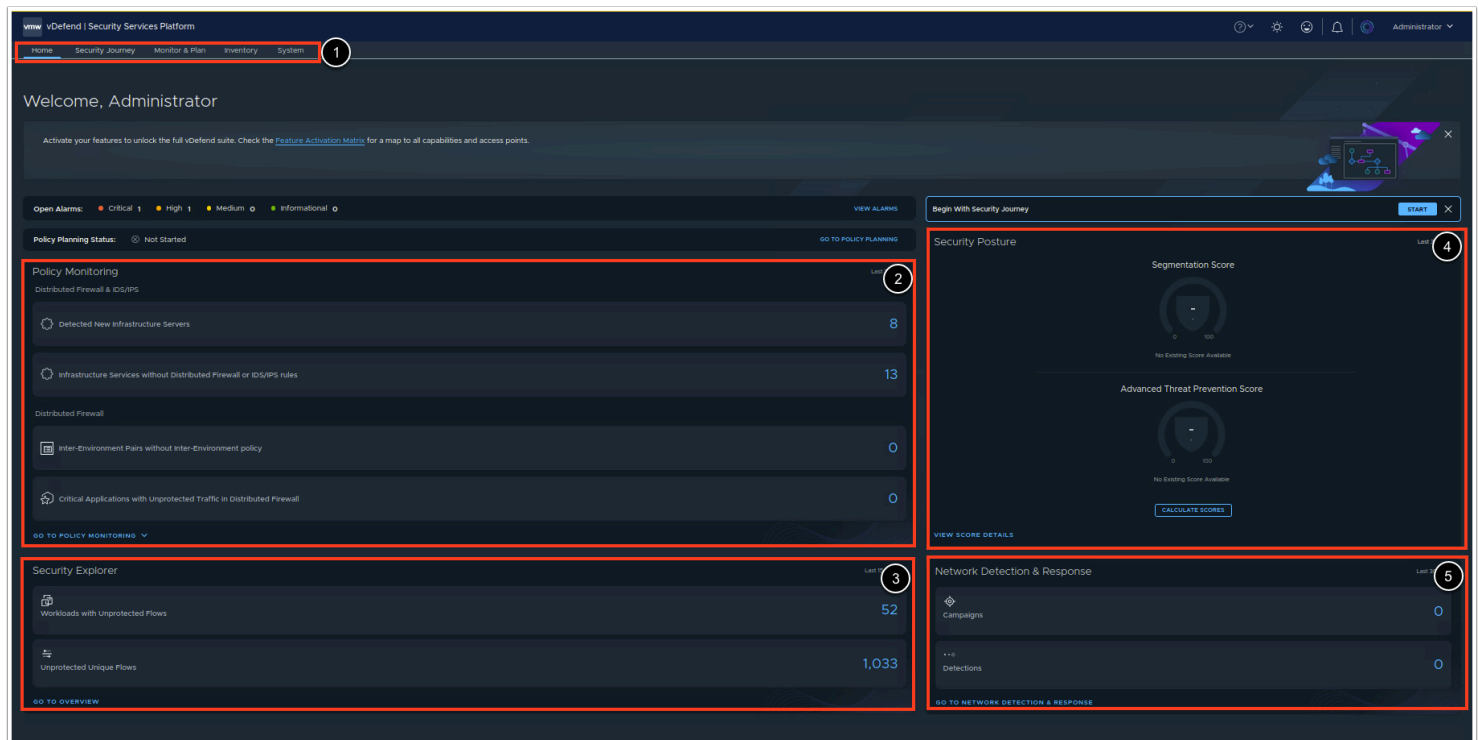
## Accessing the vDefend Security Platform Instance



**i** If you skipped the SSP Installer overview, you will need to open the Instance view from the vDefend folder. Else you should be take directly to the SSP instance without having to login and proceed to the next step.

1. **Login Method:** This should already be set to **Local Account**
2. **Username:** should be pre-populated with **admin**
3. **Password:** should be pre-populated with **VMware123!VMware123!**
4. **Click LOGIN**

# vDefend Security Services Platform Instance - Home Page



The Security Services Platform Instance landing page provides an overview of various features

1. The **Menu bar** allows you to switch between various platform functionalities.
2. **Policy Monitoring** highlights critical applications with unprotected flows in Distributed Firewall and IDS/IPS.
3. **Security Explore** provides a summary of the detected Flows.
4. **Security Posture** allows to calculate segmentation score & Advanced Threat Prevention Score.
5. **Network Detection & Response** highlights campaigns & Detections.

# Security Services Platform Instance - Platform and Features

The screenshot displays the VMware vDefend Security Services Platform interface. The top navigation bar includes 'Home', 'Security Journey', 'Monitor & Plan', 'Inventory', and 'System' (highlighted with a red box and a circled '1'). The left sidebar contains 'Overview', 'Infrastructure' (with 'Platform & Features' highlighted by a red box and a circled '2'), 'Lifecycle Management', 'Settings', and 'Administration'. The main content area is titled 'Platform & Features' (circled '3') and contains a table with instance details. Below this, the 'Features' tab (circled '4') is active, showing 'Metrics' and 'Security Intelligence' sections. The 'Metrics' section shows 'Data Collection' is 'On'. The 'Security Intelligence' section shows 'Status: UP' and 'Feature Version: 5.2.0.0.0.25665216'. A 'Network Traffic Analysis' section is also visible.

| Name                            | Version            | Features                          | Status                 | Overall Alarms            |
|---------------------------------|--------------------|-----------------------------------|------------------------|---------------------------|
| ssp                             | 5.2.0.0.0.25665216 | 3 Activated, 6 Pending Activation | STABLE                 | 0                         |
| UUID                            | Form Factor        | Deployment Information            | Platform FQDN          | Service FQDN              |
| 56cd21eb-2626-4691-bfd5-0e2b... | Large              | 3 Controllers, 6 Workers          | ssp.site-a.vcf.lab:443 | ssp-m.site-a.vcf.lab:9092 |

SERVICE SCALE OUT ⚠ Scaleout is not possible. [VIEW DETAILS](#)

Activate your features to unlock the full vDefend suite. Check the [Feature Activation Matrix](#) for a map to all capabilities and access points.

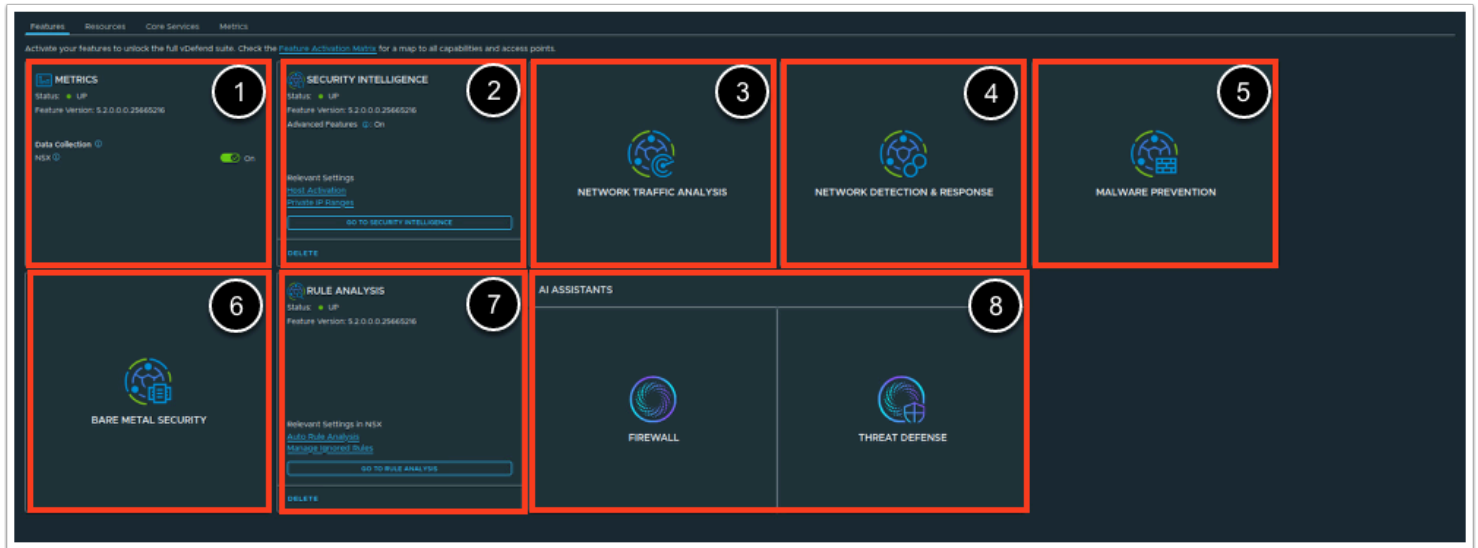
**METRICS**  
Status: UP  
Feature Version: 5.2.0.0.0.25665216  
Data Collection: NSX (On)

**SECURITY INTELLIGENCE**  
Status: UP  
Feature Version: 5.2.0.0.0.25665216  
Advanced Features: On  
Relevant Settings: [Host Activation](#), [Private IP Ranges](#)  
[GO TO SECURITY INTELLIGENCE](#)  
[DELETE](#)

**Network Traffic Analysis**  
Network Traffic Analysis (NTA) for Suspicious Traffic continuously monitors and analyzes network traffic and flow records using advanced machine learning (ML) models and statistical techniques to establish a baseline of normal network behavior.  
[ACTIVATE](#)

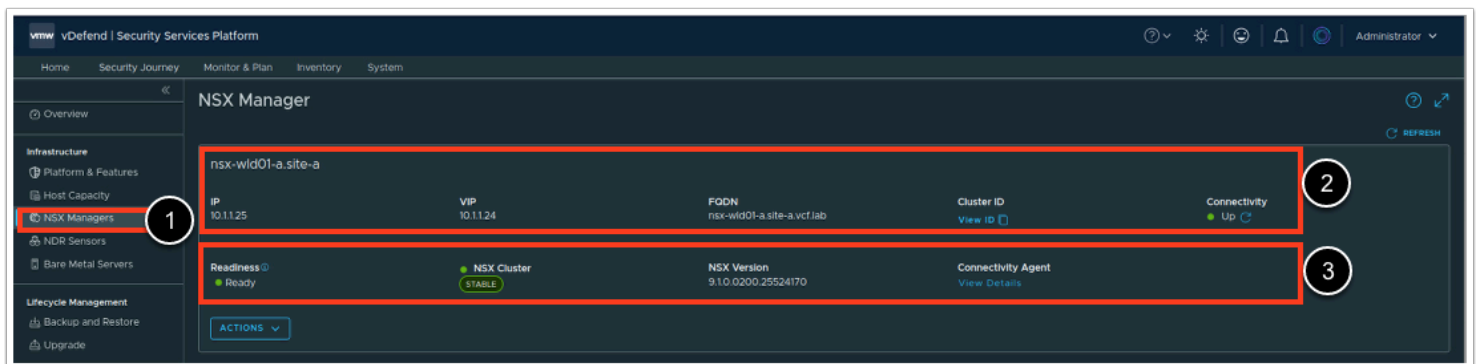
1. Click **System**
2. Click **Platform & Features**
3. **Platform & Features:** Here we find the Platform instance details and summary - similar to what we saw in the Installer.
4. **Features:** Here is where we activate the features which will run on the Security Services Platform Instance. You will notice that Metrics and Security Intelligence have already been activated.

# Security Services Platform Instance - Features



1. **Metrics** will be enabled by default when deploying the SSP instance.
2. **Security Intelligence** was pre-activated and from here you can access the feature dashboard or configure settings.
3. **Network Traffic Analysis** has not been activated yet, this will be performed later in the lab.
4. **Network Detection & Response** has not been activated yet, this will be performed later in the lab.
5. **Malware Prevention** has not been activated yet, this will be performed later in the lab.
6. **Bare Metal Security** has not been activated yet, this will be performed later in the lab.
7. **Rule Analysis** was pre-activated and from here you can access the feature dashboard or configure settings.
8. **AI Assistants** has not been activated yet.

## Security Services Platform Instance - NSX Manager

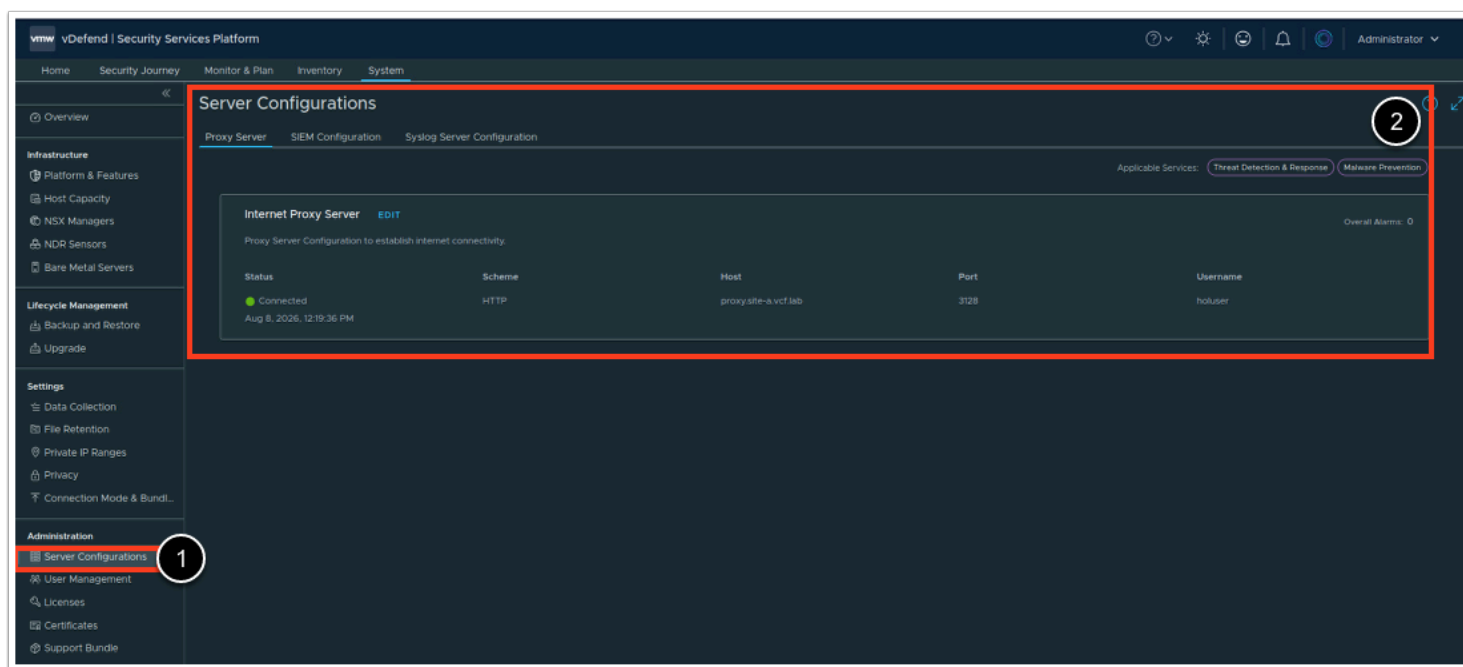


**i** In the current version of SSP 5.2 , **Each SSP Instance** maintains a strict **1:1 relationship** with a single NSX Manager cluster.

**i** In the current version of SSP 5.2 , **Single SSP Installer appliance** can manage **maximum 5 SSP instances**. You do not need to deploy a new installer VM every time you add another environment.

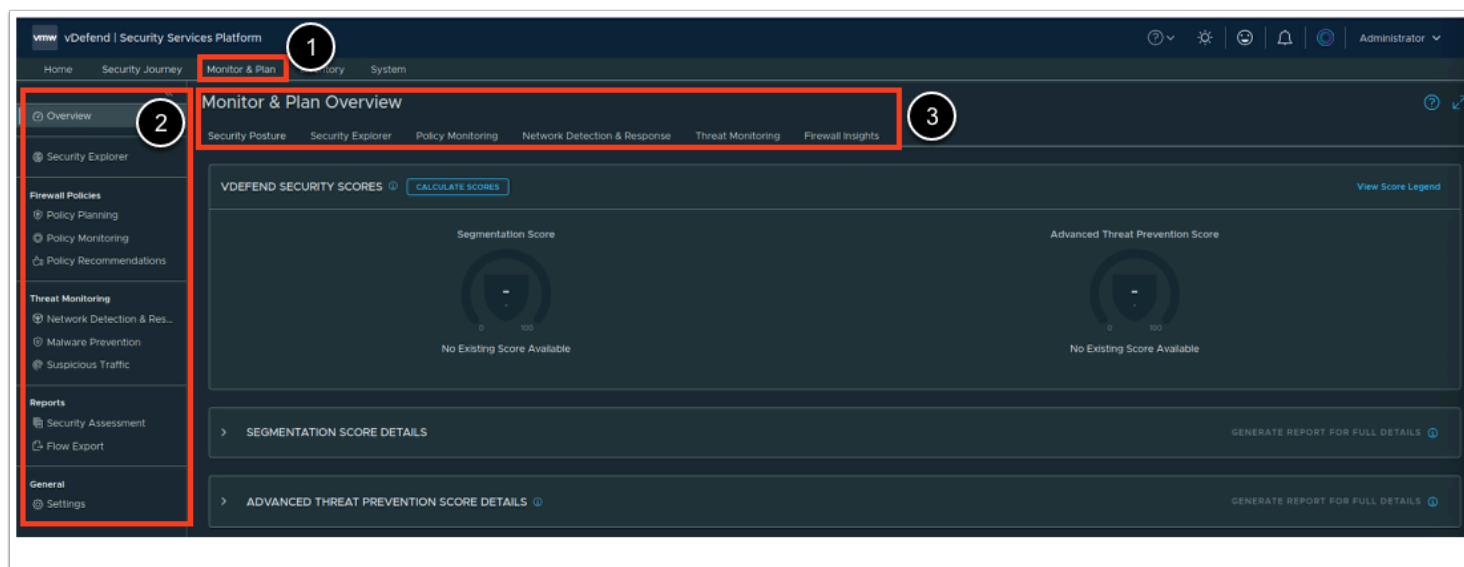
1. Click **NSX Managers**.
2. Here we see the NSX Manager that has been onboarded to this SSP instance.
3. Here we can validate the connectivity and status with the NSX Manager.

## Security Services Platform Instance - Server Configurations



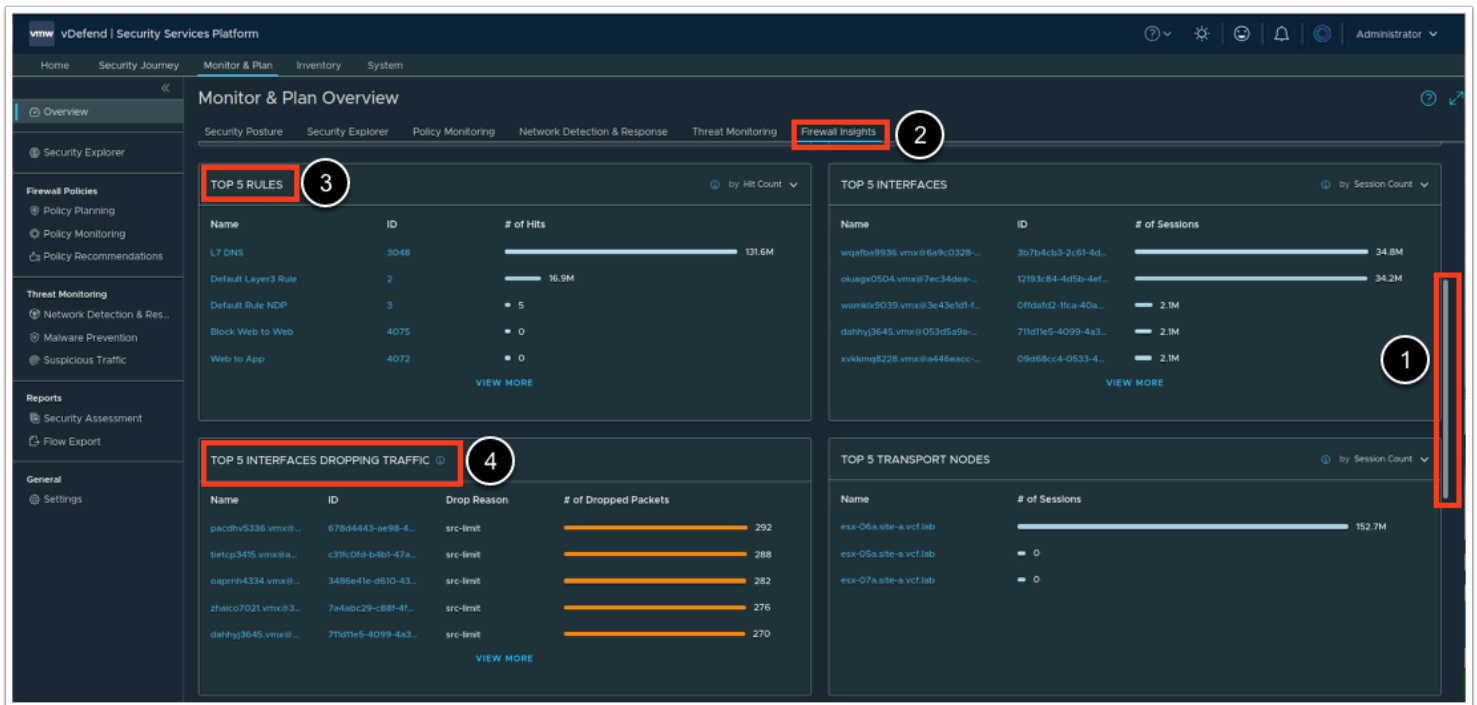
1. Click **Server Configurations**.
2. The SSP Platform supports configuring a Proxy Server, SIEM Server and Syslog Server Configurations.

# Security Services Platform Instance - Monitor & Plan



1. Click **Monitor & Plan**.
2. Here are options to review features and capabilities related to Security Intelligence.
3. Here are monitoring tabs for various features.

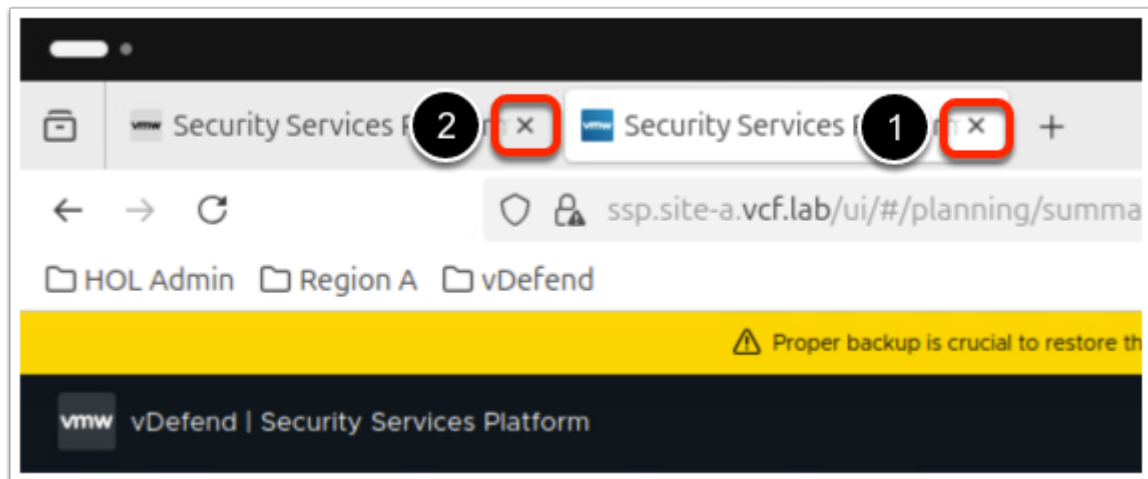
# Security Services Platform Instance - Monitor & Plan, Firewall Insights



1. Click **Firewall Insights**
2. **Click and hold** the scroll bar on the right and page down to see all the feature tiles.
3. **Top 5 Rules**; Here you see the Top 5 Firewall rules by hit count.
4. **Top 5 Interfaces**; Here you see the Top 5 VM interfaces by session count.

 This module provides a brief overview of these Security Services Platform instance dashboard, you may want click around and get familiar with other capabilities or features or continue with the next steps

## Closing the browser tabs



1. Click the **x** to close the SSP Instance page.
2. Click the **x** to close the SSP Installer page.

# vDefend Security Services Platform Summary

In this module, you explored the foundational concepts of the **vDefend Security Services Platform (SSP)** and gained hands-on experience navigating its deployment

Proceed to **Module 2** to learn about Security Segmentation Journey



# **Module 2 – Security Journey Overview**

# Module 2 Overview: Securing VCF with DFW 1-2-3-4 & ATP 1-2-3

In this module, you will embark on a guided Zero Trust security journey that combines progressive east-west micro-segmentation with multi-layered threat protection across VMware Cloud Foundation (VCF). You will learn how **vDefend DFW 1-2-3-4** systematically isolates workloads while **vDefend ATP 1-2-3** provides advanced threat detection, inspection, and response.

# Distributed Firewall (DFW) 1-2-3-4

Deploying Zero Trust across greenfield or brownfield private cloud environments requires a structured approach to securing internal east-west traffic. **vDefend Distributed Firewall (DFW) 1-2-3-4** provides an automated, prescriptive workflow for VMware Cloud Foundation (VCF). It systematically guides security teams through progressive segmentation phases from protecting core infrastructure services to achieving application-level micro-segmentation.

## Why Comprehensive East-West Segmentation is the Need of the Hour

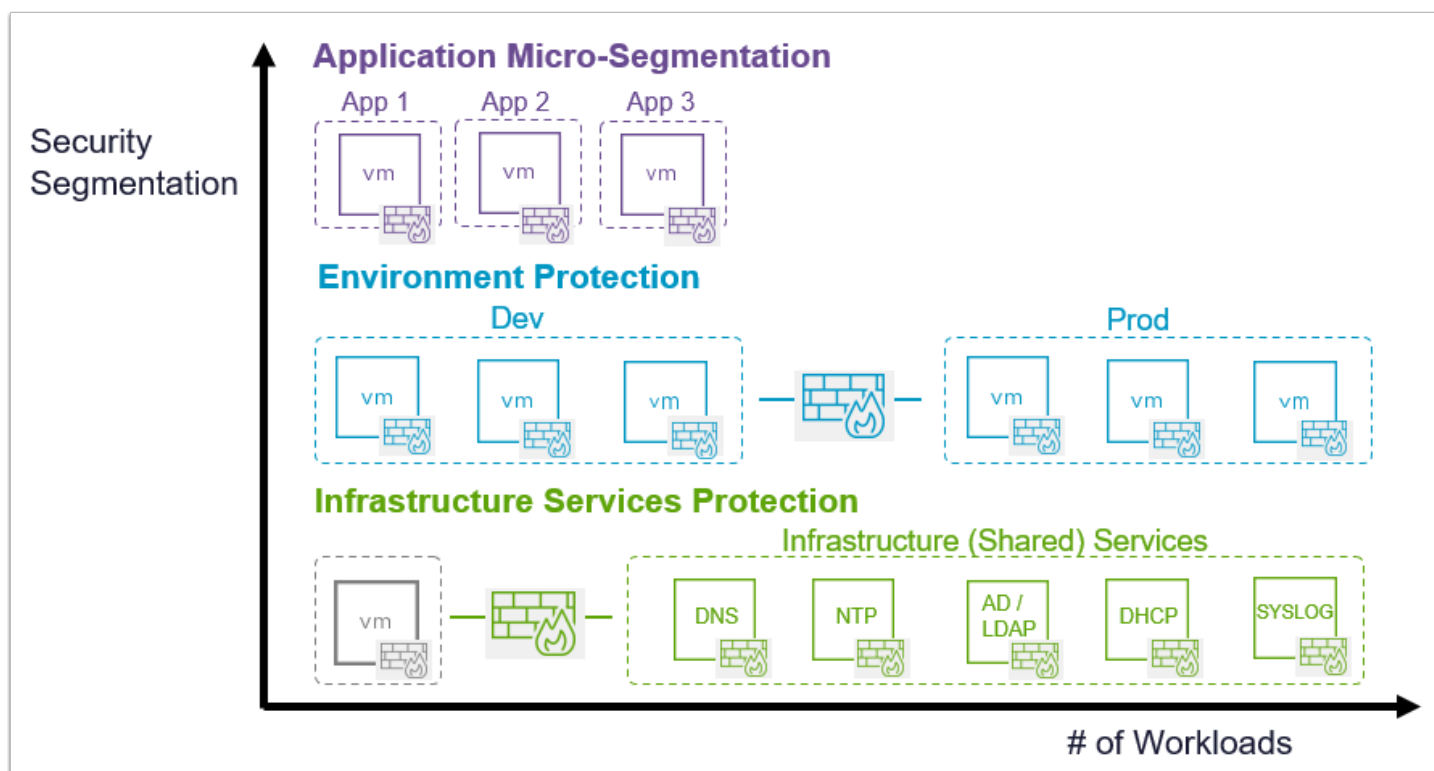
Perimeter firewalls protect only north-south entry points, leaving internal communication lines vulnerable:

- **High Traffic Volume:** East-west (lateral) traffic accounts for roughly four times the volume of perimeter traffic.
- **Lateral Movement:** Attackers exploit unprotected internal workloads to move horizontally toward high-value assets.
- **Implementation Silos:** Jumping directly to application-level micro-segmentation often fails due to a lack of visibility and team misalignment.

vDefend addresses these hurdles by auto-discovering workload communications, recommending security rules, and verifying policies without disrupting active traffic.

### Key Capabilities

- **Automated Discovery:** Analyzes traffic patterns, tags workloads, and flags unprotected communication paths.
- **Guided Rule Creation:** Recommends optimized DFW policies to eliminate manual guesswork.
- **Continuous Enforcement:** Monitors policy compliance pre- and post-deployment, alerting administrators to unauthorized changes.



| Macro-segmentation                                                                                                                                                                                              |                                                                                                                                                                                                                                                              | Micro-segmentation                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Infrastructure (Shared) Services                                                                                                                                                                                | Environment (Zone) Segmentation                                                                                                                                                                                                                              |                                                                                                         |
| <ul style="list-style-type: none"> <li>Support essential enterprise functions including Active Directory, DNS, NTP, logging, and monitoring utilities.</li> <li>Required by all enterprise workloads</li> </ul> | <ul style="list-style-type: none"> <li>Includes Prod, Staging, DMZ, PCI, etc</li> <li>Inter-environment traffic is limited with strict controls and exceptions</li> <li>Locking down inter-zone traffic is crucial to preventing lateral movement</li> </ul> | <p>App-Level segmentation:<br/>Ring-fence applications to control how they interact with each other</p> |

## vDefend DFW 1-2-3-4

The **(DFW) 1-2-3-4** streamlines Zero Trust micro-segmentation into a guided, prescriptive workflow that reduces deployment complexity and strengthens east-west security.

- **Accelerated Time-to-Value:** Replaces unstructured segmentation projects with a clear roadmap, enabling teams to secure core infrastructure in a few days rather than months.
- **Automated Policy Generation:** Leverages an internal analytics engine to discover traffic patterns and automatically tag workloads, generating firewall and IDPS rule recommendations without manual mapping.

- **Lateral Threat Containment:** Prevents attackers from moving east-west across virtualized workloads by isolating shared services, enforcing inter-zone boundaries, and ring-fencing application tiers.
- **Continuous Operational Visibility:** Provides pre- and post-deployment traffic monitoring with immediate alerts for rule changes, traffic violations, or policy drift.
- **Measurable Progress & Compliance:** Establishes security baseline scores during assessment, allowing security teams to quantify risk reduction and prove compliance at each stage.

## 4 Stage Deployment Journey

The **vDefend Distributed Firewall (DFW) 1-2-3-4** methodology provides a progressive, four-stage deployment path to transition an enterprise network to a Zero Trust architecture.

### Stage 1: Security Segmentation Assessment

- **Baseline Posture:** Discovers network communication patterns and calculates an initial security segmentation score to highlight existing exposure across east-west traffic.

### Stage 2: Infrastructure (Shared) Services Protection

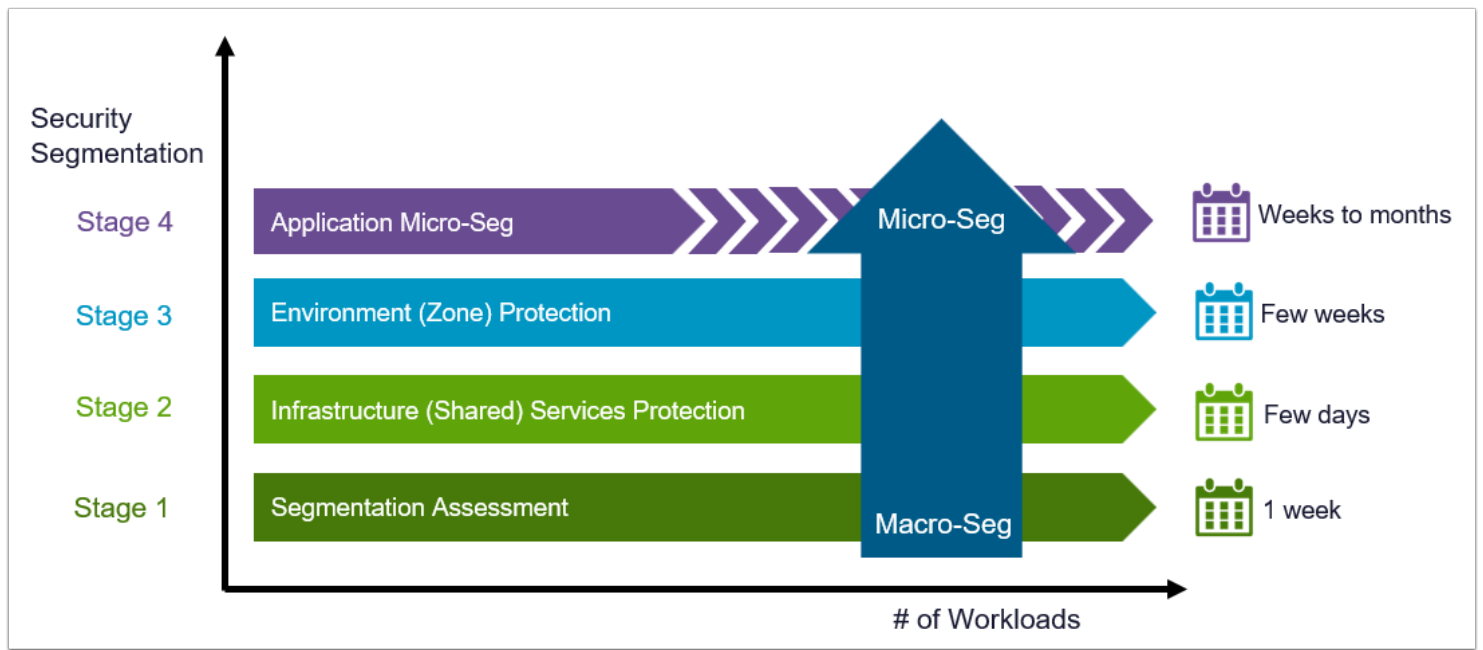
- **Core Service Isolation:** Locks down core infrastructure workloads such as Active Directory, DNS, NTP, and DHCP to prevent lateral attacks against management and identity systems.

### Stage 3: Environment (Zone) Boundaries

- **Zone Isolation:** Establishes boundaries between major environment tiers (e.g., Development vs. Production) and alerts administrators to unauthorized cross-zone traffic leakages.

### Stage 4: Application Microsegmentation

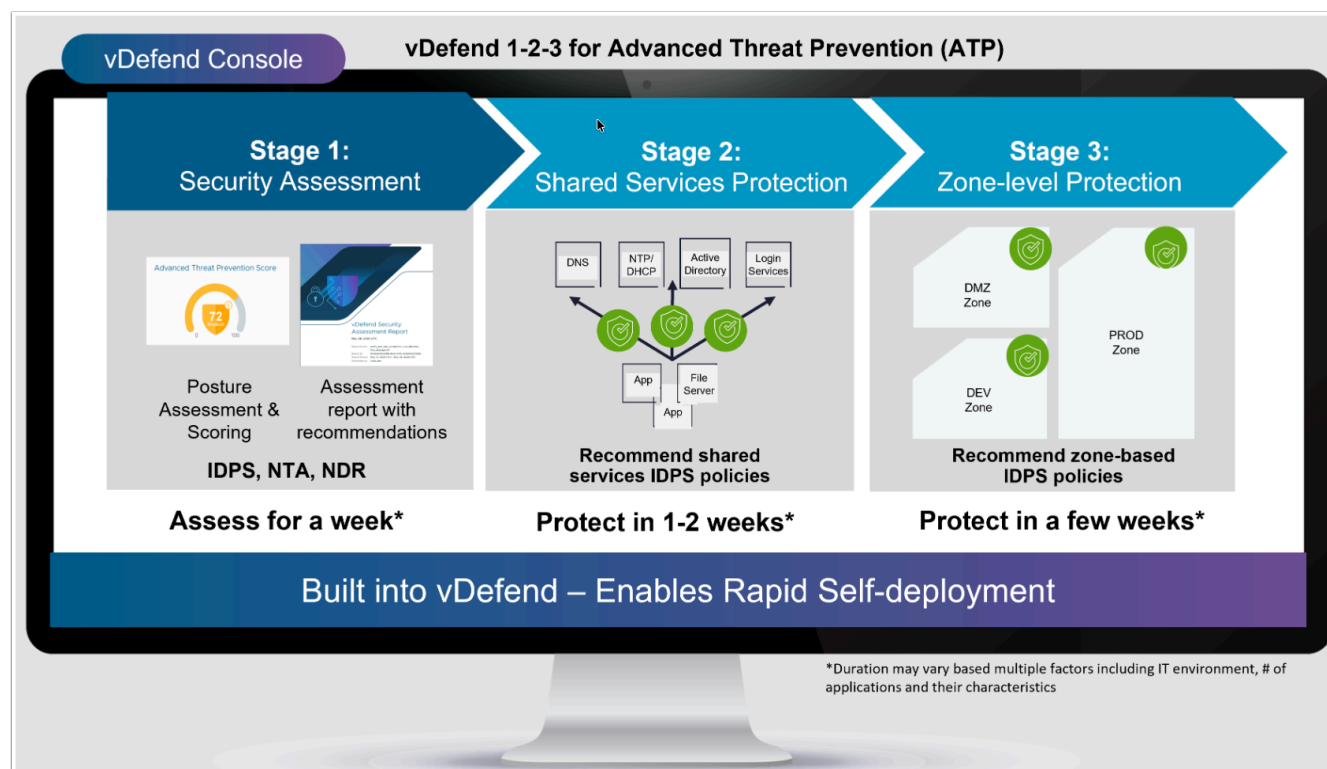
- **Ring-Fencing Workloads:** Maps workloads to specific applications, applies granular firewall policies between application tiers, and continuously fine-tunes rules based on real-time traffic flow metrics.



# Advanced Threat Protection (ATP) 1-2-3

**vDefend 1-2-3 for ATP (Advanced Threat Prevention)** is a guided, three-step deployment workflow that streamlines the implementation of Intrusion Detection and Prevention System (IDPS), Network Traffic Analysis (NTA), and Network Detection and Response (NDR) in private cloud environments, cutting rollout times down from months to weeks.

## The 3-Stage Workflow of ATP 1-2-3



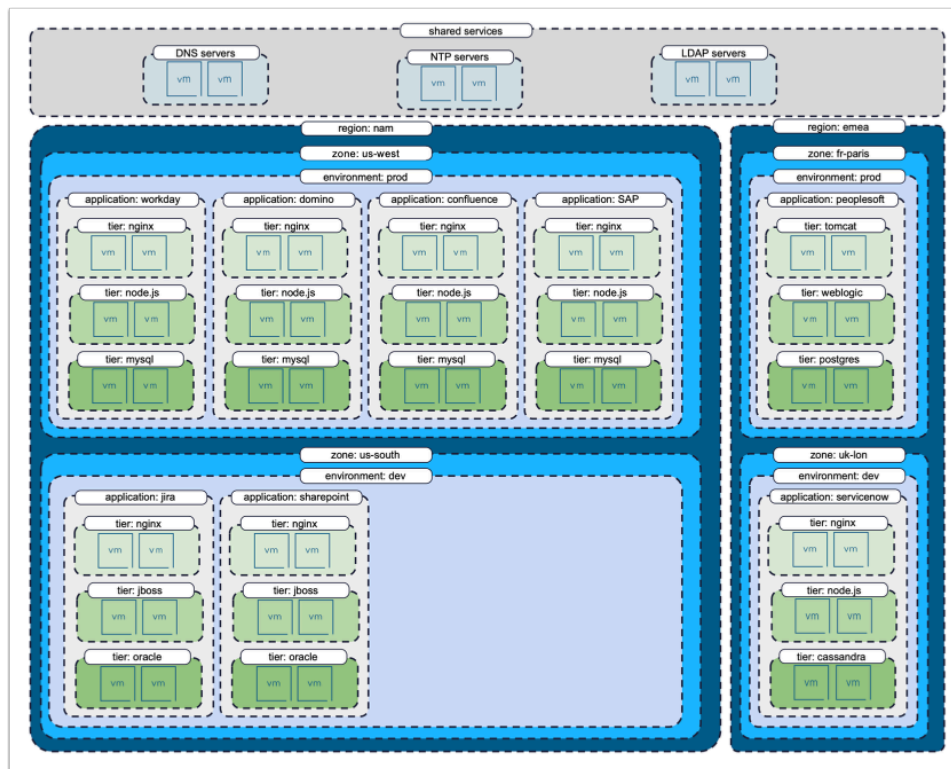
**Stage 1 (Assessment):** Conducts a threat posture assessment across IDPS, NTA, and NDR to evaluate current risks and generate tailored security recommendations.

**Stage 2 (Infrastructure Protection):** Applies recommended IDPS policies to secure core shared services infrastructure, including Active Directory, DNS, and NTP.

**Stage 3 (Zone Expansion):** Expands IDPS policies across surrounding environment zones, such as DMZ, Development, and Production.

# **Module 3 - Importing Datacenter Assets**

# Datacenter Topology: Acme



Acme has 8 applications, distributed across multiple environments, zones and regions. Each application is multi-tier and is made of a front-end, an application tier and a database tier.

The nam region is made of 2 zones: us-south and us-west. The us-south zone runs a dev environment and the us-west zone runs the prod environment. The emea region has 2 zones: fr-paris and uk-lon. The fr-paris has a single prod environment while the uk-lon zone runs a dev environment.

Some services are shared within the datacenter. Acme has some DNS servers, NTP servers and LDAP servers that are accessed from all regions/zones/applications.

# CSV Import Introduction

One critical aspect of the Security Journey workflow is to prepare a CSV file that describes the datacenter hierarchy. Once the CSV is ready, it can be imported into the vDefend Security Services Platform to proceed to the Security Journey stages.

The CSV file is made of multiple optional columns that serve different objectives:

- Identify your datacenter assets
- Push tags to identified VMs and/or networks and create hierarchical groups
- Create groups for infrastructure services

There are multiple ways to create the CSV. Here are some examples:

- Export from a CMDB: most CMDBs have an option to export the content as a CSV file. Security Journey supports extra columns that the CMDB would generate, so you don't have to delete useless columns
- Export from vCenter: using scripts, you can generate a CSV from the vCenter inventory. Information such as vSphere tags, folders, clusters, can be used as hierarchy levels
- If the VMs are deployed by an automation solution (i.e VCF Automation), scripts can be used to generate the CSV from this solution
- If the VMs are using a specific naming convention, a script can be used to generate a CSV with the right hierarchy level information
- Manually, in case of testing or small environment

## Identify Assets (1/2)

Multiple CSV columns can be used to help identify an asset from the NSX inventory.

A column can be used to set the asset type. Valid options are: "VIRTUAL\_MACHINE", "SEGMENT", "DVPG", "TIER\_0", "TIER\_1", "GROUP" and "IP". If not used, Security Journey will assume that the asset type to identify from the NSX inventory is "VIRTUAL\_MACHINE".

1 to 3 columns can be used as an asset identifier. Security Journey can use the name of the asset, its IP and/or its ID. Each row must have **at least 1 value set** for the columns used for identification. When importing the CSV, the order used to choose between these values can be configured. Typically, 1 = the asset ID, 2 = the asset IP, 3 = the asset name.

Here are a few examples:

| asset type      | asset id                             | asset ip | asset name     | comments                                                                                                                                                                                                                        |
|-----------------|--------------------------------------|----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VIRTUAL_MACHINE |                                      |          | xvkkmq8227     | Identify a VM named xvkkmq8227 from the NSX inventory                                                                                                                                                                           |
| VIRTUAL_MACHINE |                                      | 10.0.0.3 |                | Identify a VM with the IP 10.0.0.3 from the NSX inventory                                                                                                                                                                       |
| VIRTUAL_MACHINE |                                      | 10.0.0.3 | zckzgt8967     | Identify a VM with the IP 10.0.0.3 from the NSX inventory. If multiple VMs are found with IP = 10.0.0.3, then use zckzgt8967                                                                                                    |
| VIRTUAL_MACHINE | 5039de40-1280-cd02-56d5-15dd85ea43b5 | 10.0.0.4 | tyzhta1563     | Identify a VM with the IP 10.0.0.4 from the NSX inventory. If multiple VMs are found with IP = 10.0.0.4, then use tyzhta1563<br>If multiple VMs are found with name = tyzhta1563, then use 5039de40-1280-cd02-56d5-15dd85ea43b5 |
| GROUP           |                                      |          | existing-group | Identify an existing group named "existing-group" from the NSX inventory                                                                                                                                                        |
| SEGMENT         |                                      |          | prod-seg1      | Identify a segment named "prod-seg1" from the NSX inventory                                                                                                                                                                     |

## Identify Assets (2/2)

In the case you want Security Journey to create a new group for some assets external to the workload domain (see Tips and Tricks), use the "IP" asset type. For example:

| asset type | asset id | asset ip    | asset name | comments                                                |
|------------|----------|-------------|------------|---------------------------------------------------------|
| IP         |          | 10.2.0.0/22 |            | Create a group for with 10.2.0.0/22 as static IP member |

 Note: make sure the NSX inventory doesn't have VMs using IPs in that subnet if you want Security Journey to create a group with 10.2.0.0/22 as an IP member.

## Push tags and create hierarchy groups (1/5)

Once Security Journey has identified the assets, it can tag the asset to the right hierarchy level thanks to 1 or multiple columns (up to 5 hierarchy levels are currently supported). Typical hierarchy level column names are:

- region
- zone
- environment
- application
- application tier

| asset name | region | zone     | environment | application | application tier |
|------------|--------|----------|-------------|-------------|------------------|
| xvkkmq8227 | nam    | us-south | dev         | sap         | nginx            |

## Push tags and create hierarchy groups (2/4)

The CSV can use other column names as well. For example:

- BU (can be mapped to region when imported)
- Team (can be mapped to zone when imported)
- Env (can be mapped to environment when imported)
- App (can be mapped to application when imported)

| asset name | BU  | Team | env | app |
|------------|-----|------|-----|-----|
| xvkkmq8227 | ans | pm   | dev | sap |

## Push tags and create hierarchy groups (3/4)

Another example of different column names is:

- Workload Domain (can be mapped to zone when imported)
- Environment
- Application
- Application tier

| asset name | WLD  | environment | application | application tier |
|------------|------|-------------|-------------|------------------|
| xvkkmq8227 | wld1 | dev         | sap         | nginx            |

## Push tags and create hierarchy groups (4/4)

Or the column names could be as simple as

- Environment
- Application

| asset name | environment | application |
|------------|-------------|-------------|
| xvkkmq8227 | dev         | sap         |

**i** Note: if a tier column is used, then an application column is mandatory

## Infrastructure Services

Security Journey can detect the infrastructure services leveraging the flows sent to the vDefend Security Services Platform. However, these services can also be specified in the CSV so infrastructure groups and rules can be created even if flows are not seen yet. Optionally, a CSV column can be used to specify 1 or multiple infrastructure services.

The expected format is <name of service>:<protocol>/<port>

Multiple services can be specified, separated with ;

As an example:

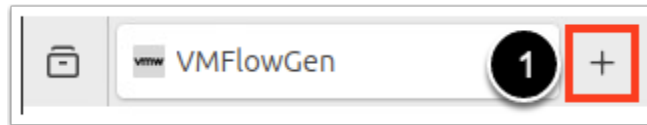
| asset type      | asset ip  | asset name | infrastructure service |
|-----------------|-----------|------------|------------------------|
| VIRTUAL_MACHINE | 10.0.0.20 | lnavti4788 | ntp:udp/123            |
| VIRTUAL_MACHINE | 10.0.0.21 | rqdnnu5719 | ntp:udp/123            |
| VIRTUAL_MACHINE | 10.0.0.22 | wqafba9935 | dns:udp/53             |
| VIRTUAL_MACHINE | 10.0.0.23 | oiuagx0503 | dns:udp/53;ntp:udp/123 |
| VIRTUAL_MACHINE | 10.0.0.24 | yyhgza8638 | ntp:udp/123            |

**i** Note: Security Journey doesn't push specific tags to infrastructure servers, but creates groups with static membership.

## Tips and Tricks for CSV Import

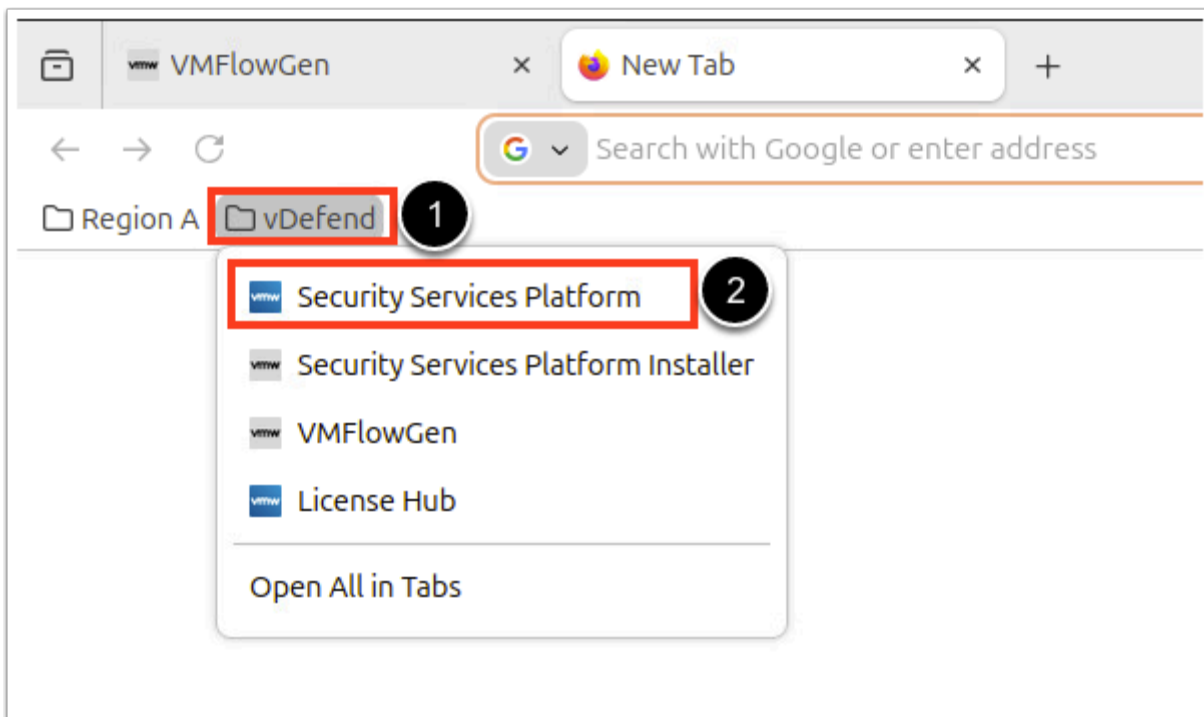
- As explained previously, most of the CSV columns are optional.
- Security Journey supports 1 to 5 hierarchy levels. We recommend to use at least environment, application and tier levels to leverage the Security Journey monitoring and segmentation features.
- When defining the hierarchy in the “Segmentation Planning - Prepare for Analysis” page, remove any pre-selected CSV column that you don’t use. For example, if you plan to use only environment, application and tier, you can remove the placeholder values for region and zone.
- The region, zone and environment levels are used to implement segmentation strategies between security zones. The goal is to isolate the traffic between these security zones, and to implement some exceptions if required. For example, emea\_dev to emea\_prod = DROP
- The application level is used for 2 purposes:
  - implement application security policies, to achieve application ring fencing or full micro-segmentation
  - implement exception rules between security zones. For example, emea\_dev\_workday to emea\_prod\_peoplesoft = ALLOW. So, even if application segmentation is not required, it is useful to use this hierarchy level to have proper environment segmentation.
- The application tier level is used to implement proper application security policies. For example, for the emea\_dev\_workday application:
  - emea\_dev\_workday\_nginx to emea\_dev\_workday\_nodejs = ALLOW
  - emea\_dev\_workday\_nodejs to emea\_dev\_workday\_mysql = ALLOW
  - default rule = DROP
- We recommend importing a CSV that describes only the assets to be used by the NSX onboarded with the vDefend Security Services Platform instance. For example, assuming the Security Services Platform is onboard with the NSX of a VCF Workload domain A:
  - Import a CSV that includes all VMs running in VCF WLD A
  - Use the zone to identify the workload domain (for example: zone = WLD A)
  - In the CSV, use an “IP” asset row to define the subnet used by other workload domains (for example: 172.20.0.0/22 for WLD B VMs). Assign this IP asset to another zone (for example: zone = WLD B)
  - Thanks to this configuration, a single group will be created for WLD B, saving NSX A resources
- Multiple CSVs can be imported, so you can start with a subset of your asset inventory
- In case of errors, generated groups can be manually deleted from NSX
- In case of duplicated tags (same scope but different value. i.e: environment=prod and environment=dev), Security Journey highlights these tags in the “Remove Tags” tab. You can review the tags and remove the duplicates from that page

# Login to Security Services Platform (SSP)



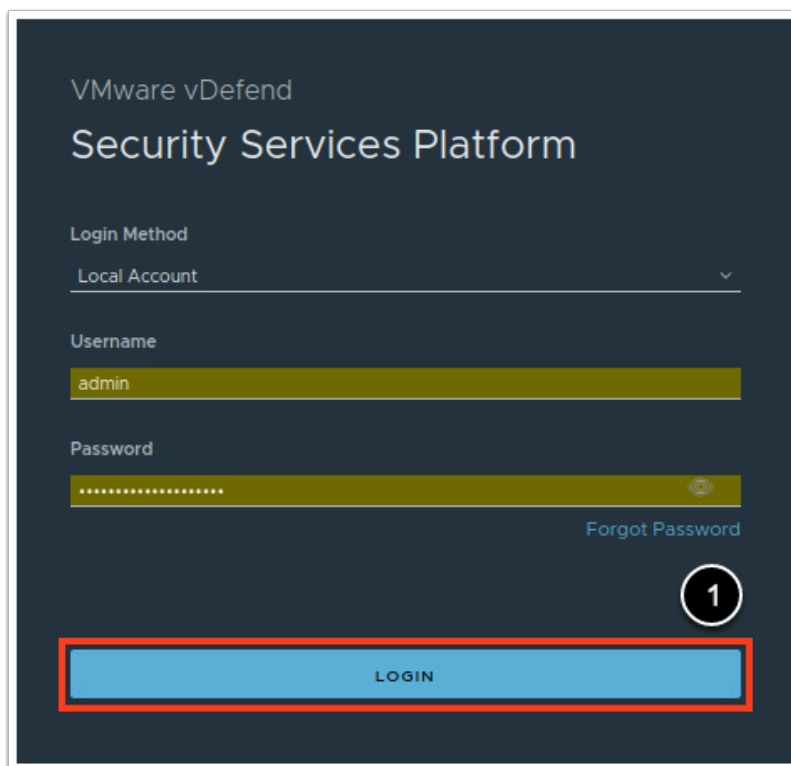
1. Click the + at the top of the browser window to open a new tab

## Open Security Services Platform Console



1. Click on the **vDefend** bookmark bar folder
2. Click **Security Services Platform** to launch **SSP**

## Login to Security Services Platform Console



VMware vDefend  
Security Services Platform

Login Method  
Local Account

Username  
admin

Password  
.....

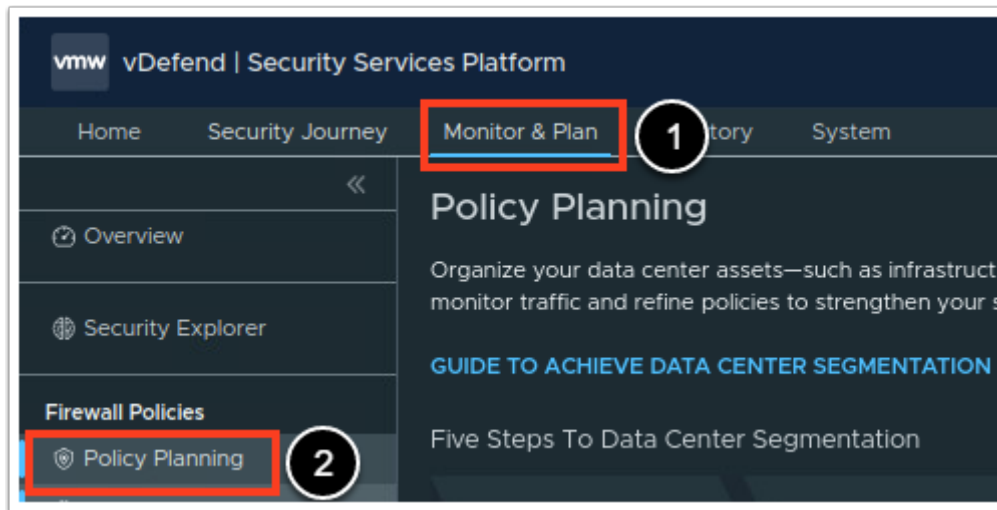
[Forgot Password](#)

1

LOGIN

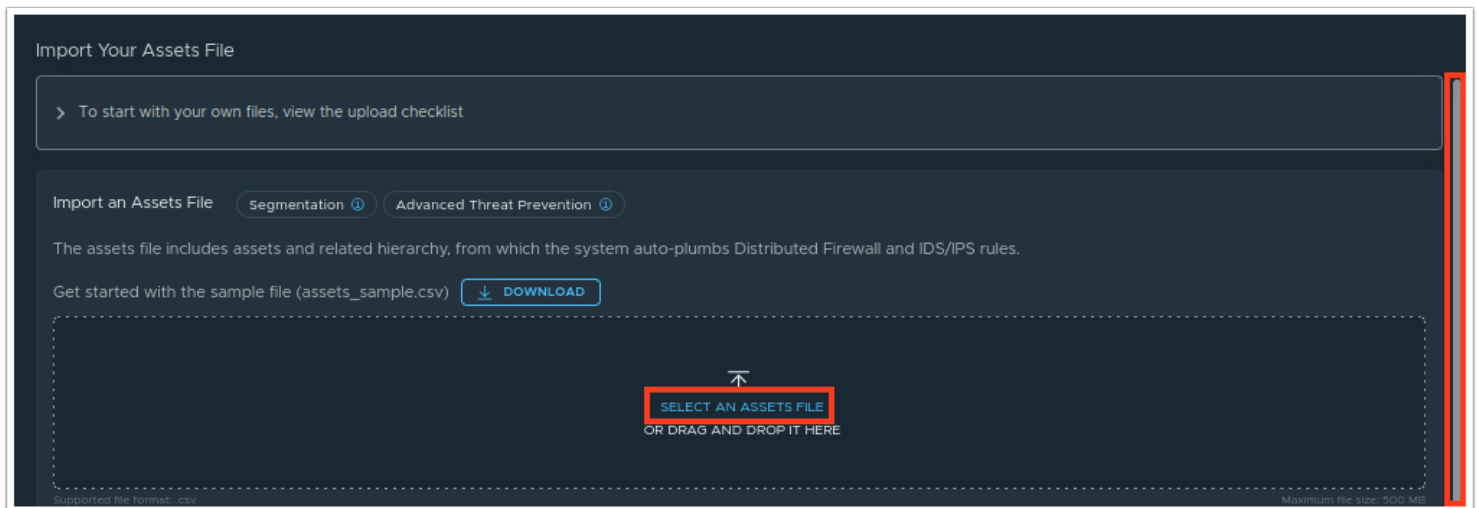
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Import the CSV



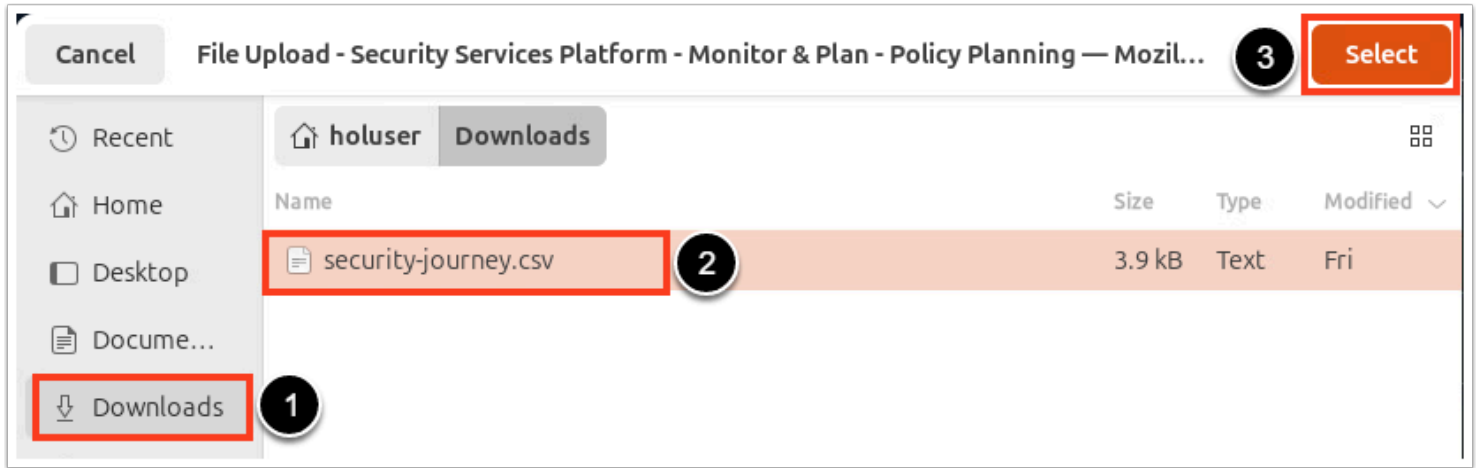
1. Click **Monitor & Plan**
2. Click **Policy Planning**

## Open File Selection Window



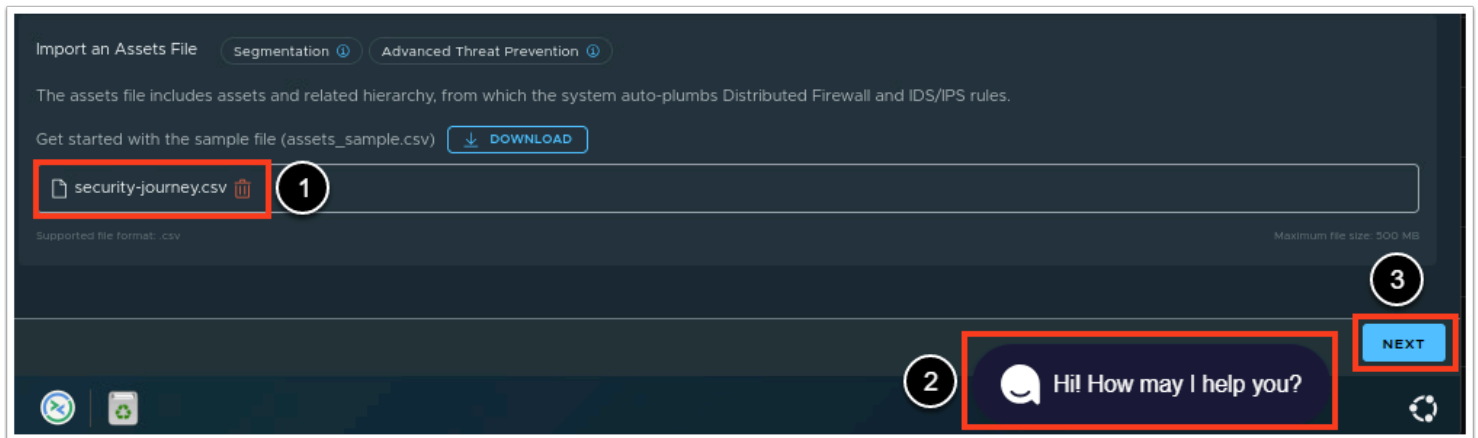
1. Use the **scroll bar** if needed to page down to the Import an Assets File section
2. Click **SELECT AN ASSETS FILE** to open the file selection window

## Select the CSV



1. Click **Downloads**
2. Click the **security-journey.csv**
3. Click **Select**

## Import the File



1. Ensure the security-journey.csv is now in the asset file section.
2. Click and drag the **Virtual Assistant** to the top left to uncover the NEXT button
3. Click **NEXT**

# Map Data to Assets

Policy Planning - Prepare For Analysis

Map Data To Assets

Map data in the imported file to the required fields. If auto-populated mappings are incorrect, you can change them.

| Required Input                                                                                                                      | Typical Example Values ⓘ                                              | 1 | Select CSV Column                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Asset Type</b> (Optional)<br>If Asset Type is not specified, the system automatically classifies the asset as a Virtual Machine. | VIRTUAL_MACHINE<br>GROUP<br>TIER_0<br>TIER_1<br>IP<br>SEGMENT<br>DVPG |   | <div>asset type ✓</div> <div>This column defines your inventory asset types</div>                                                                                                                                                                                                      |
| <b>Asset Identifier</b>                                                                                                             | UUID<br>IP<br>NAME                                                    |   | <div>asset id ✓<br/>Used as the first choice to identify assets. (Required Field)</div> <div>asset name ✓<br/>Used if the primary column has empty values. (Optional Field)</div> <div>asset ip ✓<br/>Used only if both of the above columns have empty values. (Optional Field)</div> |
| <b>Infrastructure Services</b> (Optional)                                                                                           | DNS:udp/53<br>FTP                                                     |   | <div>infrastructure service ✓</div>                                                                                                                                                                                                                                                    |

1. The CSV column names that are used in this export are asset type, asset id, asset name, asset ip, infrastructure service, region, zone, environment, application and application tier, so the “map data to assets” is automatic.
2. Use the **scroll bar** to scroll down and continue viewing the asset mapping

# View Asset Hierarchy

**Define Hierarchy (Optional)**

Define up to five nested hierarchy scopes to organize your data center; at least one scope is required. You can skip levels, and any values in the CSV will be ignored. This section is mandatory if you are defining assets beyond infrastructure servers. [Learn more about defining hierarchy](#)

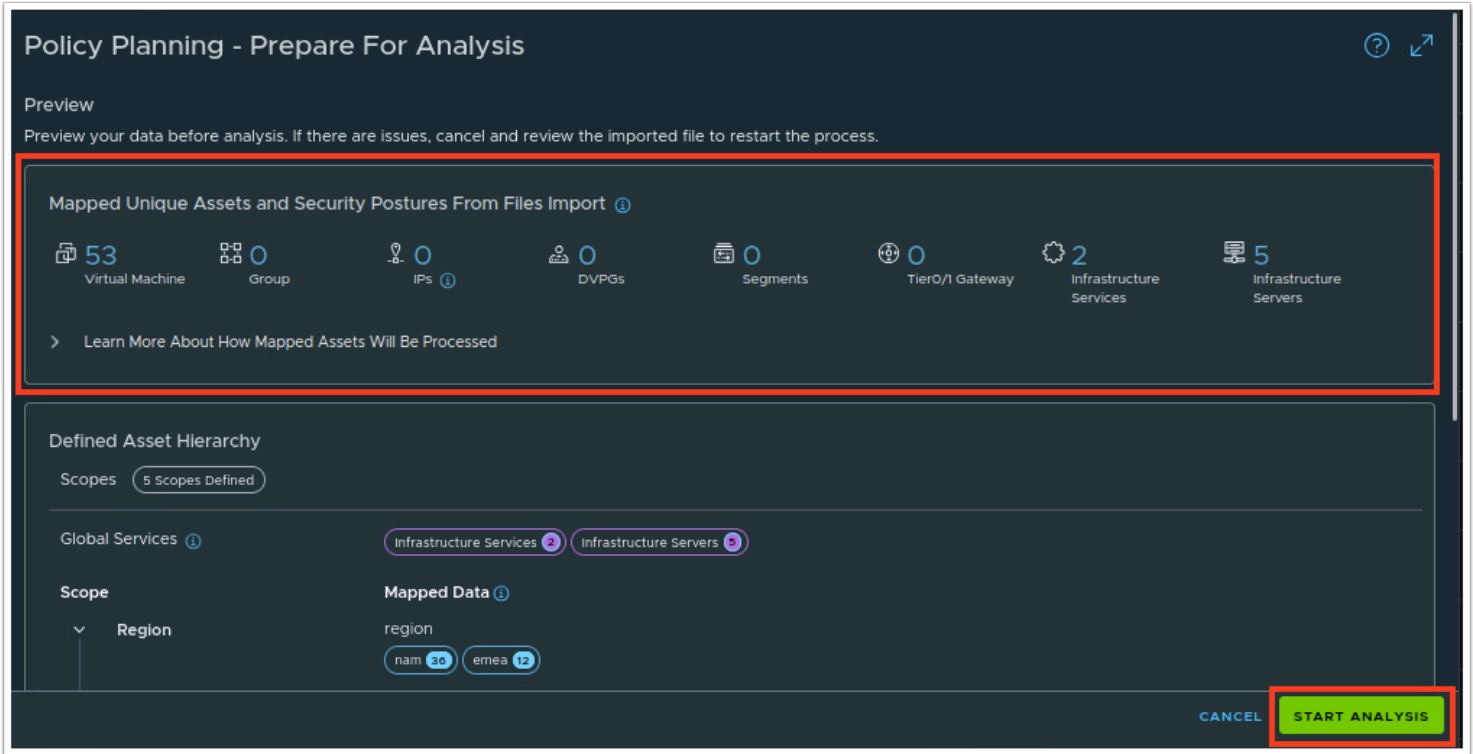
| Scope                                | Select CSV Column  |
|--------------------------------------|--------------------|
| Region                               | region ✓           |
| Zone                                 | zone ✓             |
| Environment <span>Recommended</span> | environment ✓      |
| Application <span>Recommended</span> | application ✓      |
| Tier <span>Recommended</span>        | application tier ✓ |

"Tier" requires its parent scope: "Application" to also be defined.

**PREVIEW**

1. View the hierarchy scopes that will be created from the CSV
2. Click **PREVIEW**

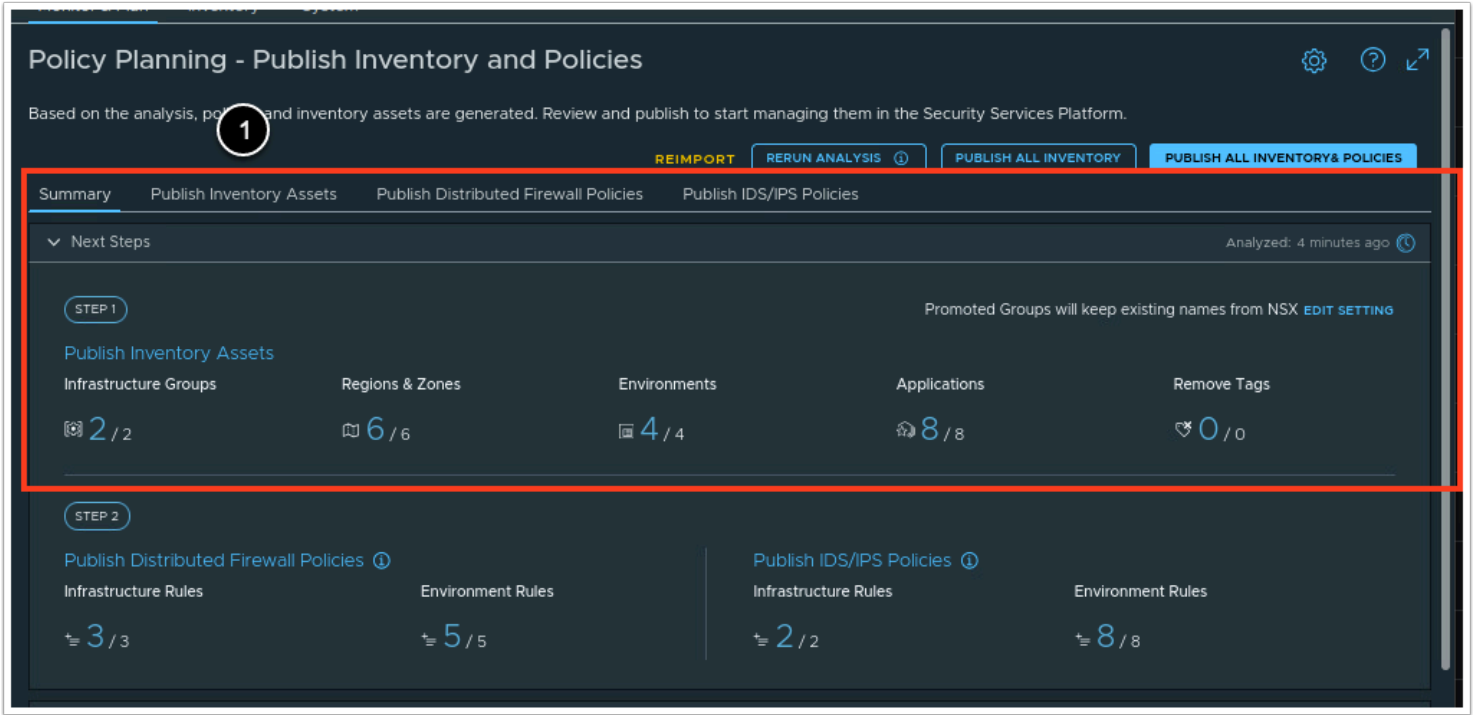
# Start Analysis



- 1. View the Mapped Assets From CSV
- 2. Click **Start Analysis**

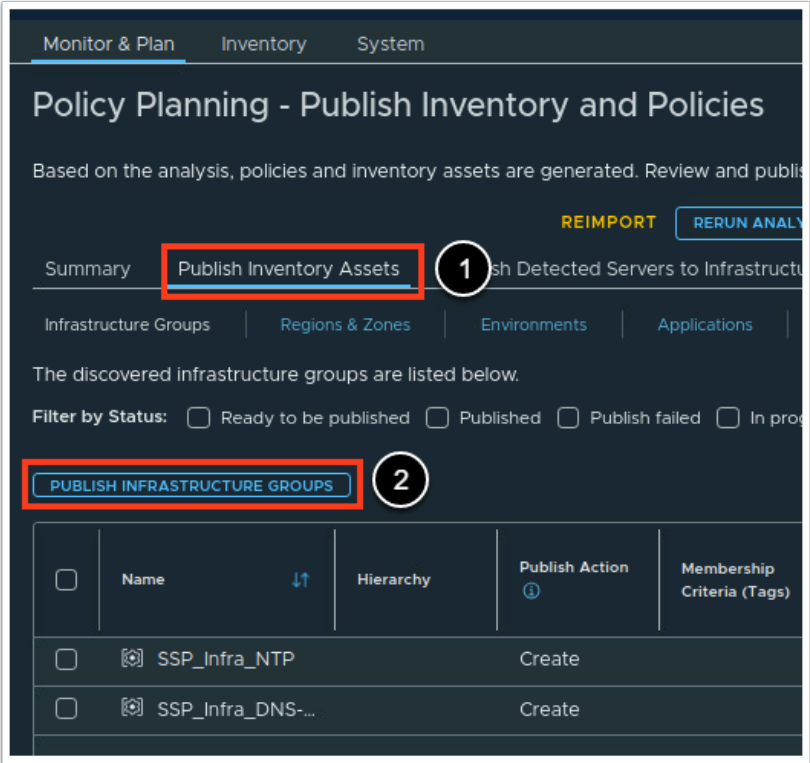
**i** Analysis will take a few minutes to complete.

# Publish Assets



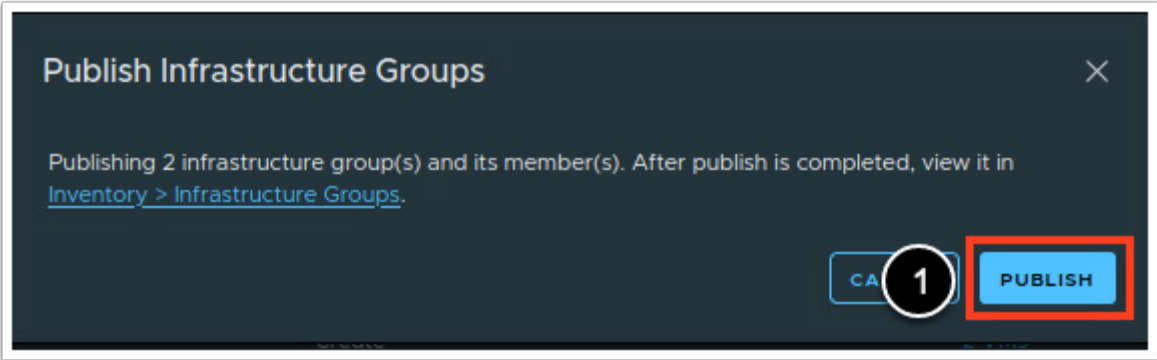
1. Review the inventory assets

# Publish Infrastructure Assets



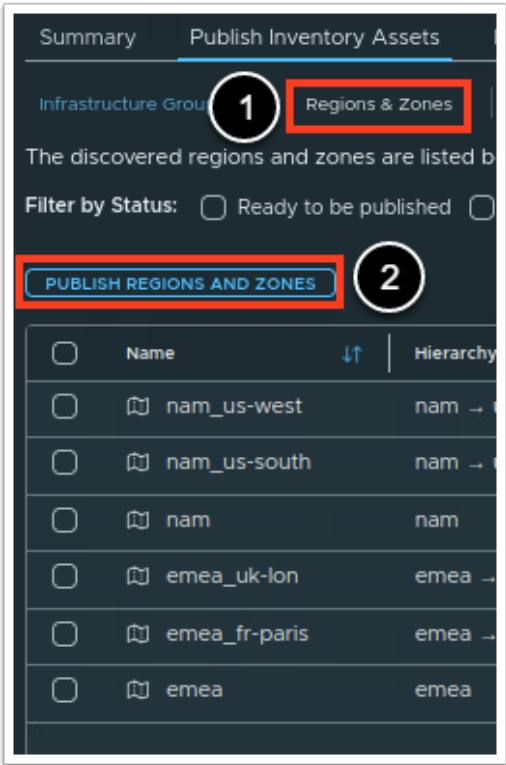
- 1. Click **Publish Inventory Assets**
- 2. Click **PUBLISH INFRASTRUCTURE GROUPS**

# Confirm Infrastructure Asset Publish



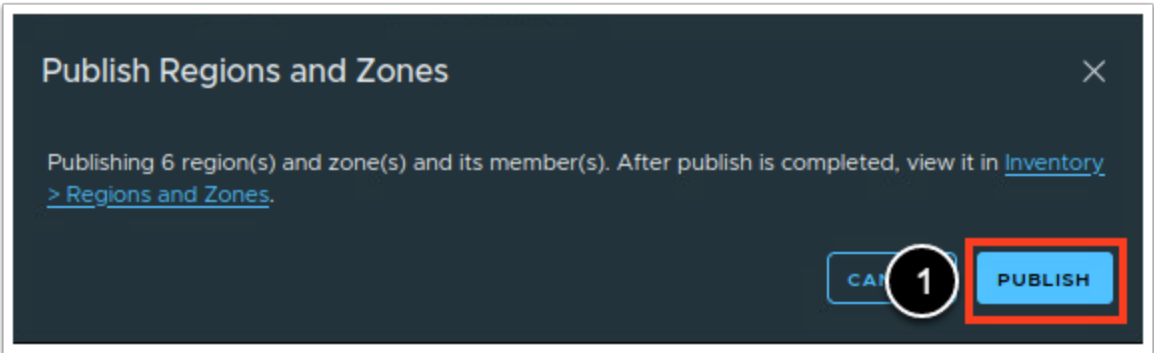
- 1. Click **PUBLISH**

# Publish Regions & Zones



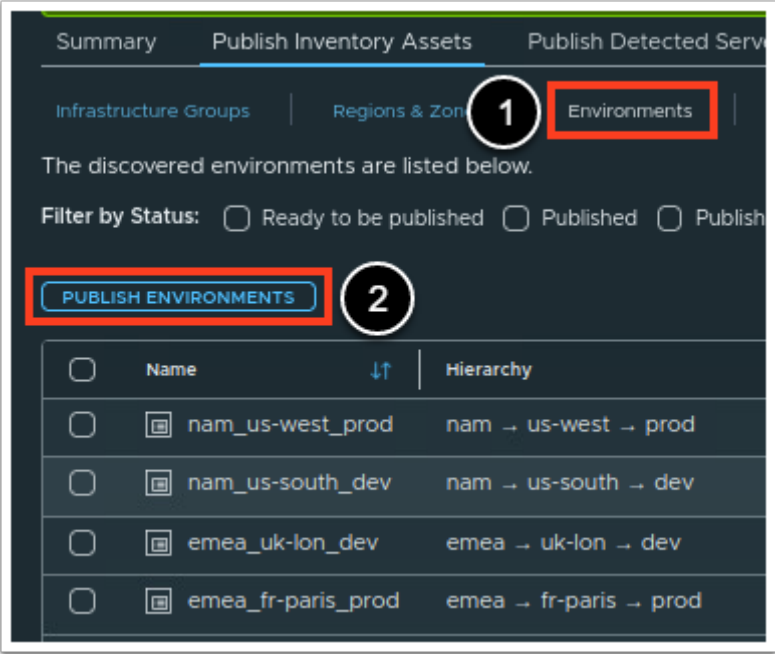
- 1. Click **Regions & Zones**
- 2. Click **PUBLISH REGIONS AND ZONES**

## Confirm Region & Zone Publish



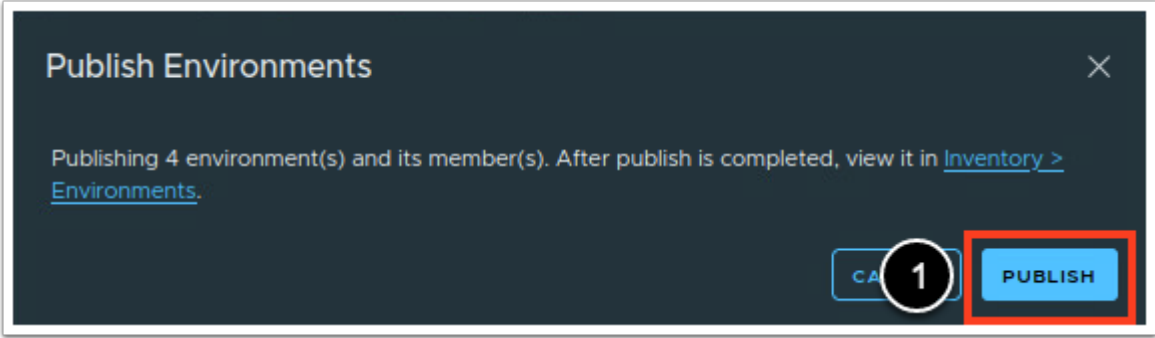
- 1. Click **PUBLISH**

# Publish Environments



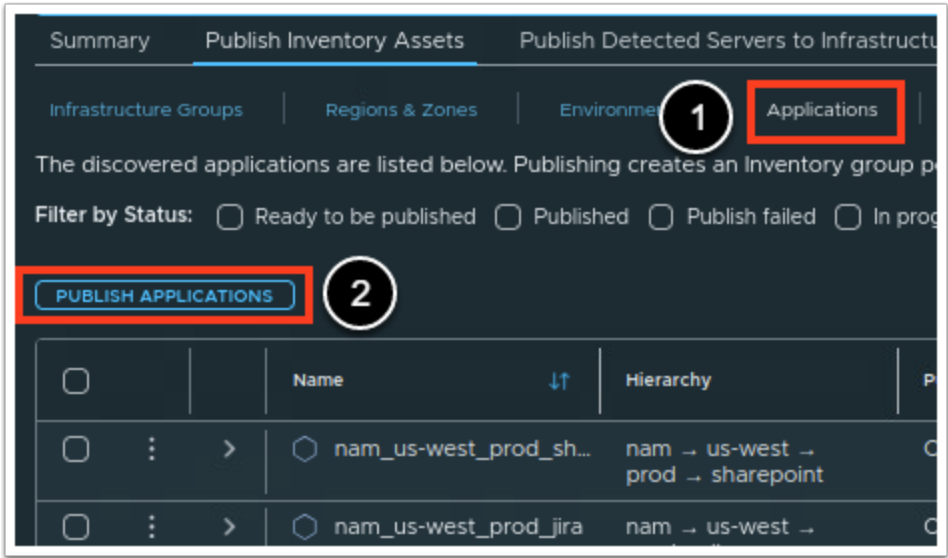
- 1. Click **Environments**
- 2. Click **PUBLISH ENVIRONMENTS**

# Confirm Environment Publish



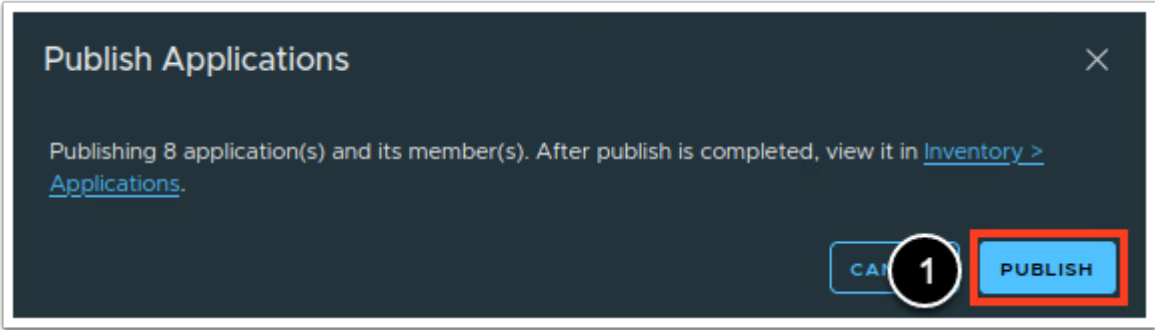
- 1. Click **PUBLISH**

# Publish Applications



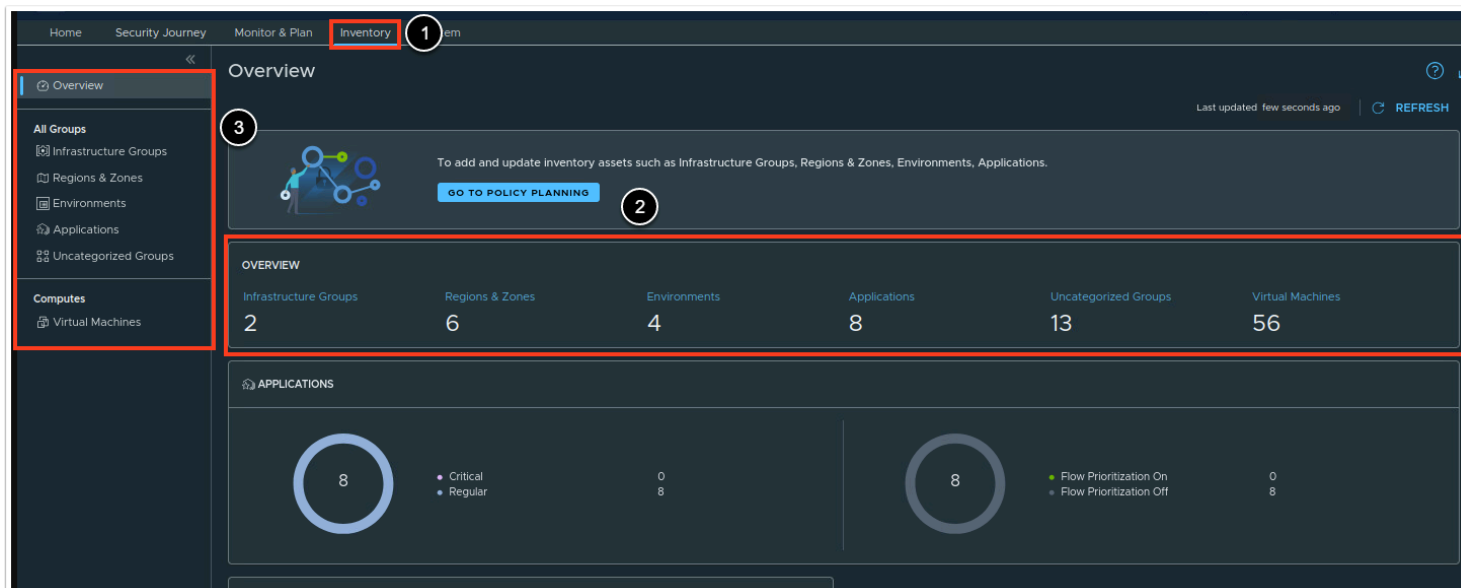
- 1. Click **Applications**
- 2. Click **PUBLISH APPLICATIONS**

## Confirm Application Publish



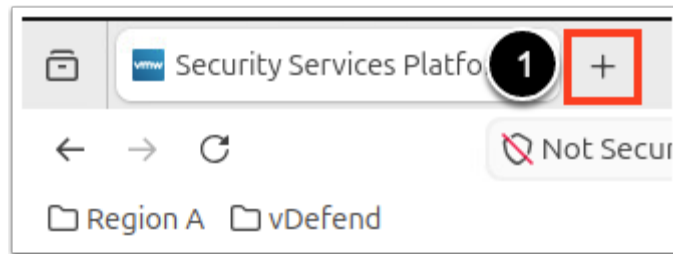
- 1. Click **PUBLISH**

# Validate Published Assets in SSP



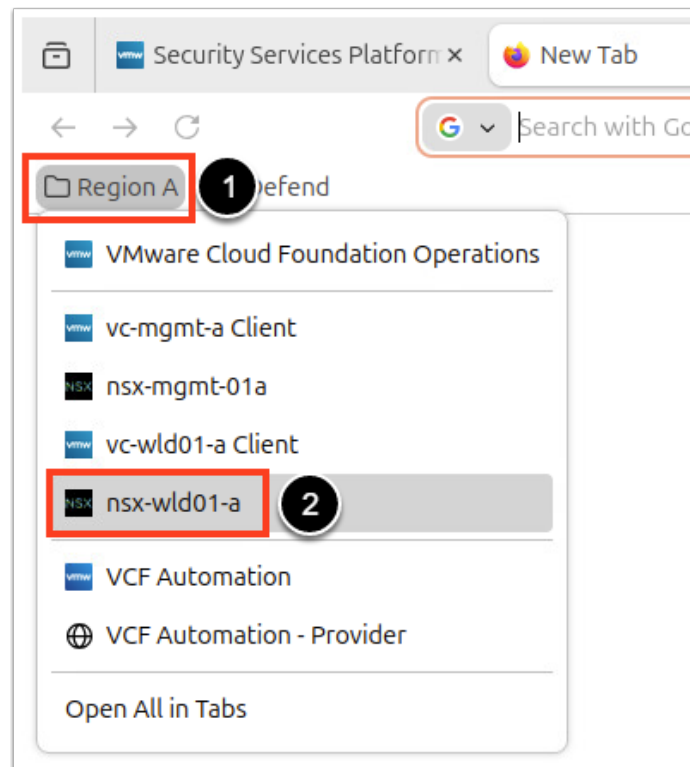
1. Click **Inventory**
2. The Overview page will show a quick count of the asset inventory
3. Navigate to each section to view more details on the published assets

# Login to NSX



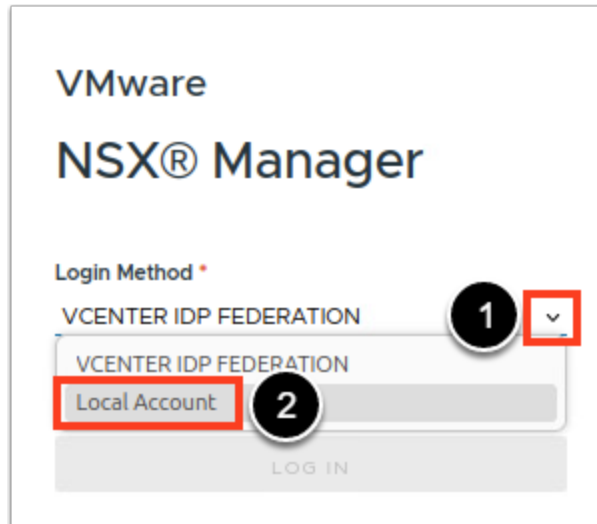
1. Click the + at the top of the browser window to open a new tab

## Open NSX Manager



1. Click on the **Region A** bookmark bar folder
2. Click **nsx-wld01-a** to launch NSX Manager

## Select Login Method



VMware  
NSX® Manager

Login Method \*

VCENTER IDP FEDERATION

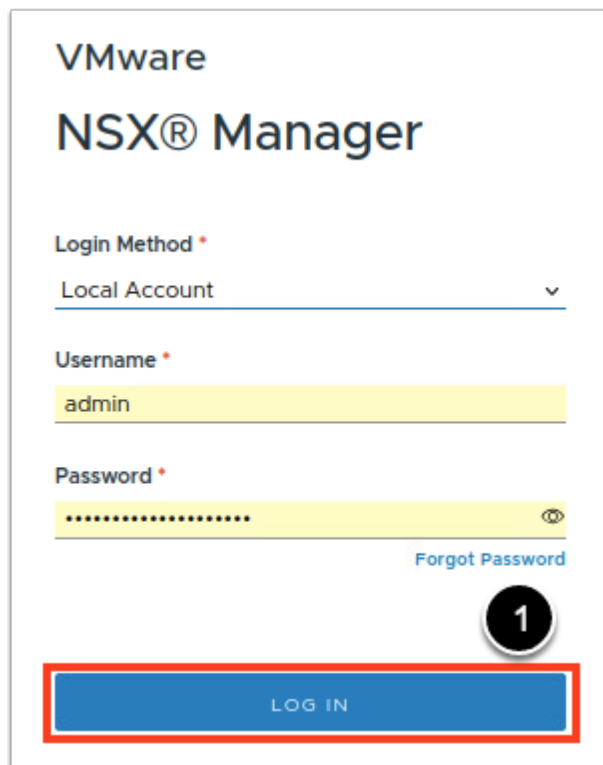
VCENTER IDP FEDERATION

Local Account

LOG IN

1. Click the **arrow** to expand the drop-down menu
2. Click **Local Account**

## Login to NSX Manager



VMware  
NSX® Manager

Login Method \*

Local Account

Username \*

admin

Password \*

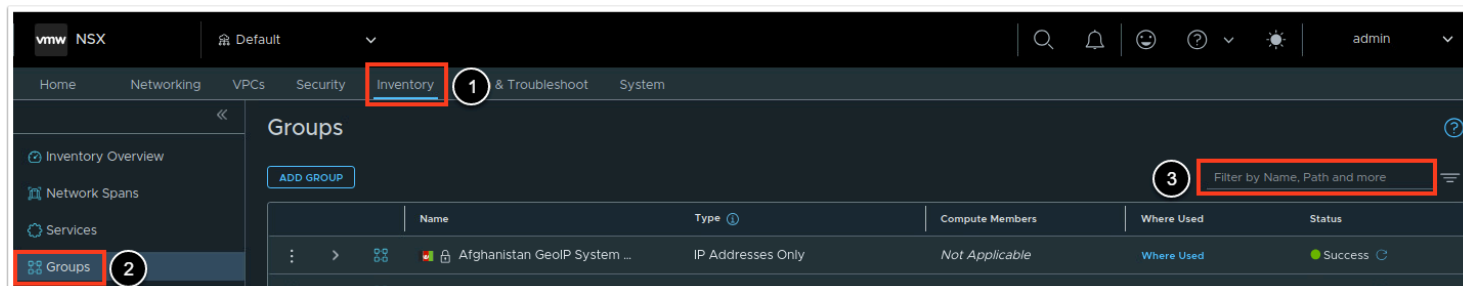
.....

[Forgot Password](#)

LOG IN

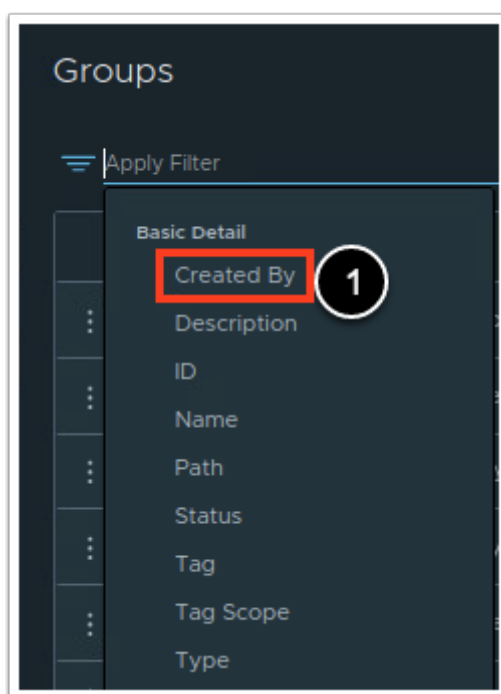
- 
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Validate Published Assets in NSX



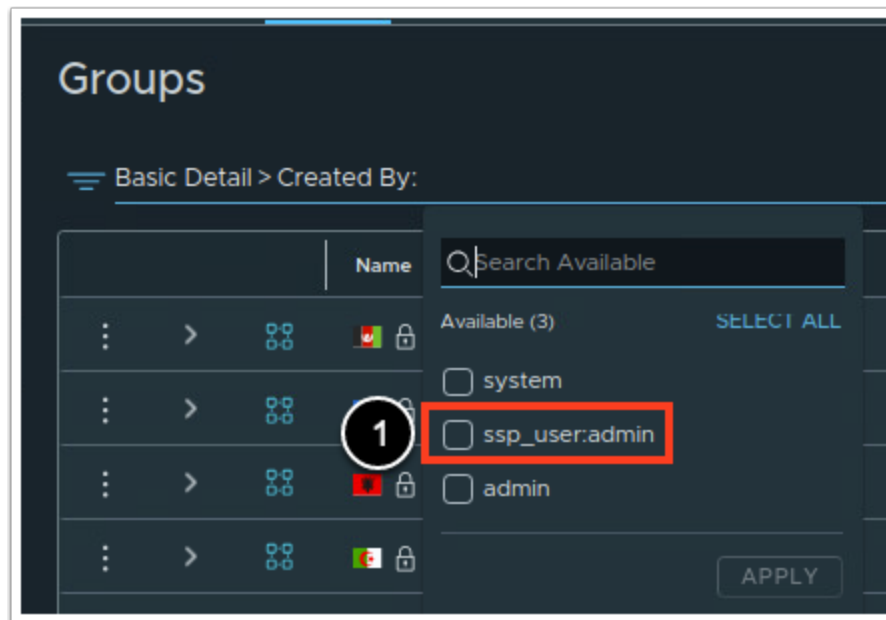
1. Click **Inventory**
2. Click **Groups**
3. Click in the **filter box**

## Select "Created By" Filter



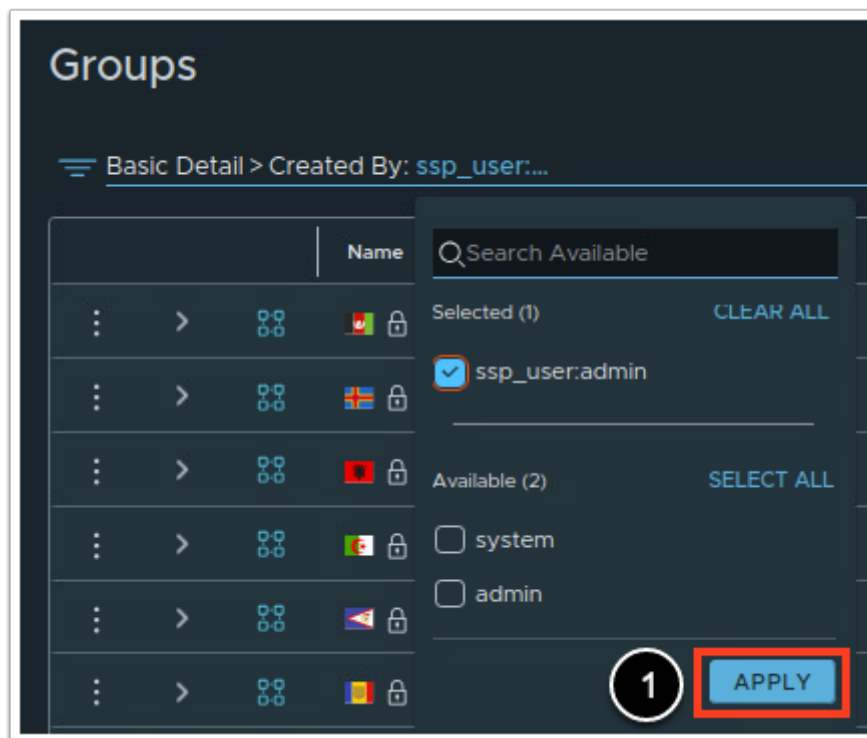
1. Click **Created By** to choose filter

## Select SSP Admin User



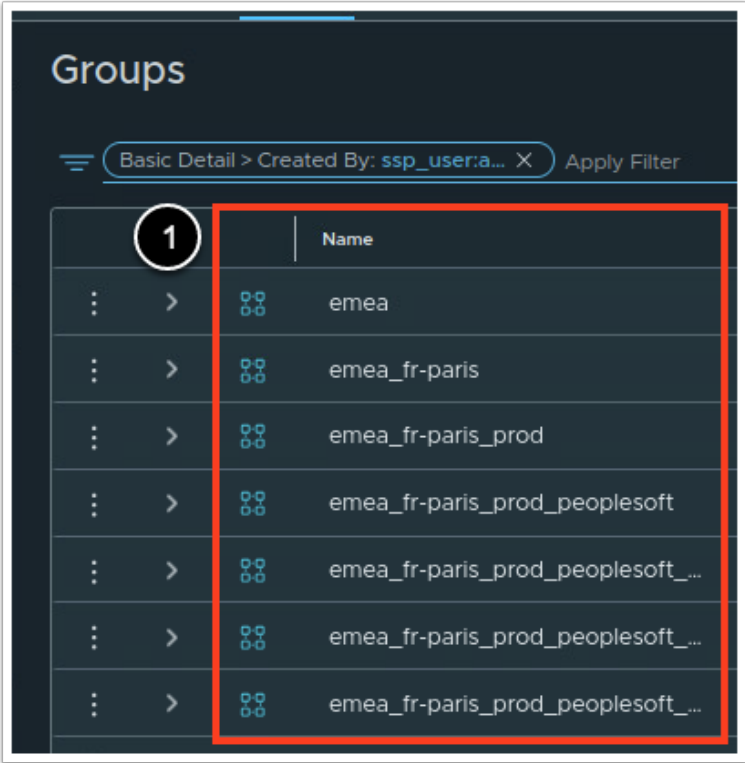
1. Click the **box next to user:admin** to choose the SSP admin user as a filter object

## Apply Search Filter



1. Click **APPLY** to apply the 'Created by SSP Platform' filter to the group list

# View Groups Published by SSP



1. The list is now showing groups that were created by SSP

# Open Group Membership View



1. Click **View Members** for the emea\_fr-paris\_prod\_peoplesoft\_postgres group

## View Group Membership

View Members | emea\_fr-paris\_prod\_peoplesoft\_postgres

To edit below content, click [EDIT](#)

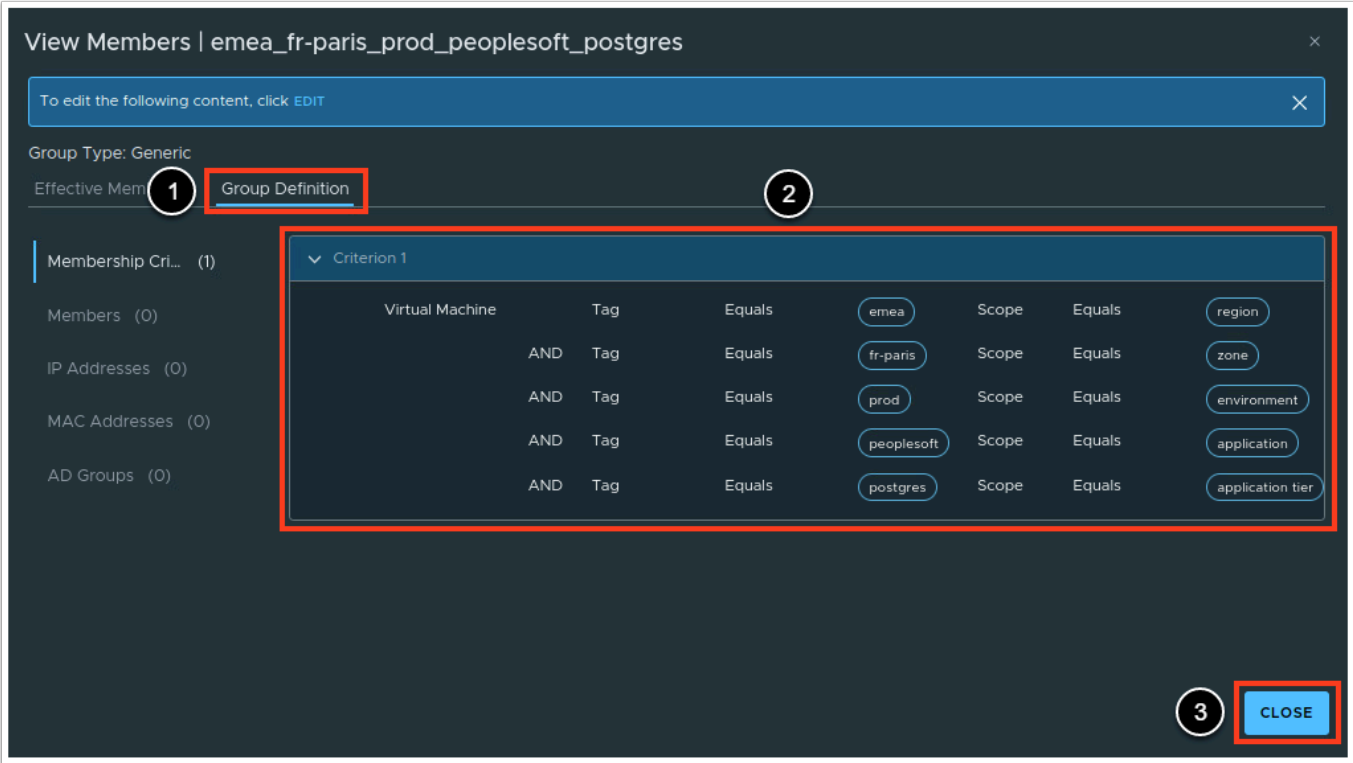
Group Type: Generic

Effective Members   Group Definition

|                      |            |
|----------------------|------------|
| Virtual Machines (2) | Name       |
| IP Addresses (2)     | jtgcoy1735 |
| NSX Segments (0)     | xvytsw0504 |
| Segment Ports (2)    |            |

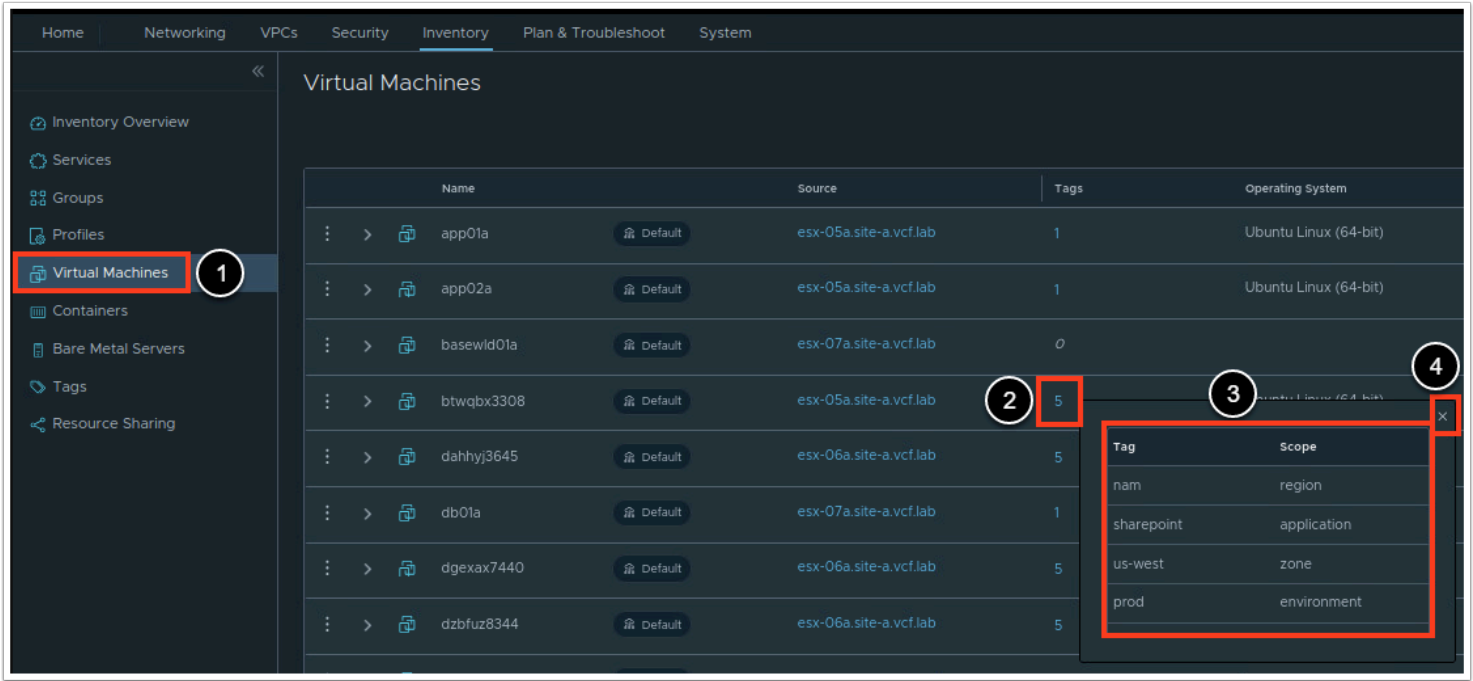
1. There are 2 VMs in this group

# View Group Definition



1. Click **Group Definition**
2. The Group criteria are tags that were published by SSP from the CSV import
3. Click **CLOSE** to close the View Members window

# View Tags Applied to VMs from CSV



1. Click **Virtual Machines**
2. Click the **5** next to the VM named btwqbx3308 to view the tags assigned to it
3. View the SSP created and applied tags
4. Click the **X** to close the tag window

# Conclusion

In this module the concept of the CSV datacenter asset import to Security Services Platform was introduced. An example CSV was exported from VMFlowGen that modeled what a CMDB export CSV may look like in a customer environment. The CSV was imported to Security Services Platform, and assets that were identified in the import process were published. Finally, published assets were validated in Security Services Platform and NSX.

At this stage, metadata information included in the CSV has been used to tag VMs and create groups for each hierarchy level - now we can proceed to the first stage in the Security Journey.



# **Module 4 – Stage 1: Security Segmentation Assessment & Report**

# Security Segmentation Assessment & Report

## Introduction

Assessing your current network segmentation provides essential visibility into your security posture and highlights key vulnerabilities across internal workloads. This lab section focuses on assessing existing east-west communication flows and generating actionable compliance and threat reports. Combining these analytical insights with **vDefend Distributed Firewall** and **vDefend Advanced Threat Prevention** enables organizations to systematically design, validate, and enforce zero-trust micro segmentation strategies.

The Security Segmentation Assessment report gives an overall view of your security posture and provides recommendations for leveraging the capabilities of vDefend Distributed Firewall and vDefend Advanced Threat Prevention to enhance your organizations security posture:

- A **Segmentation Score** represents your environments current segmentation state. Each category has criteria and sub-factors - points are awarded only when the rules and configuration meet the requirements.
- The **Advanced Threat Prevention (ATP) Score** evaluates your organizations overall security posture by measuring how effectively key security controls are deployed. This score delivers a unified view of your data center's utilization of core threat features, including Distributed Intrusion Detection and Prevention (IDS/IPS), Network Traffic Analysis (NTA), and Network Detection and Response (NDR).

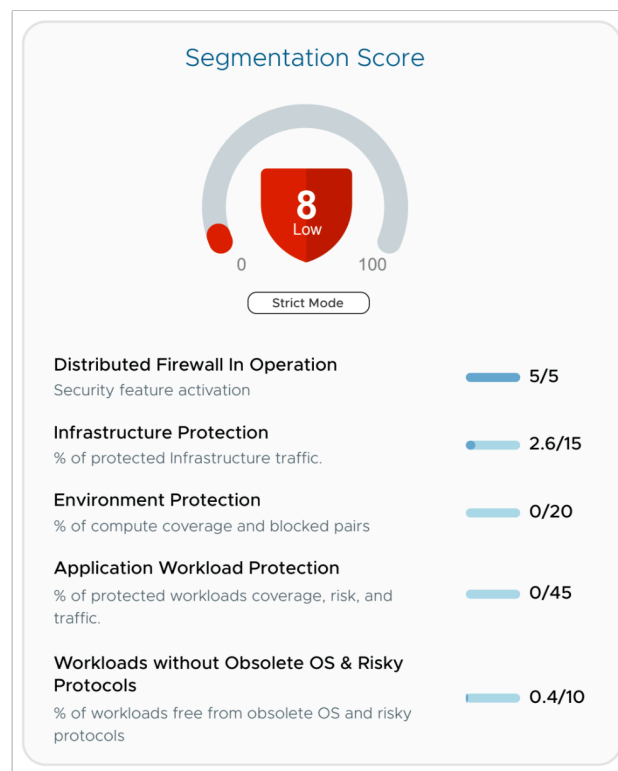
 Note: For score calculation and report generation, at least 1000 unique flows are required over the selected time range, for VMs that exist at the time of report generation. Only those flows completed up to an hour before report generation are taken into account during score calculation and report generation.

 Note: Security Assessment for vDefend Advanced Threat Prevention is supported for NSX4.2.1 and later and for VCF 9.0 and later.

 Note: To ensure the DVPG-connected VMs are included in the Security Assessment Report, enable NSX on DVPG from the NSX Manager. Enabling NSX on DVPG may cause some traffic to drop.

**i** Note: VMs present in the firewall exclusion list are not included in the Security Assessment Report.

## Security Segmentation Score



The Segmentation Score is computed using multiple factors, such as:

- **Distributed Firewall In Operation (5 Points Total)**
  - Stateful Distributed Firewall Activated (2.5 Points): Verifies that the vDefend DFW is enabled to enforce session-aware, stateful inspection, blocking out-of-order, spoofed, or hijacked packets.
  - Malicious IP Protection Activated (2.5 Points): Requires Auto-Update to be active alongside at least two top-priority rules denying inbound and outbound malicious IP traffic in the Infrastructure category.
- **Infrastructure Protection (15 Points)**
  - Evaluates whether critical infrastructure workloads (DNS, NTP, DHCP, Syslog, SNMP) are explicitly protected by non-default allow rules using group memberships and API-discovered servers.

- **Environment Protection (20 Points)**

- Environment Pair Isolation (10 Points): Measures the percentage of environment pairs (e.g., Dev vs. Prod) in blocked mode via rules published from the Security Services Platform.
- Compute Coverage (10 Points): Evaluates the percentage of total compute workloads covered by at least one environment-level rule, highlighting gaps where inter-environment policies are missing.

- **Application Protection in Strict Mode (45 Points)**

- Application Workload Coverage (6.5 Points): Measures the percentage of application workloads protected by custom Application-category rules rather than default rules.
- Application Traffic Coverage (6.5 Points): Tracks the volume of protected network traffic across the workloads identified in Application Workload Coverage.
- Application Lock-Down (16 Points): Evaluates the percentage of Application-category policies enforcing a default-deny rule, with higher weight given to policies protecting designated critical applications and applied-to groups.
- Data Center Lock-Down (16 Points): Requires the last rule in the Default Layer 3 section to be explicitly set to Deny.

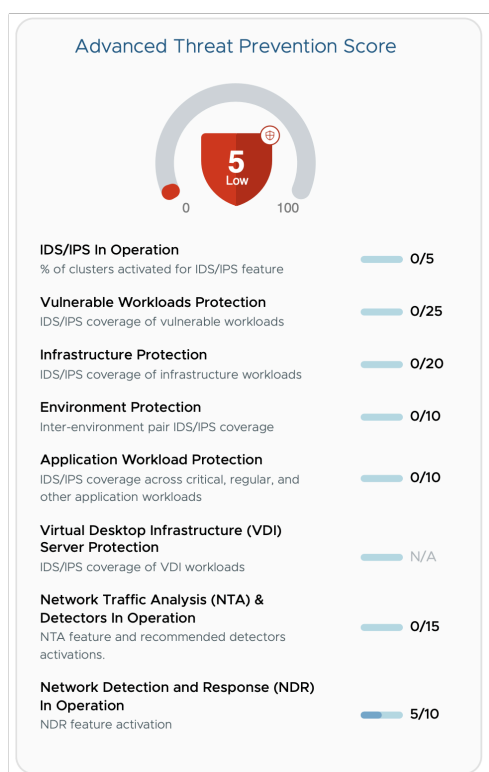
- **Workloads without Obsolete OS and Risky Protocols (10 Points)**

- Measures the percentage of workloads free from outdated, unsupported operating systems and unencrypted or high-risk network protocols.



Note: As a result of these score factors, the maximum score value is 95.

# Advanced Threat Prevention (ATP) Score



The Advanced Threat Prevention (ATP) Score is computed using multiple factors, such as:

- **IDS/IPS In Operation (5 Points)**
  - Points are awarded based on the proportion of host clusters with active transport nodes that have the Distributed IDS/IPS feature enabled.
- **IDS/IPS Workload Coverage (55 Points)**
  - The largest overall scoring category, evaluating how extensively workloads are protected by active IDS/IPS rules and threat signatures. The 55 points allocated to workload coverage are dynamically distributed based on active workload types. If an environment lacks a specific workload category, its point weight is automatically redistributed among the remaining active categories.
- **Environment Protection (10 Points)**
  - Measures the percentage of environment boundaries (e.g., Production to Development) guarded by IDS/IPS rules or DFW block rules with zero flow leakage.
- **NTA In Operation (15 Points)**
  - Earned by enabling eight recommended behavioral detectors, such as DNS tunneling and Domain Generation Algorithm (DGA) detection.
- **NDR In Operation (10 Points)**

- Split evenly between active NDR feature activation (5 points) and integrating at least one SIEM platform to receive NDR event logs (5 points).



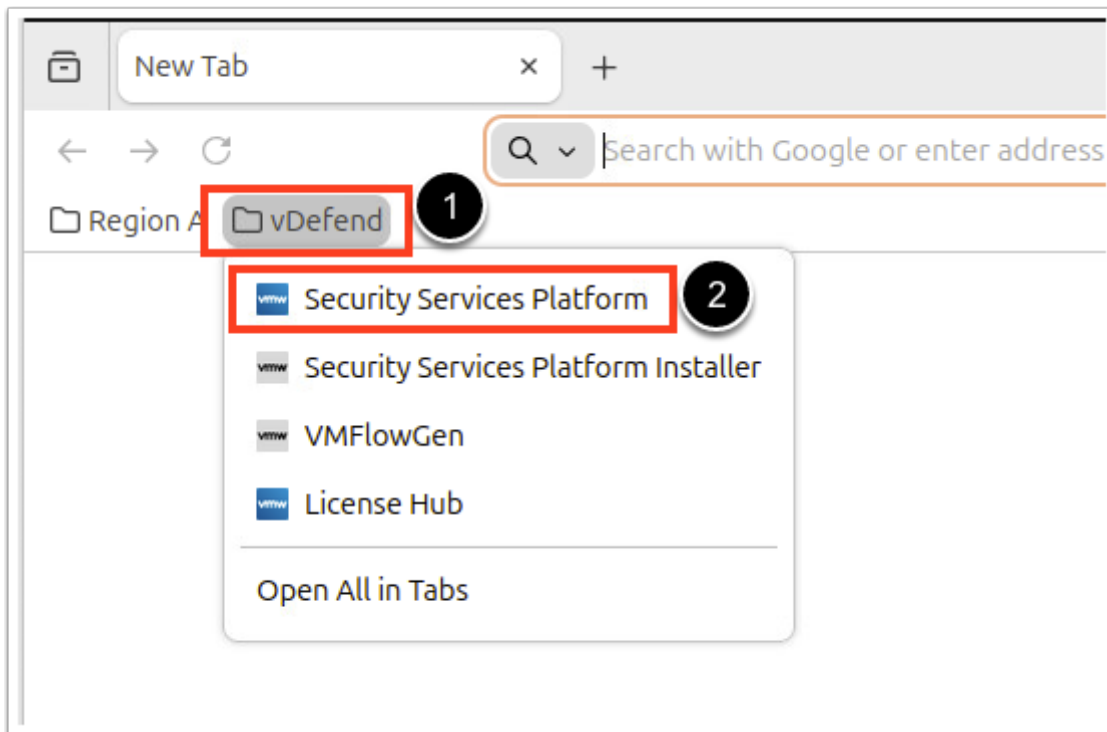
Note: As a result of these score factors, the maximum score value is 95.

# Login to Security Services Platform (SSP)



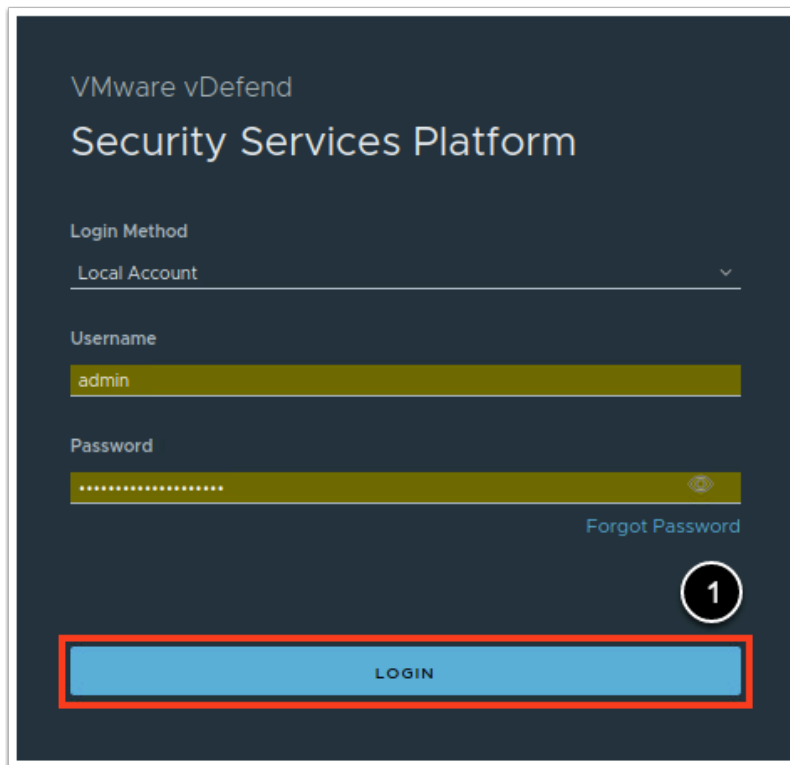
1. Click the **Firefox icon** in the taskbar

## Open Security Services Platform Console



1. Click on the **vDefend** bookmark bar folder
2. Click **Security Services Platform** to launch **SSP**

## Login to Security Services Platform Console



VMware vDefend  
Security Services Platform

Login Method  
Local Account

Username  
admin

Password  
.....

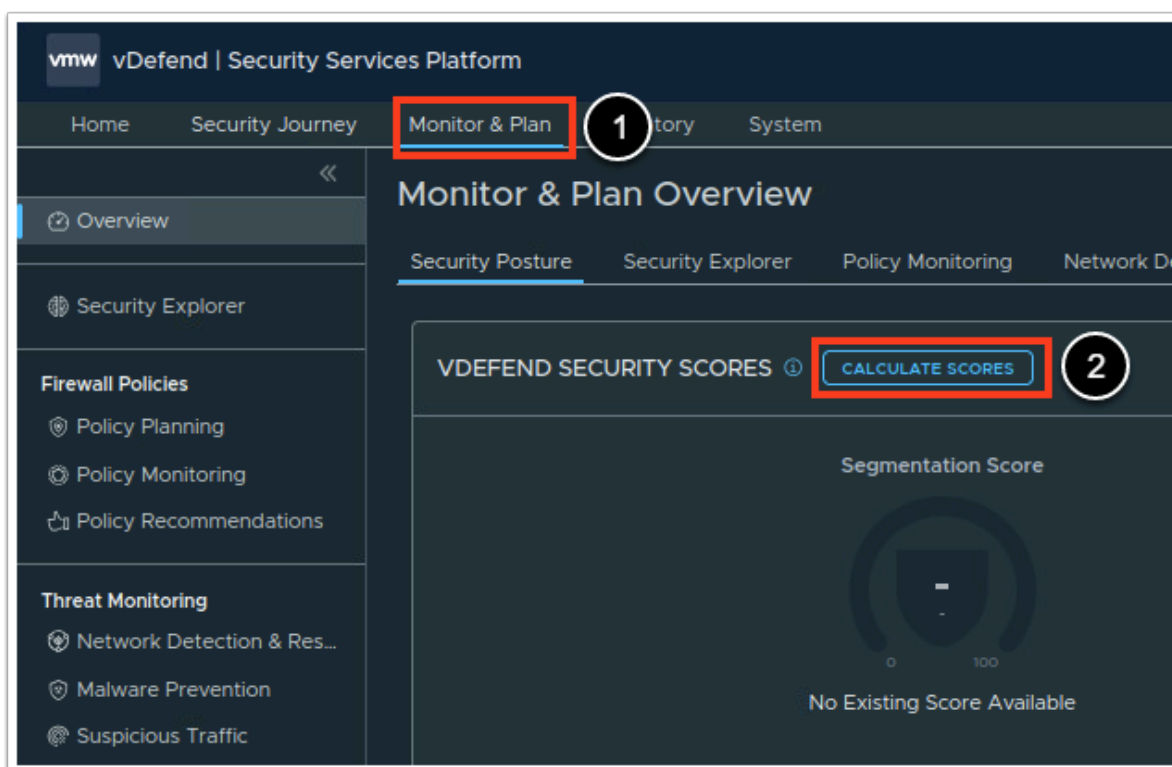
[Forgot Password](#)

1

LOGIN

1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Calculate Security Segmentation Score



1. Click **Monitor & Plan**
2. Click **CALCULATE SCORES**

## Select Security Segmentation Score mode

Calculate Security Segmentation Score

Select Security Segmentation Score mode to calculate your score in.

☒ Strict **1**

☐ Relaxed

**2** ☒ Learn about Security Segmentation Score Modes

**Strict**  
Customers looking for a score that accurately reflects their data center security posture should use Strict mode. This mode highlights achieved security while applying stricter penalties for any allowed unidentified traffic.

**Relaxed**  
Customers creating security policies for their data center workloads may be cautious about denying traffic for fear of disrupting production applications. Relaxed mode emphasizes progress in rule creation rather than strict enforcement.

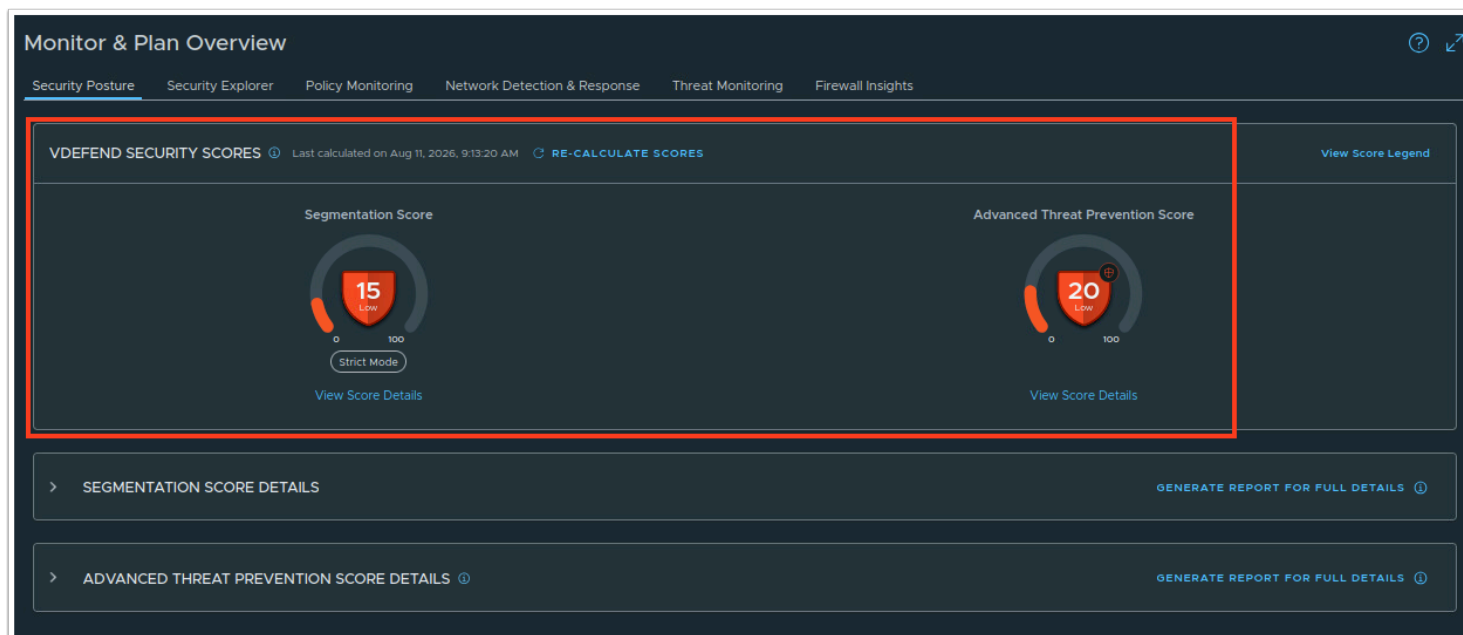
**3**

1. Select **Strict** as the mode (this should be selected by default)
2. Click the **arrow** to learn more about the two Security Segmentation Score Modes (optional)
3. Click **CALCULATE**



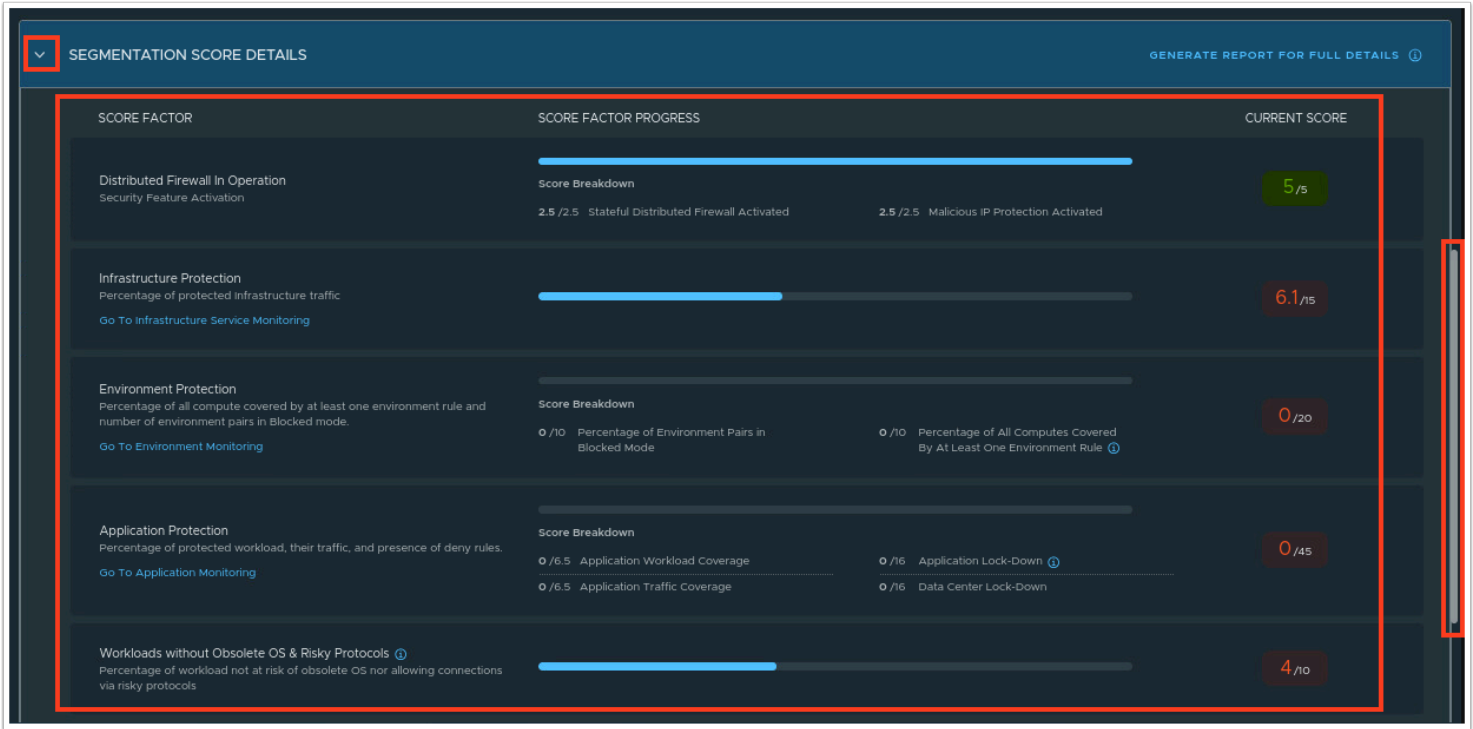
Note: Calculating the Security Segmentation Score will take a few minutes

# View vDefend Security Scores



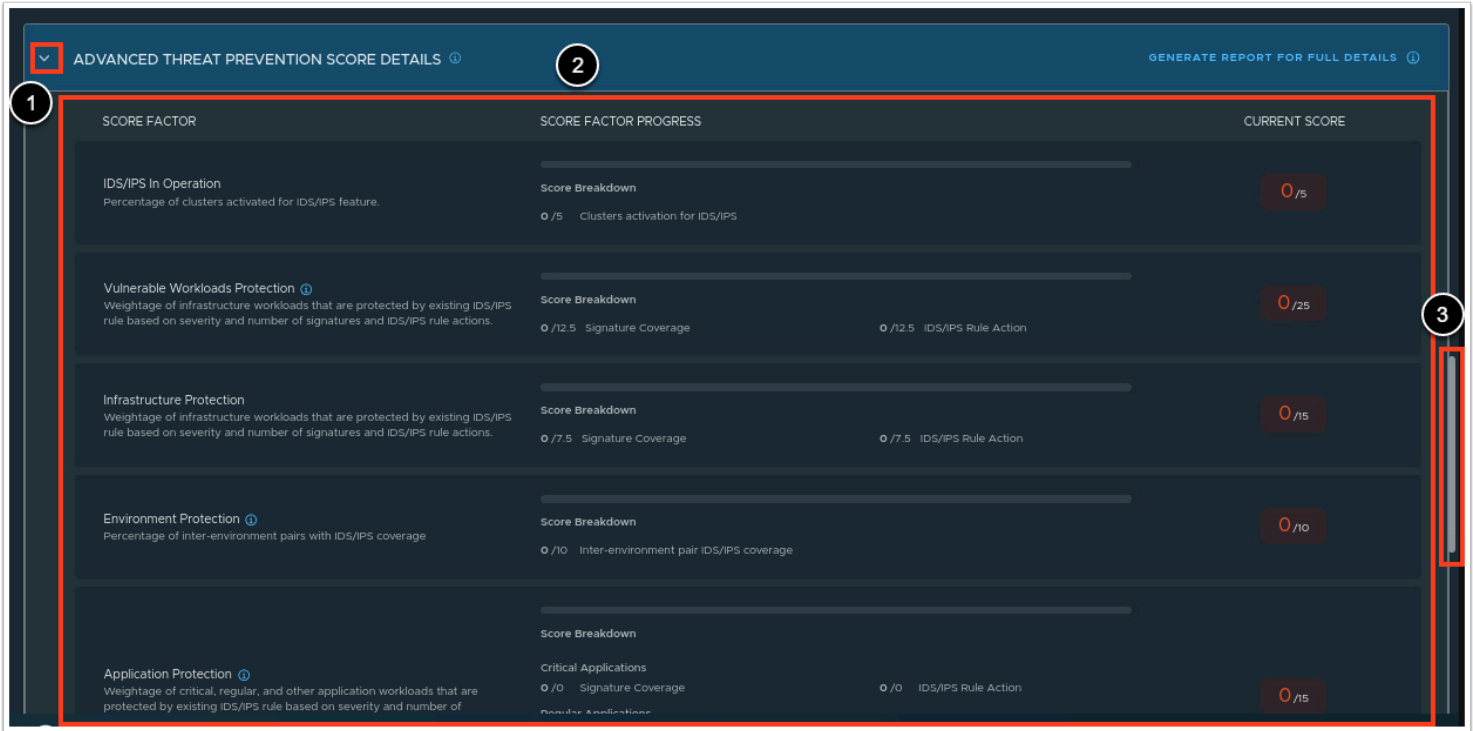
1. View the calculated Security Segmentation Score and the Advanced Threat Protection Score - the scores are expected to be low in this instance since the Security Journey has just been started

# View Segmentation Score Details



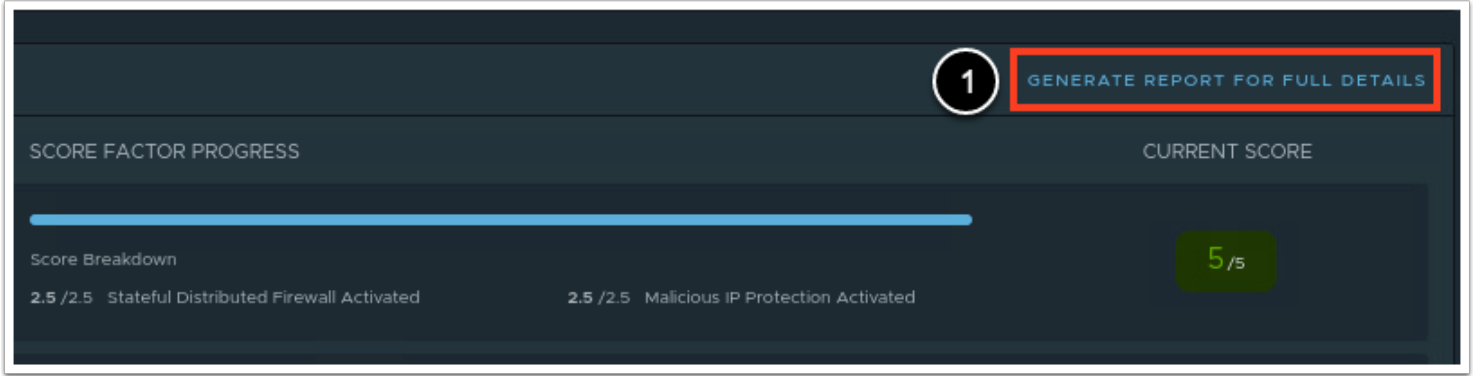
1. Click the **arrow** next to SEGMENTATION SCORE DETAILS to expand score details
2. Score factor, factor progress, and current score are shown
3. Use the scroll bar to continue viewing the Segmentation Score Details

# View Advanced Threat Prevention Score Details



1. Click the **arrow** next to ADVANCED THREAT PREVENTION SCORE DETAILS to expand score details
2. Score factor, factor progress, and current score are shown
3. Use the scroll bar to continue viewing the Advanced Threat Prevention Score Details

# Navigate to Security Segmentation Reports



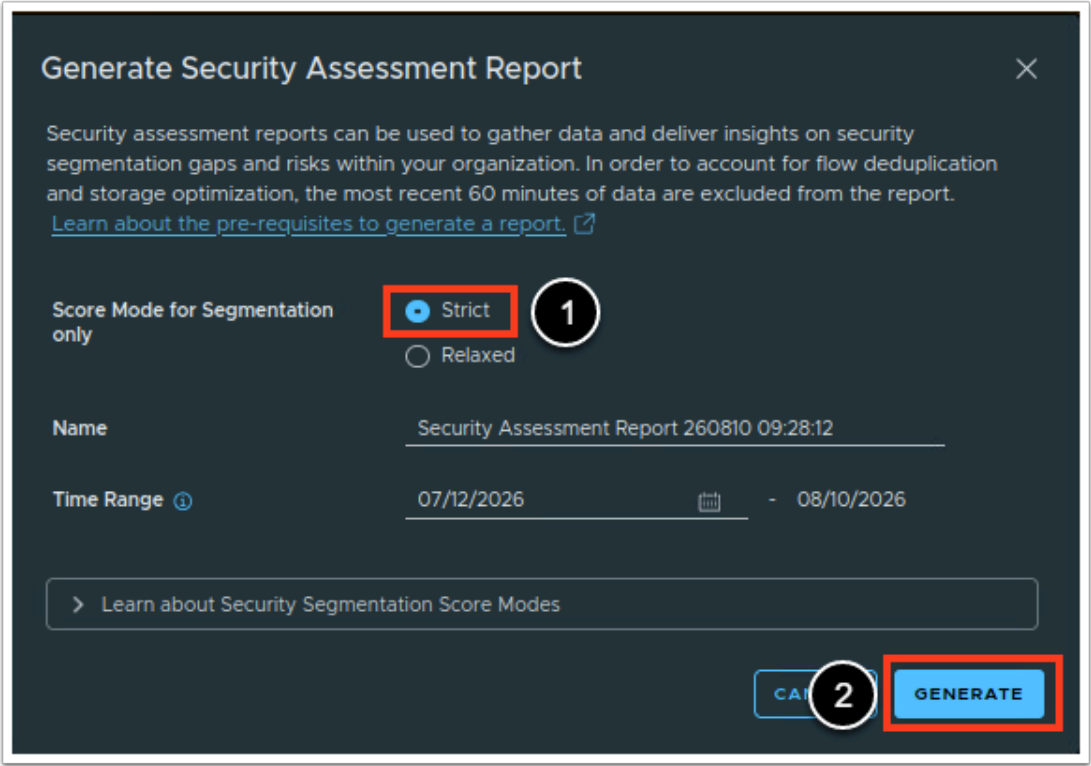
1. Click **GENERATE REPORT FOR FULL DETAILS** to navigate to the Security Security Assessment Report page

# Choose Security Assessment Report Parameters



- 1. Click **GENERATE REPORT**

## Generate Security Assessment Report



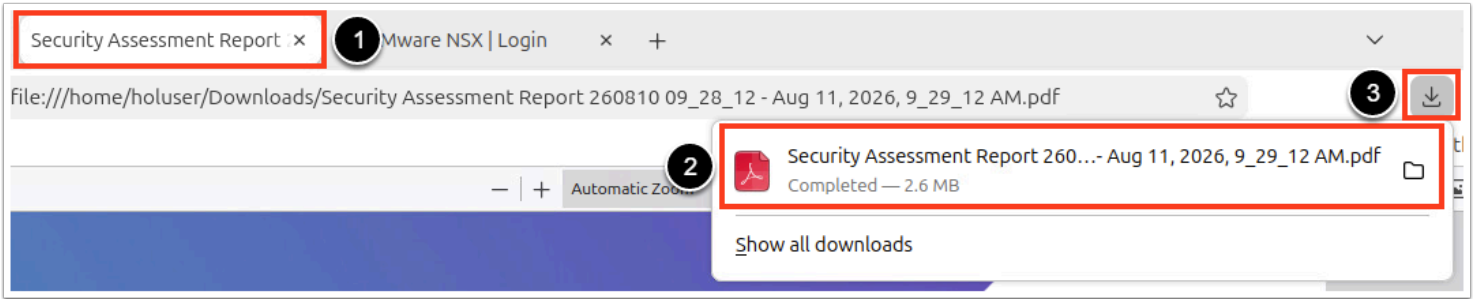
- 1. Confirm **Strict** Segmentation Score Mode is selected
- 2. Click **GENERATE**

# Download Security Assessment Report

|                   |                |               |      |                                            |              |                          |          |           |
|-------------------|----------------|---------------|------|--------------------------------------------|--------------|--------------------------|----------|-----------|
| GENERATE REPORT ⓘ |                |               |      |                                            |              |                          | Filter   | Sort By ▾ |
|                   | Seg. Score ⓘ ↑ | ATP Score ⓘ ↑ | Name | Mode for Segmentation                      | Generated on | Status                   |          |           |
| ⋮                 | >              | 15            | 20 ⓘ | Security Assessment Report 260810 09:28:12 | Strict Mode  | Aug 11, 2026, 9:29:12 AM | 1        |           |
|                   |                |               |      |                                            |              | 2 Success                | Download |           |

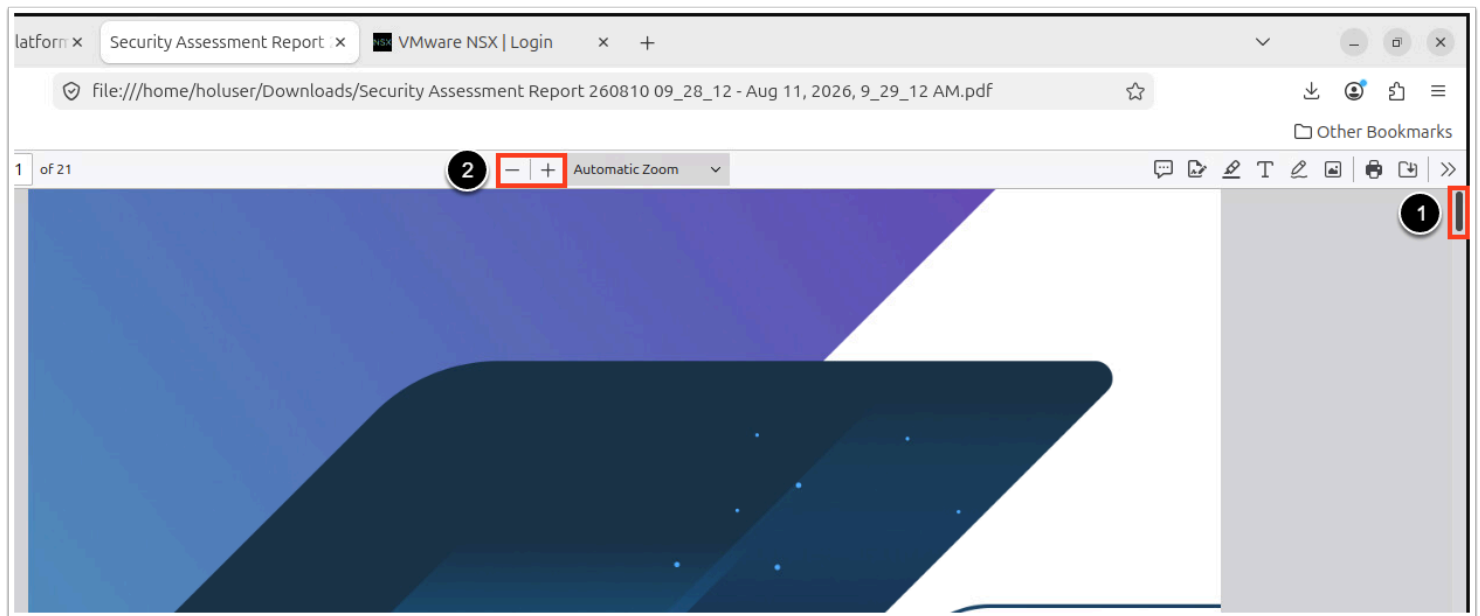
1. The Security Assessment Report Status will cycle through "In Queue", "In Progress", and finally "Success" status
2. Click **Download** once the status shows "Success"

# Confirm Report Download



1. The Security Assessment Report will automatically open in a new browser tab
2. The browser download window will show the downloaded report
3. Click the **browser download icon** to close the download window

## View the Security Assessment Report



1. Use the browser **scroll bar** to navigate the Security Assessment Report
2. Use the **zoom buttons** to adjust the zoom for the report for easier viewing

# View Security Assessment Report

## vDefend Security Assessment Report

August 11, 2026

Report Name: Security Assessment Report 260810 09:28:12

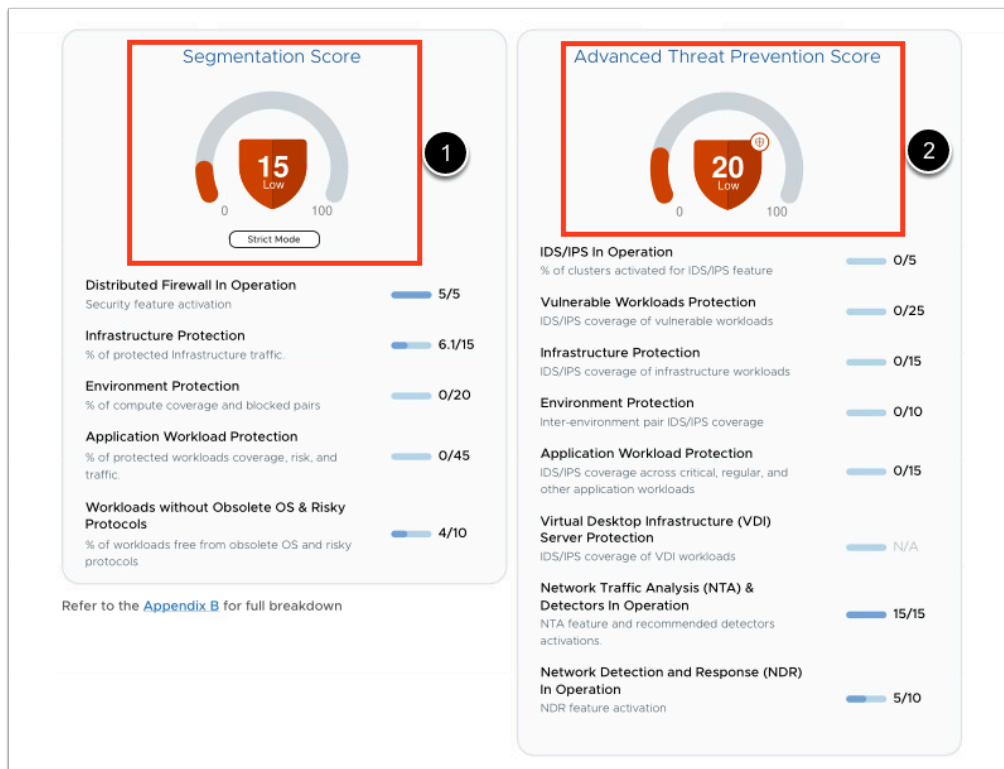
Report ID: c98dd50a-95a1-11f1-ab2f-52cd01d7600e

Report Period: Jul 12, 2026 PDT - Aug 11, 2026 PDT

Generated by: admin

### 1. Executive Summary (Page 3)

This section summarizes the environment's security posture over a 30-day assessment period, evaluating **1.14k unique flows** across **55 VMs / 50 IPs**. Both security pillars currently show a **Low** rating due to minimal active rule enforcement across workloads.



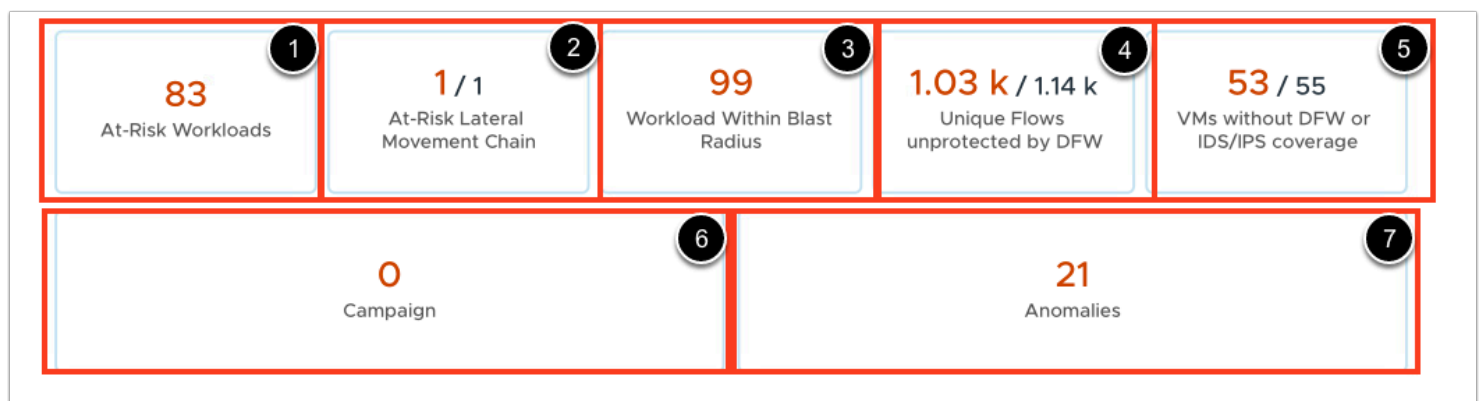
- Segmentation Score: 15 (Low - Strict Mode)** Evaluates how effectively the Distributed Firewall (DFW) isolates network traffic and isolates workload tiers.
  - Distributed Firewall In Operation (5/5 Points): Core DFW functionality is fully active across the deployment.
  - Infrastructure Protection (6.1/15 Points): Partial rules exist to explicitly permit and isolate core services (e.g., DNS, DHCP, NTP).
  - Environment Protection (0/20 Points): Zero isolation policies exist between environment boundaries (e.g., Production vs. Development).
  - Application Workload Protection (0/45 Points): Application workloads lack microsegmentation rules and default-deny policies.
  - Workloads without Obsolete OS & Risky Protocols (4/10 Points): A significant portion of VMs run legacy operating systems or clear-text protocols (e.g., HTTP, Telnet, RDP).
- Advanced Threat Prevention Score: 20 (Low)** Evaluates how active threat detection mechanisms (IDS/IPS, NTA, NDR) protect east-west data center traffic.
  - IDS/IPS In Operation (0/5 Points): Distributed IDS/IPS features are not enabled across host clusters.
  - Workload Protection Breakdown (0 Points Total): Because IDS/IPS is inactive, no workloads have active signature coverage:
    - Vulnerable Workloads: 0/25 Points
    - Infrastructure Protection: 0/15 Points
    - Environment Protection: 0/10 Points
    - Application Workload Protection: 0/15 Points

- *VDI Server Protection*: N/A (No VDI instances detected; weight automatically redistributed)
- Network Traffic Analysis (NTA) & Detectors (15/15 Points): Behavioral detectors (e.g., DNS tunneling, DGA detection) are fully enabled and actively monitoring.
- Network Detection and Response (NDR) (5/10 Points): NDR features are active, but points are docked due to missing SIEM logging integration.

## Recommended Projects (Pages 4-5)

These pages detail specific remediation projects to raise your overall security scores. They categorize gaps across active workloads, environmental boundaries, and threat detection engines.

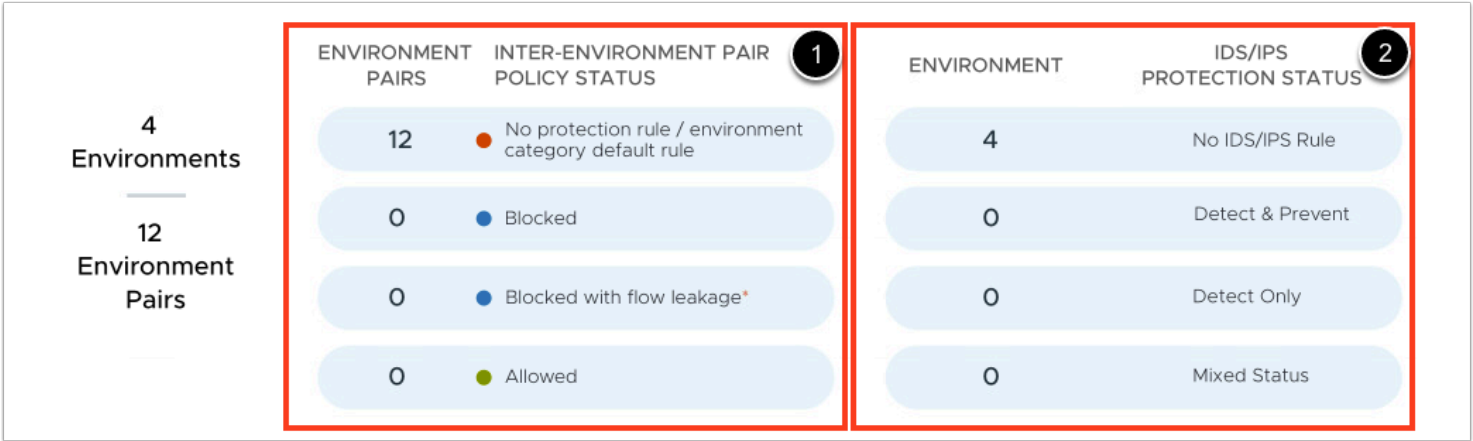
## 2. Overview of Current Security Posture (Page 6)



This section outlines the data center's current vulnerability landscape, traffic exposure, and blast radius metrics across 105 total workloads and 1.14k unique traffic flows.

1. 83 At-Risk Workloads: 83 out of 105 total workloads are exposed due to missing firewalls, lack of IDS/IPS, or risky protocol usage.
2. 1 Workload in Lateral Movement Chain: Identified as a primary entry point capable of facilitating lateral threat progression.
3. 99 Workloads in Blast Radius: Unrestricted east-west paths allow a breach on an exposed VM to potentially spread across 99 total systems.
4. 1.03k / 1.14k Unprotected Flows: 90%+ of all observed network traffic lacks Distributed Firewall (DFW) enforcement.
5. 53 / 55 Unprotected VMs: 96% of virtual machines operate without DFW or IDS/IPS rule coverage.
6. 0 Campaigns: During the 30-day assessment window no campaigns were detected.
7. 21 Behavioral Anomalies: All 21 detected anomalies stem from Uncommonly Used Ports (affecting 1 Infrastructure server and 2 Regular Application workloads). Zero active threat campaigns or IDS events were logged.

Environment & Policy Status (Page 6)



1. 12 / 12 Environment Pairs Unprotected: All 12 environment boundaries (e.g., Prod, Dev, Test) lack inter-environment isolation rules or default-deny policies.
2. 0% IDS/IPS Coverage: No environment pairs have active IDS/IPS inspection rules (configured to Detect or Detect & Prevent).

Workloads At-Risk Workloads & Blast Radius (Page 7)

1. Infrastructure Servers (6 Total / 5.7% of Environment)
  - At-Risk Workloads: 1 server flagged for elevated security risk due to protocol exposure.
  - Traffic Exposure: 147 internal traffic flows operate completely unprotected by DFW.
  - Coverage Gaps: 4 out of 6 infrastructure VMs lack any DFW or IDS/IPS rule coverage.
2. Application Workloads (99 Total / 94.3% of Environment)
  - At-Risk Workloads: 82 workloads flagged for missing security policies or risky communications.
  - Blast Radius Exposure: 99 total application workloads fall within the potential lateral movement blast radius.
  - Traffic Exposure: 1.03k internal traffic flows lack DFW protection.
  - Coverage Gaps: 49 application VMs operate entirely without DFW or IDS/IPS security coverage.

Risky & Unprotected Traffic (Page 8)

This section analyzes network traffic risk factors across 1.14k unique flows and breaks down active behavioral anomalies flagged by NTA.

3. Recommended Project Details (Page 8)

This section breaks down the specific remediation projects required across core infrastructure, environmental boundaries, application workloads, and behavior-based threat detections.

### 3.1 Project : Secure Infrastructure Services

This section focuses on isolating core operational servers (DNS, NTP, Directory Services) and remediating clear-text protocol exposures.

- Scope & Traffic: 6 internal infrastructure servers handle 250 internal flows (100% internal). 147 flows (58.8%) are completely unprotected by DFW.
- Protocol Breakdown: NTP (3 servers / 144 flows) is 100% unprotected; DNS-UDP (2 servers / 103 flows) is 100% allowed via DFW; LDAP (1 server / 3 flows) is unprotected.
- Target Vulnerable Workloads: VM oyoicr6976 is the highest priority due to clear-text LDAP usage. All 5 key infrastructure VMs (oyoicr6976, lnnavti4789, yyhgza8639, oiugax0504, wqafba9936) currently lack active IDS/IPS rule coverage.

### 3.2 Project: Secure Environments (Page 11)

- Scope & Exposure: 4 environments with 12 inter-environment pairs containing 48 total workloads. 32 workloads are at-risk, and 22 out of 22 inter-environment flows lack DFW protection.
- Policy Status: All 12 environment pairs lack DFW or IDS/IPS protection rules; 0 pairs are currently configured in Block mode.

### 3.3 Project: Secure Application Workloads (Pages 12-14)

- Massive Traffic Exposure: 1.03k out of 1.13k total unique flows are completely unprotected by DFW.
- Blast Radius & Workloads: 82 at-risk workloads with 49 VMs lacking DFW/IPS coverage, creating a potential lateral movement blast radius of 99 workloads.
- Top Protocol Risks: SSL (68 workloads / 68.7%), HTTP (28 workloads / 28.3%), and LDAP (3 workloads / 3.0%) drive the primary application risks. 32 of 82 vulnerable application workloads lack IDS/IPS rules.
- Target VMs: reyhth5902, eaineg0296, btwqbx3308, ugiurx6833, and ldnjqc8541 are the top priority VMs needing active IDS/IPS enforcement.

### 3.4 Project: Assess Anomalies (Page 14)

This section highlights machine-learning-based Network Traffic Analysis (NTA) detections across zero-day and behavioral threats.

- Primary Anomaly Driver: Uncommonly Used Port is the sole active detector triggering alerts on the top-ranked workloads.
- MITRE ATT&CK Mapping: All flagged events are mapped to the Command and Control (C2) tactic.
- Severity & Impact: Every recorded incident carries a Low (3) impact score.
- Key Target Workload: Infrastructure server oyoicr6976 accounts for all top 5 significant anomaly events recorded between August 8 and August 10, 2026.

## 4. Appendix

### • **Appendix A: Terminology Key Takeaways (Pages 15-18)**

- Traffic Classifications: Blast Radius measures potential lateral threat spread; Unprotected Traffic matches DFW default allow rules; Protected Traffic is actively covered by policy.
- Workloads & Entities: Workloads are NSX-managed or IP-represented entities; At-Risk Workloads exhibit public, obsolete OS, or risky protocol exposure; Entry Points are compromised VMs used for lateral movement.
- Scoring Modes: Strict Mode heavily penalizes allowed unidentified traffic; Relaxed Mode focuses on rule creation progress over strict enforcement.
- Protocol Categories: Defines standard infrastructure protocols (DNS, NTP, LDAP, AD), VDI protocols (BLAST, PCoIP, CTRXCGP), and App ID deep-packet inspection.

### • **Appendix B: Security Segmentation Score Breakdown (Page 19)**

- This page details the mathematical scoring breakdown for the overall 15 / 100 Security Segmentation Score:
- Distributed Firewall in Operation (5 / 5 pts): Fully achieved; stateful DFW (2.5/2.5) and Malicious IP Protection (2.5/2.5) are active.
- Infrastructure Protection (6.1 / 15 pts): Partial score (40.7% progress) based on the percentage of infrastructure traffic covered by non-default allow rules.
- Environment Protection (0 / 20 pts): Zero points awarded; 0% of environment pairs are in Block mode (0/10) and 0% of computes have environment rules (0/10).
- Application Protection (0 / 45 pts): Zero points awarded; 0% application workload coverage (0/6.5), 0% traffic coverage (0/6.5), 0% application lock-down (0/16), and 0% data center lock-down (0/16).
- Workloads without Obsolete OS & Risky Protocols (4 / 10 pts): Low score (40% progress) due to active workloads running unencrypted or risky protocols.

### • **Appendix C: Advanced Threat Prevention Score Breakdown (Page 20)**

- IDS/IPS in Operation (0 / 5 pts): 0% progress; host clusters lack host-level activation for IDS/IPS.
- Vulnerable Workload Protection (0 / 25 pts): Zero signature coverage (0/12.5) or rule actions (0/12.5) for identified vulnerable VMs.
- Infrastructure Protection (0 / 15 pts): Zero signature coverage (0/7.5) or rule actions (0/7.5) applied to infrastructure servers.
- Environment Protection (0 / 10 pts): Zero inter-environment pairs have active IDS/IPS inspection.
- Application Workload Protection (0 / 15 pts): Zero signature coverage or rule actions across Regular (0/10) and Other Application VMs (0/5).
- Virtual Desktop Infrastructure Protection (N/A): Excluded from scoring due to zero VDI workloads detected.

- Network Traffic Analysis (15 / 15 pts): Fully achieved (100%); all 8 recommended machine-learning detectors are active.
- Network Detection & Response (5 / 10 pts): Partial score (50%); NDR feature is activated (5/5), but no SIEM configuration is active (0/5).

# Conclusion

In this module, you learned how to use the Security Services Platform (SSP) to generate a Security Segmentation Score and review a detailed Security Segmentation Report. Re-running this assessment at regular intervals allows you to measure security posture improvements and track progress toward a Zero Trust architecture.

**You've Finished Stage 1!** Congratulations on completing Stage 1 of Module 4!

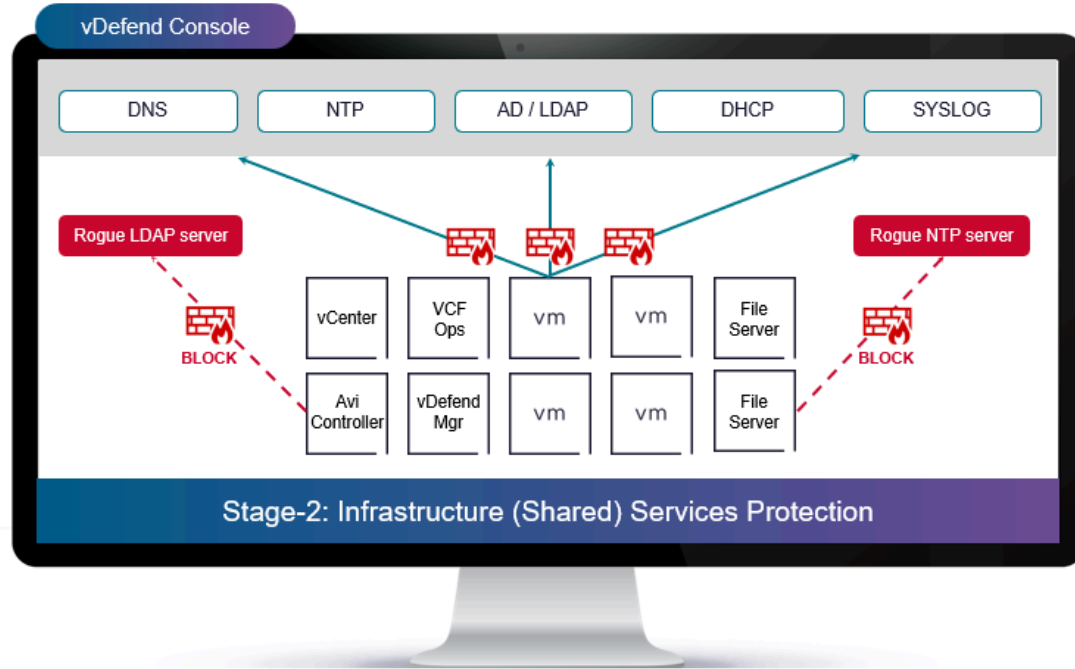
In the next module, **Infrastructure (Shared) Services Segmentation**, you will begin configuring segmentation policies to protect shared infrastructure services across your deployment.



# **Module 5 – Stage 2: Infrastructure (Shared) Services Segmentation**

# Infrastructure (Shared) Services Segmentation Introduction

## Stage 2: Securing Infrastructure (Shared) Services



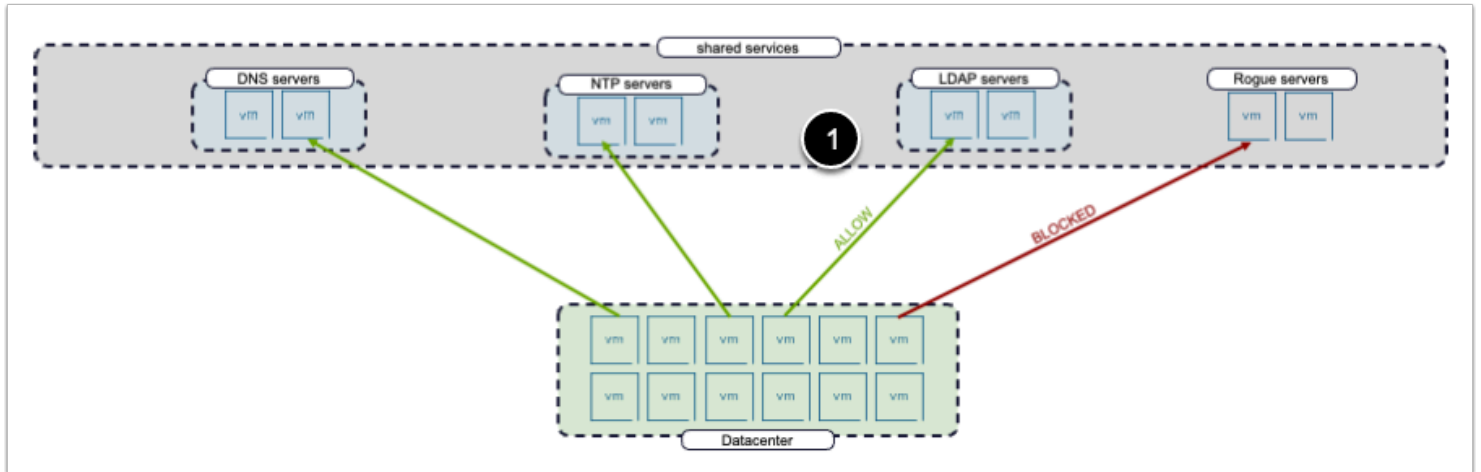
 Few days

In the vDefend DFW & ATP 1-2-3-4 deployment workflow, Stage 2 focuses on securing the core Infrastructure (Shared) Services such as DNS, NTP, Syslog, SNMP, DHCP, and LDAP/LDAPs.

DFW 1-2-3-4 automatically discovers infrastructure services to identify service endpoints and allows the user to validate and automatically create protection rules to these services.

Alternatively, users can feed their known infrastructure service endpoints via CSV file for the system to add infrastructure services.

## Stage 2 Objective



**Core Objective:** Enforce explicit access controls between datacenter workloads and shared infrastructure, ensuring traffic is strictly limited to verified management and identity services.

**Authorized Traffic (Allowed):** Datacenter workloads are explicitly permitted to communicate with essential core infrastructure components specifically **DNS**, **NTP**, and **LDAP** servers.

**Unauthorized Traffic (Blocked):** Any attempt to connect to unapproved or rogue servers within the shared services zone is automatically blocked.

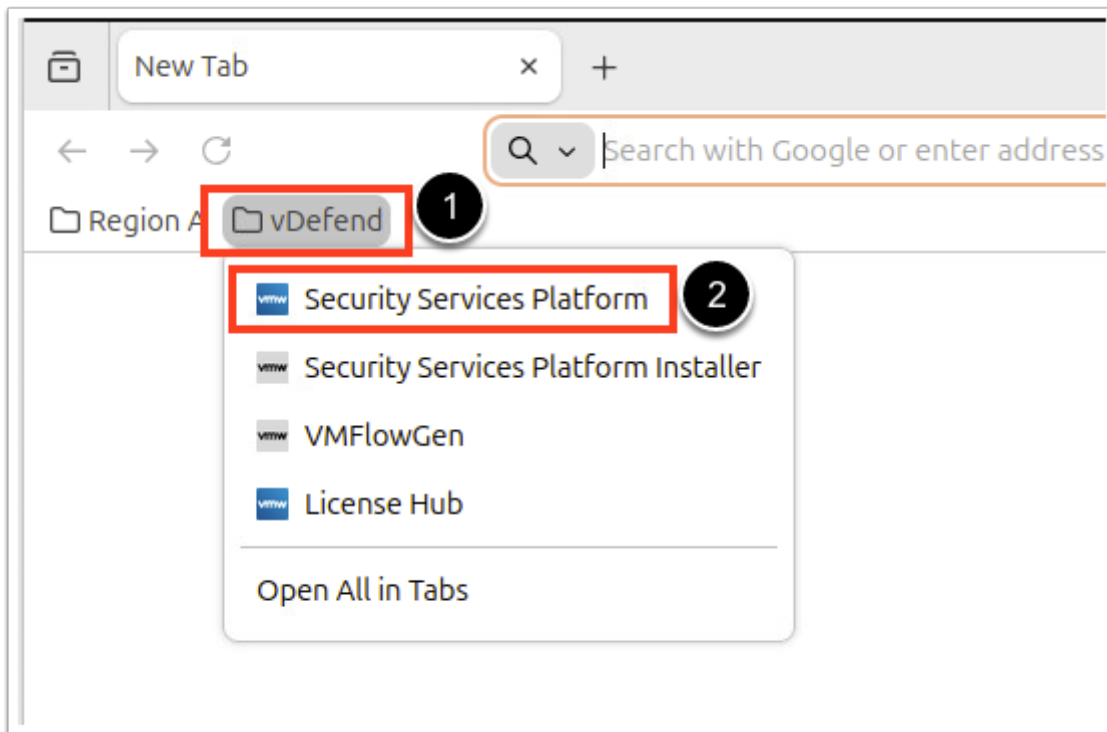
**Key Benefit:** Eliminates unauthorized lateral movement and prevents attackers from leveraging rogue endpoints for covert Command & Control (C2) or data exfiltration, all without impacting standard operational dependencies.

# Login to Security Services Platform (SSP)



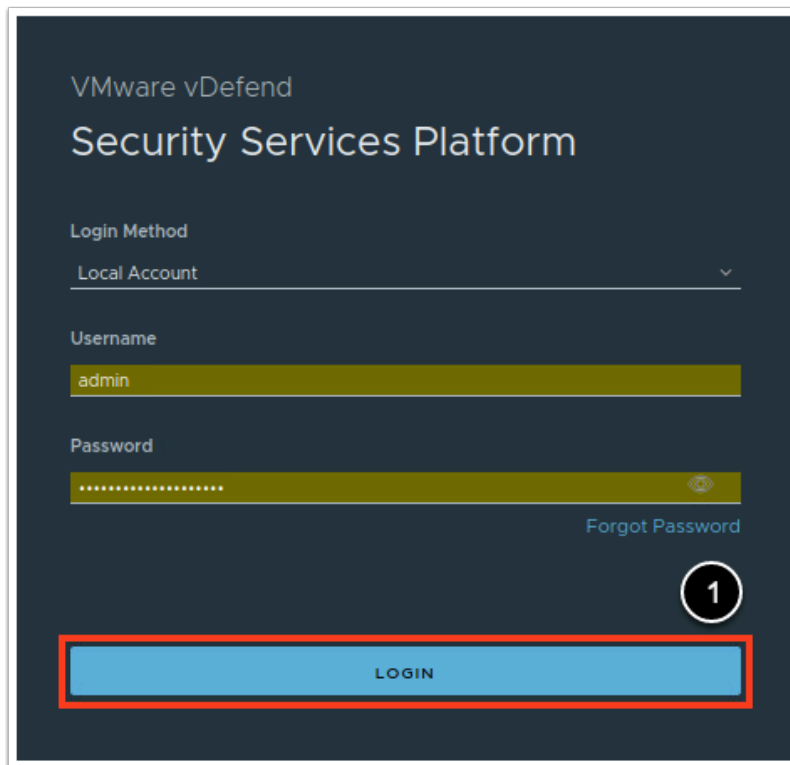
1. Click the **Firefox icon** in the taskbar

## Open Security Services Platform Console



1. Click on the **vDefend** bookmark bar folder
2. Click **Security Services Platform** to launch **SSP**

## Login to Security Services Platform Console



VMware vDefend  
Security Services Platform

Login Method  
Local Account

Username  
admin

Password  
.....

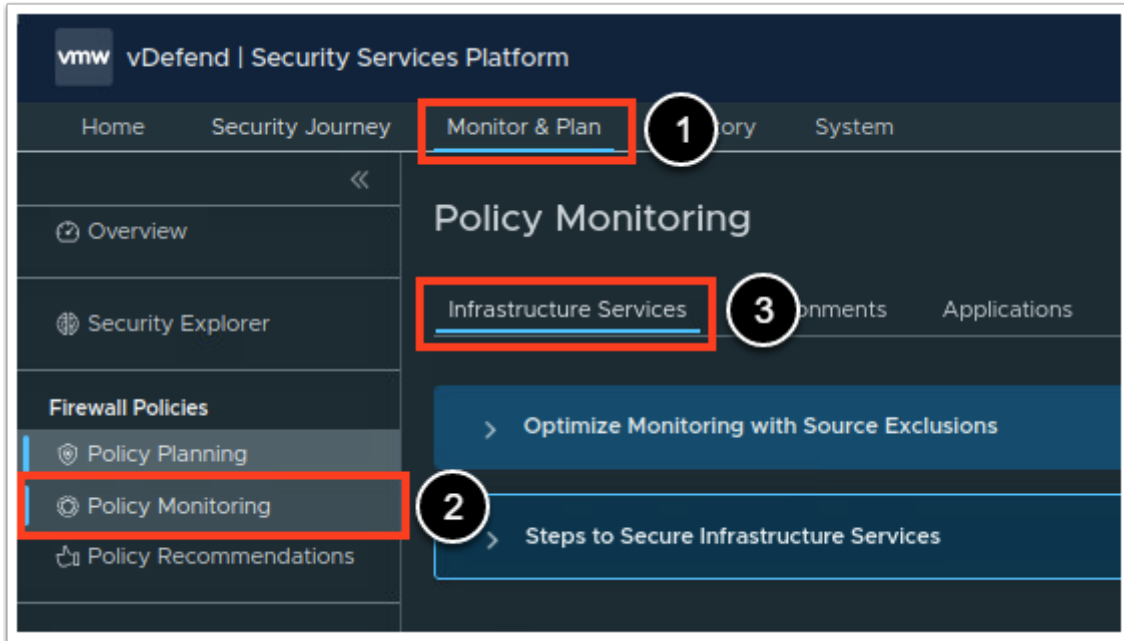
[Forgot Password](#)

1

LOGIN

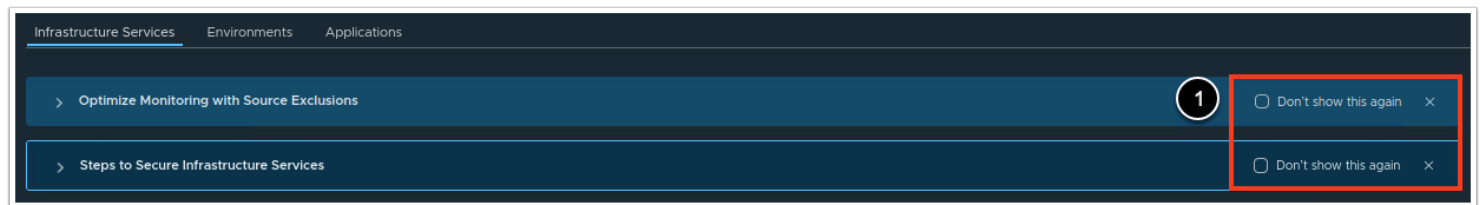
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Publish Infrastructure Services Rules to Distributed Firewall & IDS/IPS



1. Click **Monitor & Plan**
2. Click **Policy Monitoring**
3. Confirm **Infrastructure Services** is selected

## (Optional) Clear Banners From Screen



1. Click the **check boxes** and the **close buttons** for the info banners to gain more screen space

# View Infrastructure Services

| Infrastructure Service | Infrastructure Group | Protocols/Ports | New Detected   Suppressed Servers | Security Rules                         |        |
|------------------------|----------------------|-----------------|-----------------------------------|----------------------------------------|--------|
| > LDAPS                | None                 | TCP/636         | 1   0                             | No Rules <a href="#">Add</a>           | ACTION |
| > LDAP-UDP             | None                 | UDP/389         | 1   0                             | No Rules <a href="#">Add</a>           | ACTION |
| > LDAP                 | None                 | TCP/389         | 1   0                             | No Rules <a href="#">Add</a>           | ACTION |
| > NTP                  | [8] SSP_Infra_NTP    | UDP/123         | 0   0                             | Incomplete Ruleset <a href="#">Add</a> | ACTION |

- 1. The infrastructure assets imported from the CSV file are listed under the “Infrastructure Group” column
- 2. Look at the “New Detected Servers” column to review servers discovered from flows
- 3. Use the **scroll bar** to continue viewing the infrastructure services

# View Discovered Infrastructure Assets from Flows

Policy Monitoring

Infrastructure Services   Environments   Applications

3 / 13 SERVICES AVAILABLE FOR UPDATE

Total Detected Servers ⓘ

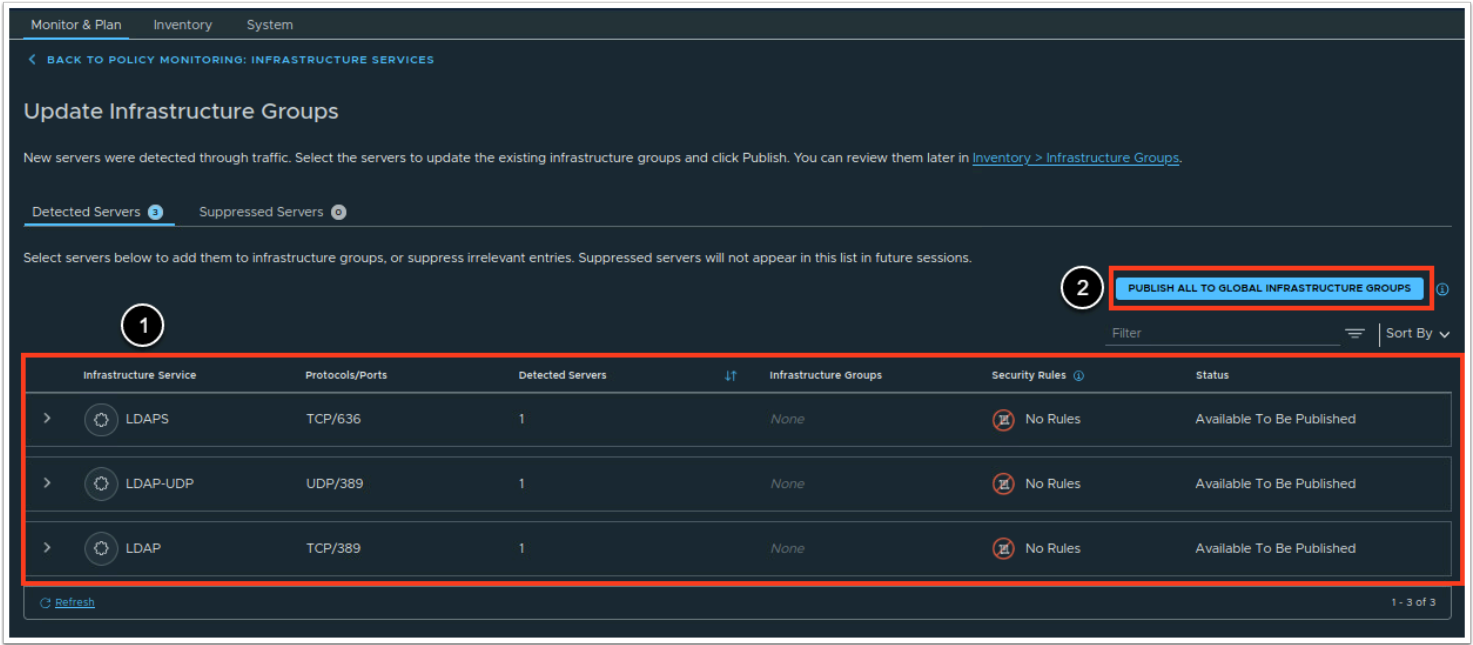
3

UPDATE INFRASTRUCTURE GROUPS

Last Refresh: 8/11/26, 9:42 AM

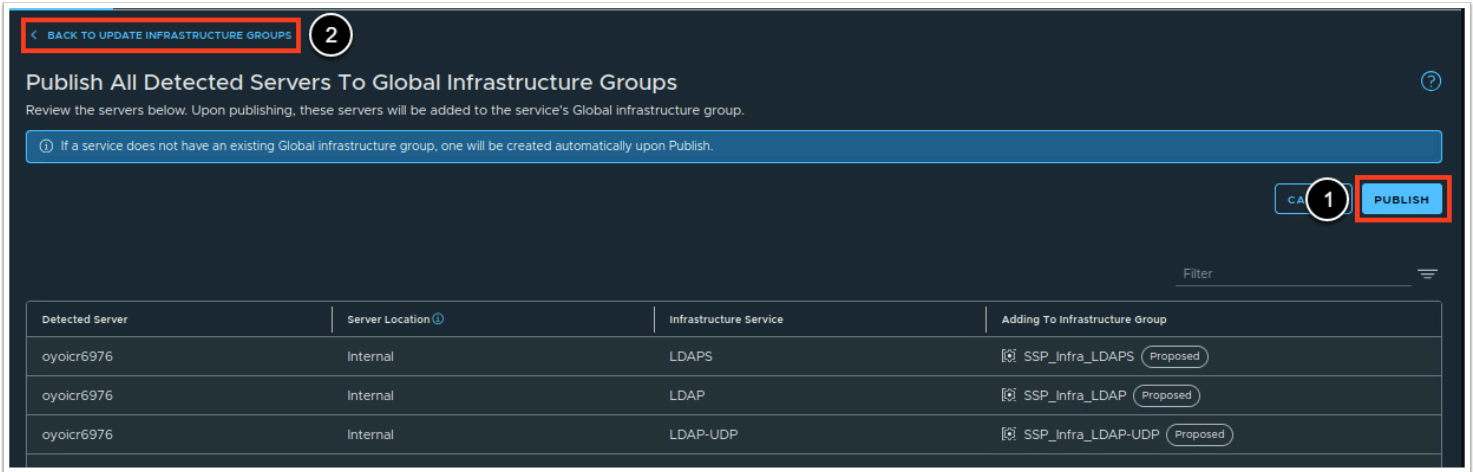
- 1. Click **UPDATE INFRASTRUCTURE GROUPS** in the Infrastructure Services Policy Overview tile to review servers discovered from flows

# Publish Infrastructure Assets Updates



1. View the available updates discovered through traffic flows
2. Click **PUBLISH ALL TO GLOBAL INFRASTRUCTURE GROUPS**

# Confirm Infrastructure Group Publish



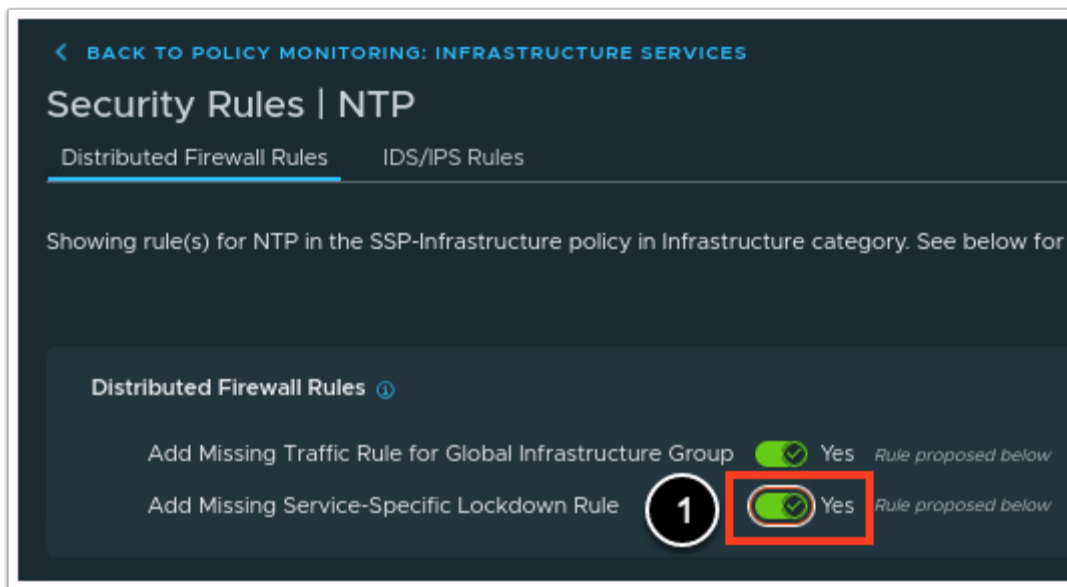
1. Click **PUBLISH**
2. Click **BACK TO POLICY MONITORING: INFRASTRUCTURE SERVICES** to return to the previous page

## View Recommended Infrastructure Service Rule

| Infrastructure Service | Infrastructure Group | Protocols/Ports | New Detected   Suppressed Servers | Security Rules                                                                                                                                  |
|------------------------|----------------------|-----------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| > NTP                  | SSP_Infra_NTP        | UDP/123         | 0   0                             | <div> <span>Incomplete</span> <span>1 Add</span> </div> <div> <span>2 Add Distributed Firewall Rule</span> <span>Add IDS/IPS Rule</span> </div> |
| > LDAPS                | SSP_Infra_LDAPS      | TCP/636         | 0   0                             |                                                                                                                                                 |

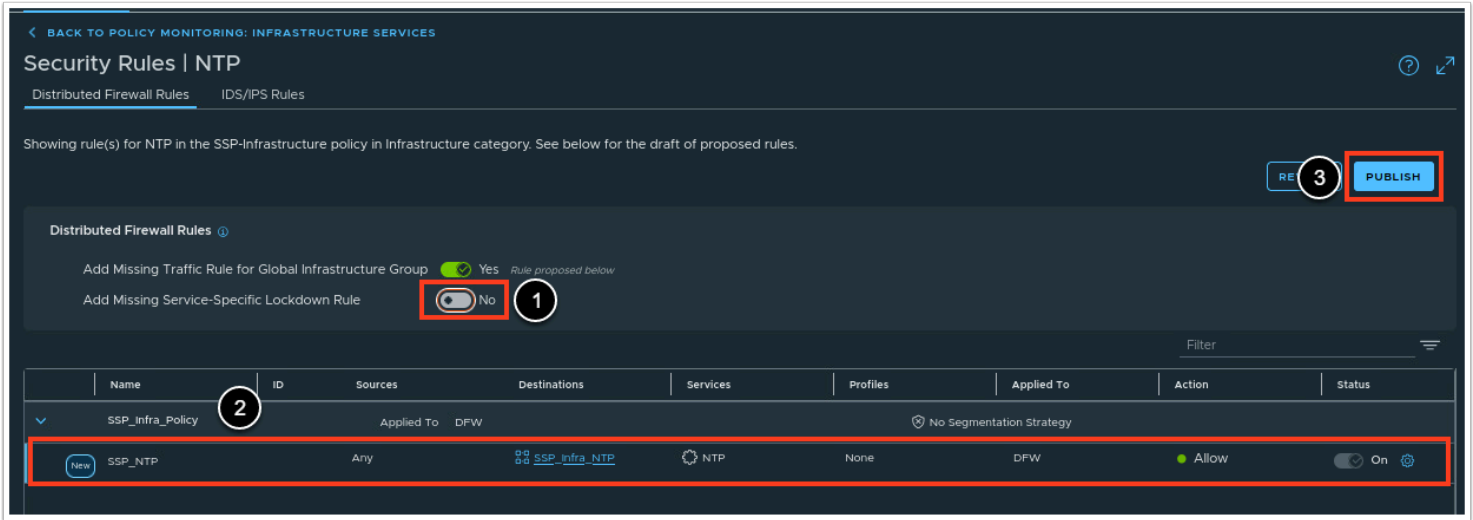
1. Click **Add** for the first Infrastructure Service in the list
2. Click **Add Distributed Firewall Rule**

## Deselect "Add Unpublished Lockdown Rule"



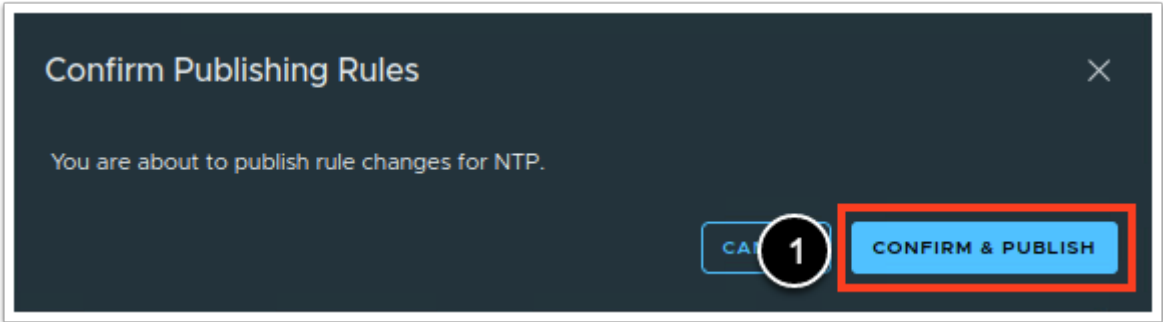
1. Click the **slider** to change "Add Missing Service-Specific Lockdown Rule" to No

# Publish Recommended Infrastructure Service Rules



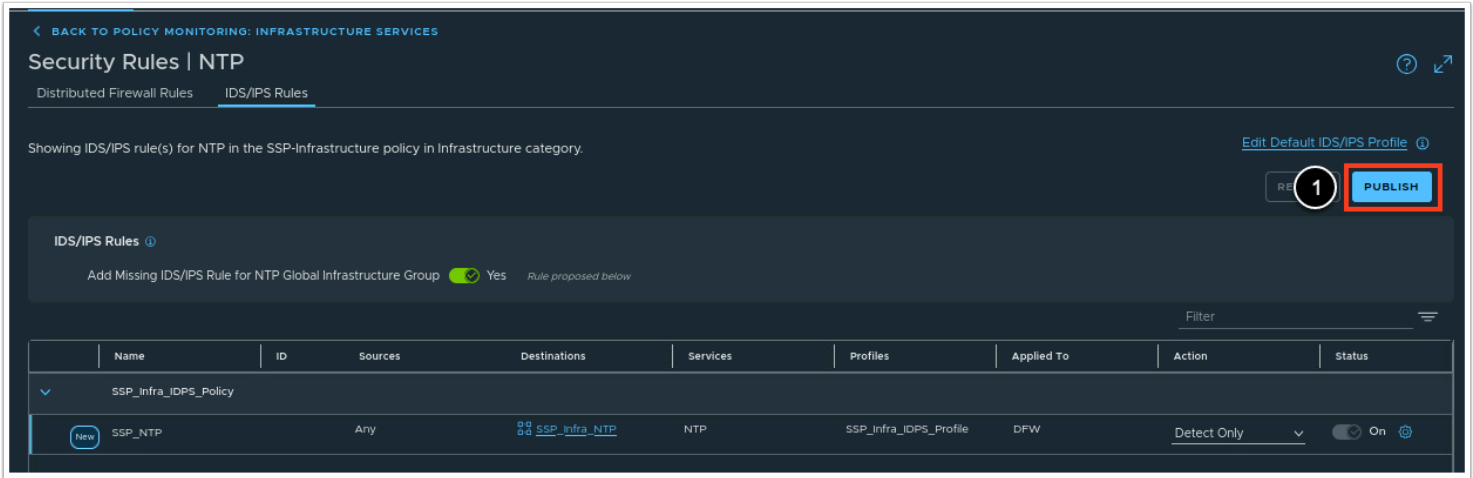
1. Confirm that "Add Missing Service-Specific Lockdown Rule" is set to "No"
2. View the recommended rule that will be published to the Distributed Firewall
3. Click **Publish** to publish the rule to the Distributed Firewall

# Confirm Publish of Distributed Firewall Rules



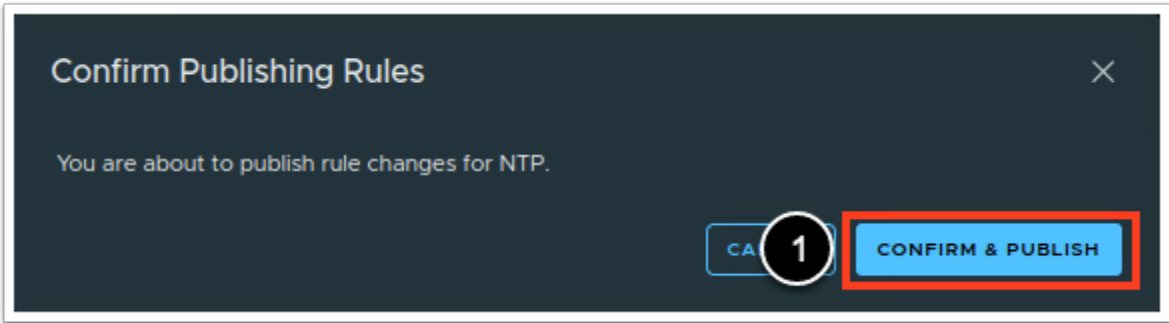
1. Click **CONFIRM & PUBLISH**

# View and Publish Recommended IDS/IPS Rules



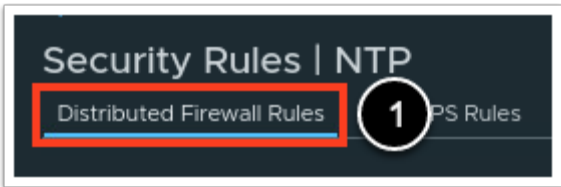
- 1. Click **IDS/IPS Rules** to view the recommended IDS/IPS rule for the Infrastructure Service
- 2. Click **PUBLISH** to publish the recommended IDS/IPS rule

## Confirm Publish of IDS/IPS Rules



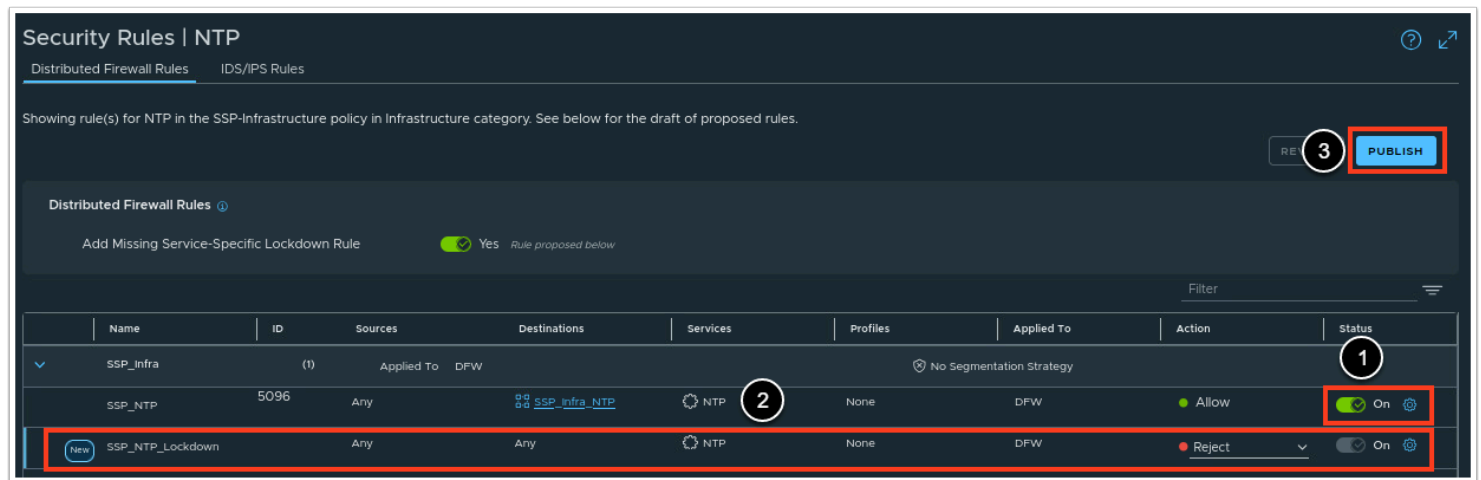
- 1. Click **CONFIRM & PUBLISH**

## Return to Distributed Firewall Rules



1. Click **Distributed Firewall Rules**

## View Lockdown Rule



Security Rules | NTP

Distributed Firewall Rules | IDS/IPS Rules

Showing rule(s) for NTP in the SSP-Infrastructure policy in Infrastructure category. See below for the draft of proposed rules.

RE 3 PUBLISH

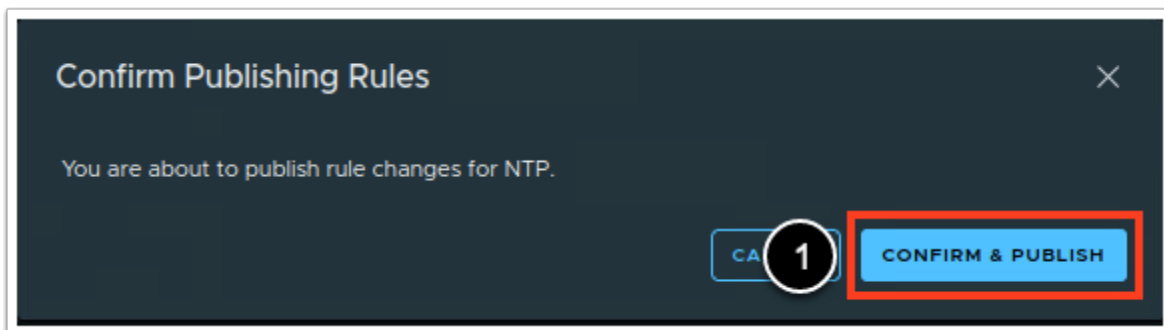
Distributed Firewall Rules

Add Missing Service-Specific Lockdown Rule ☒ Yes Rule proposed below

| Name                 | ID   | Sources    | Destinations  | Services | Profiles                 | Applied To | Action | Status |
|----------------------|------|------------|---------------|----------|--------------------------|------------|--------|--------|
| SSP_Infra            | (1)  | Applied To | DFW           |          | No Segmentation Strategy |            |        | 1      |
| SSP_NTP              | 5096 | Any        | SSP_Infra_NTP | NTP      | 2                        | DFW        | Allow  | On     |
| New SSP_NTP_Lockdown |      | Any        | Any           | NTP      |                          | DFW        | Reject | On     |

1. Under Security Rules, the traffic rule is published and activated
2. View the recommended Lockdown rule - note it is not published
3. Confirm that "Add Missing Service-Specific Lockdown Rule" is set to "Yes"
4. Click **PUBLISH**

## Confirm Publish of Distributed Firewall Lockdown Rule



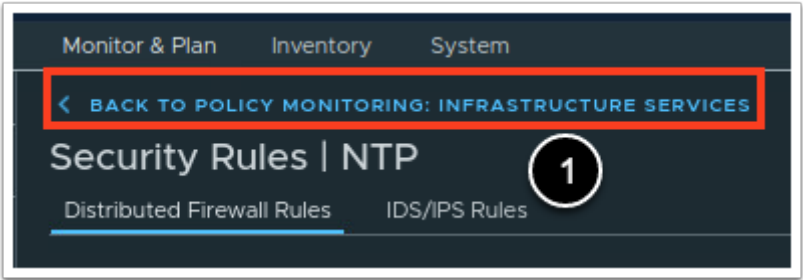
Confirm Publishing Rules

You are about to publish rule changes for NTP.

CANCEL 1 CONFIRM & PUBLISH

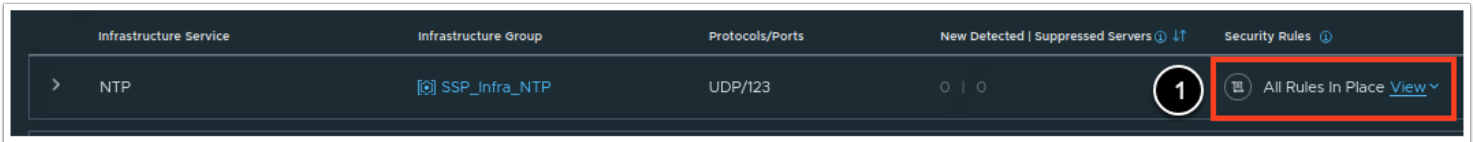
1. Click **CONFIRM & PUBLISH**

# Return to Policy Monitoring Page



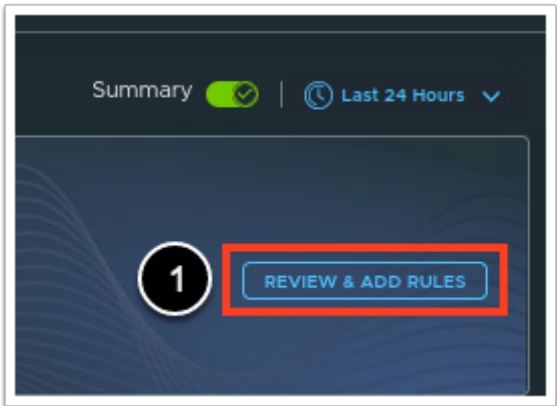
1. Click **BACK TO POLICY MONITORING: INFRASTRUCTURE SERVICES** to return to the Policy Monitoring page

# Validate Infrastructure Service Rules are Published



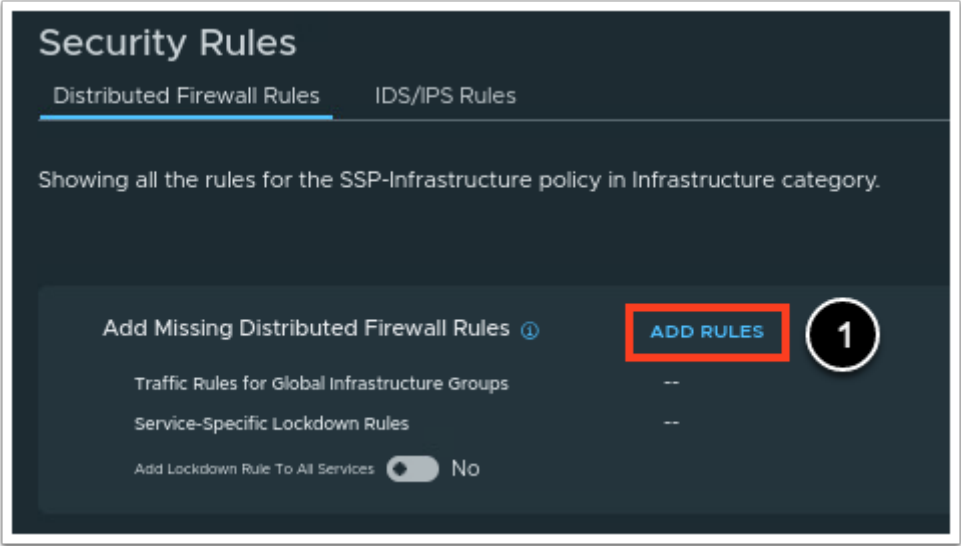
1. Now under Security Rules "All Rules in Place" shows for the NTP Infrastructure Service

# Review & Add Infrastructure Services Rules



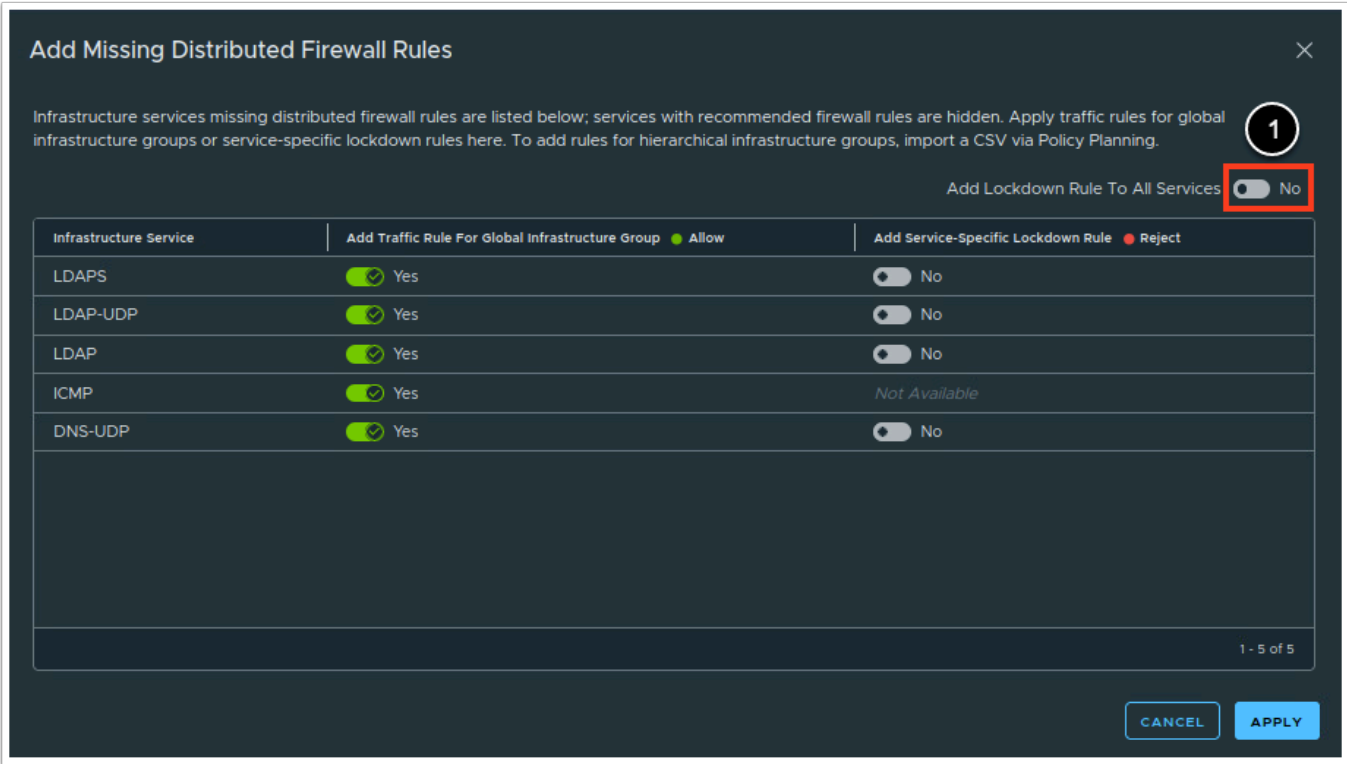
1. Click **REVIEW & ADD RULES** in the summary tile

# Add Missing Distributed Firewall Rules



- 1. Click **ADD RULES**

# Add Lockdown Rule to All Services



1. Click the **slider** to change "Add Lockdown Rule To All Services" to Yes

## Apply All Traffic and Lockdown Distributed Firewall Rules

**Add Missing Distributed Firewall Rules**

Infrastructure services missing distributed firewall rules are listed below; services with recommended firewall rules are hidden. Apply traffic rules for global infrastructure groups or service-specific lockdown rules here. To add rules for hierarchical infrastructure groups, import a CSV via Policy Planning.

| Infrastructure Service | Add Traffic Rule For Global Infrastructure Group <span>● Allow</span> | Add Lockdown Rule To All Services <span>● Yes</span> | Add Service-Specific Lockdown Rule <span>● Reject</span> |
|------------------------|-----------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| LDAPS                  | <span>✓ Yes</span>                                                    | <span>✓ Yes</span>                                   | <span>✓ Yes</span>                                       |
| LDAP-UDP               | <span>✓ Yes</span>                                                    | <span>✓ Yes</span>                                   | <span>✓ Yes</span>                                       |
| LDAP                   | <span>✓ Yes</span>                                                    | <span>✓ Yes</span>                                   | <span>✓ Yes</span>                                       |
| ICMP                   | <span>✓ Yes</span>                                                    | <span>✓ Yes</span>                                   | <i>Not Available</i>                                     |
| DNS-UDP                | <span>✓ Yes</span>                                                    | <span>✓ Yes</span>                                   | <span>✓ Yes</span>                                       |

1 - 5 of 5

**APPLY**

1. Confirm that Lockdown Rule sliders are set to **Yes**
2. Click **APPLY**

## Confirm Applying Distributed Firewall Rules

**Confirm Applying Rules**

You are about to add Distributed Firewall rules for infrastructure services. This will add:

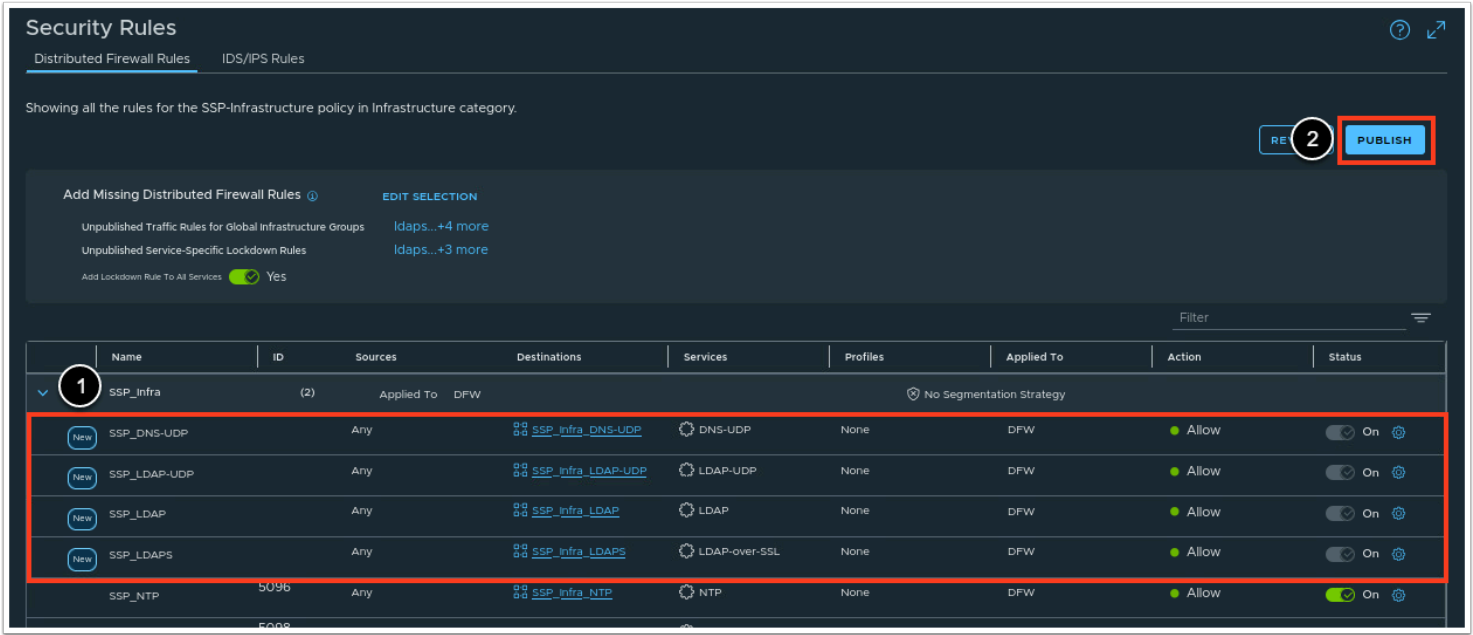
- Traffic rules 'allow' for 5 services.
- Lockdown rules 'reject' for 4 services.

Once added to the Infrastructure policy table, you can proceed to publish the rules.

**CONFIRM & APPLY**

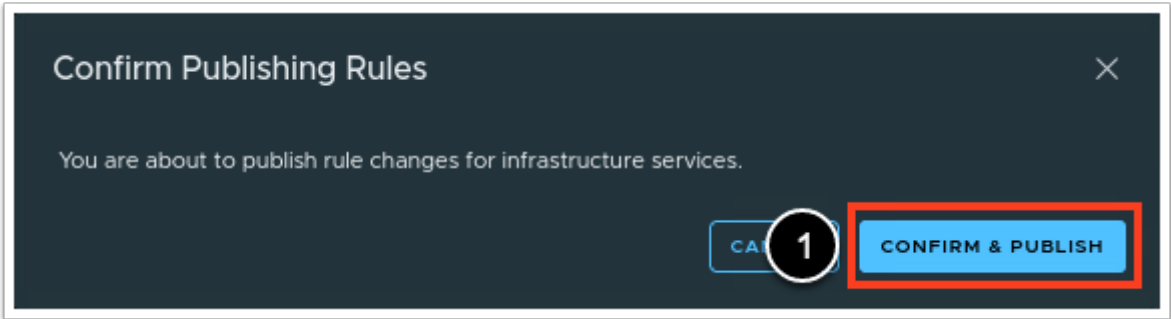
1. Click **CONFIRM & APPLY**

# View and Publish All Distributed Firewall Rules



- 1. The remaining rules are added to the Distributed Firewall Rules list
- 2. Click **PUBLISH** to apply these rules to the Distributed Firewall

# Confirm Publishing Distributed Firewall Rules



- 1. Click **CONFIRM & PUBLISH** to apply the rules to the Distributed Firewall

## Add Missing IDS/IPS Rules



1. Click **IDS/IPS Rules**
2. Click **ADD RULES**

## Apply All IDS/IPS Rules

**Add Missing IDS/IPS Rules**

Infrastructure services missing IDS/IPS rules for global infrastructure groups are listed below. Apply IDS/IPS rules for global infrastructure groups here. To add rules for hierarchical infrastructure groups, import a CSV via Policy Planning.

| Infrastructure Service | Add IDS/IPS Rules For Global Infrastructure Group |
|------------------------|---------------------------------------------------|
| LDAPS                  | <input checked="" type="checkbox"/> Yes           |
| LDAP-UDP               | <input checked="" type="checkbox"/> Yes           |
| LDAP                   | <input checked="" type="checkbox"/> Yes           |
| DNS-UDP                | <input checked="" type="checkbox"/> Yes           |

1 - 4 of 4

**APPLY**

1. Confirm sliders for all Infrastructure Services are set to Yes
2. Click **APPLY**

## Confirm Applying IDS/IPS Rules

**Confirm Applying Rules**

You are about to add IDS/IPS rules for the global infrastructure groups. This will add:

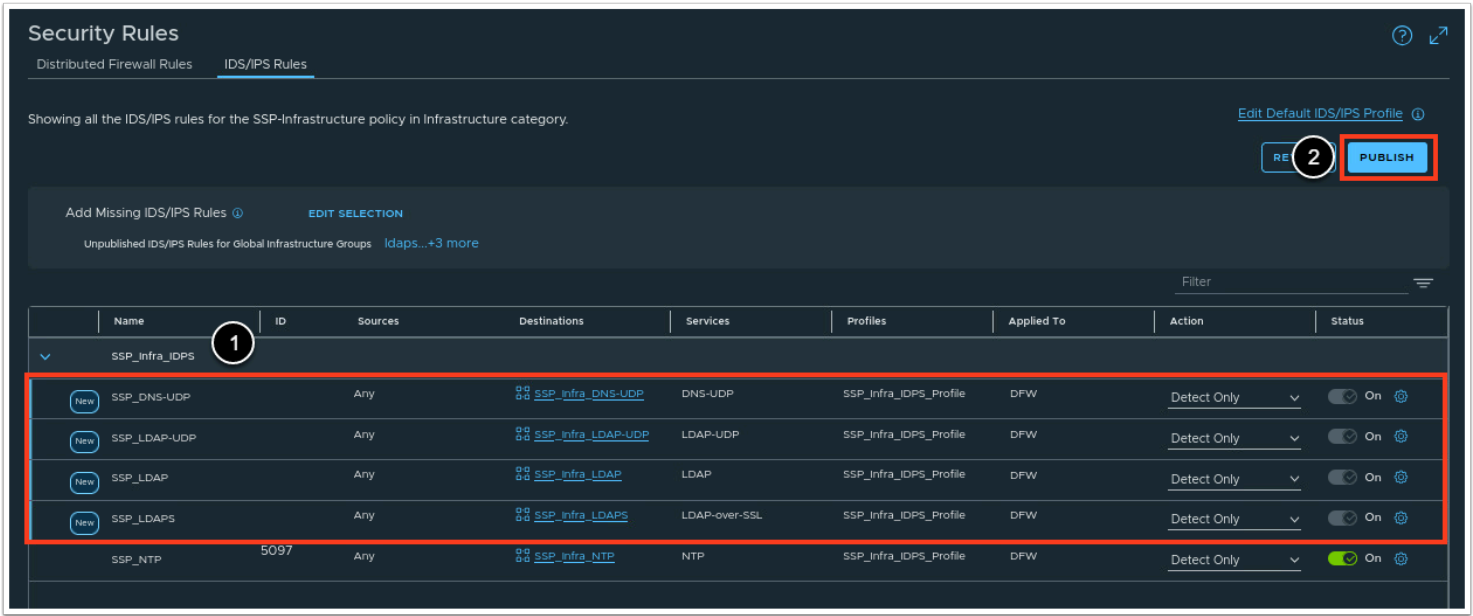
- IDS/IPS rules for 4 services.

Once added to the Infrastructure policy table, you can proceed to publish the rules.

**CONFIRM & APPLY**

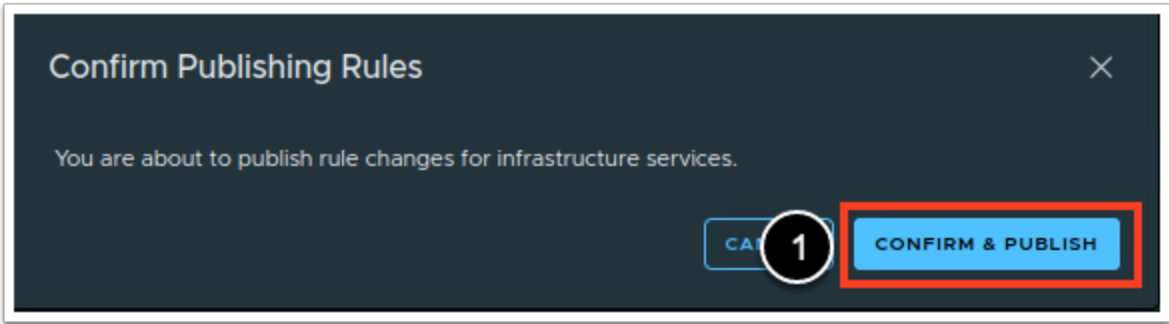
1. Click **CONFIRM & APPLY**

# View and Publish All IDS/IPS Rules



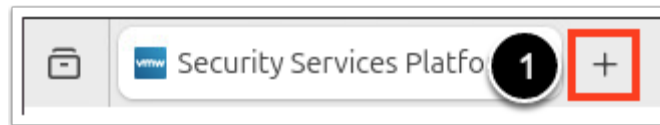
1. The remaining rules are added to the IDS/IPS Rules list
2. Click **PUBLISH** to apply these rules to the IDS/IPS

## Confirm Publishing IDS/IPS Rules



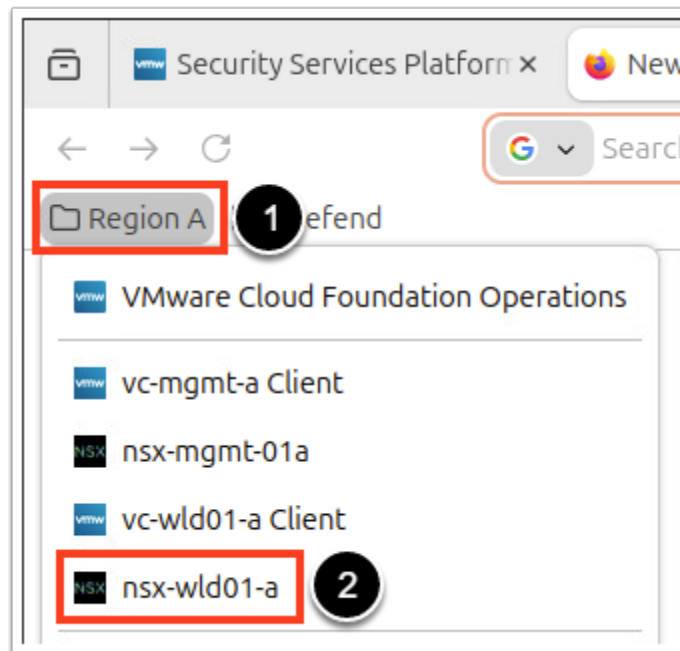
1. Click **CONFIRM & PUBLISH** to apply the rules to the IDS/IPS

# Login to NSX



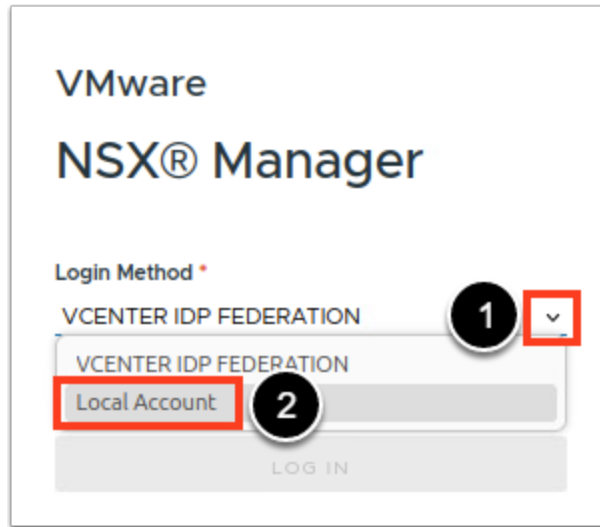
1. Click the + at the top of the browser window to open a new tab

## Open NSX Manager



1. Click on the **Region A** bookmark bar folder
2. Click **nsx-wld01-a** to launch NSX Manager

## Select Login Method



VMware  
NSX® Manager

Login Method \*

VCENTER IDP FEDERATION

VCENTER IDP FEDERATION

Local Account

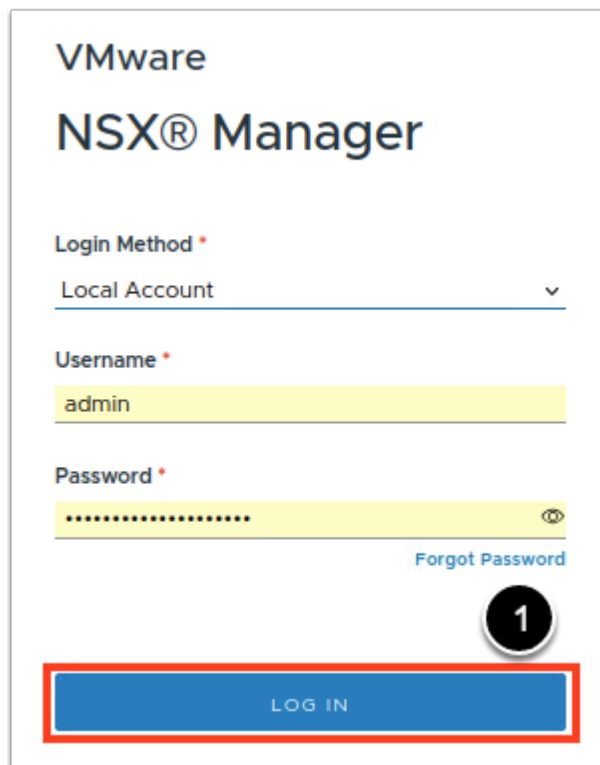
LOG IN

1

2

1. Click the **arrow** to expand the drop-down menu
2. Click **Local Account**

## Login to NSX Manager



VMware  
NSX® Manager

Login Method \*

Local Account

Username \*

admin

Password \*

.....

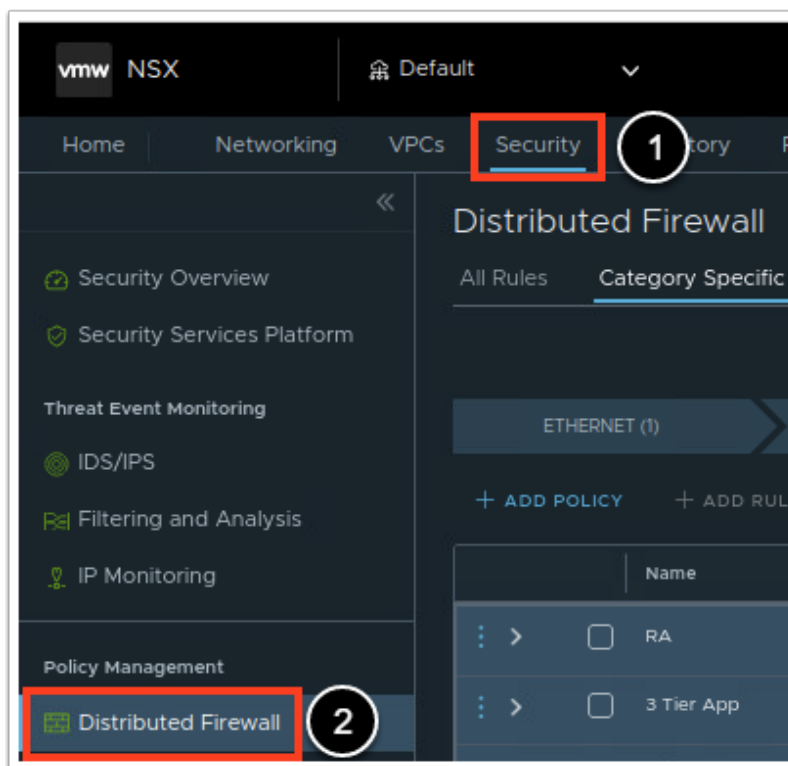
Forgot Password

LOG IN

1

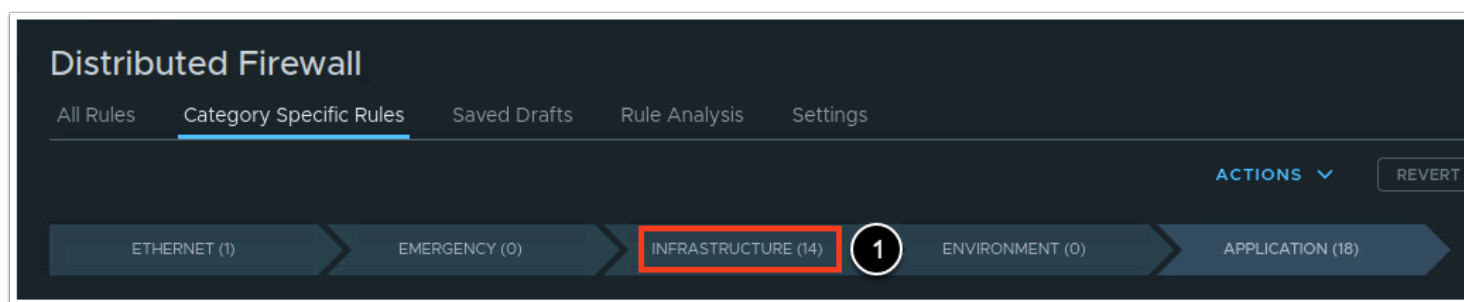
- 
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Validate Published Infrastructure Services Rules in NSX



1. Click **Security**
2. Click **Distributed Firewall**

## View Infrastructure Section of Distributed Firewall



1. Click **INFRASTRUCTURE** to navigate to the Infrastructure section of the DFW

# View Published Infrastructure Services Rules

ETHERNET (1)

EMERGENCY (0)

INFRASTRUCTURE (14)

ENVIRONMENT (0)

APPLICATION (18)

+ ADD POLICY

+ ADD RULE

CLONE

DELETE

UNDO

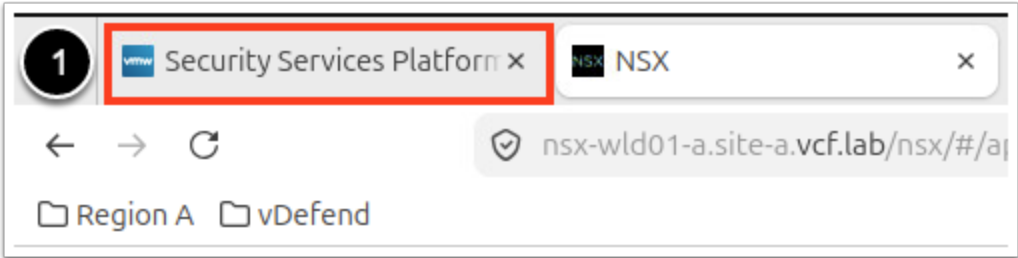
...

Filter by Name, Path and more

|   | Name            | ID   | Sources      | Destinations      | Services    | Context Profiles | Applied To | Action  |  |
|---|-----------------|------|--------------|-------------------|-------------|------------------|------------|---------|--|
| 1 | Default ... (3) |      | Applied: DFW |                   |             |                  | Stateful   | Success |  |
| 2 | SSP_Infra (11)  |      | Applied: DFW | App Strategy: N/A |             |                  | Stateful   | Success |  |
|   | SSP_DNS-UDP     | 4076 | Any          | SSP_Infra...      | DNS-UDP     | None             | DFW        | Allow   |  |
|   | SSP_LDAP        | 4079 | Any          | SSP_Infra...      | LDAP        | None             | DFW        | Allow   |  |
|   | SSP_LDAP-U...   | 4080 | Any          | SSP_Infra...      | LDAP-UDP    | None             | DFW        | Allow   |  |
|   | SSP_LDAPS       | 4082 | Any          | SSP_Infra...      | LDAP-ove... | None             | DFW        | Allow   |  |
|   | SSP_NTP         | 4085 | Any          | SSP_Infra...      | NTP         | None             | DFW        | Allow   |  |

1. Click the **arrow** to the left of the "SSP Infra" policy to expand the policy ruleset
2. View the published rules pushed from SSP
3. Use the scroll bar to view all of the rules in the ruleset

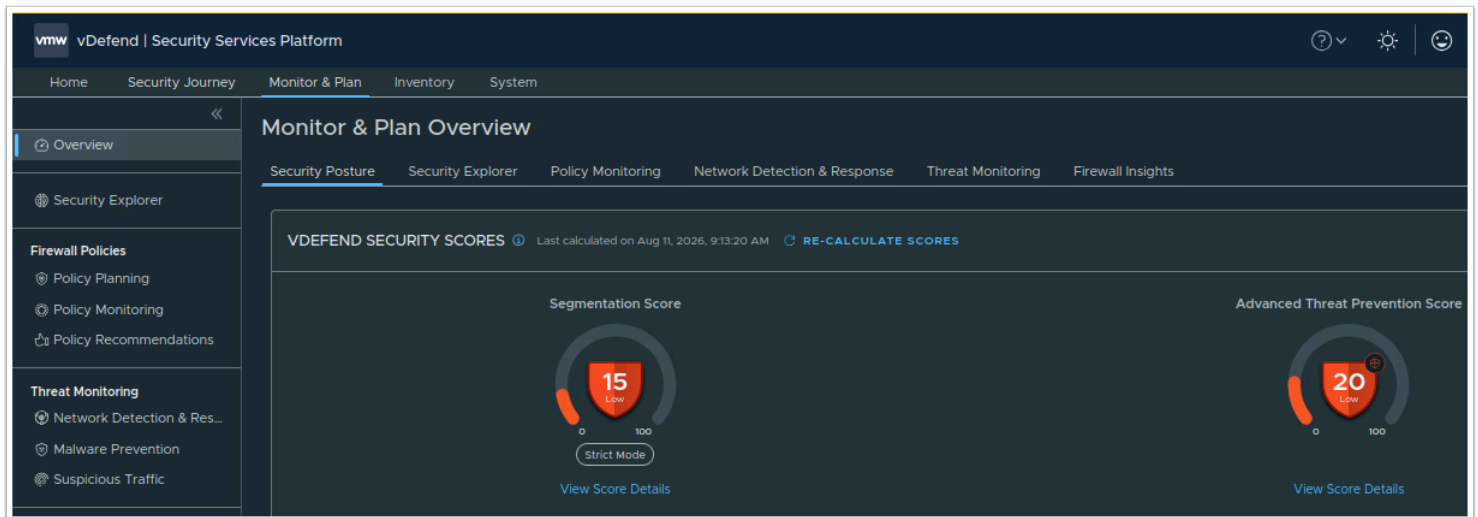
# Re-calculate Security Segmentation Score



1. Click on the Security Services Platform tab in the browser to return to Security Services Platform

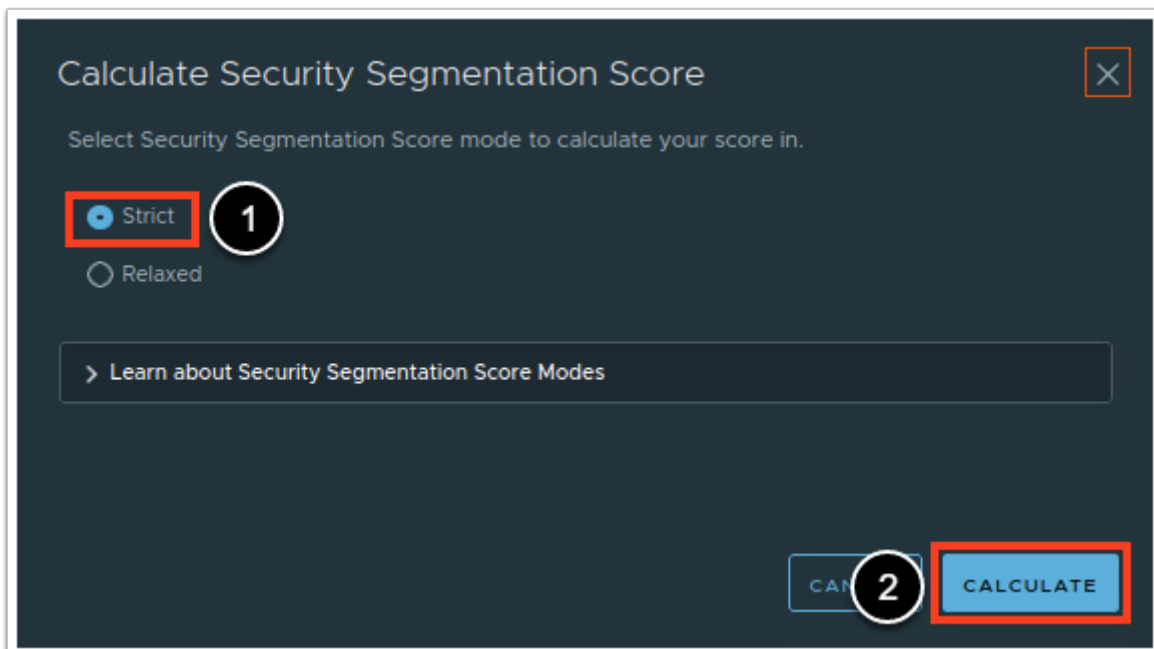
## Re-calculate Security Segmentation Score





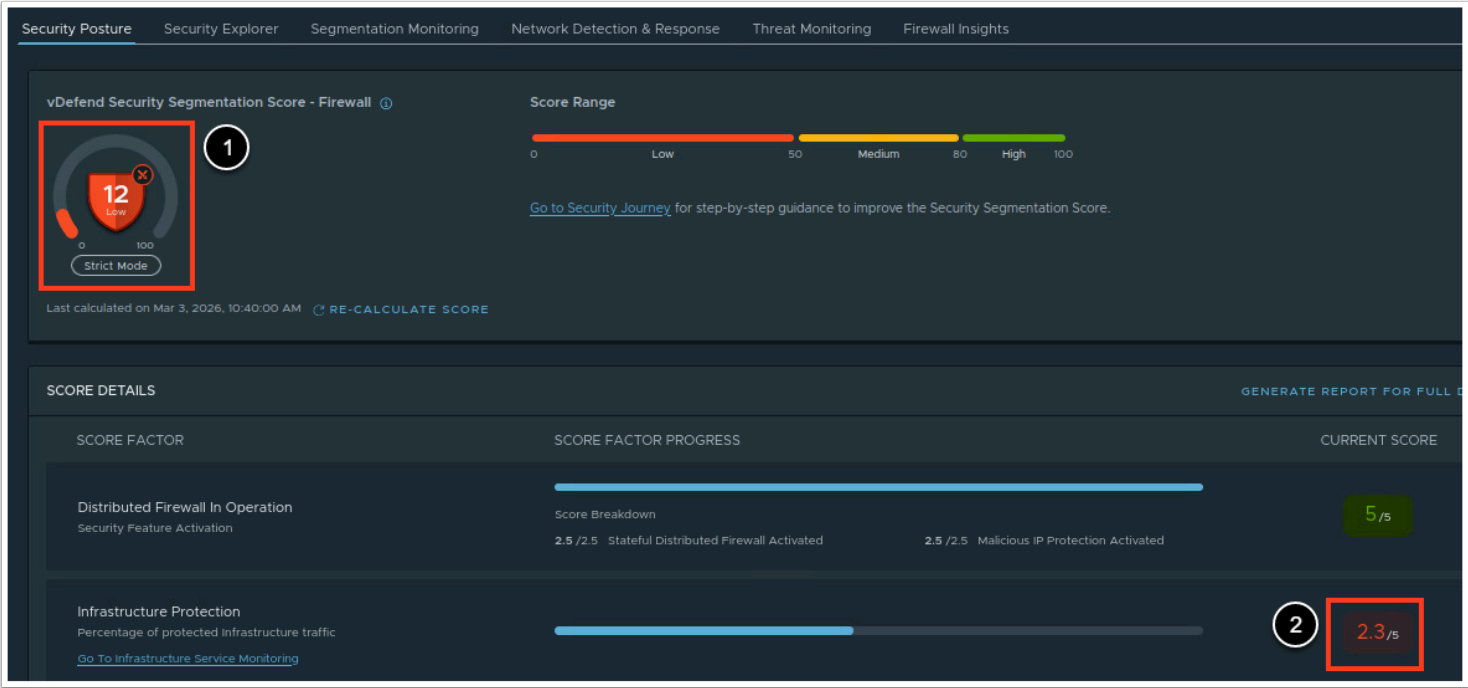
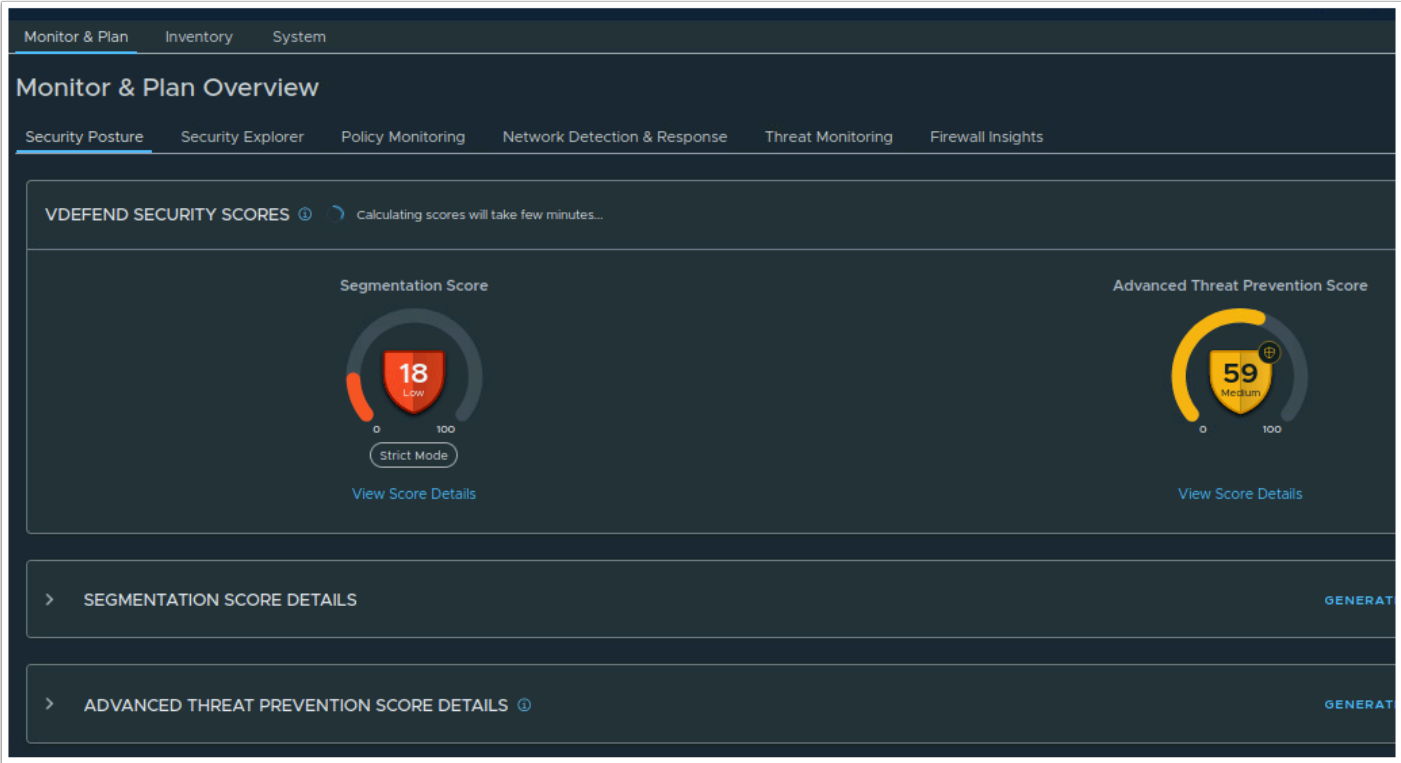
1. Click **Monitor & Plan**
2. Click **Overview**
3. Click **RE-CALCULATE SCORE**

## Choose Security Segmentation Score Calculation Mode and Calculate



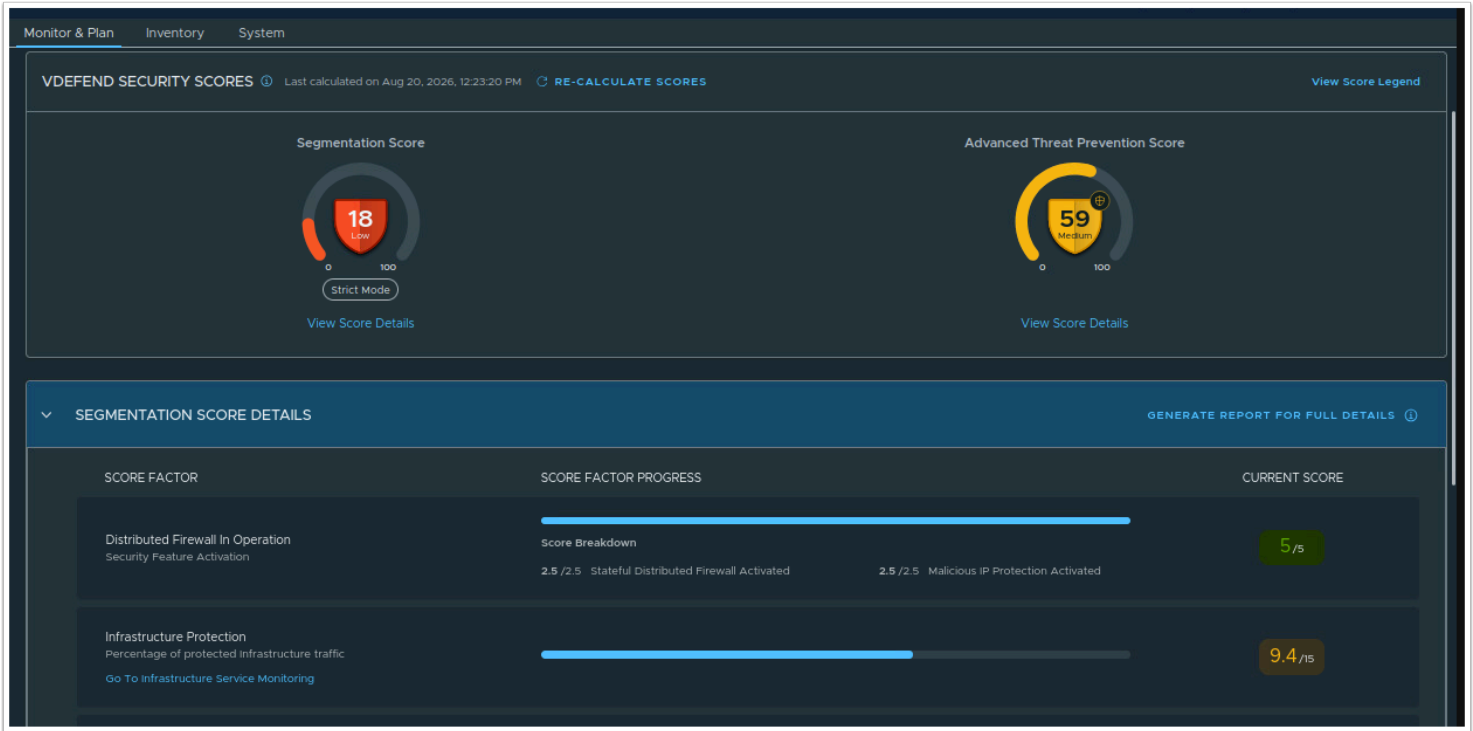
1. Confirm **Strict** mode is selected
2. Click **Calculate**

# View Updated Security Score



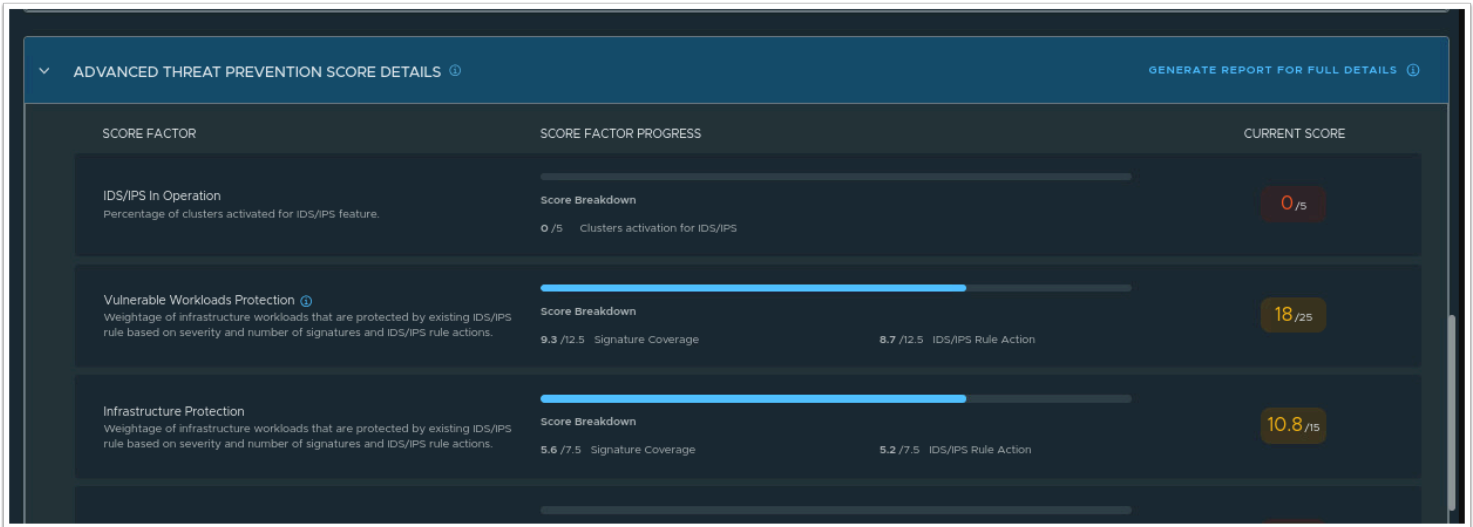
1. The Security Segmentation Score has increased
2. The Infrastructure Protection Score has increased

# View Segmentation Score Details



1. Click the arrow next to SEGMENTATION SCORE DETAILS
2. Use the scroll bar if needed to see the details
3. The Infrastructure Protection score has increased

# View Advanced Threat Prevention Score Details



- 
1. Click the arrow next to ADVANCED THREAT PREVENTION SCORE DETAILS
  2. Use the scroll bar if needed to see the details
  3. The Infrastructure Protection score has increased

# Conclusion

In this module, you updated the infrastructure assets discovered via traffic flows in the Security Services Platform (SSP). You then published recommended infrastructure security rules directly to the NSX Distributed Firewall and recalculated your Security Segmentation Score to reflect your improved posture.

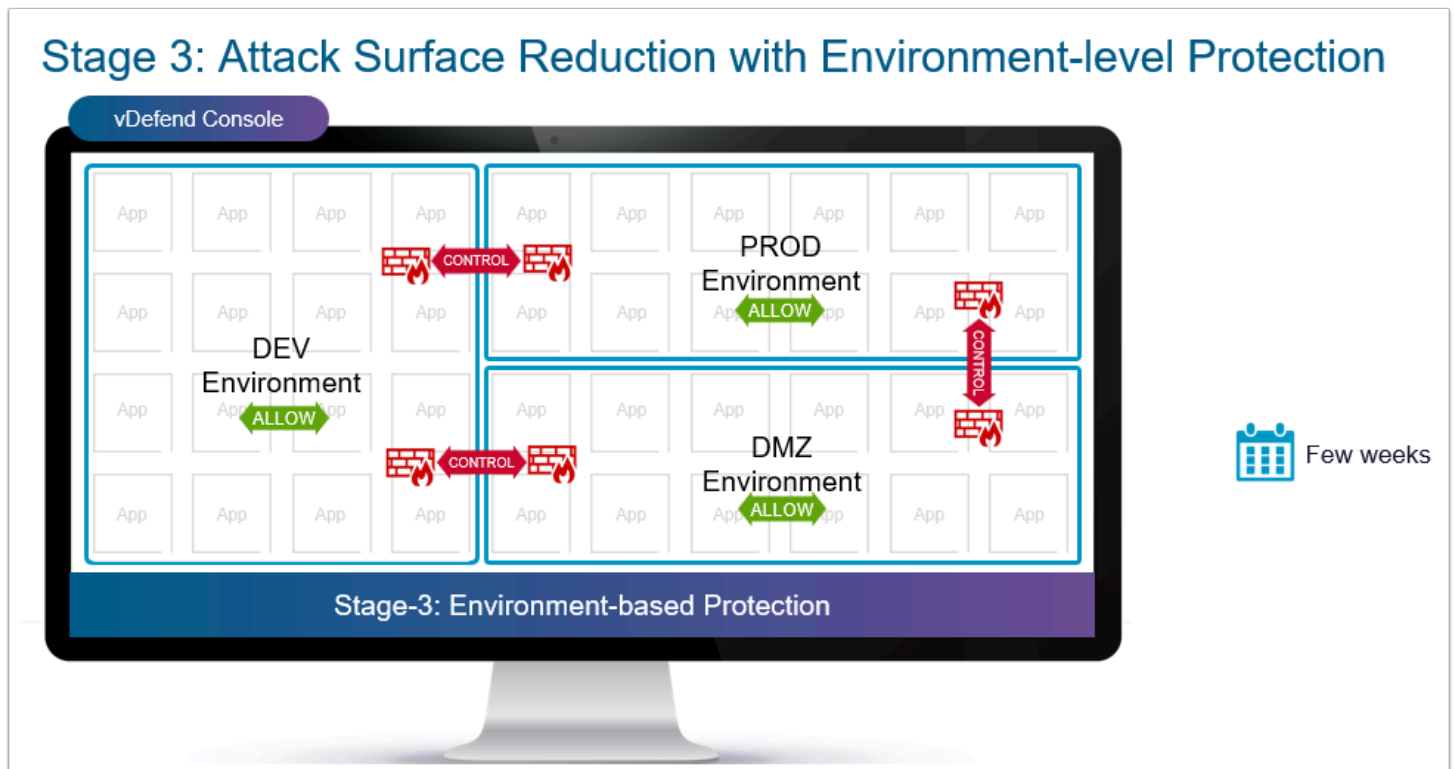
**You've Finished Stage 2!** Congratulations on completing Module 5!

In the next stage, **Environment (Zone) Segmentation**, you will learn how to define and enforce security rules across different workload environment zones.



# **Module 6 – Stage 3: Environment (Zone) Segmentation**

# Environment (Zone) Segmentation Introduction



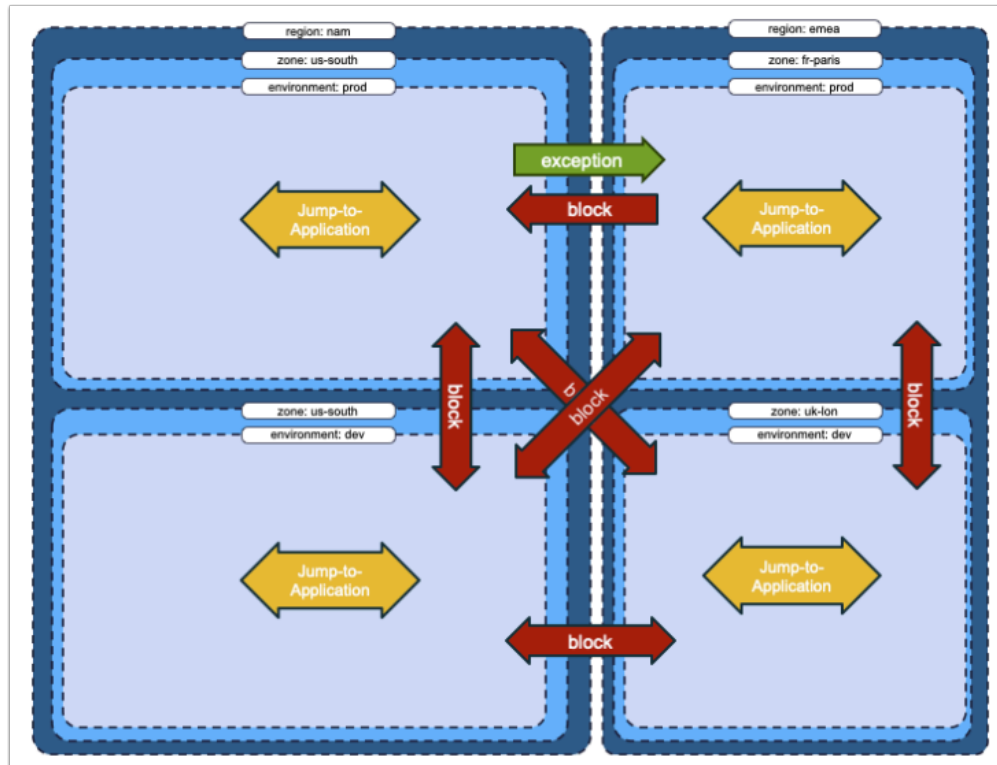
Once infrastructure (shared) services are protected, users can define environment (zone) boundaries (e.g., Development and Production).

## Key Capabilities

- **Metadata Import:** Import workload metadata via CSV files exported from CMDB systems (e.g., ServiceNow), vCenter, or a spreadsheet template provided by DFW 1-2-3-4.
- **Tagging & Policies:** Automatically assign security tags, validate workload relationships, and enforce default environment-level rules using DFW.
- **Leakage Monitoring:** Continuously track cross-zone traffic leakages, generate recommended rule exceptions, and receive real-time alerts for newly discovered exposure risks.

This phase ensures environments remain isolated, minimizing cross-environment exposure and strengthening overall security posture.

## Stage 3 Objective



The aim of this stage is to secure the communications between environments by allowing only required traffic as exceptions, and by blocking everything else. This isolation between security zones immediately provides a better security posture by reducing threats blast radius without the need for a deep application traffic knowledge.

We will proceed to Stage 3 following these steps:

1. Generate rules for controlling intra-zone traffic. These rules have a specific action: "Jump to Application", which creates a dynamic policy chain, allowing for the delegation of traffic inspection and enforcement. So the traffic that matches these rules in the Environment category policy will continue to be inspected by the Application category policies and rules.
2. Monitor the traffic between environments
3. For each environment pair (i.e: from Prod to Dev), decide if some exception should be created
4. Lockdown environment pairs

# Login to Security Services Platform (SSP)

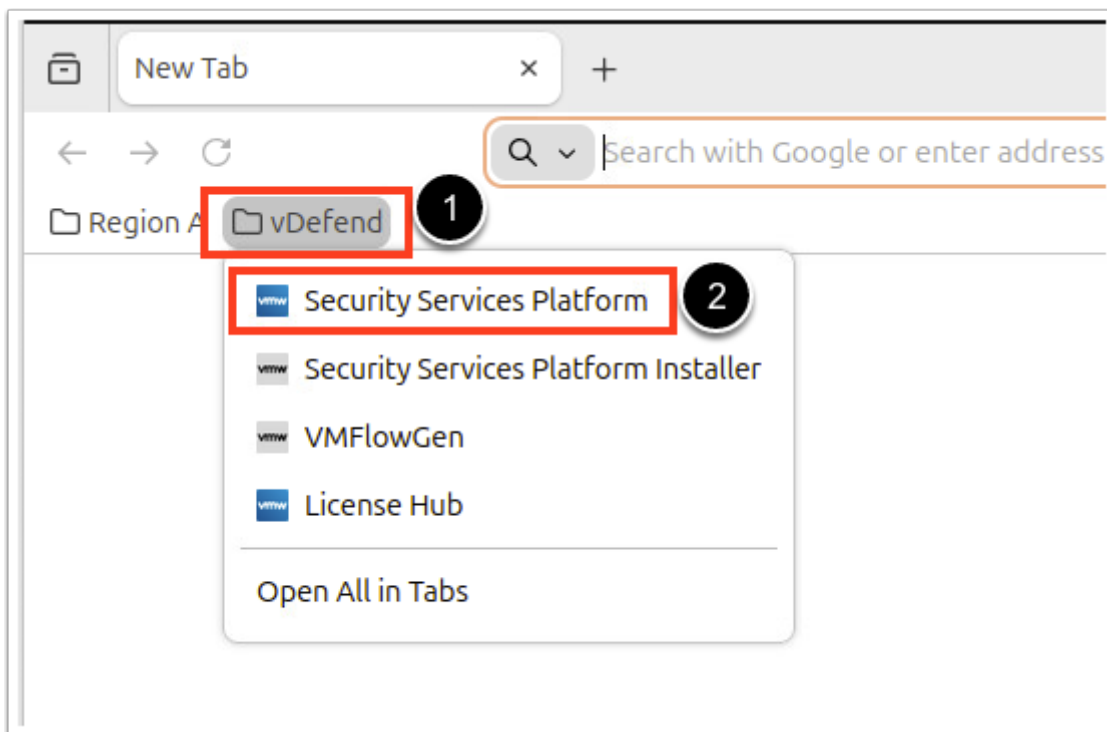
Log into the vDefend Security Services Platform (SSP) console to begin configuring and monitoring environment segmentation policies for this stage of the lab.

## Start Firefox



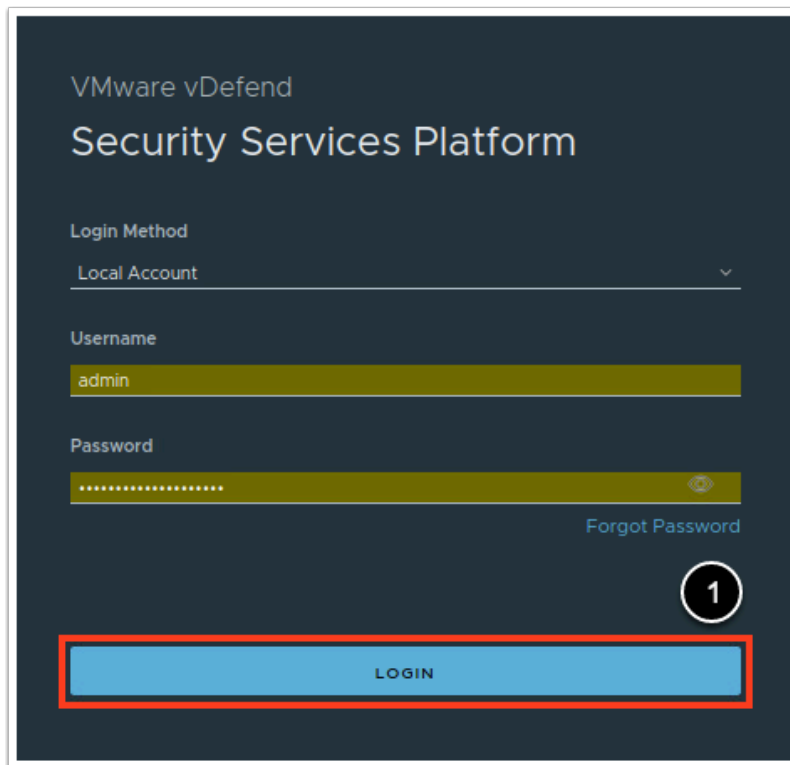
1. Click the **Firefox icon** in the taskbar

## Open Security Services Platform Console



1. Click on the **vDefend** bookmark bar folder
2. Click **Security Services Platform** to launch **SSP**

## Login to Security Services Platform Console



VMware vDefend  
Security Services Platform

Login Method  
Local Account

Username  
admin

Password  
.....

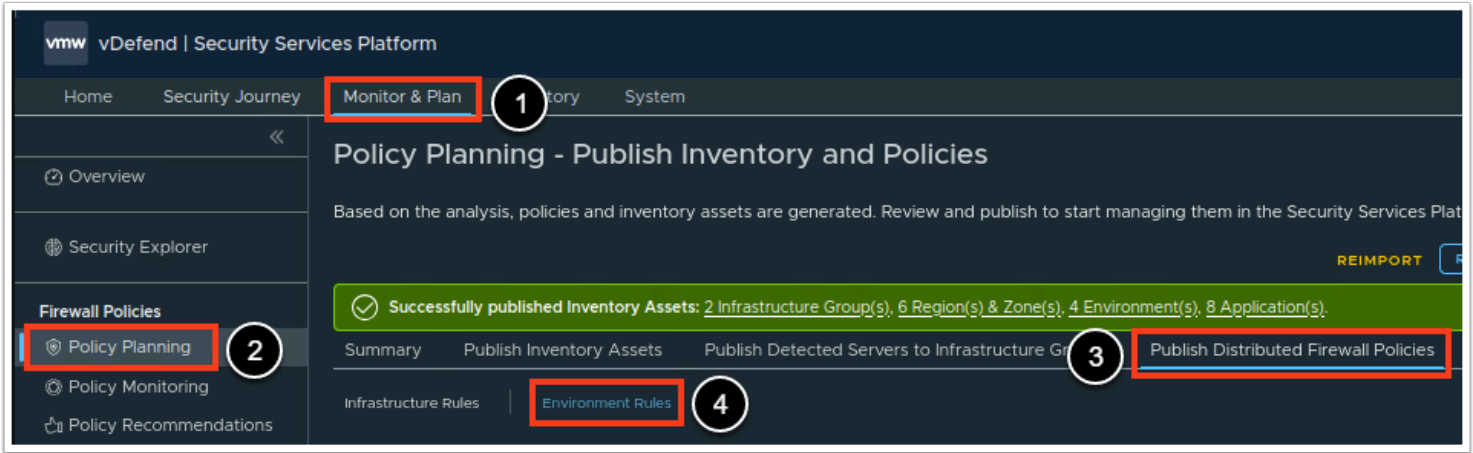
[Forgot Password](#)

1

LOGIN

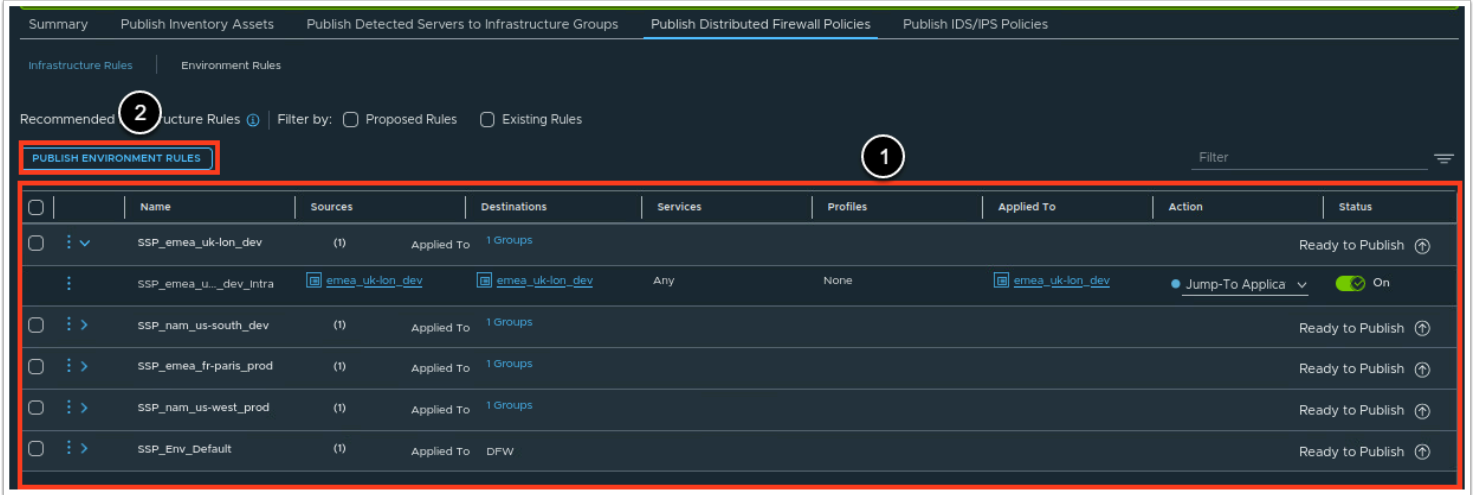
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Generate and Publish Default Environment Policies



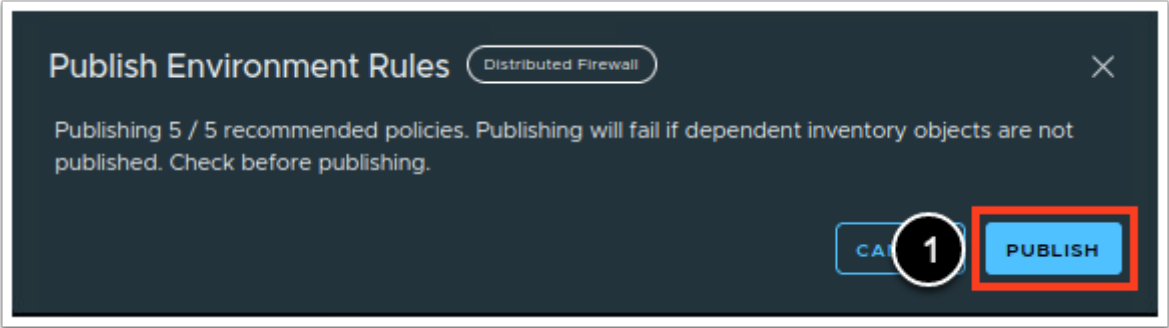
1. Click **Monitor & Plan**
2. Click **Policy Planning**
3. Click **Publish Distributed Firewall Policies**
4. Click **Environment Rules**

## View and Publish Default Environment Distributed Firewall Policies



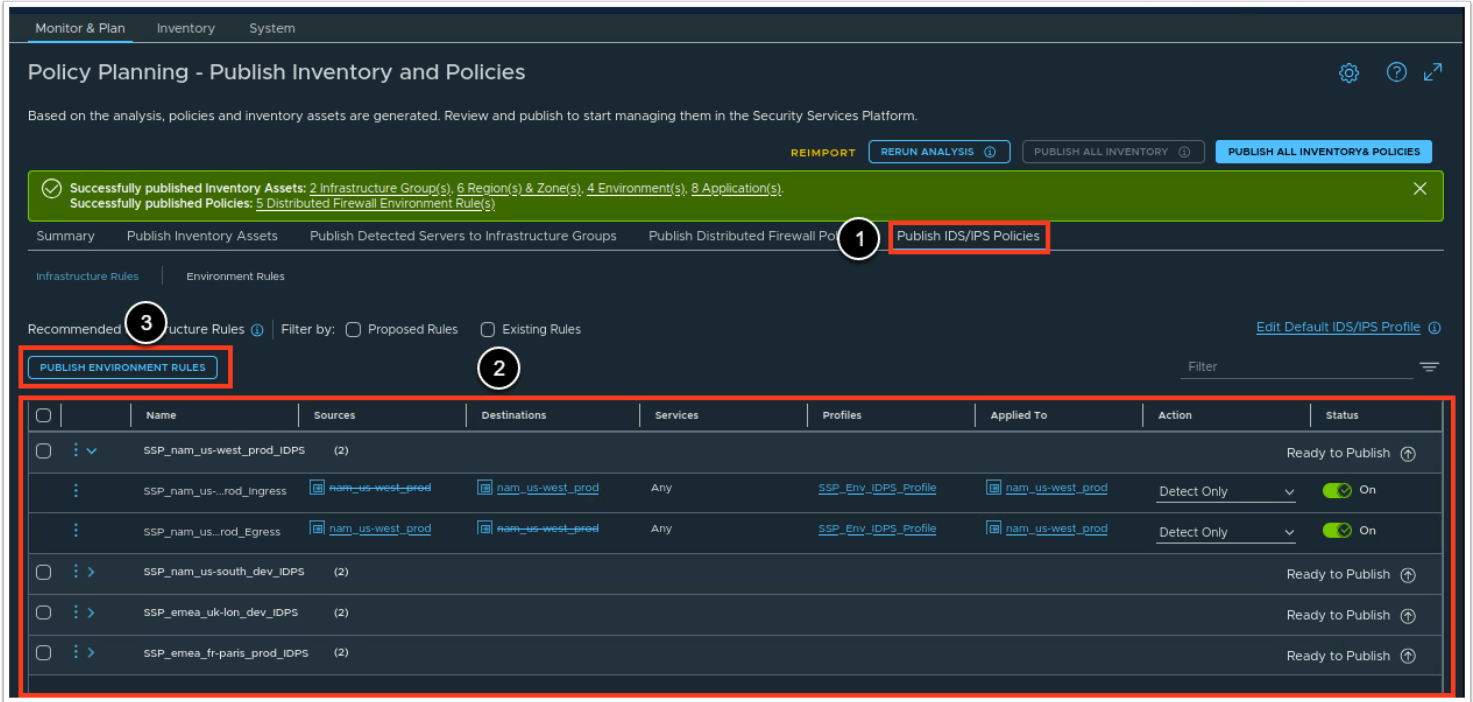
1. View the recommended environment rules
2. Click **PUBLISH ENVIRONMENT RULES**

# Confirm Environment Rule Publish



1. Click **PUBLISH**

# View and Publish Default Environment IDS/IPS Policies



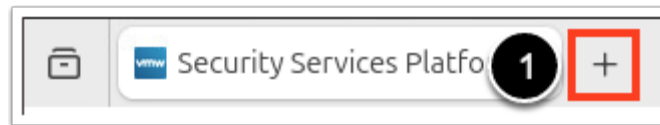
1. Click **Publish IDS/IPS Policies**
2. View the recommended environment IDS/IPS rules
3. Click **PUBLISH ENVIRONMENT RULES**

## Confirm Environment Rule Publish



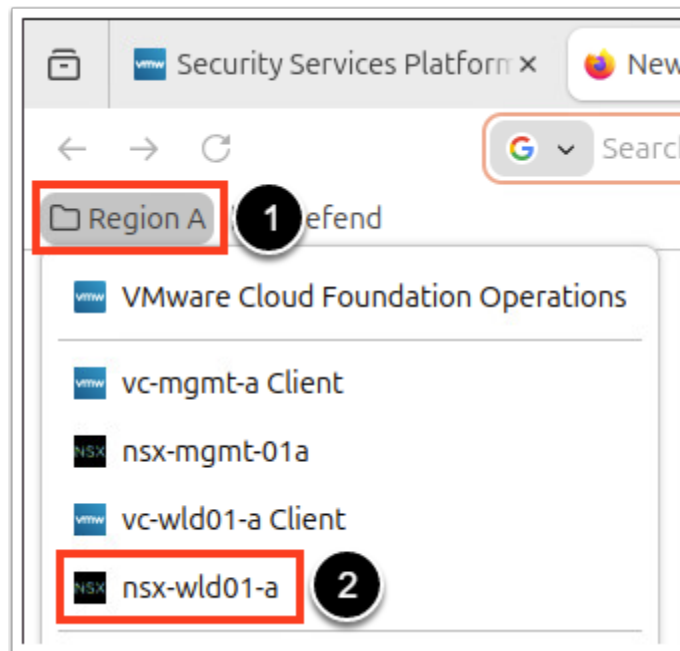
1. Click **PUBLISH**

# Login to NSX



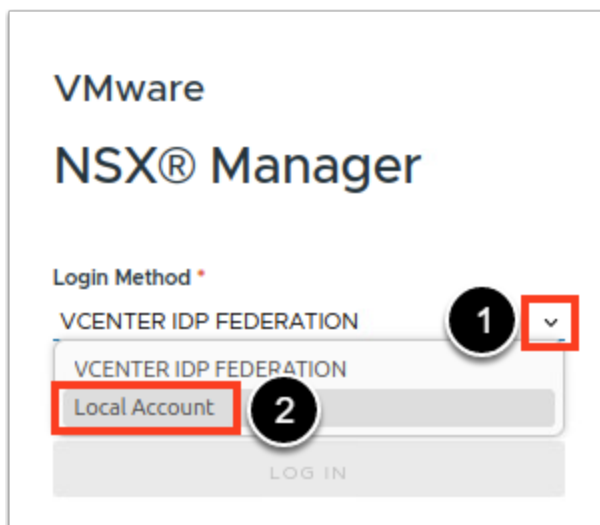
1. Click the + at the top of the browser window to open a new tab

## Open NSX Manager



1. Click on the **Region A** bookmark bar folder
2. Click **nsx-wld01-a** to launch NSX Manager

## Select Login Method



VMware  
NSX® Manager

Login Method \*

VCENTER IDP FEDERATION

VCENTER IDP FEDERATION

Local Account

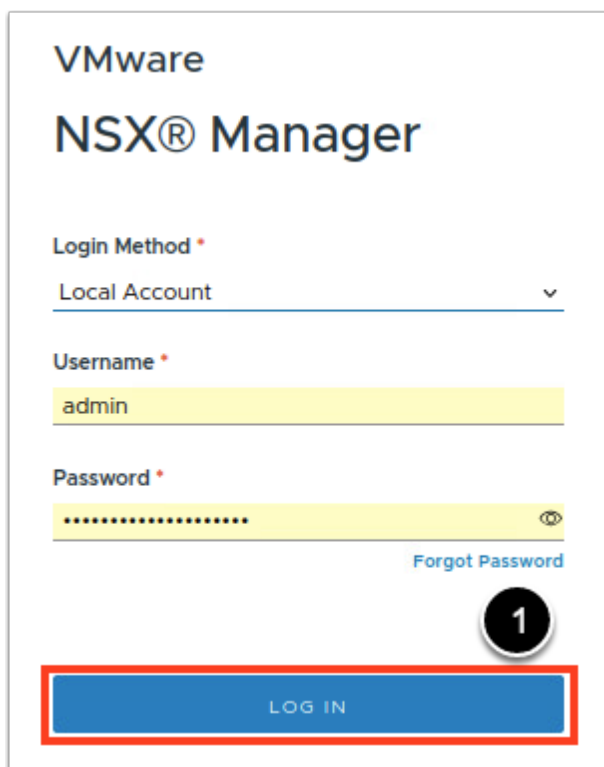
LOG IN

1

2

1. Click the **arrow** to expand the drop-down menu
2. Click **Local Account**

## Login to NSX Manager



VMware  
NSX® Manager

Login Method \*

Local Account

Username \*

admin

Password \*

.....

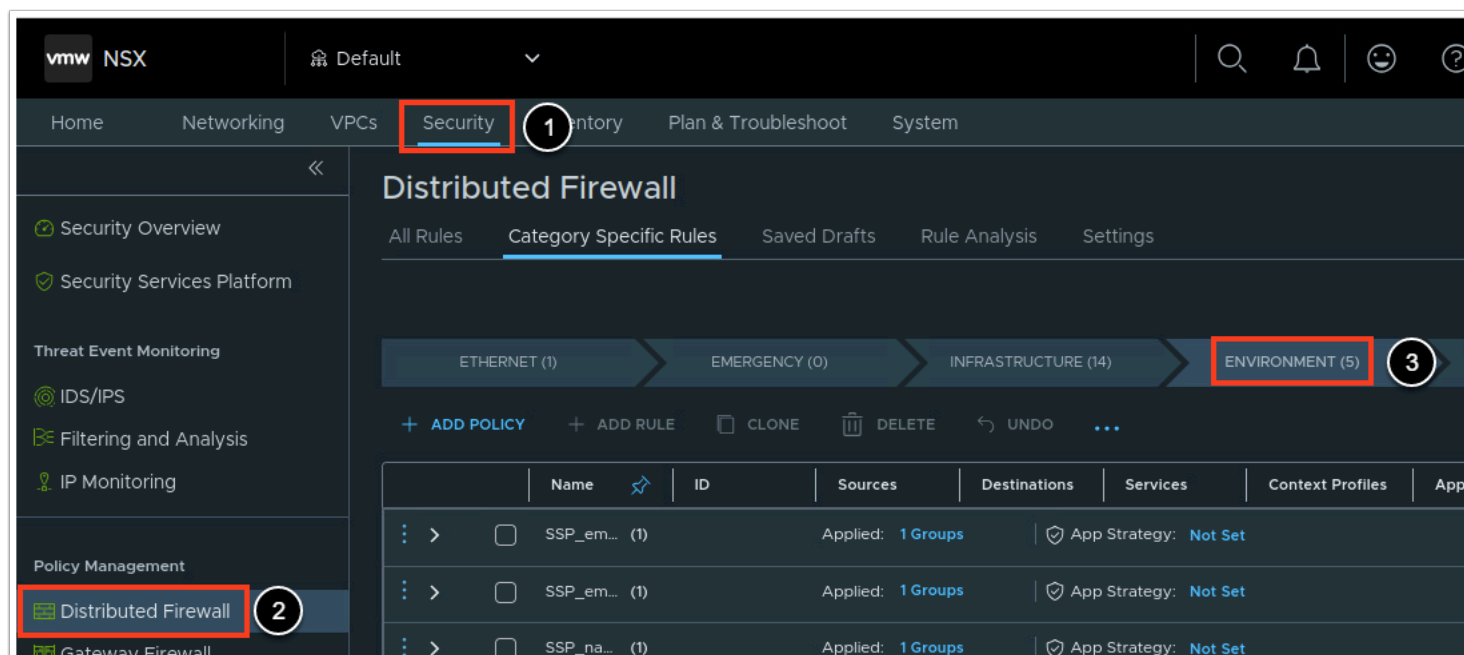
Forgot Password

LOG IN

1

- 
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Validate Environment Rule Publish in NSX



1. Click **Security**
2. Click **Distributed Firewall**
3. Click **Environment** to look at the Environment section of the DFW

## View Published Rules

Distributed Firewall

All Rules Category Specific Rules Saved Drafts Rule Analysis Settings

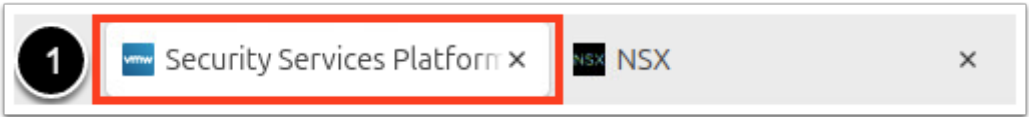
ETHERNET (1) EMERGENCY (0) INFRASTRUCTURE (14) ENVIRONMENT (5) APPLICATION (18)

+ ADD POLICY + ADD RULE 1 ONE DELETE UNDO ... Filter by Name, Path and more

|   | Name                   | ID   | Sources           | Destinations          | Services | Context Profiles | Applied To         | Action |
|---|------------------------|------|-------------------|-----------------------|----------|------------------|--------------------|--------|
| 2 | SSP_emea_uk-lo... (1)  |      | Applied: 1 Groups | App Strategy: Not Set | 3        | Stateful         | Success            |        |
|   | SSP_emea_uk-lon_dev... | 4095 | emea_uk...        | emea_uk...            | Any      | None             | emea_uk... Jump to |        |
|   | SSP_emea_fr-pa... (1)  |      | Applied: 1 Groups | App Strategy: Not Set |          | Stateful         | Success            |        |
|   | SSP_nam_us-sou... (1)  |      | Applied: 1 Groups | App Strategy: Not Set |          | Stateful         | Success            |        |
|   | SSP_nam_us-we... (1)   |      | Applied: 1 Groups | App Strategy: Not Set |          | Stateful         | Success            |        |
|   | SSP_Env_Default (1)    |      | Applied: DFW      | App Strategy: N/A     |          | Stateful         | Success            |        |

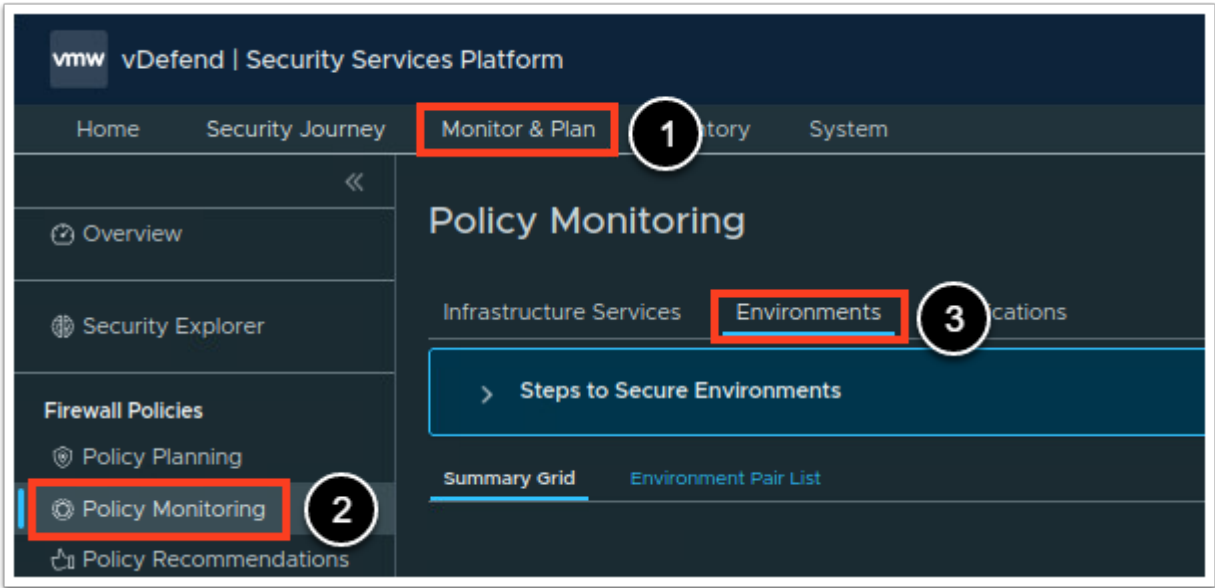
1. If needed, **drag** the column header separator to expand the Name column
2. Click the **arrow** to the left of a policy name to expand its ruleset
3. View the rule - note the Action is "Jump to Application" - meaning when this rule is matched in the rule table, instead of the flow exiting the firewall, it will then continue to the application section of the firewall to be evaluated against those rules

# Monitor Inter-Environment Traffic



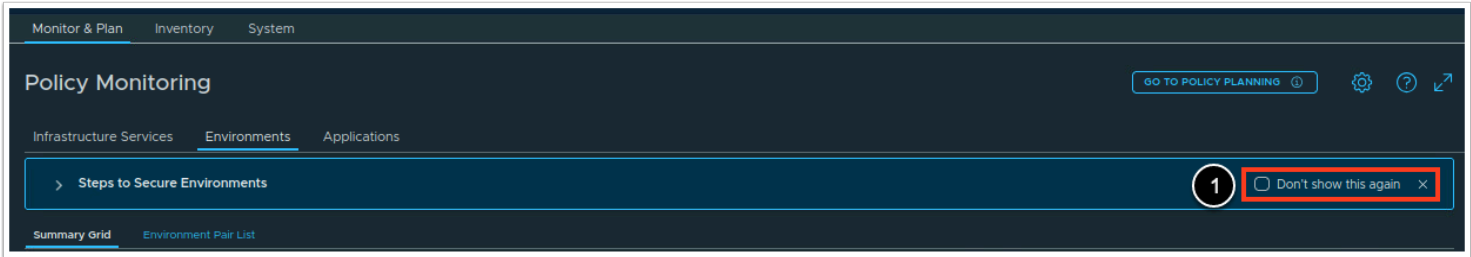
1. Click on the **Security Services Platform** tab to return to SSP

## Navigate to Policy Monitoring for Environments



1. Click **Monitor & Plan**
2. Click **Policy Monitoring**
3. Click **Environments**

## (Optional) Clear Banners From Screen



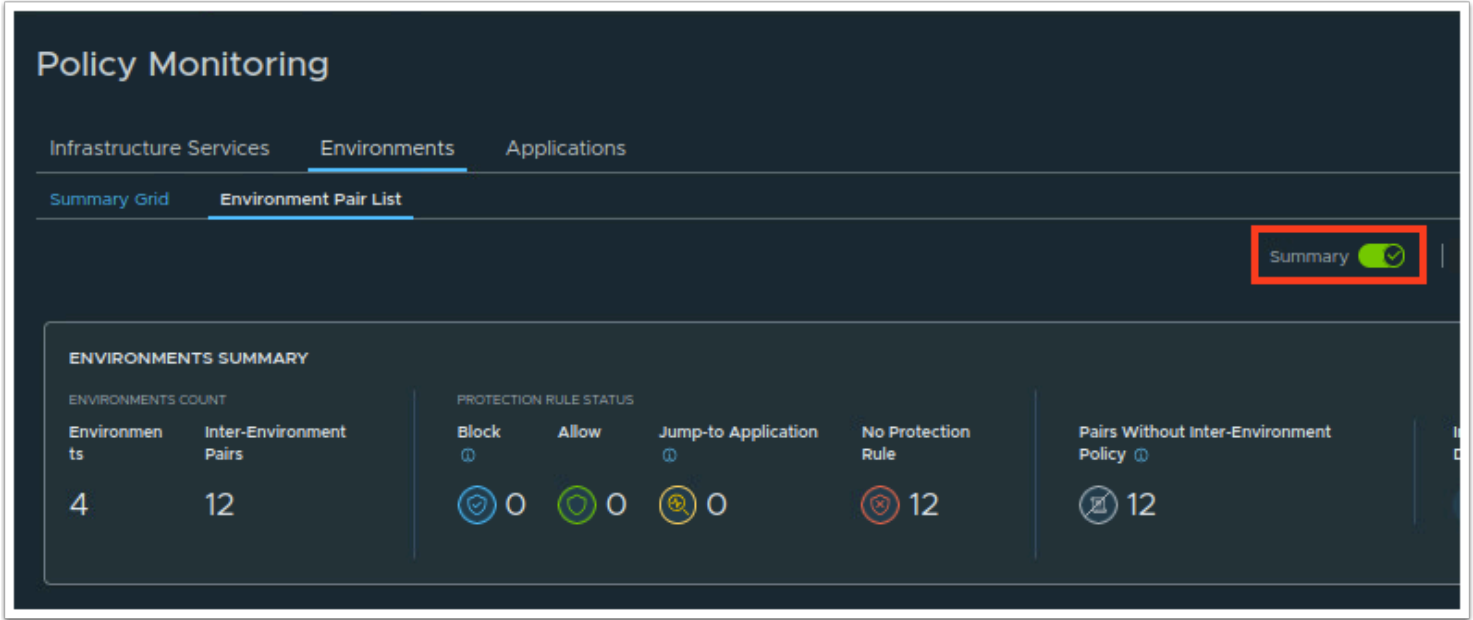
1. Click the **check box** and the **close button** for the info banner to gain more screen space

# View Environment Pair Monitoring

The screenshot displays the 'Policy Monitoring' dashboard. At the top, there are tabs for 'Infrastructure Services', 'Environments', and 'Applications'. The 'Environments' tab is active, and the 'Environment Pair List' is selected, indicated by a red box and a circled '1'. Below the tabs, there's a 'Summary Grid' section with a 'Summary' toggle and a 'Last 24 Hours' filter. A red box labeled '2' highlights the 'ENVIRONMENTS SUMMARY' section, which includes statistics for 'ENVIRONMENTS COUNT' (4 environments, 12 inter-environment pairs), 'PROTECTION RULE STATUS' (0 Block, 0 Allow, 0 Jump-to Application, 12 No Protection Rule, 12 Pairs Without Inter-Environment Policy), 'Inter-Environment Traffic Detected' (6), and 'Environment Category Default Rule Action' (Jump-to Application). Below this, there's a 'Filter by' section with options for 'Security Policy' and 'Traffic Flows'. A red box labeled '3' highlights the 'Environment Pairs' table, which lists pairs like 'nam\_us-west\_prod -> nam\_us-south\_dev' and 'nam\_us-west\_prod -> emea\_uk-lon\_dev'. A red box labeled '4' highlights the 'Distributed Firewall Security Policies' section, which shows 'No Inter-Environment Policy' for each pair. A scroll bar is visible on the right side of the table, indicating that there are more pairs to view.

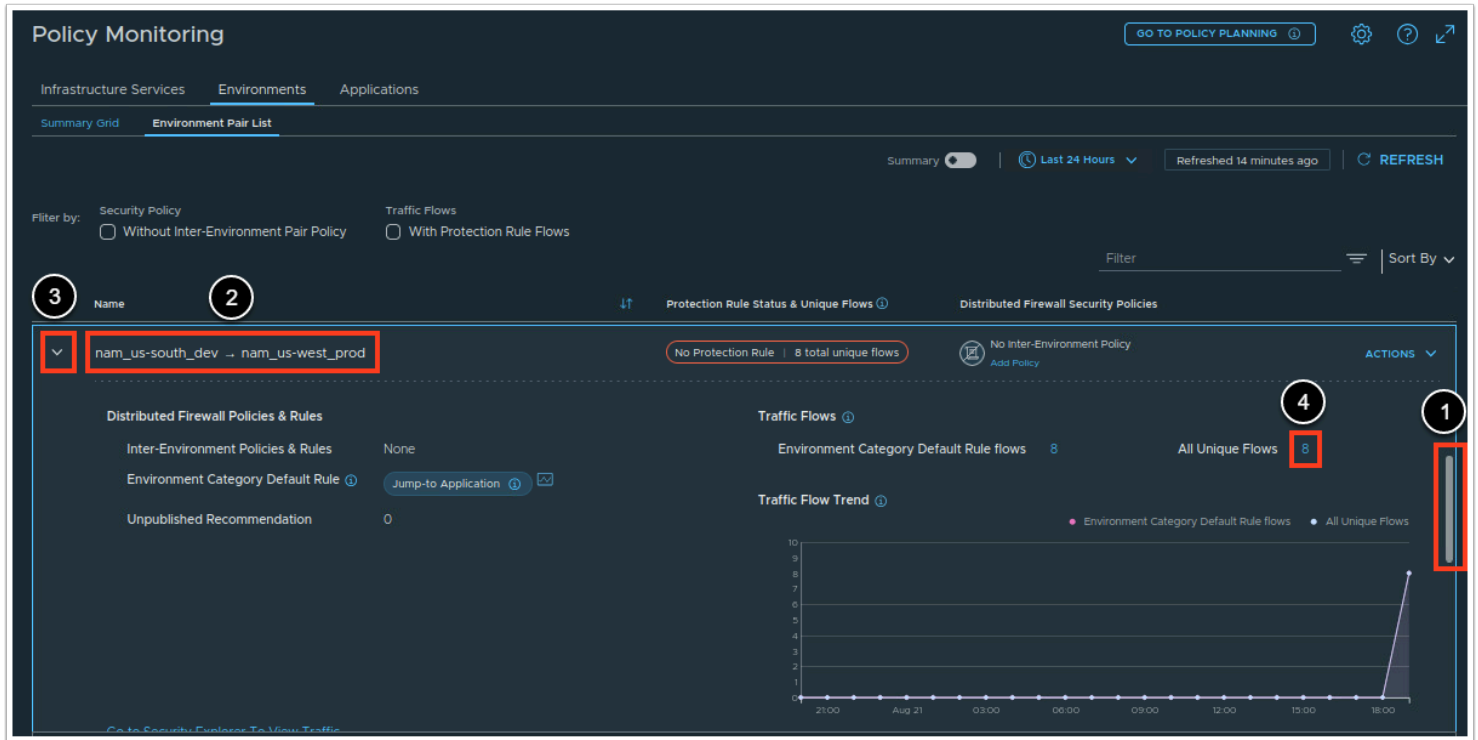
1. Click **Environment Pair List**
2. View the Environments Monitoring Summary
3. View Environment Pairs
4. Use the **scroll bar** to view all the Environment Pairs in the list

# Collapse Summary Pane



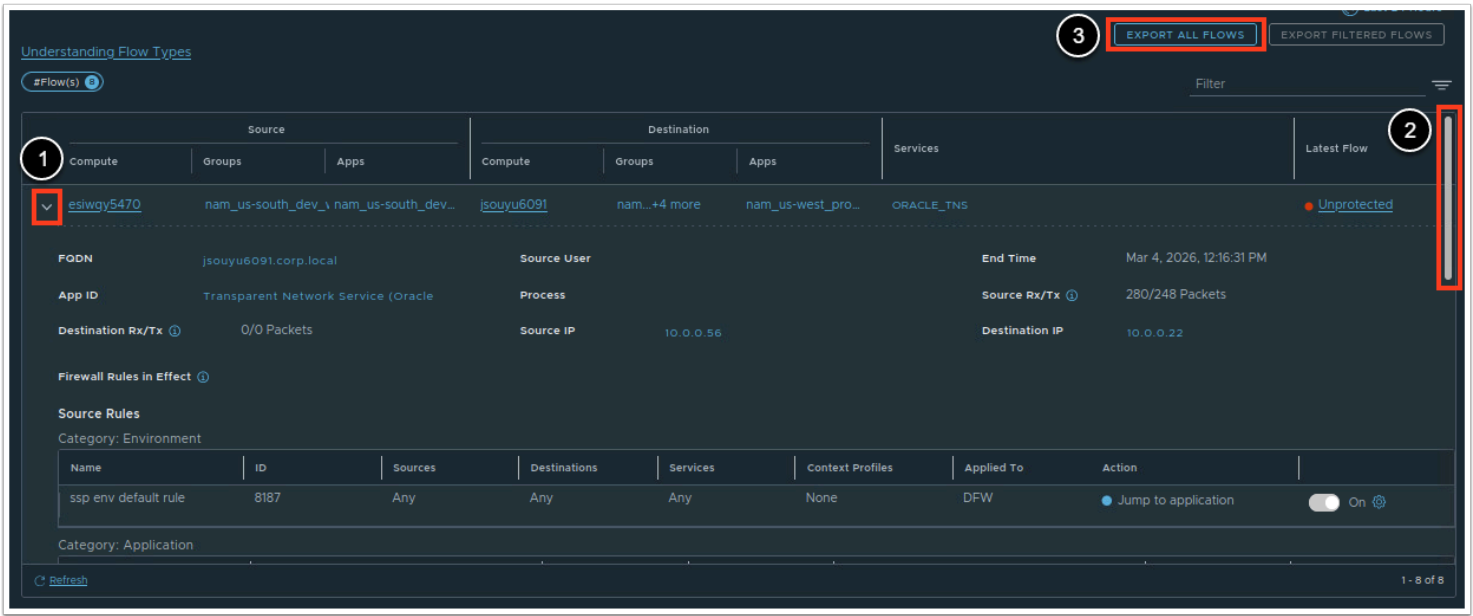
1. Click the **Summary** slider to toggle the summary off - this will provide more screen space to view details for Environment Pairs

# View Environment Pair Details



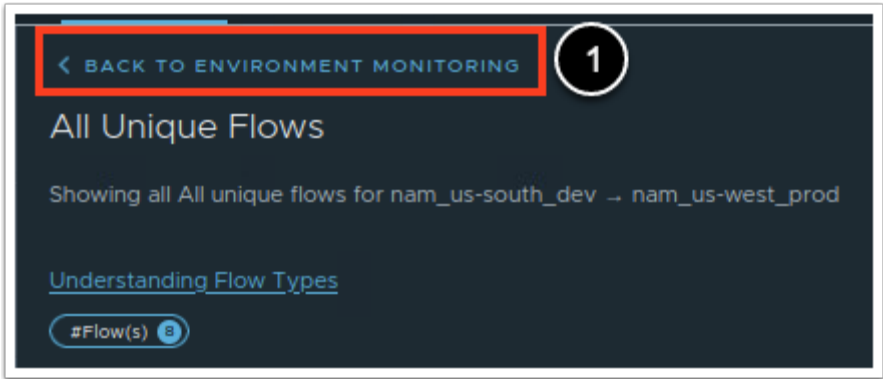
1. Use the **scroll bar** to scroll down the list of Environment Pairs
2. Locate the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
3. Click the **arrow** to the left of the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair to expand its details
4. Click the "**8**" to the right of "All Unique Flows" to view the details of the flows for this Environment Pair

# View Environment Pair Flow Details



1. Click the **arrow** to the left of one of the flows in the list to view details about the flow
2. Use the **scroll bar** to view more details about the flow and to view more flows in the list
3. There is an option to also export the flows as a CSV if required. This is not a requirement for this lab

# Return to Environment Monitoring

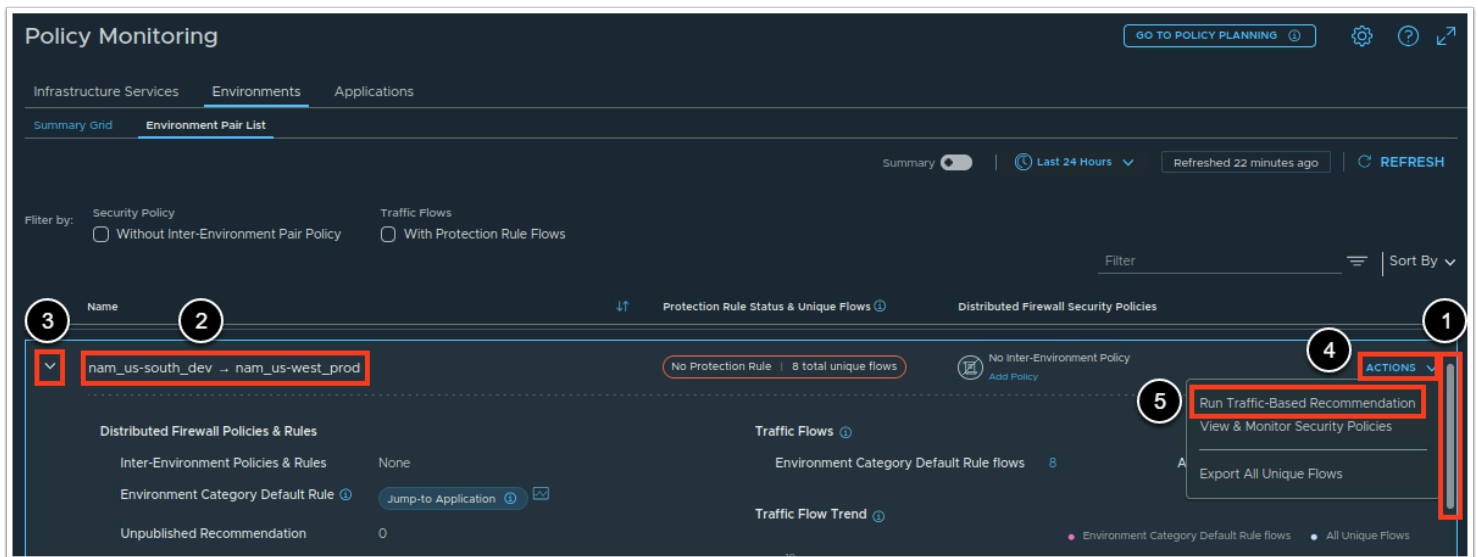


1. Click **BACK TO ENVIRONMENT MONITORING** to return to Environment Monitoring page

# Create Exceptions

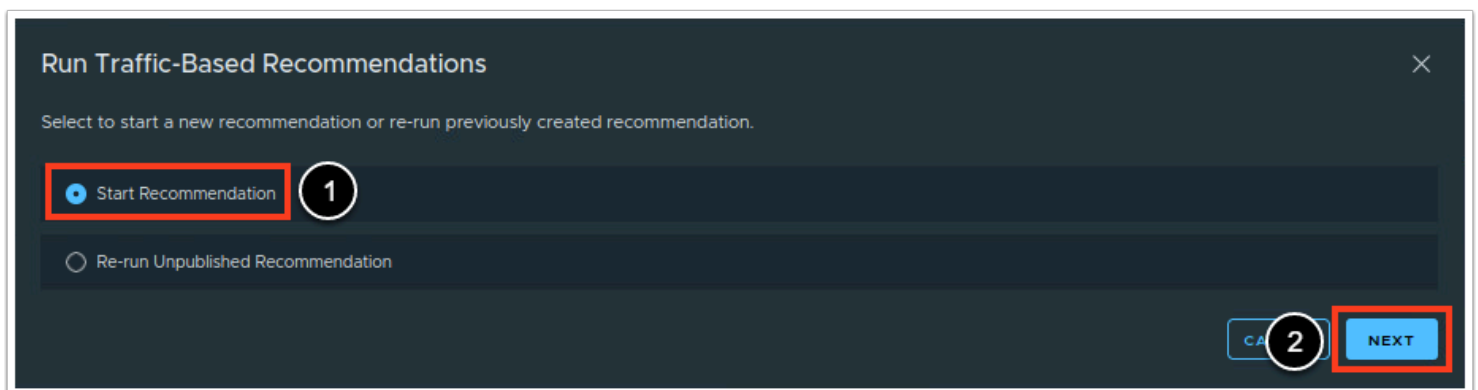
Some flows should be legitimately permitted for an environment pair. A rule recommendation can be run for an Environment Pair and can then be published directly to the Distributed Firewall.

## Run Rule Recommendation for Environment Pair



1. Use the **scroll bar** to scroll down in the list of Environment Pairs
2. Locate the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
3. Click the **arrow** next to the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
4. Click **ACTIONS**
5. Click **Run Traffic-Based Recommendation**

## Start New Recommendation



- 1. Start Recommendation should be selected by default
- 2. Click **NEXT**

## Define Rule Recommendation Parameters

Run Recommendation | nam\_us-south\_dev To nam\_us-west\_prod

1 Recommendation Details

2 Advanced Settings

Recommendation Name

REC 260821 08:14:19 - nam\_us-south\_dev to nam\_us-west\_prod

Note: This will be used as a suffix when naming new recommended groups and rules

Description

Recommendation created on 8/21/26, 8:14 PM

Selected Environments

nam\_us-south\_dev → nam\_us-west\_prod

Recommendation will create a new DFW section.

Distributed Firewall Category

Environment Category

Protection Rule

1

Jump To Application

This is action for Protection Rule

| Name                 | Source           | Destination |
|----------------------|------------------|-------------|
| protection-rule_n... | nam_us-south_dev | nam_us-we   |

- 1. Confirm "**Jump To Application**" is selected for the Protection Rule - this allows for monitoring to continue for any inter-environment traffic leakage before proceeding to lockdown.

## Define Flow Data Time Range for Recommendation

Protection Rule

Jump To Application

This is action for Protection Rule

| Name                 | Source           | Destination      | Services | Profile | Applied To | Action              |
|----------------------|------------------|------------------|----------|---------|------------|---------------------|
| protection-rule_n... | nam_us-south_dev | nam_us-west_prod | Any      | None    | 2 groups   | Jump To Application |

Time Range for Traffic

1

Last 1 Week

2

Last 1 Hour

Last 12 Hours

Last 24 Hours

Last 1 Week

Last 2 Weeks

Last 1 Month

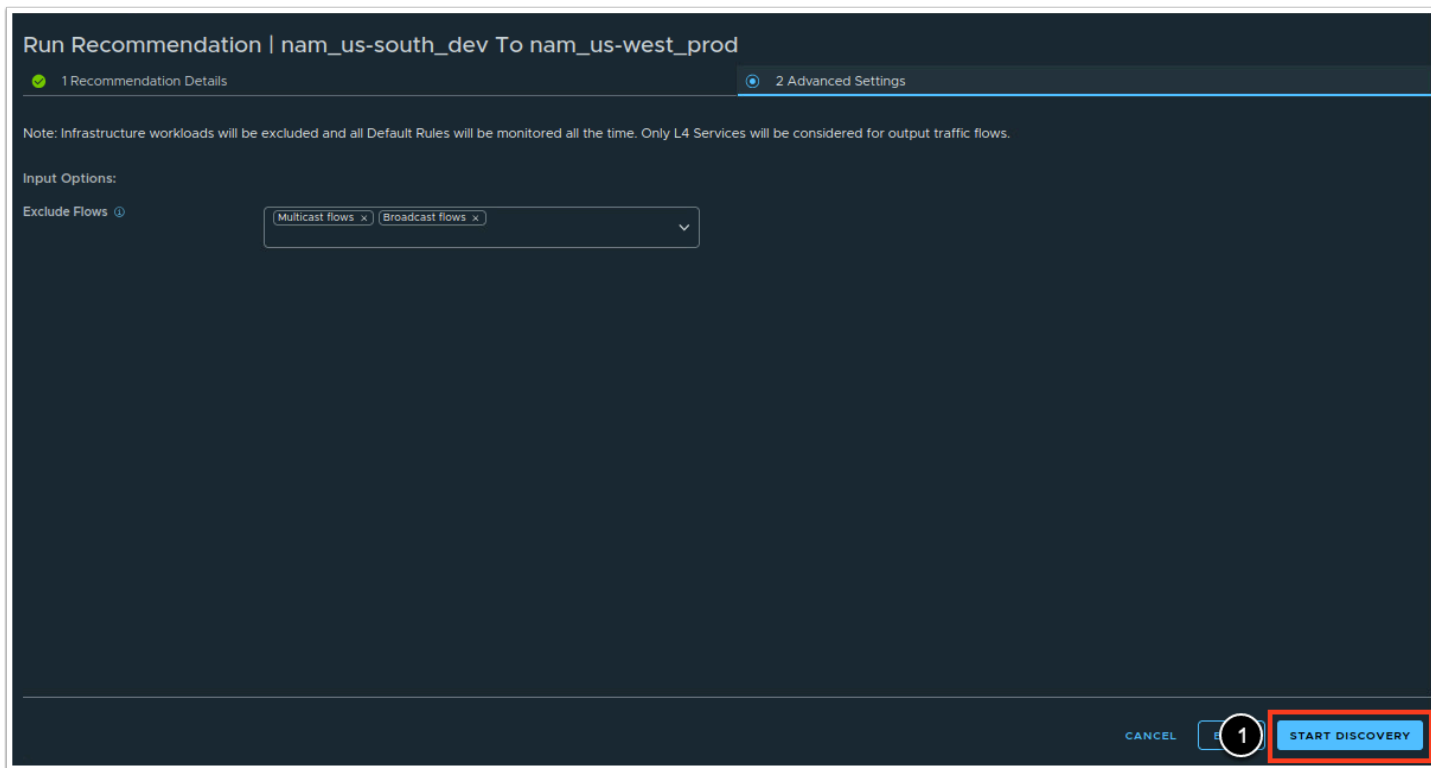
3

NEXT

- 1. Click the "Time Range for Traffic" **drop down menu**

2. Choose "**Last 1 Hour**" for the Flow Data Time Range
3. Click **NEXT** to proceed

## Start Discovery



Run Recommendation | nam\_us-south\_dev To nam\_us-west\_prod

1 Recommendation Details 2 Advanced Settings

Note: Infrastructure workloads will be excluded and all Default Rules will be monitored all the time. Only L4 Services will be considered for output traffic flows.

Input Options:

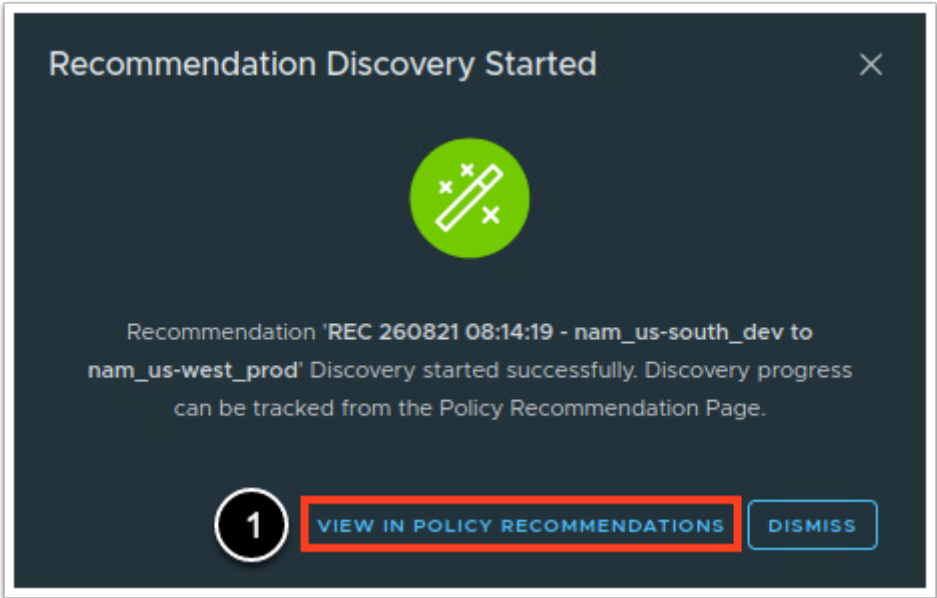
Exclude Flows ⓘ

Multicast flows x Broadcast flows x

CANCEL 1 START DISCOVERY

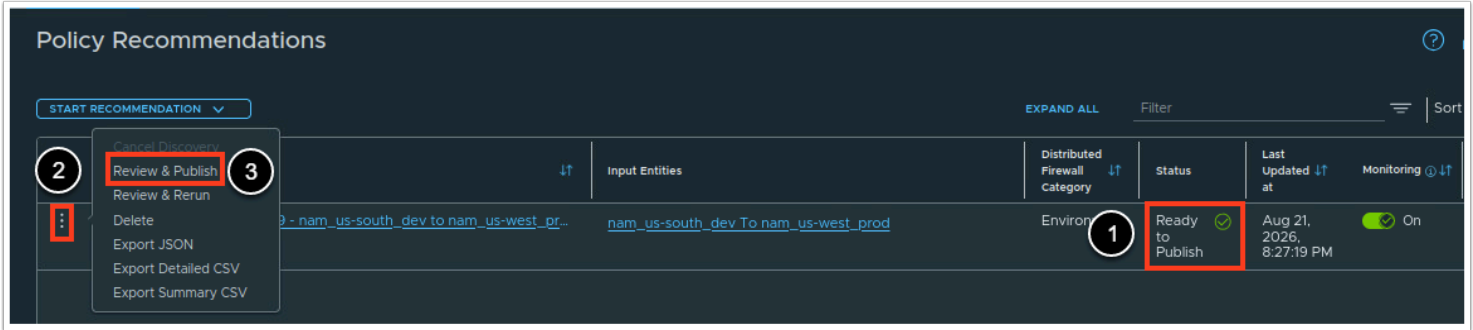
1. Click **START DISCOVERY** to begin rule recommendation process

# View in Recommendations Tab



- 1. Click **VIEW IN POLICY RECOMMENDATIONS**

# Open Recommended Rules Page



- 1. Wait for the "**Ready to Publish**" Status - this may take a few minutes
- 2. Click the **3 dots** to the left of the recommendation name
- 3. Click **Review & Publish**

 The recommendation may take a few minutes to complete

# Review Recommended Rules

Recommended Policies Rules (2) Groups (0) Services (0) Category: Environment

| Name                              | Sources                                                                                     | Destinations                                   | Services | Profiles | Applied To                           | Action              | Status |
|-----------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------|----------|----------|--------------------------------------|---------------------|--------|
| nam_us-south_d_m_us-west_prod (2) | Applied To 2 Groups                                                                         |                                                |          |          |                                      |                     |        |
| New Rule-1 (nam...-west_prod)     | nam_us-s..._workday<br>nam_us-s..._s_domino<br>nam_us-s..._influence<br>nam_us-s..._dev_sap | nam_us-we..._prod_jira<br>nam_us-w..._arepoint | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Application | On     |
| New protection...s-west_prod      | nam_us-south_dev                                                                            | nam_us-west_prod                               | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Application | On     |

CANCEL SAVE PROCEED

< REC 260821 08:14:19 - nam\_us-south\_dev to nam\_us-west\_prod

Review Recommendations > Sequence & Publish

LEARN MORE

Show Graph

Recommended Policies Rules (3) Groups (0) Services (0) Category: Environment

REPLACE DELETE

| Name                            | Sources                                                                             | Destinations          | Services | Profiles | Applied To                           | Action         | Status |
|---------------------------------|-------------------------------------------------------------------------------------|-----------------------|----------|----------|--------------------------------------|----------------|--------|
| Policy_nam_u...us-west_prod (3) | Applied To 2 Groups                                                                 |                       |          |          |                                      |                |        |
| New Rule_1 (na...west_prod)     | nam_us..._workday<br>nam_us-s..._influence<br>nam_us..._domino<br>nam_us..._dev_sap | nam_us-w..._prod_jira | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Applic | On     |
| New Rule_2 (na...west_prod)     | nam_us..._workday<br>nam_us..._dev_sap                                              | nam_us-w..._arepoint  | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Applic | On     |
| New protection...s-west_prod    | nam_us-south_dev                                                                    | nam_us-west_prod      | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Applic | On     |

CANCEL SAVE PROCEED

## 1. Review the recommended rules

- Please note that:
  - source and destination are using Applications when feasible. This is why it is recommended to have completed the "Segmentation Planning" with your application and application tier information so the vDefend Security Services Platform can leverage this information for optimized rules

- If no application can be selected in Source or Destination, existing groups might be reused. If the vDefend Security Services Platform doesn't consider any existing group as a valid option, it will recommend the creation of new groups
- Rules action is "Jump-to-Application", so the final decision about traffic hitting these rules is taken in the Application category. These rules are required to allow locking down the environment pair
- The environment pair default rule is set to "Jump-to-Application", but can be directly changed to Drop or Reject if immediate lockdown without any further monitoring is required
- The Services column is ANY and Profiles column is None: finer security rules are expected to be defined in the Application category

2. Click **PROCEED**

## Publish the Recommended Rules

< REC 260821 08:14:19 - nam\_us-south\_dev to nam\_us-west\_prod

Review Recommendations > Sequence & Publish

Recommended Rules are placed at the top of existing Firewall rules by default.

① **Sequencing Tip:** To rearrange the sequence of newly recommended policies, first select the policies you wish to move. You can then click the Actions " icon to place them quickly above or below an existing section, or you may use drag and drop to precisely sequence the selected policies within the existing Firewall rule sections.

Category: Environment Recommended Rules: 3 Recommended Groups: 0 Recommended Services: 0

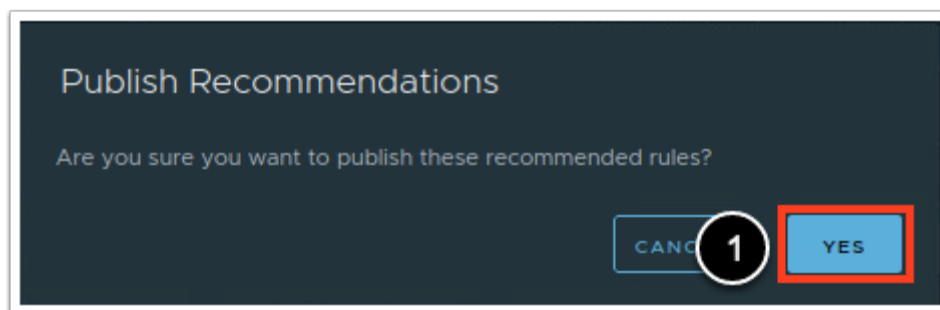
|     | Name                                | Sources                                                                                | Destinations          | Services | Profiles | Applied To                           | Action              | Status |
|-----|-------------------------------------|----------------------------------------------------------------------------------------|-----------------------|----------|----------|--------------------------------------|---------------------|--------|
| ☐   | Policy_nam_u...us-west_prod (3)     | Applied To 2 Groups                                                                    |                       |          |          |                                      |                     |        |
| ⋮   | <b>New</b> Rule_1 (na...west_prod)  | nam_us-s...workday<br>nam_us-to...onfluence<br>nam_us-s...domino<br>nam_us-s...dev_sap | nam_us-west_prod_jira | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Application | On     |
| ⋮   | <b>New</b> Rule_2 (na...west_prod)  | nam_us-s...workday<br>nam_us-s...dev_sap                                               | nam_us-we...harepoint | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Application | On     |
| ⋮   | <b>New</b> protection...s-west_prod | nam_us-south_dev                                                                       | nam_us-west_prod      | Any      | None     | nam_us-west_prod<br>nam_us-south_dev | Jump-To Application | On     |
| ⋮ > | SSP_emea_fr-paris_prod (1)          | Applied To 1 Groups                                                                    |                       |          |          |                                      |                     |        |
| ⋮ > | SSP_emea_uk-lon_dev (1)             | Applied To 1 Groups                                                                    |                       |          |          |                                      |                     |        |
| ⋮ > | SSP_nam_us-south_dev (1)            | Applied To 1 Groups                                                                    |                       |          |          |                                      |                     |        |

3 of 5 Policies

CANCEL BACK **1** **PUBLISH**

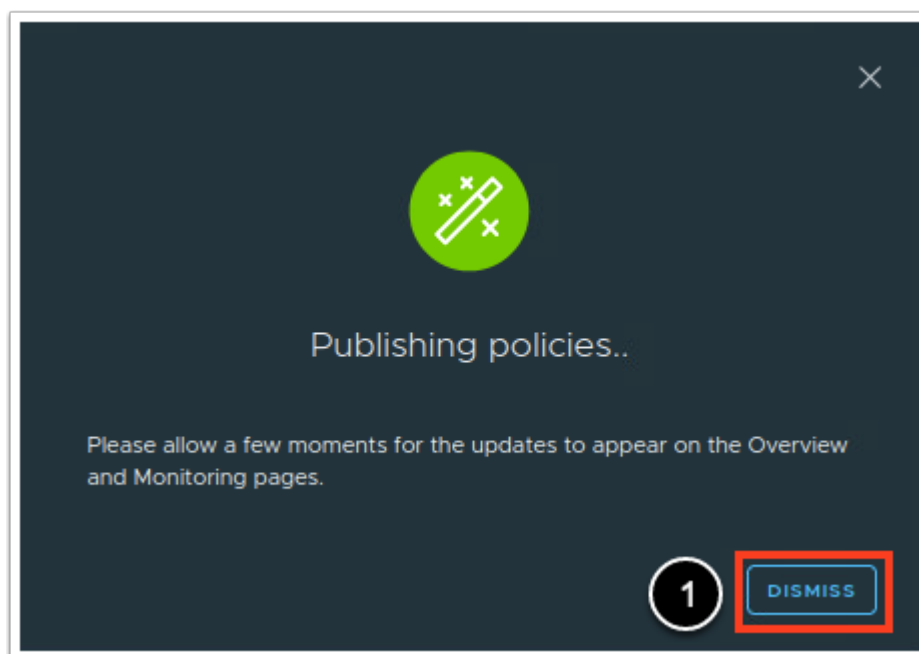
1. Click **PUBLISH** to publish the recommended rules to the Distributed Firewall

## Confirm Rule Publish



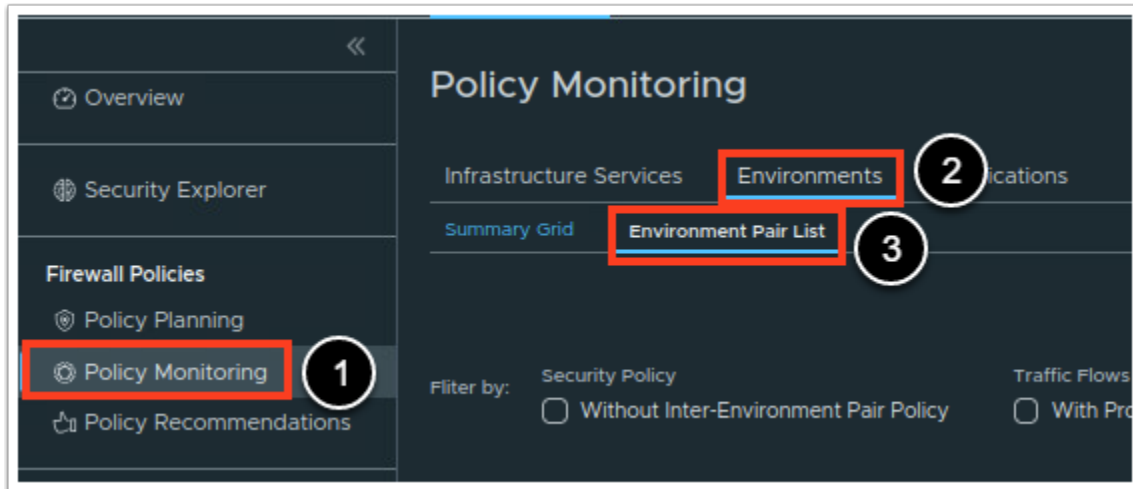
1. Click **YES** to confirm rule publish

## Close Rule Publish



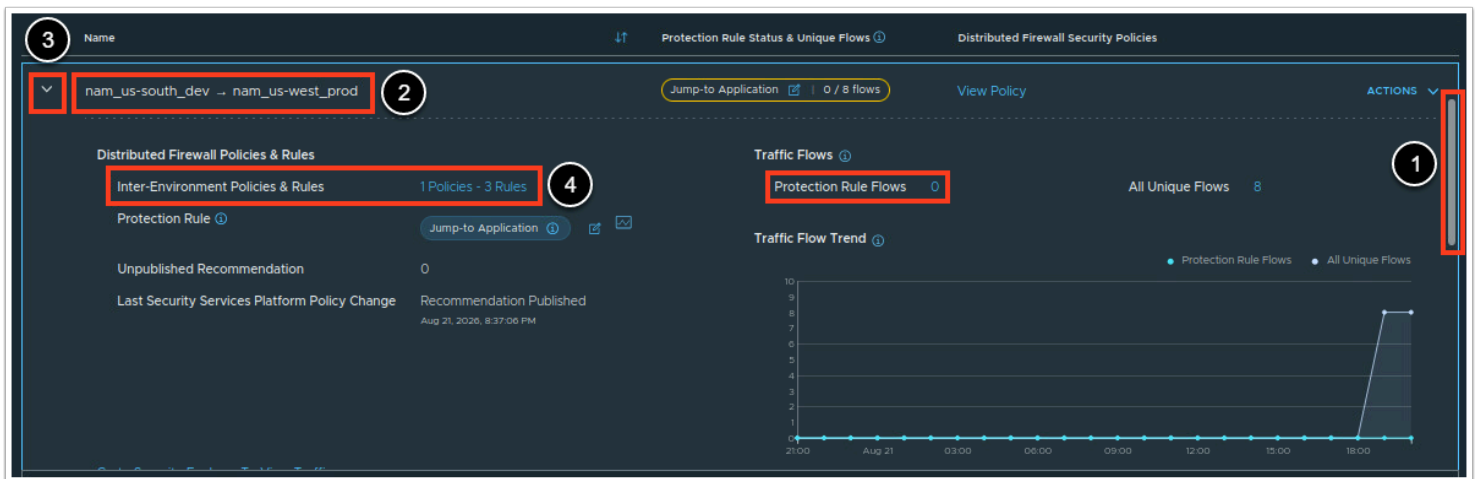
1. Click **DISMISS** to close the rule publish window

## Return to Policy Monitoring - Environments



1. Click **Policy Monitoring**
2. Click **Environments**
3. Click **Environment Pair List**

## View Updated Environment Pair Monitoring



1. Use the **scroll bar** to scroll down in the list of Environment Pairs
2. Locate the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
3. Click the **arrow** next to the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
4. Note the Inter-Environment Policies & Rules section now shows 1 Policy and 3 Rules

5. After the recommendation is published, you can continue to monitor the environment pair you're focusing on. Here the "**Protection Rule Flow: 0**" is a key indicator to check if you still have traffic leakage for this environment pair.

 In a production environment the recommendation is to monitor each environment pair for days before proceeding to lockdown. If new flows are detected and considered as legitimate, you can rerun a recommendation to have additional rules be recommended and published.

# Lockdown Environment Pairs

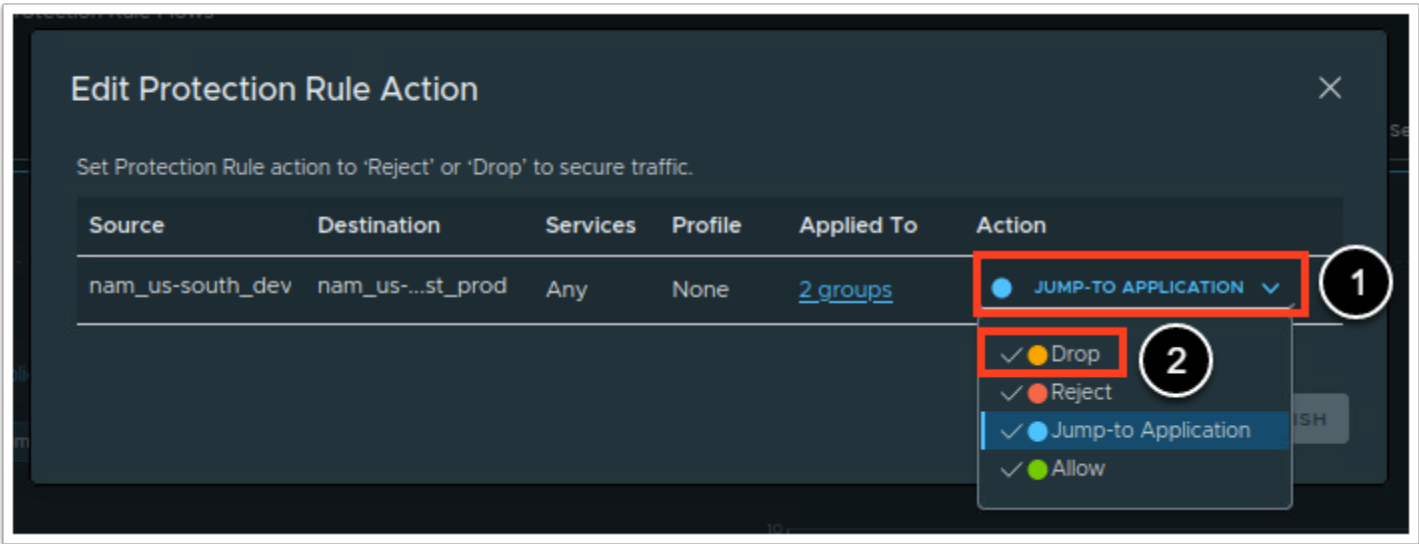
When it has been determined that all legitimate environment pair leakage traffic has been addressed, the next step is to proceed to the environment pair lockdown.

## Lockdown Environment Pair



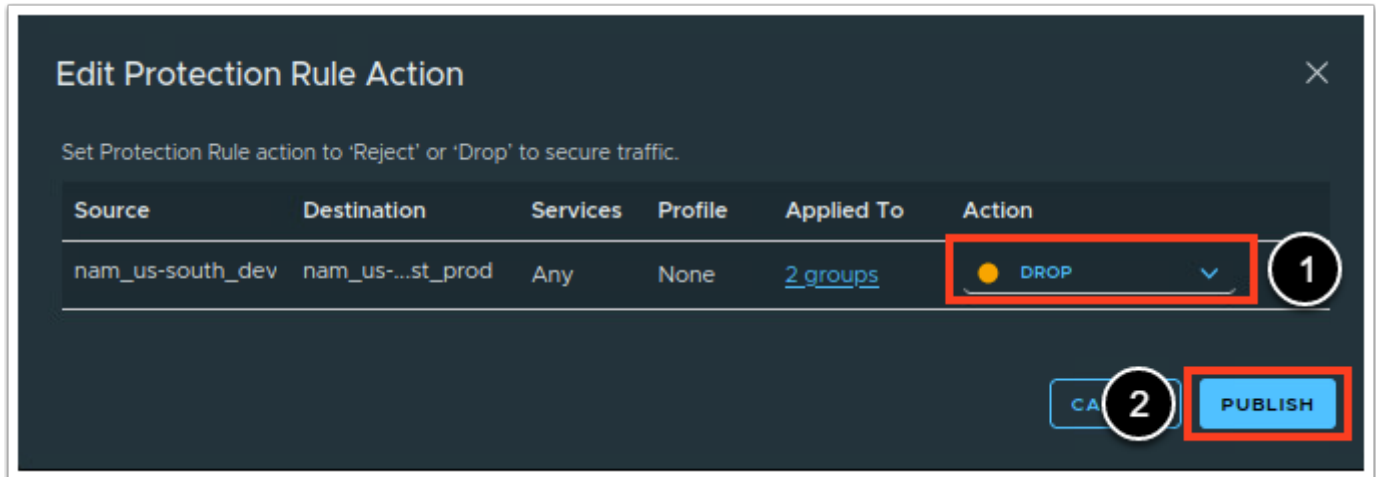
1. Use the **scroll bar** to scroll down in the list of Environment Pairs
2. Locate the "nam\_us-south\_dev -> nam\_us-west\_prod" Environment Pair
3. Click the **arrow** next to the "nam\_us-south\_dev -> nam\_us-west\_prod" Environment Pair
4. Click the **Change Protection Rule Action** button

## Change Protection Rule Action to Drop



1. Click the Action **drop-down menu**
2. Select **Drop** as the Action

## Publish Protection Rule Change



1. Validate that the protection rule action is "Drop"
2. Click **PUBLISH**

## View Environment Pair Information After Publish



1. Use the **scroll bar** to scroll down in the list of Environment Pairs
2. Locate the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
3. Click the **arrow** next to the "**nam\_us-south\_dev -> nam\_us-west\_prod**" Environment Pair
4. Note the Block Protection Rule Status and Unique Flows
5. The Protection Rule now displays "Drop"

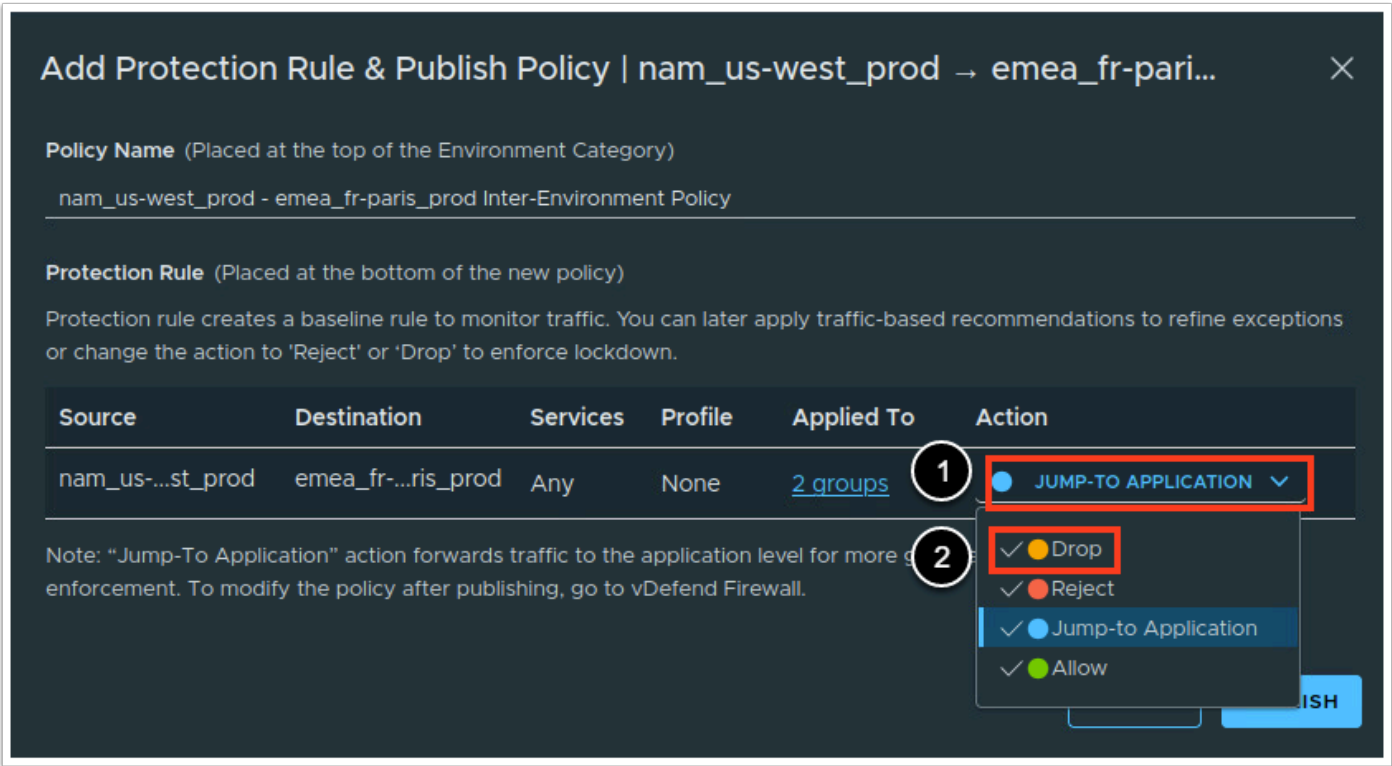
6. The traffic flow trend updates in the Environment Pair list

## OPTIONAL: Create Lockdown Rule for Environment Pairs with 0 Unique Flows (1/3)



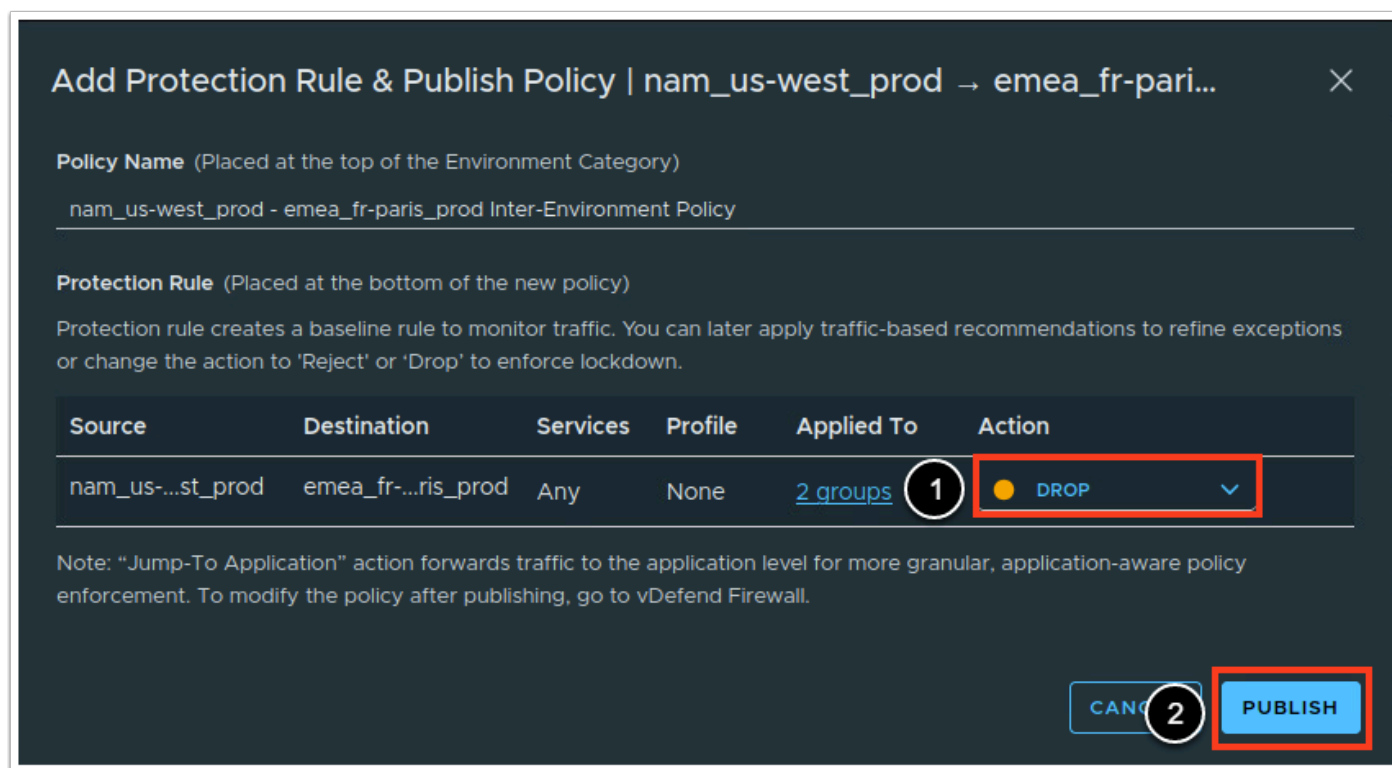
- 1. Locate an Environment Pair that has **0 total unique flows**
- 2. Click **Add Policy**

## OPTIONAL: Create Lockdown Rule for Environment Pairs with 0 Unique Flows (2/3)



- 1. Click the Action **drop-down menu**
- 2. Select **Drop** as the Action

## OPTIONAL: Create Lockdown Rule for Environment Pairs with 0 Unique Flows (3/3)



**Add Protection Rule & Publish Policy** | nam\_us-west\_prod → emea\_fr-pari... ×

**Policy Name** (Placed at the top of the Environment Category)  
nam\_us-west\_prod - emea\_fr-paris\_prod Inter-Environment Policy

**Protection Rule** (Placed at the bottom of the new policy)  
Protection rule creates a baseline rule to monitor traffic. You can later apply traffic-based recommendations to refine exceptions or change the action to 'Reject' or 'Drop' to enforce lockdown.

| Source            | Destination         | Services | Profile | Applied To               | Action                                            |
|-------------------|---------------------|----------|---------|--------------------------|---------------------------------------------------|
| nam_us-...st_prod | emea_fr-...ris_prod | Any      | None    | <a href="#">2 groups</a> | <span>1</span> <span>● DROP</span> <span>▼</span> |

Note: "Jump-To Application" action forwards traffic to the application level for more granular, application-aware policy enforcement. To modify the policy after publishing, go to vDefend Firewall.

CANCEL 2 PUBLISH

1. Validate that the protection rule action is "Drop"
2. Click **PUBLISH**

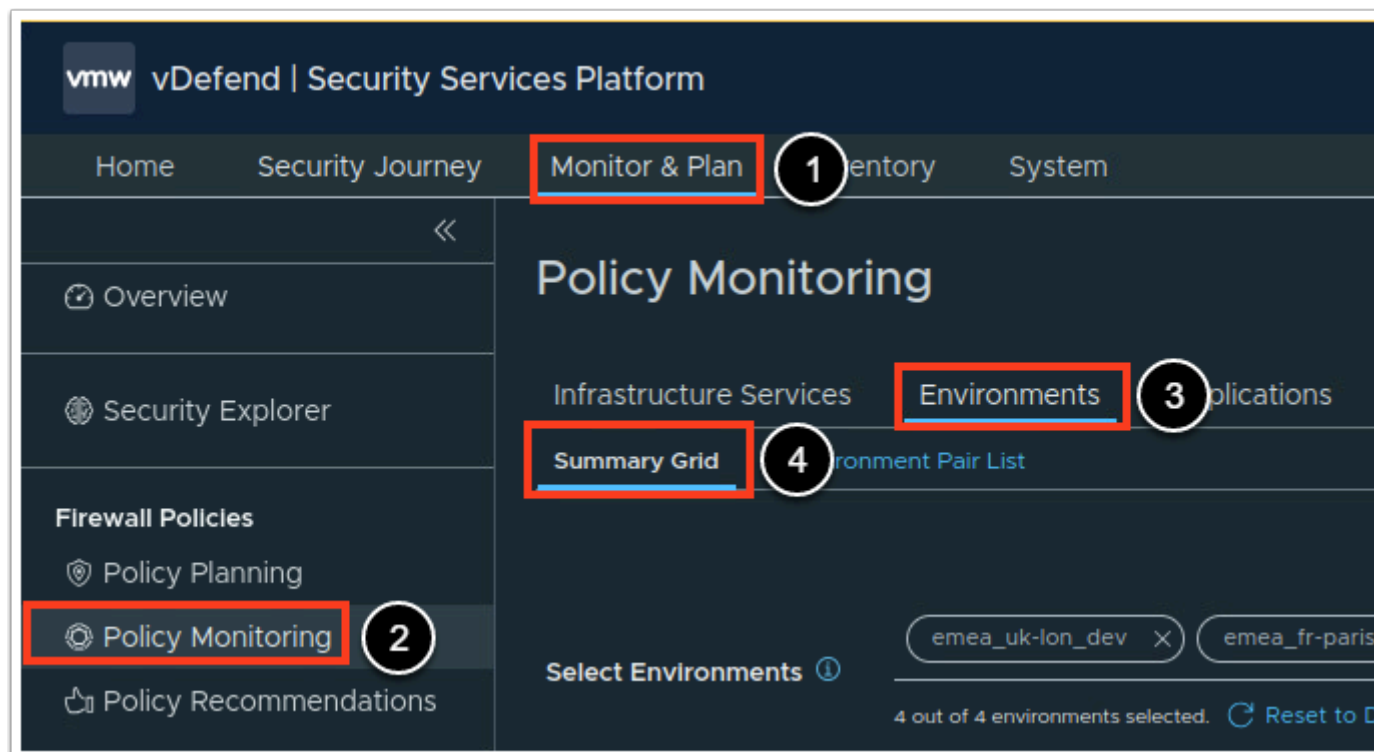
## OPTIONAL: Create Exceptions and Lockdown Additional Environment Pairs

Repeat the process for the remaining Environment Pairs:

1. Run & Publish Recommendation for each Environment Pair
  - To speed up this process you can select "Drop" as the "Protection Rule" in the Recommendation Parameters (there are no unknown/new flows that will populate in this lab - so no need to monitor and then lockdown)
  - Remember to change "Time Range for Traffic" to "Last 1 Hour"
  - Do not run a Recommendation for Environment Pairs that have 0 unique flows - if you do, the recommendation will complete with a status of "Nothing to Recommend". Instead, follow the above steps to create a lockdown rule for these Environment Pairs.
2. If you want your lab to look exactly the same as this lab manual:

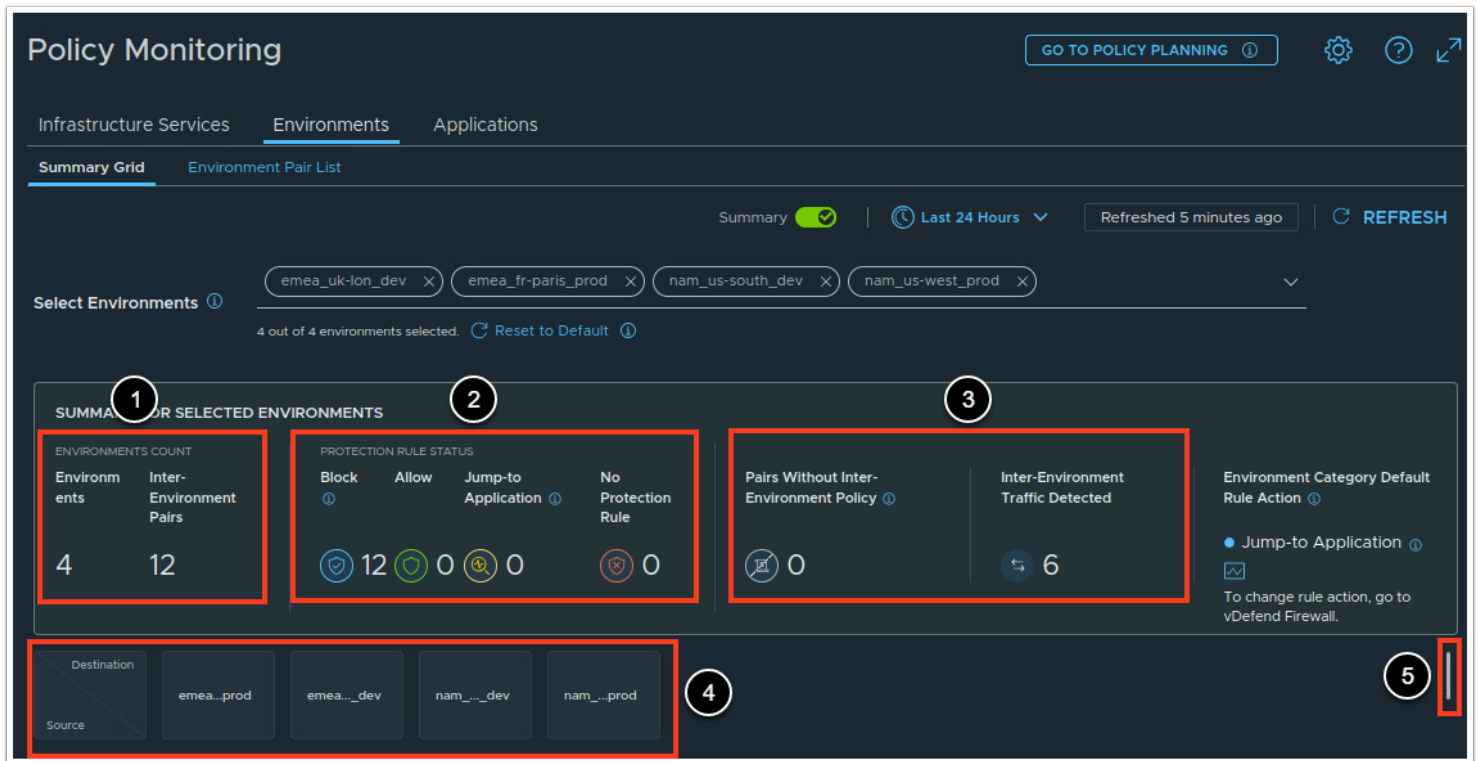
- For Environment Pairs with >0 unique flows: Run recommendation with "Drop" as the "Protection Rule" and "Time Range for Traffic" selected with "Last 1 Hour" as the time range. Publish recommendations.
- For Environment Pairs with 0 unique flows: Follow above instructions and lock down environment pairs with "Drop" as the Protection Rule Action.

# View Environment Pair Protection Progress with SSP



1. Click **Monitor & Plan**
2. Click **Policy Monitoring**
3. Click **Environments**
4. Click **Summary Grid**

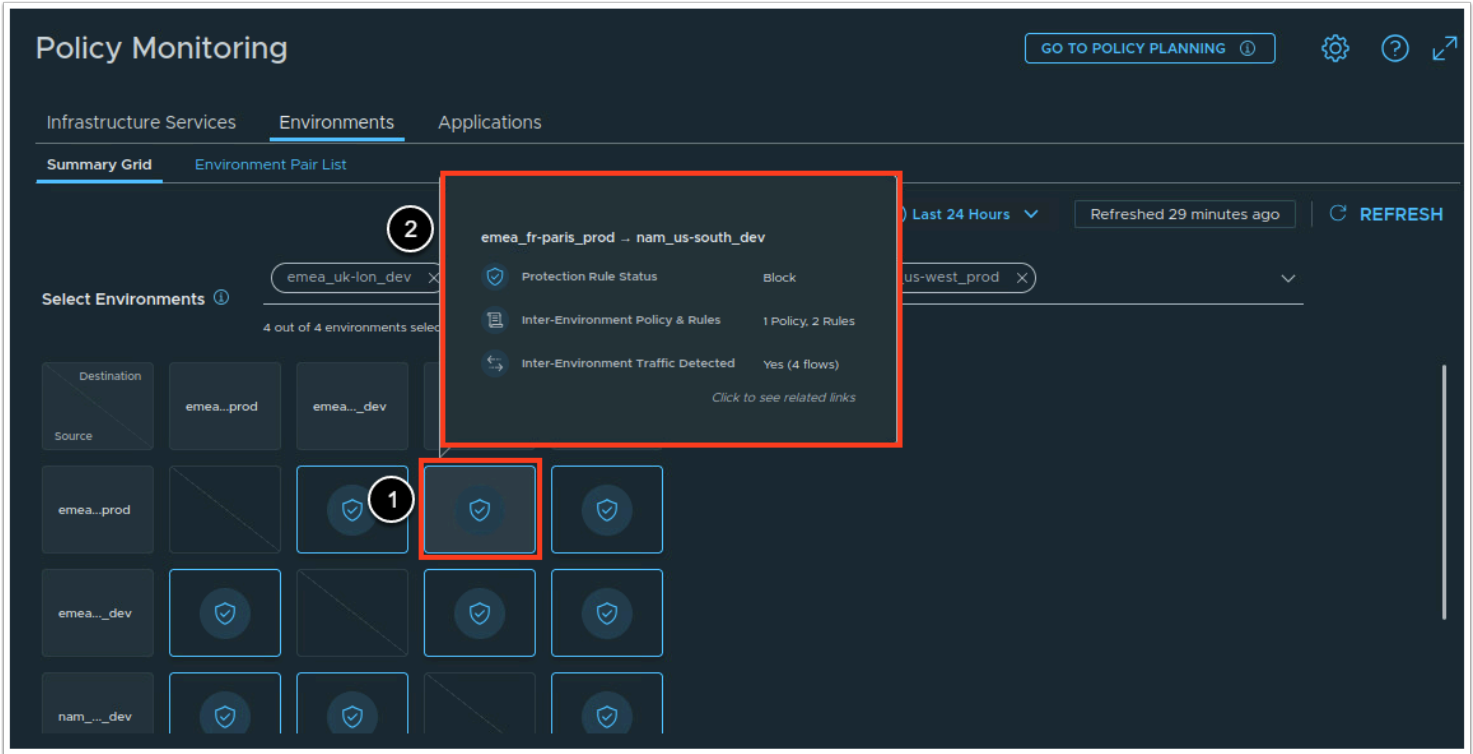
# View Environment Pair Protection Overview



1. Environments Count: displays count of Environments and Environment Pairs
2. Protection Rule Action Summary: a quick overview of all Protection Rule Actions for Environment Pairs
3. Quick view of Environment Pairs without Inter-Environment Policy & Rules and Inter-Environment Traffic Detected.
4. Visual display of the Environment Pairs
5. Use the **scroll bar** to view more of the Environment Pair tiles

**i** The numbers and icons may look different depending on whether the optional steps to secure additional Environment Pairs were taken earlier in this lab.

# View Environment Pair Tile



1. Hover the mouse over one of the Environment Pair tiles that has icons in it to view the details
2. View the details in the pop-up window

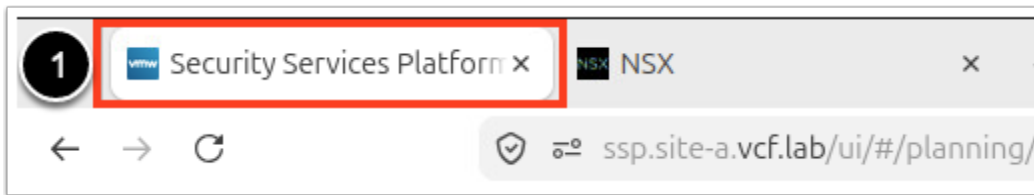
## View Environment Pair Tile Links

The screenshot displays the 'Policy Monitoring' dashboard. At the top, there are tabs for 'Infrastructure Services' and 'Environments'. Below these, there are sub-tabs for 'Summary Grid' and 'Environment Pair List'. A search bar labeled 'Select Environments' is present, with a dropdown showing 'emea\_uk-lon\_dev'. Below the search bar, a grid of environment pair tiles is shown. The first tile in the grid is highlighted with a red box and a circled '1'. A second red box and a circled '2' highlight the pop-up window that appears when the first tile is clicked. The pop-up window displays the following information:

- Environment Pair: `emea_fr-paris_prod` → `nam_us-south_dev`
- Protection Rule Status: Block [icon]
- Inter-Environment Policy & Rules: 1 Policy, 2 Rules [refresh icon]
- Inter-Environment Traffic Detected: Yes (4 flows)
- Links: [Go To Environment Monitoring](#), [View Traffic in Security Explorer](#), [Run Recommendation](#)

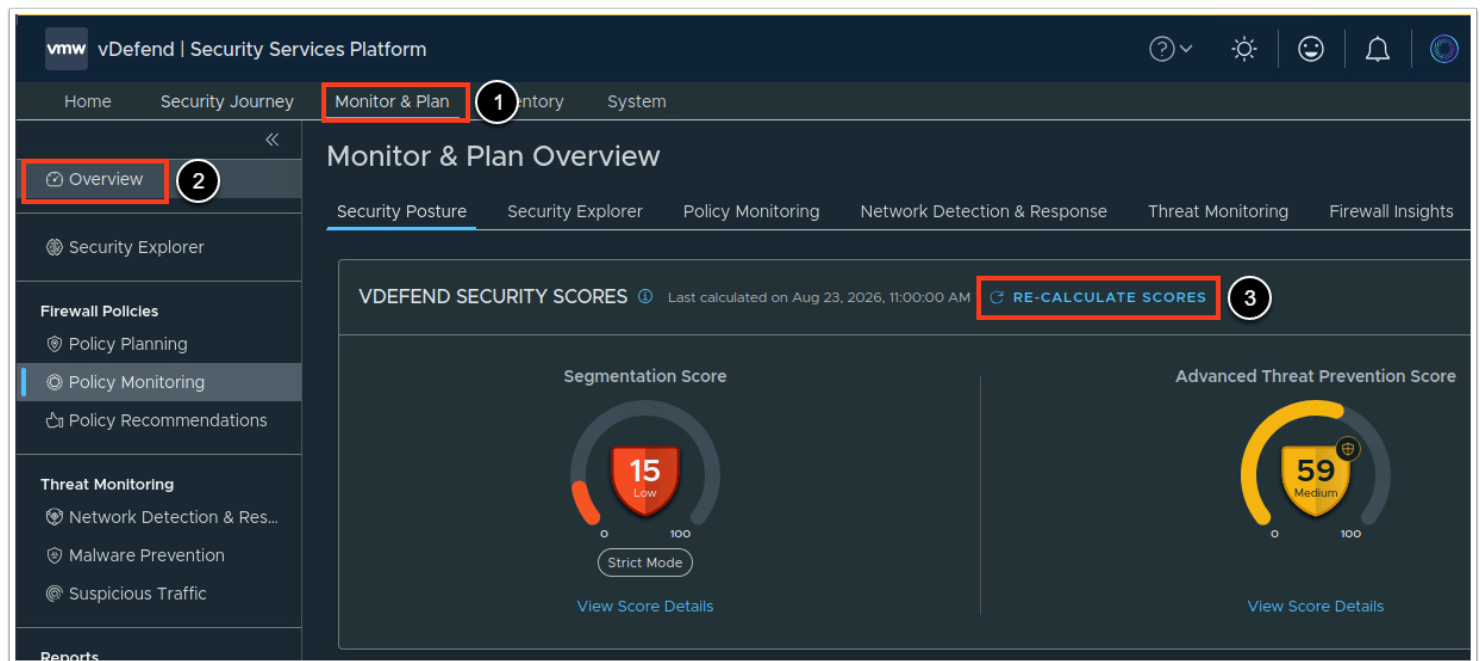
1. Click the Environment Pair **tile**
2. In the pop-up, shortcut links now appear for this Environment Pair

# Re-calculate Security Segmentation Score



1. Click on the Security Services Platform tab in the browser to return to Security Services Platform

## Re-calculate Security Segmentation Score



1. Click **Monitor & Plan**
2. Click **Overview**
3. Click **RE-CALCULATE SCORE**

 Your Security Segmentation Scores may differ based on whether you published the optional additional recommended rules earlier in this lab.

## Choose Security Segmentation Score Calculation Mode and Calculate

Calculate Security Segmentation Score

Select Security Segmentation Score mode to calculate your score in.

☒ Strict **1**

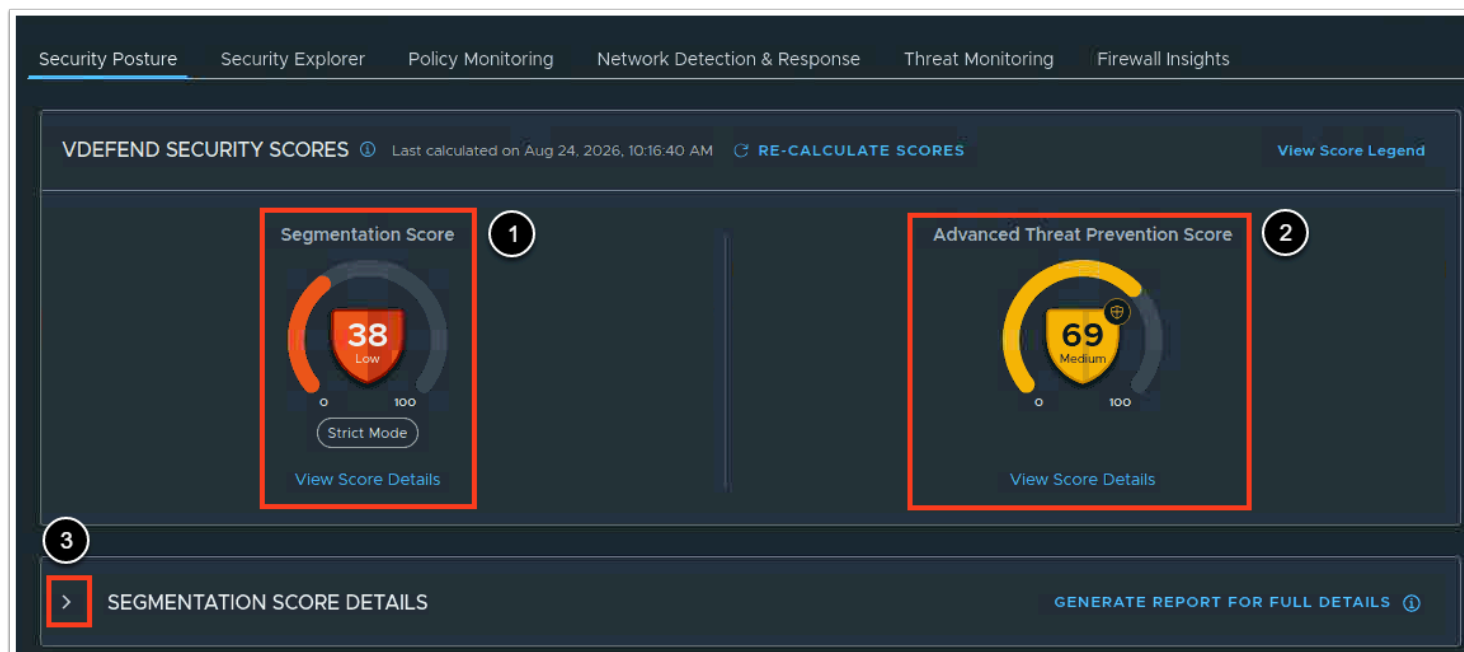
☐ Relaxed

[> Learn about Security Segmentation Score Modes](#)

**2** **CALCULATE**

1. Confirm **Strict** mode is selected
2. Click **CALCULATE**

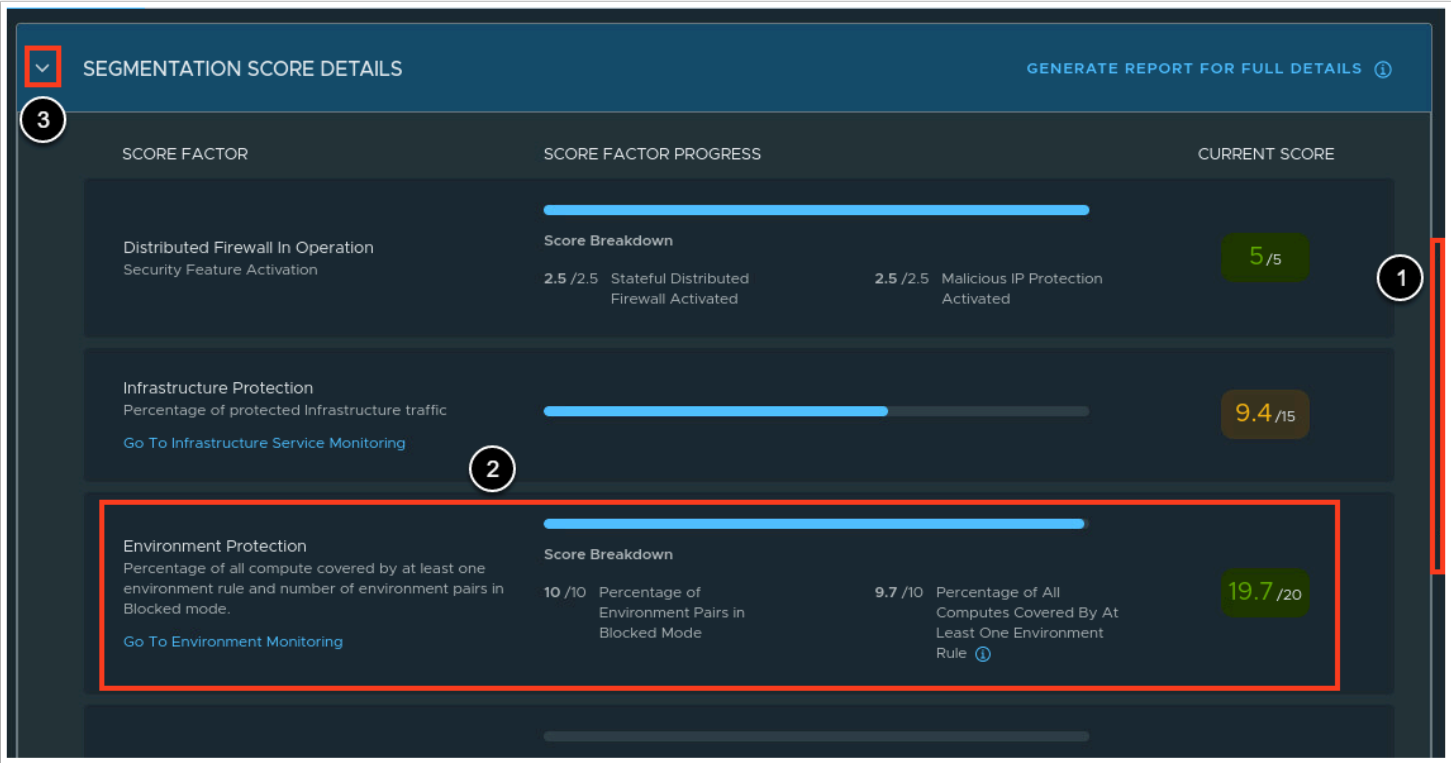
## View Updated vDefend Security Scores



1. The Segmentation Score has increased
2. The Advanced Threat Prevention Score has increased
3. Click the arrow next to SEGMENTATION SCORE DETAILS to expand

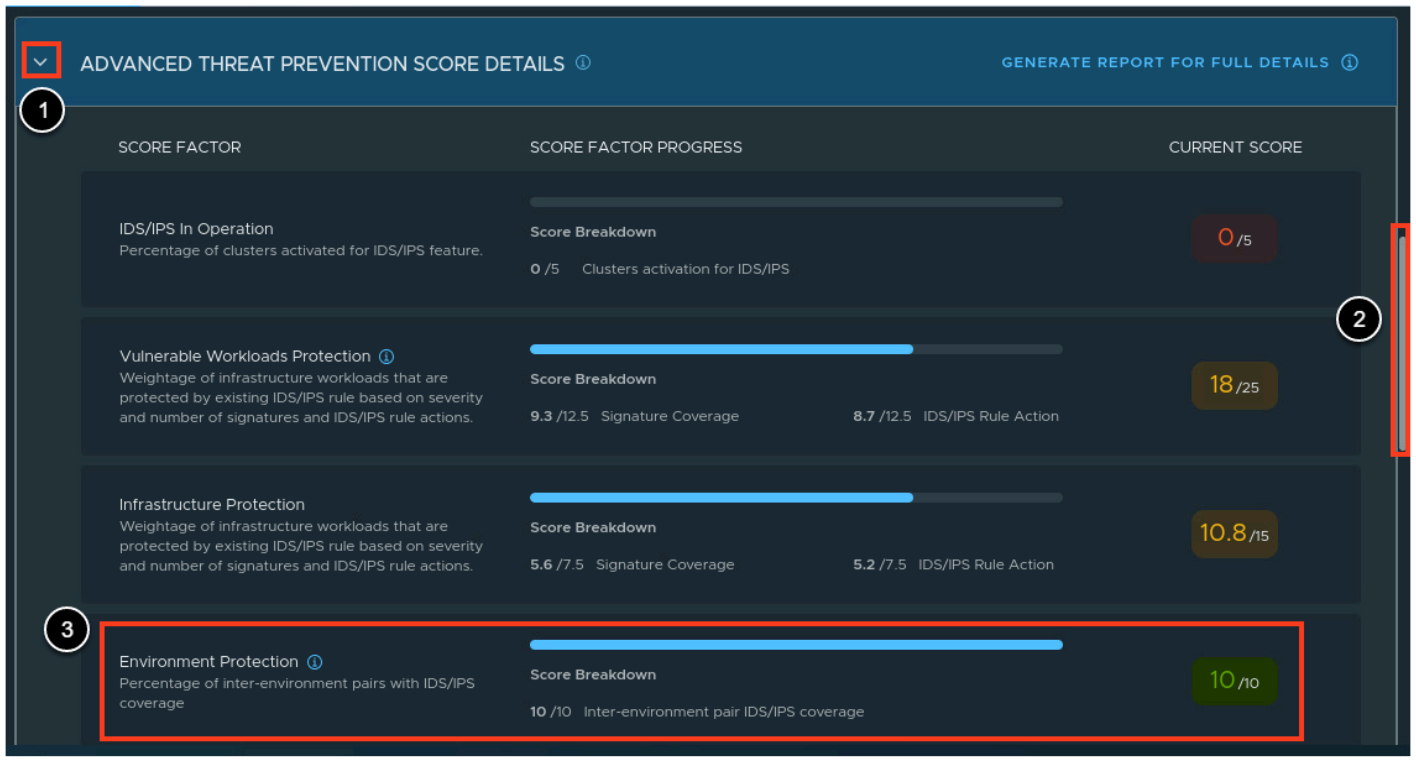
**i** Your Security Segmentation Scores may differ based on whether you published the optional additional recommended rules earlier in this lab.

# View Segmentation Score Details



1. Use the scroll bar to view the Segmentation Score Details
2. Note the Environment Protection score has increased
3. Click the arrow next to SEGMENTATION SCORE DETAILS to collapse once viewing is complete

# View Advanced Threat Prevention Score Details



1. Click the arrow next to ADVANCED THREAT PREVENTION SCORE DETAILS to expand and view details
2. Use the scroll bar to view the Advanced Threat Prevention Score Details
3. Note the Environment Protection score has increased

# Conclusion

Congratulations! You have completed **Module 6 Stage 3: Environment (Zone) Segmentation**.

## Key Takeaways

- **Published & Validated Default Policies:** Applied default environment policies to the Distributed Firewall (DFW) and verified their deployment within the NSX Environment section.
- **Monitored Inter-Environment Traffic:** Applied the concept of Environment Pairs to contextualize, track, and analyze traffic flows and potential leakage between defined zones.
- **Generated Rule Recommendations:** Leveraged the Security Services Platform (SSP) to analyze collected flow data and generate tailored rule recommendations for Environment Pairs.
- **Enforced Zone Lockdown:** Updated Protection Rule actions for Environment Pair policies and published the final lockdown rules to DFW to secure environment boundaries.

In the next module, you will build upon these zone boundaries by implementing granular application-level micro-segmentation.



# **Module 7 – Stage 4: Application Microsegmentation**

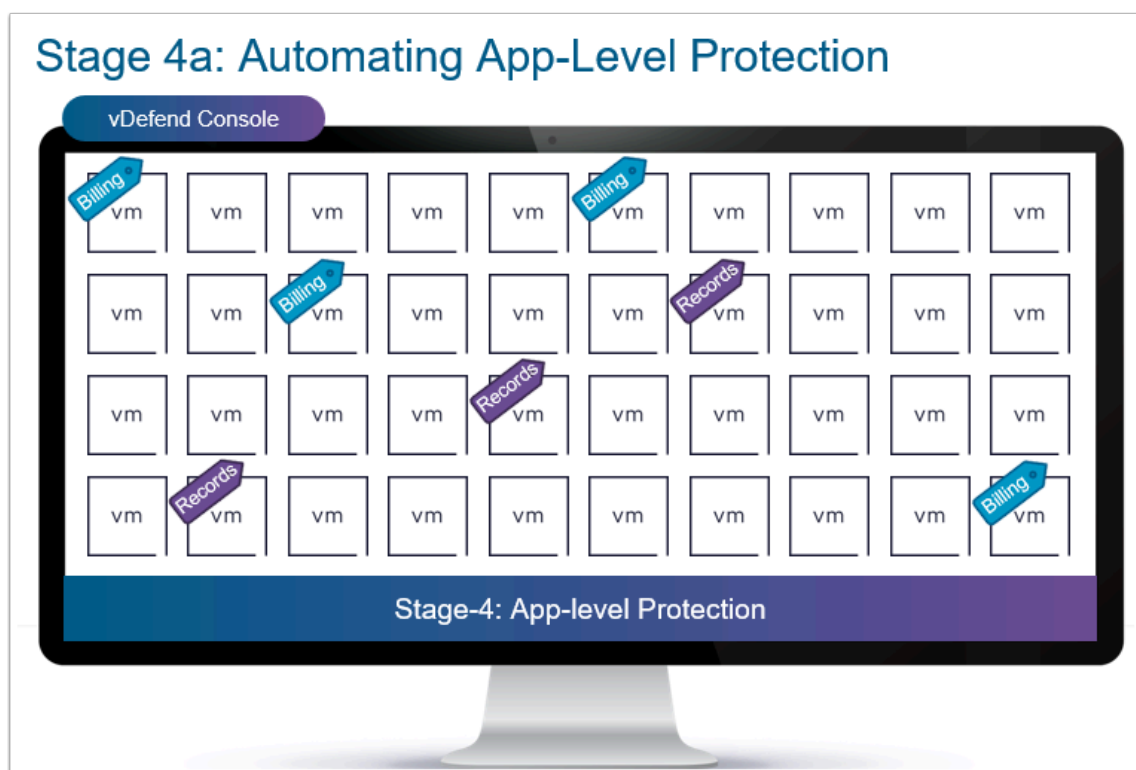
# Application Micro-segmentation Introduction

Zero Trust for datacenter traffic requires defining controls for each application. In this stage, there are three steps:

- Defining application boundaries used to convert them into tags and groups;
- Defining application ring-fencing controls that control over which ports and protocols communication is allowed;
- Defining microsegmentation by defining controls within each application across tiers (web front end, application server and database).

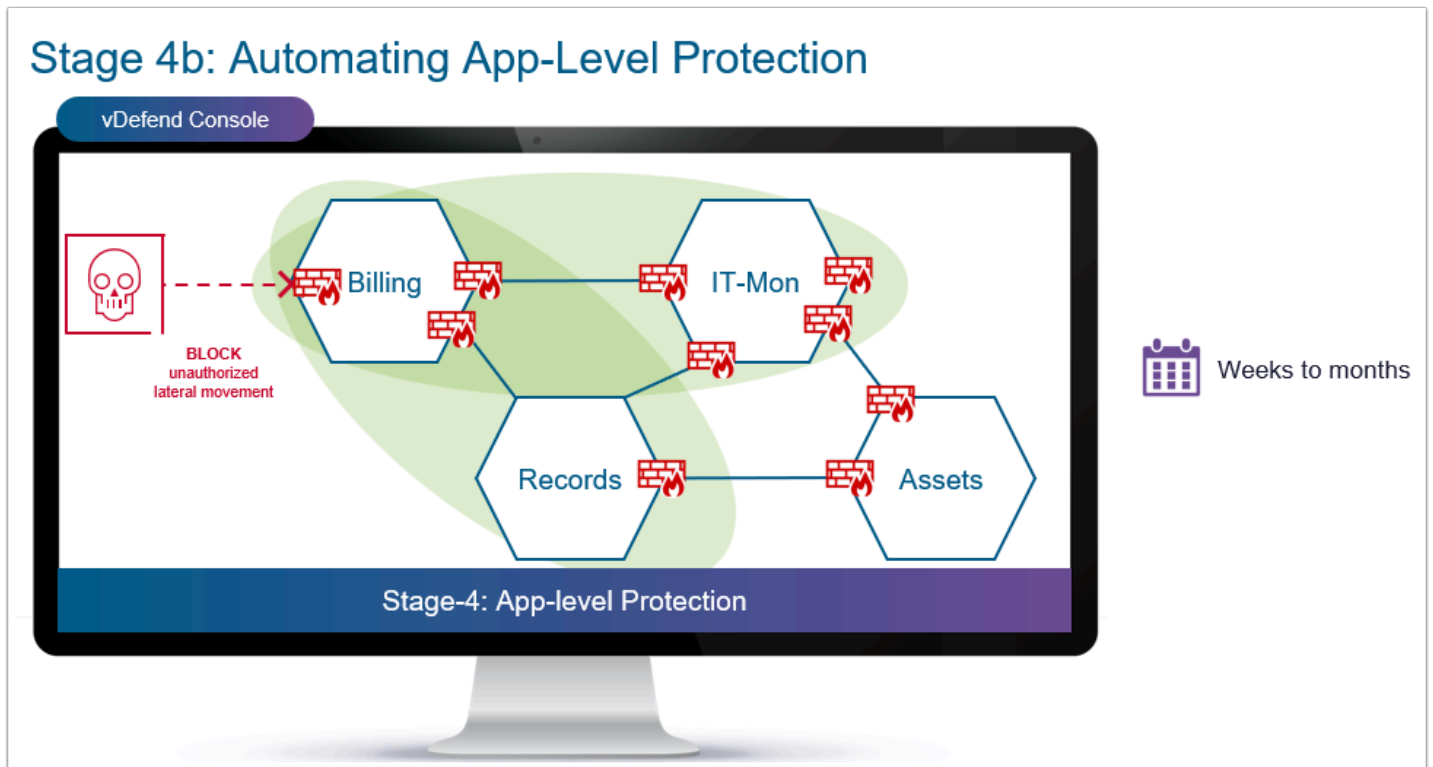
This fine-grained segmentation not only enforces least privilege but also strengthens resilience against east-west threats.

## Stage 4a: Workload to Application Mapping



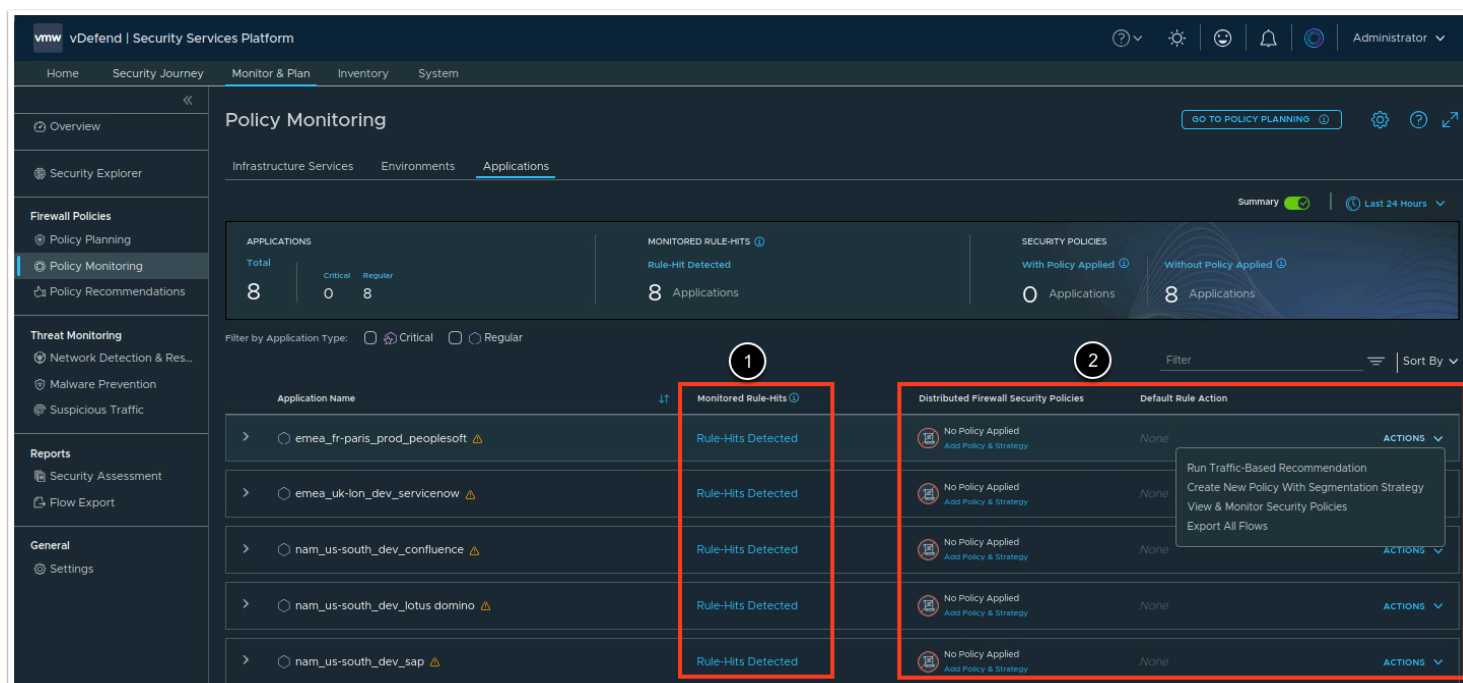
Users can upload into the system via a CSV file, VM-to-application mapping. DFW 1-2-3-4 will then auto-tag and create these application groups. These application groups can then be subsequently used for monitoring and defining DFW rules.

## Stage 4b: Defining Application Ring-fencing Controls



DFW 1-2-3-4 can now monitor these tagged applications, and the system recommends application-specific firewall controls that allow communications only between permitted entities while locking down the applications.

## Stage 4c: Continuous Monitoring of Application Traffic and Fine-Tuning Microsegmentation Controls for Application Tiers



1. DFW 1-2-3-4 continues to monitor each application, both before and after rule publishing.
2. The system continues to track application flow metrics and security posture for rules in real-time. Users can fine-tune rules for application tiers to progressively harden their microsegmentation posture.

## Stage 4 Objective

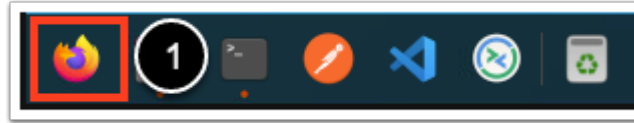
The goal of this phase is to protect every individual application across the environment. Due to the high volume of applications and the necessary alignment with application owners, this stage generally requires more time to complete. To simplify application microsegmentation, the Security Journey offers key capabilities including traffic monitoring, automated policy recommendations, and tailored connectivity strategies.

We will proceed to Stage 4 following these steps:

1. Define Application Criticality
2. Monitor Applications
3. Protect Applications leveraging an Application Connectivity Strategy
4. Lockdown Applications

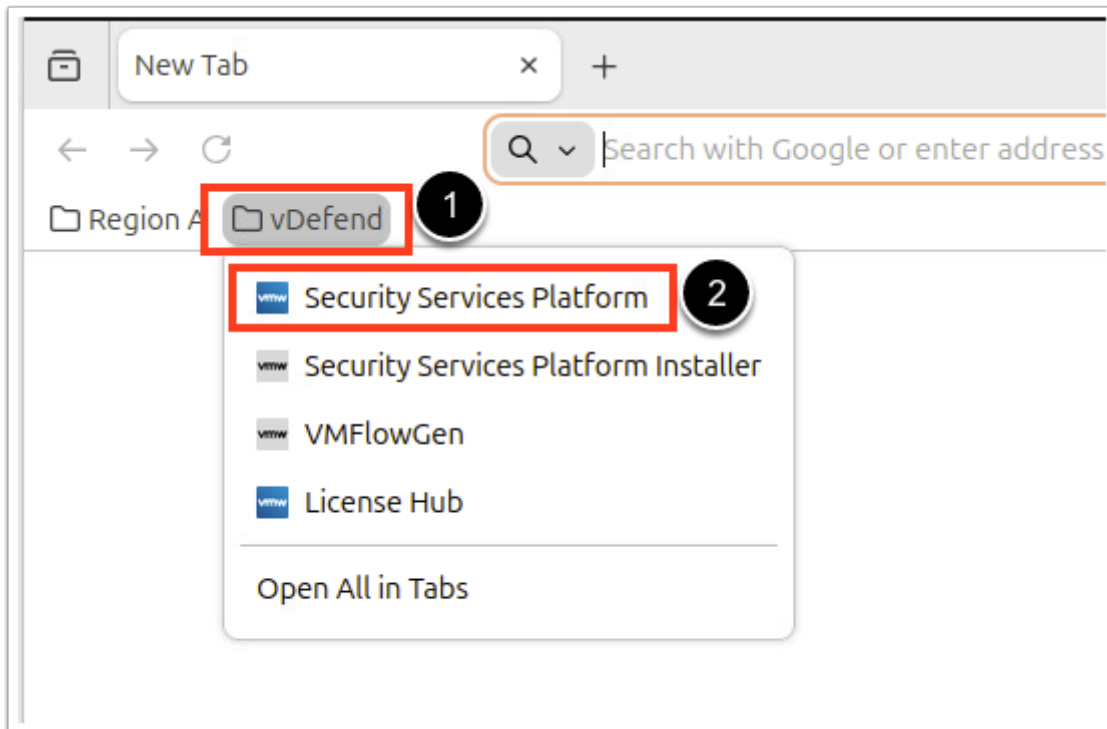
# Login to Security Services Platform (SSP)

Log into the vDefend Security Services Platform (SSP) console to begin configuring and monitoring environment segmentation policies for this stage of the lab.



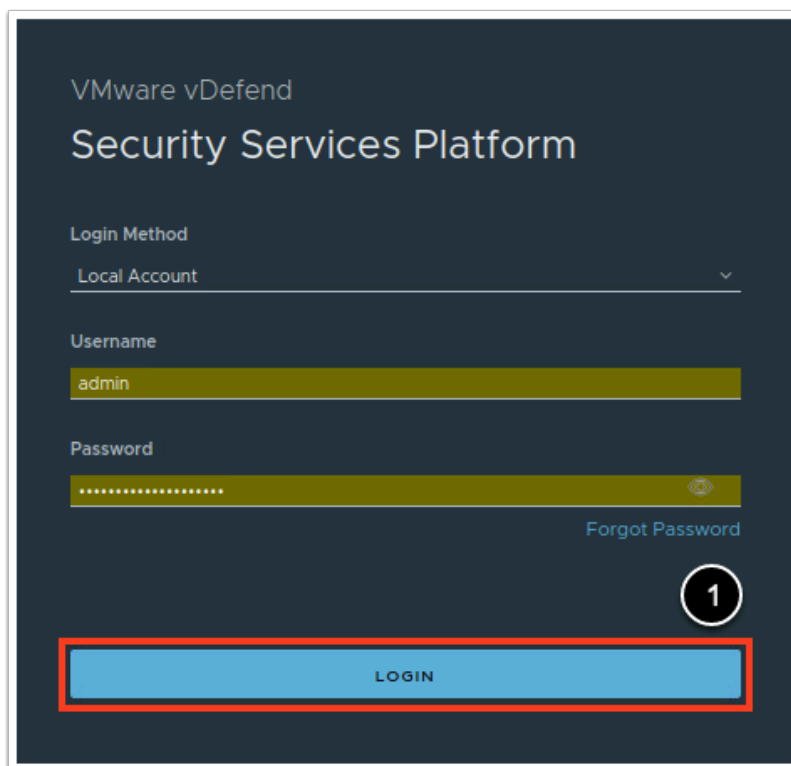
1. Click the **Firefox icon** in the taskbar

## Open Security Services Platform Console



1. Click on the **vDefend** bookmark bar folder
2. Click **Security Services Platform** to launch **SSP**

## Login to Security Services Platform Console



VMware vDefend  
Security Services Platform

Login Method  
Local Account

Username  
admin

Password  
.....

[Forgot Password](#)

1

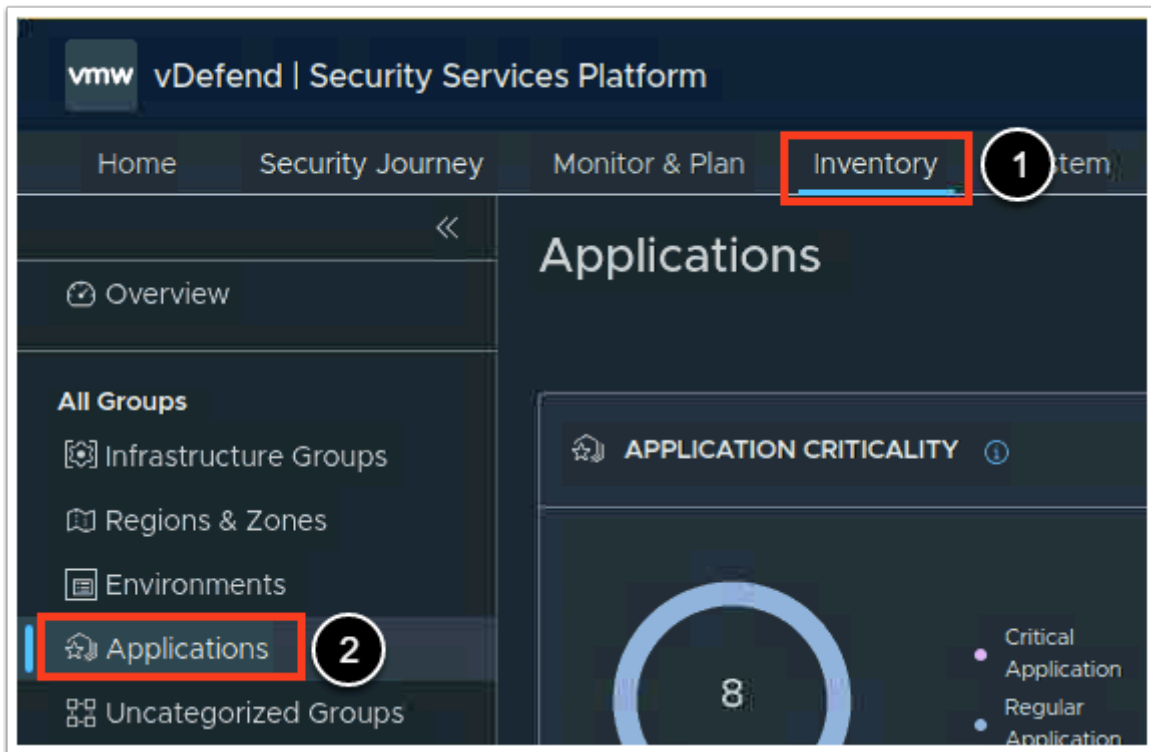
LOGIN

1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Define Application Criticality

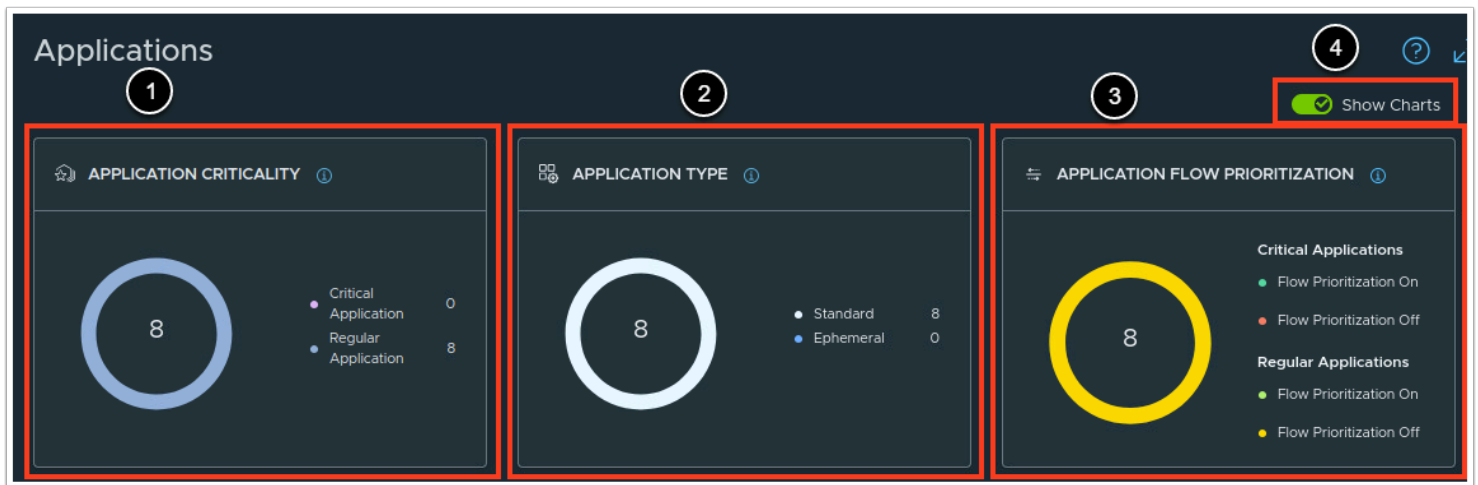
Before starting the Application Segmentation process, you can mark your most critical applications as “critical”. An “critical” application has more weight in the Security Score Computation and gets its flow collection prioritized in case of flow collection congestion.

## Navigate to Application Inventory



1. Click **Inventory**
2. Click **Applications**

# View Application Inventory Charts



These charts show a quick overview of your Application Criticality, Application Type, and Application Flow Prioritization

1. Application Criticality:
  - Critical Application: Flows are automatically prioritized and carry more weight in Security Posture scores
  - Regular Application: Requires manual flow prioritization enablement within the Inventory
2. Application Type: determines how Policy Recommendation leverages network traffic to generate policies
  - Standard: For stable workloads (VMs that exist for days), Recommendations use flows for current members only
  - Ephemeral: For dynamic workloads (VMs and Kubernetes Pods that are periodically created), Recommendations use flows for both current and deleted compute members
3. Application Flow Prioritization:
  - You can use this feature to prioritize the ingestion of flows from specific applications. It is useful when resources are limited and cannot handle all flows. In such cases, when flows exceed the platform capacity, the non-prioritized application flows are excluded from data collection. The main purpose of this feature is to prioritize flows for the application currently designated for lockdown. After an application is locked down, flow prioritization can be turned off for that application, allowing other application flows to be prioritized as needed.
4. Click the **"Show Charts" slider** to collapse charts and gain more screen space for the next step

# Select Critical Applications

Applications

2

EDIT APPLICATION CRITICALITY

EDIT APPLICATION TYPE

EDIT FLOW PRIORITIZATION

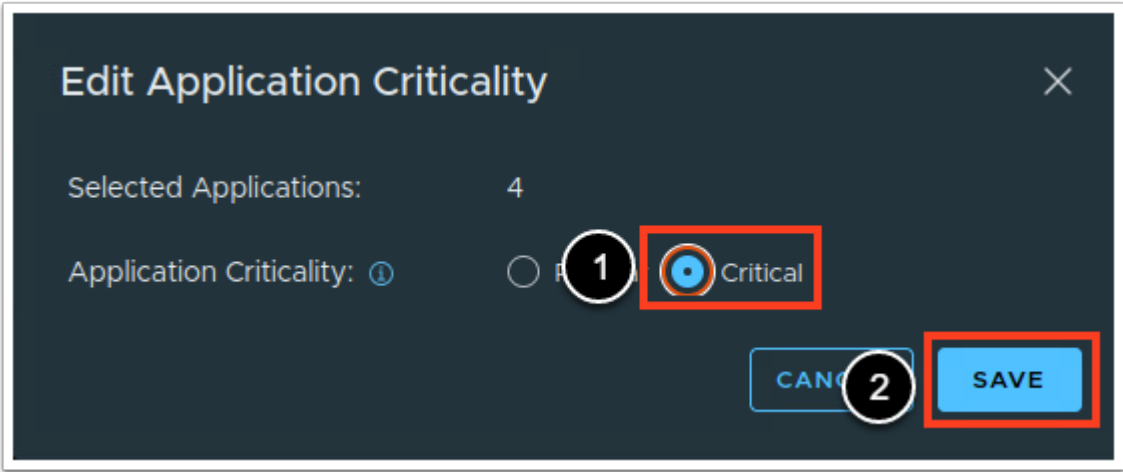
Filter

| <input type="checkbox"/>            |   |   | Name                     | Hierarchy                         | Tiers | Effective Members | Flow Prioritization | Status  | Links |
|-------------------------------------|---|---|--------------------------|-----------------------------------|-------|-------------------|---------------------|---------|-------|
| <input checked="" type="checkbox"/> | : | > | nam_us-west...sharepoint | nam → us-west → prod → sharepoint | 3     | View Members      | Off                 | Success | GO TO |
| <input checked="" type="checkbox"/> | : | > | nam_us-west_prod_jira    | nam → us-west → prod → jira       | 3     | View Members      | Off                 | Success | GO TO |
| <input checked="" type="checkbox"/> | : | > | nam_us-sou...ev_workday  | nam → us-south → dev → workday    | 3     | View Members      | Off                 | Success | GO TO |
| <input checked="" type="checkbox"/> | : | > | nam_us-south_dev_sap     | nam → us-south → dev → sap        | 3     | View Members      | Off                 | Success | GO TO |

4 Refresh 1 - 8 of 8

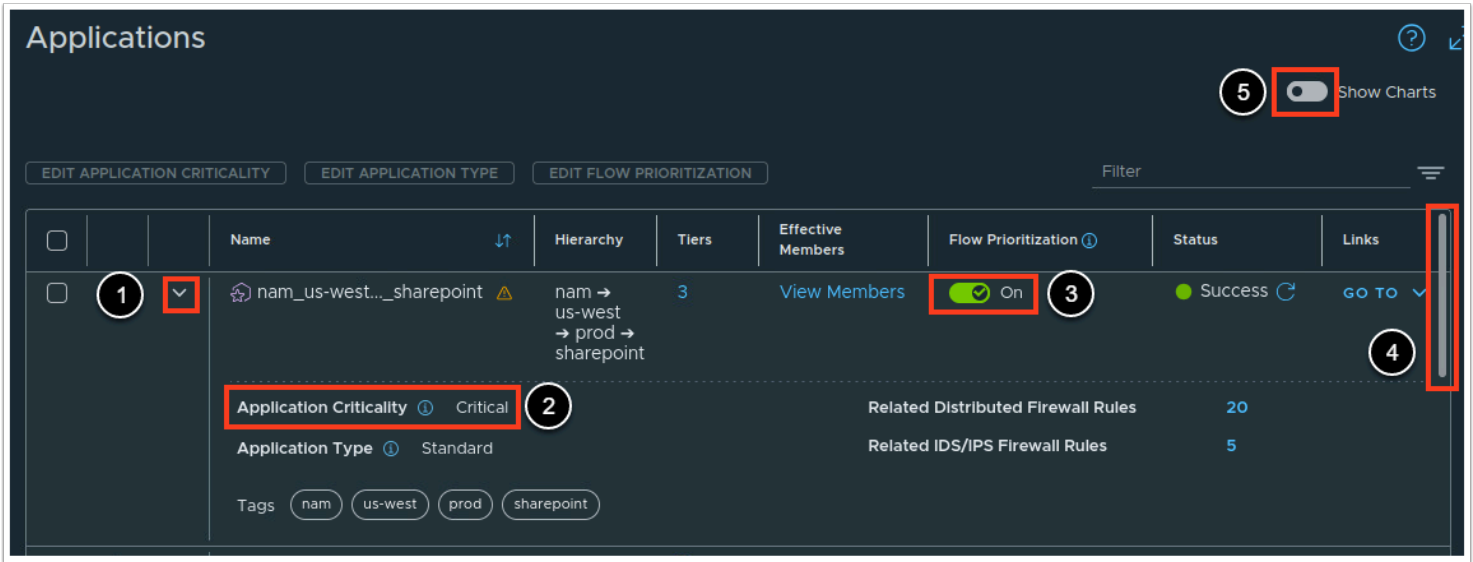
1. Use the **check boxes** to select the top 4 applications in the list. They should be:
- nam\_us-west\_prod\_sharepoint
  - nam\_us-west\_prod\_jira
  - nam\_us-south\_dev\_workday
  - nam\_us-south\_dev\_sap
2. Click **EDIT APPLICATION CRITICALITY**

# Edit Application Criticality



- 1. Click **Critical** for the Application Criticality
- 2. Click **SAVE** to save changes

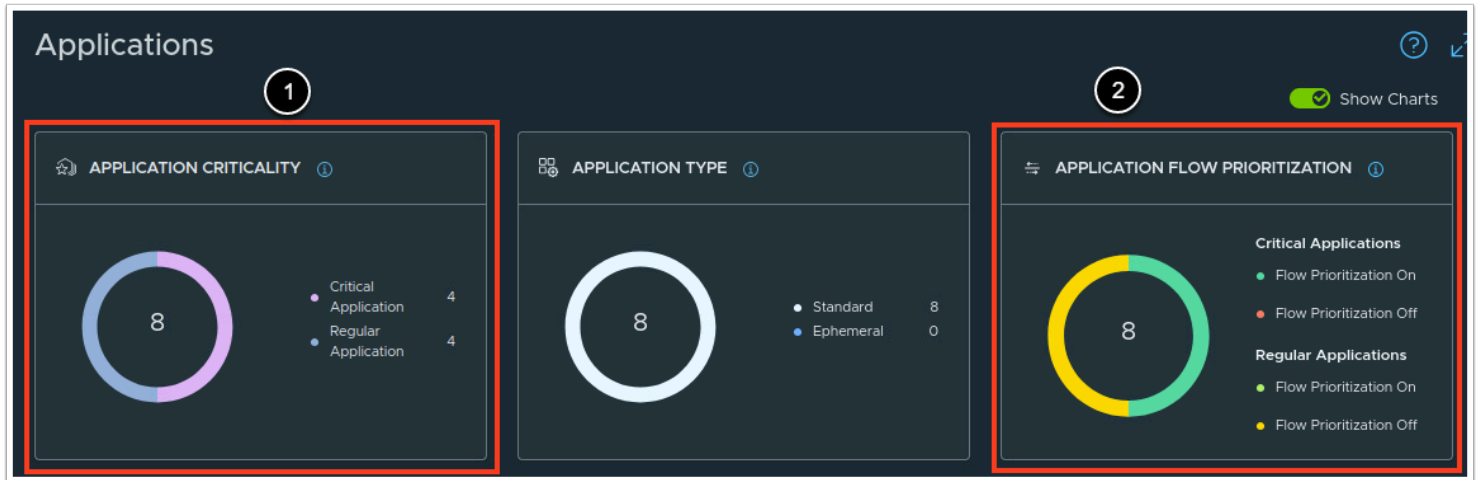
# View Application Criticality for Application



- 1. Click the **arrow** next to one of the applications that were changed to Critical
- 2. The Application Criticality is now listed as **Critical**
- 3. Flow Prioritization for the application is **On**
- 4. Use the **scroll bar** to continue viewing Applications

5. Click the **Show Charts slider** to enable and view the updated Application Inventory Charts

## View Application Inventory Chart Changes

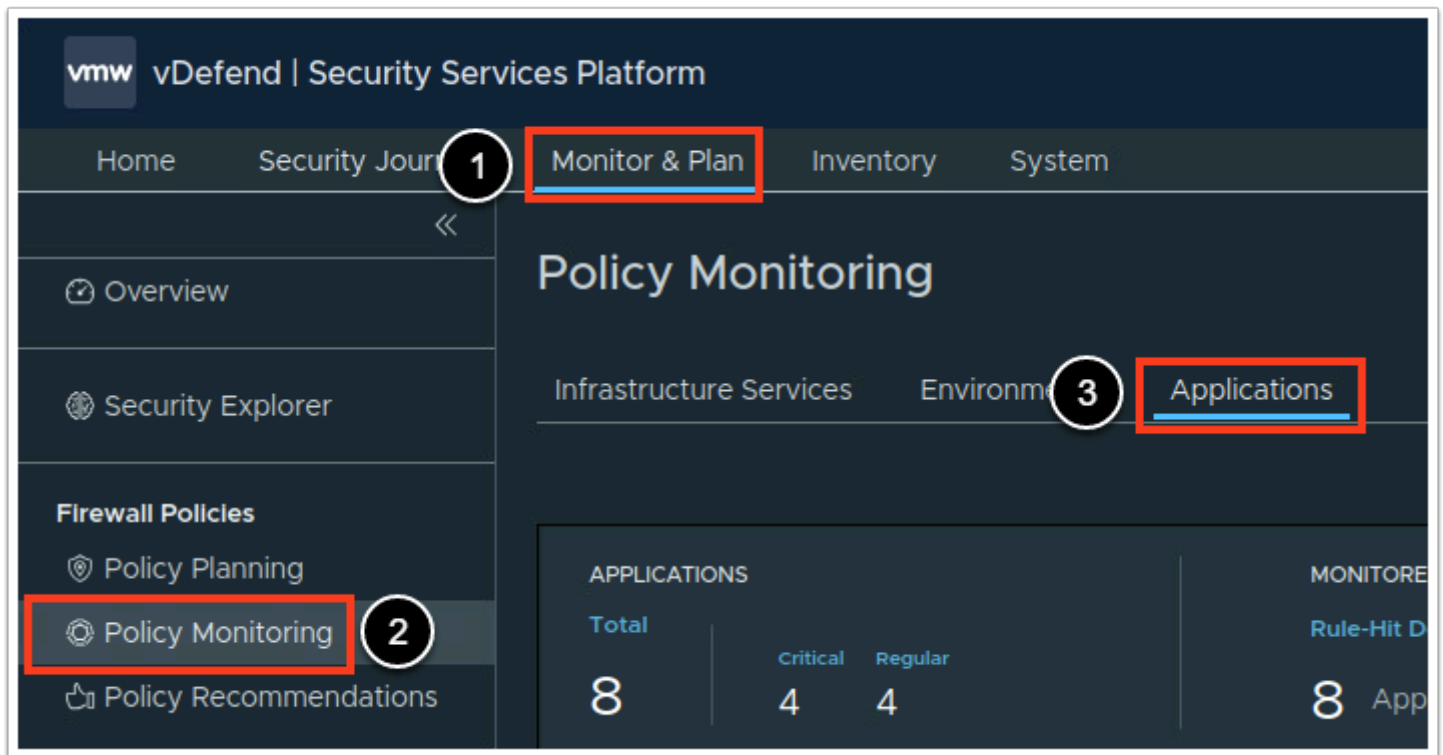


1. There are now 4 critical applications under Application Criticality
2. There are 4 critical applications with Flow Prioritization on, and 4 regular applications with Flow Prioritization off

Changing the Application Criticality to "Critical" has 2 impacts:

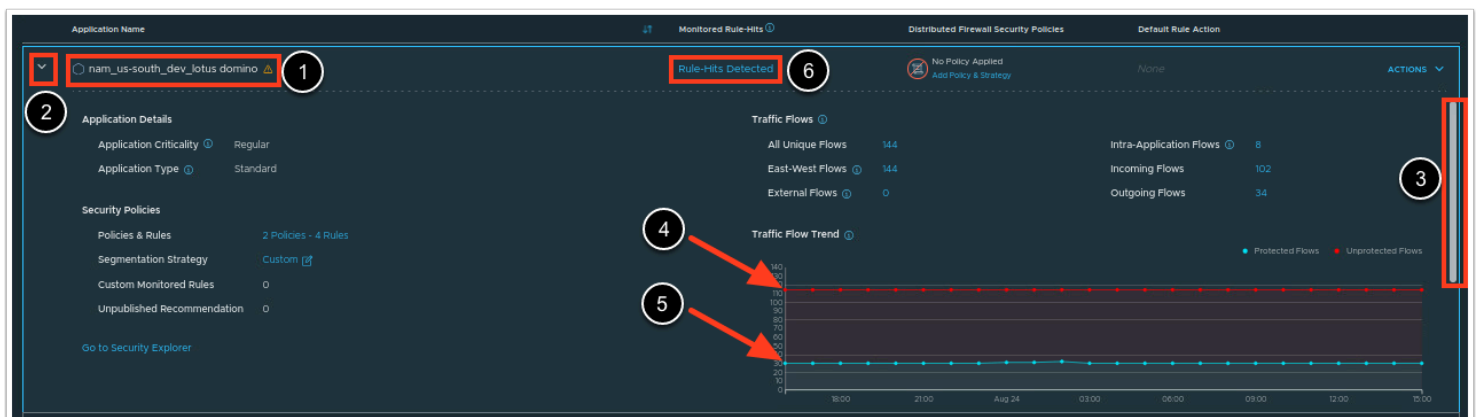
- The Security Score will not give points to regular application lock-down if ALL critical applications are not locked down
- A critical application has "Flow prioritization" enabled by default (this setting can be overridden using the "Flow Prioritization toggle". An application with "Flow prioritized" gets its flows collected with priority in case of flow collection congestion.

# Monitor Applications



1. Click **Monitor & Plan**
2. Click **Policy Monitoring**
3. Click **Applications**

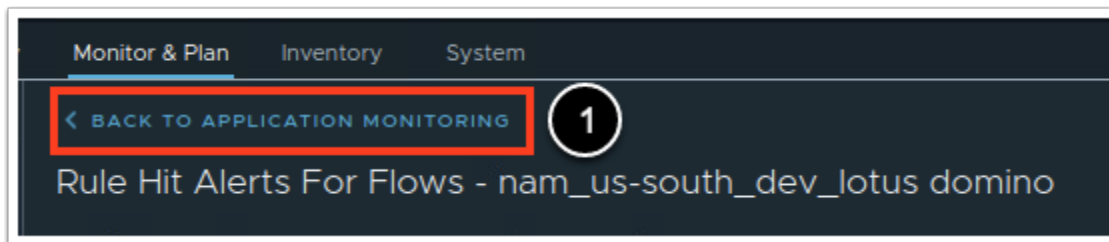
## View Application Monitoring Details



1. Locate the "**nam\_us-south\_dev\_lotus domino**" application in the application list
2. Click the **arrow** to the left of the "**nam\_us-south\_dev\_lotus domino**" application to expand the monitoring details

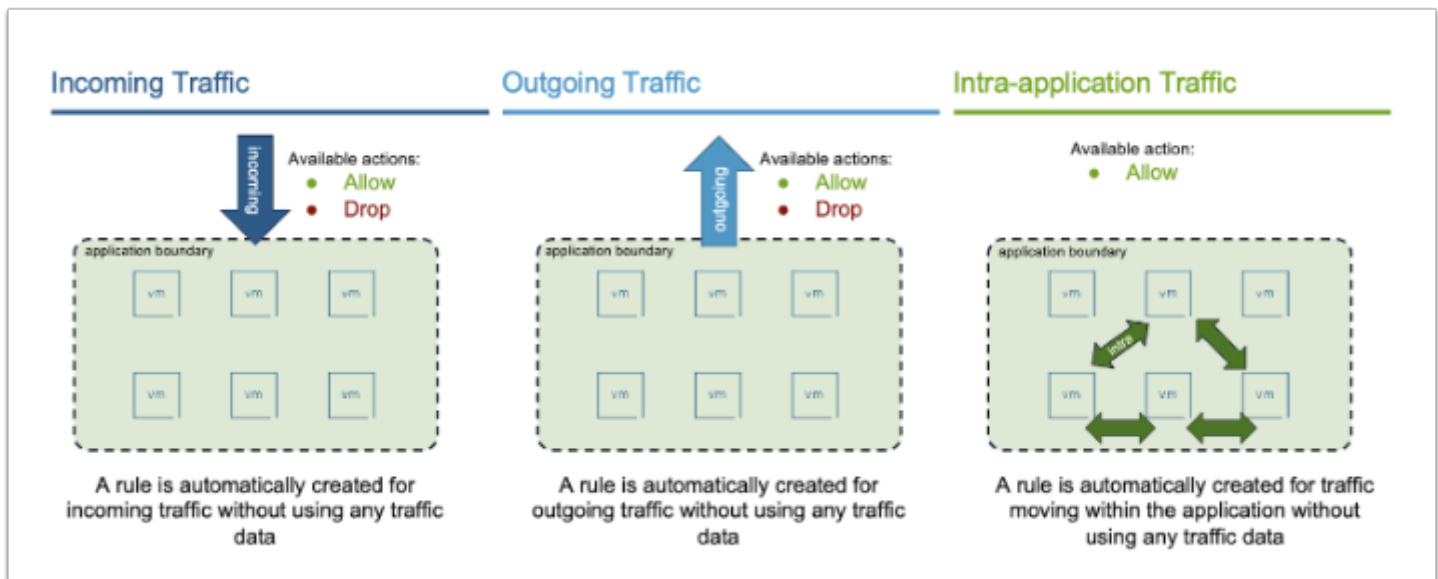


## Return to Application Monitoring Page



1. Click **BACK TO APPLICATION MONITORING** to return to Application Monitoring page

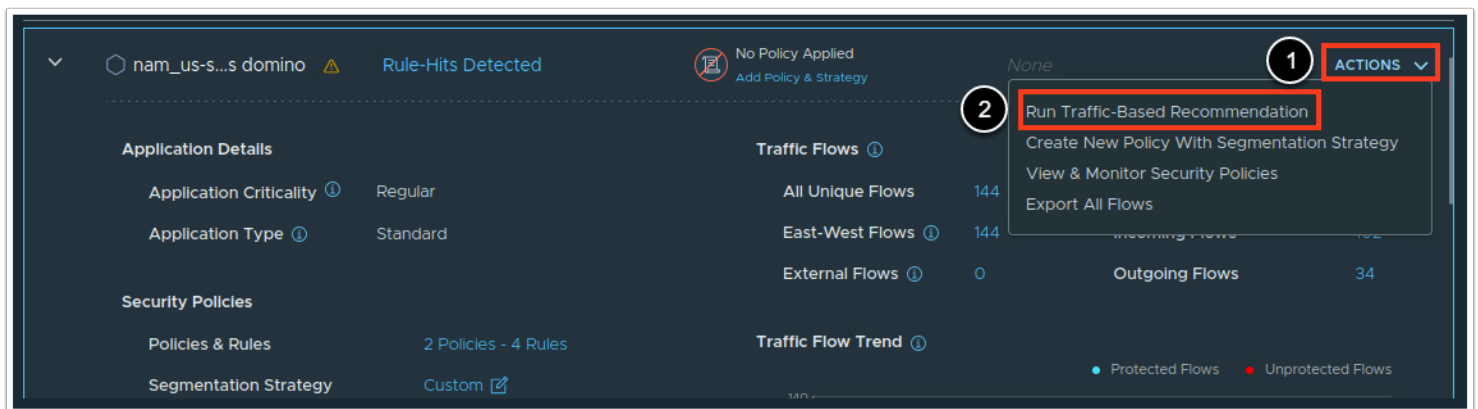
# Protect Applications



One way to speed and to ease application segmentation is to leverage an Application Connectivity Strategy to implement Application Ring Fencing instead of full Micro-segmentation.

An Application Connectivity Strategy is a specific default rule that allows or drops incoming, outgoing or intra-application traffic.

## Begin Traffic-Based Recommendation



1. For the "nam\_us-south\_dev\_lotus domino" application, click the **ACTIONS** dropdown
2. Click **Run Traffic-Based Recommendation**

# Start Recommendation

Run Traffic-Based Recommendation

×

Select to start a new recommendation or re-run previously created recommendation.

Start Recommendation

1

Re-run Unpublished Recommendation

CANCEL

2

NEXT

- 1. Click **Start Recommendation**
- 2. Click **NEXT**

# Recommendation Details

Run Recommendation | nam\_us-south\_dev\_lotus domino

1 Recommendation Details

2 Select Segmentation Strategy (Optional)

3 Advanced Settings

Recommendation Name

REC 260824 06:34:35 - nam\_us-south\_dev\_lotus domino

Note: This will be used as a suffix when naming new recommended groups and rules

Description

Recommendation created on 8/24/26, 6:34 PM

Selected Entities In Scope

nam\_us-south\_dev\_lotus domino

Application Type

Last 1 Hour

Last 12 Hours

Last 24 Hours

2

Distributed Firewall Category

Last 1 Week

Last 2 Weeks

Last 1 Month

Time Range for Traffic

Last 24 Hours

1

If no traffic is detected within the selected time range, no recommendation will be made.

CANCEL

3

NEXT

1. Click the **Time Range for Traffic** drop-down menu
2. Select **Last 1 Hour**
3. Click **NEXT**

## Select Segmentation Strategy

Run Recommendation | nam\_us-south\_dev\_lotus domino

1 Recommendation Details    2 Select Segmentation Strategy (Optional)    3 Advanced Settings

bottom of the new policy.



### Ringfencing

Allow intra-application traffic by default and start monitoring incoming & outgoing traffic to run granular rule recommendations for them.

---

Intra-application Traffic ● Allow

Default Rule ⓘ ● Allow ▼



### Microsegmentation

Monitor all traffic (incoming, intra, & outgoing) to run granular rule recommendations for them.

---

Default Rule ⓘ ● Allow ▼



### Custom

Select the traffic you want to allow or block by default, the remaining flows will be monitored to generate granular rule recommendations.

---

☐ Incoming Traffic ● Allow ▼

☐ Outgoing Traffic ● Allow ▼

☐ Intra-application Traffic ● Allow

☐ Default Rule ⓘ ● Allow ▼

CANCEL BACK NEXT

1. Click **Ringfencing** to select the Ringfencing Segmentation Strategy for this application

# Preview Auto-Plumbed Rules for the Selected Strategy

Run Recommendation | nam\_us-south\_dev\_lotus domino

1 Recommendation Details

2 Select Segmentation Strategy (Optional)

3 Advanced Settings

Default Rule ⓘ ● Allow ▾

☐ Outgoing Traffic ● Allow ▾

☐ Intra-application Traffic ● Allow

☐ Default Rule ⓘ ● Allow ▾

Review the rule actions and monitoring configurations below to confirm the settings for all plumbed rules.

▼ Preview Auto-Plumbed Rules for the Selected Strategy

ⓘ A 'Default Rule' is proposed based on your selection. This policy has broad implications for your network's security and connectivity. Review carefully before publishing to prevent unintended access or service disruption.

| Source                        | Destination                   | Service | Context Profile | Action  | Monitoring | Logging |
|-------------------------------|-------------------------------|---------|-----------------|---------|------------|---------|
| nam_us-south_dev_lotus domino | nam_us-south_dev_lotus domino | Any     | None            | ● Allow | ● Off      | ⏻ Off   |
| Any                           | Any                           | Any     | None            | ● Allow | ● On       | ⏻ Off   |

CANCEL

BA 3

NEXT

1. Use the **scroll bar** to view the auto plumbed rules for the Ringfencing Strategy for this application
2. The rules will allow the application group members to communicate with each other, and have a default allow rule with monitoring on
3. Click **NEXT**

# Start Recommendation Discovery

Run Recommendation | nam\_us-south\_dev\_lotus domino

1 Recommendation Details 2 Select Segmentation Strategy (Optional) 3 Advanced Settings

Input Options:

Protocol and Ports Criteria  Enter up to 15 entries of single ports, ranges of ports, or protocols.

Exclude Flows

Additional Rules To Consider ☐ Include the custom monitored rules as additional rules to consider.

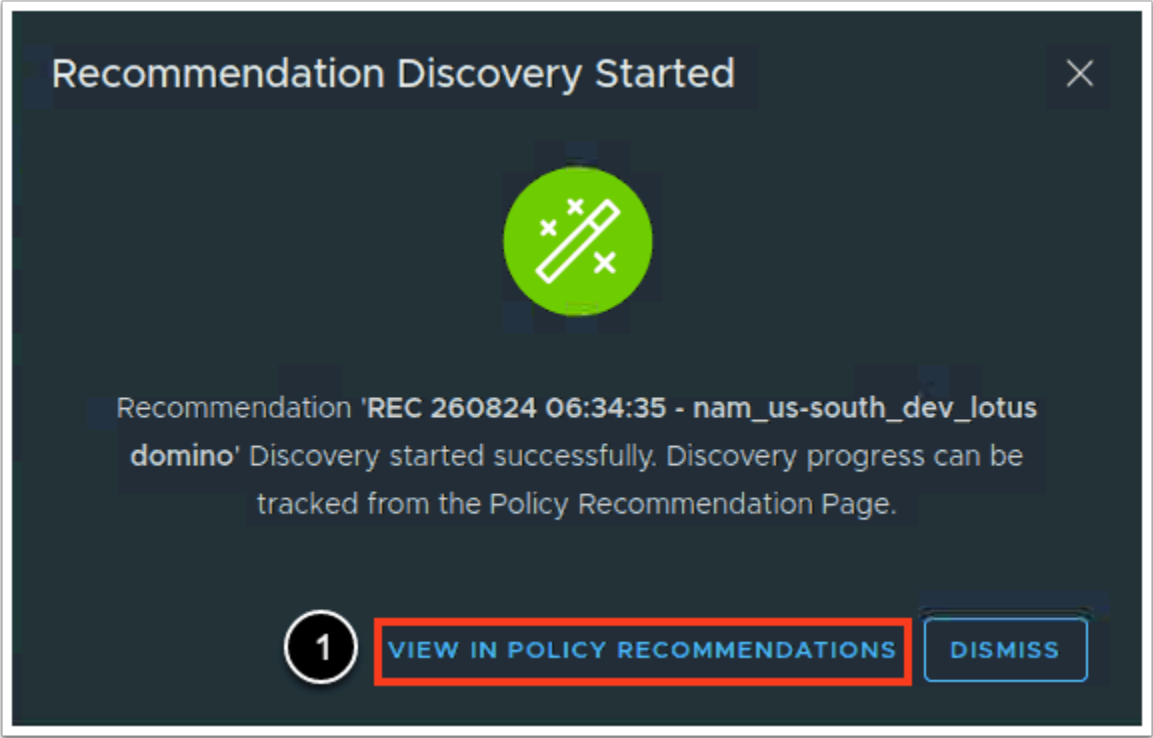
Output Options:

Recommendation Service Type ☒ L4 Services ☐ L7 Context Profiles 1

CANCEL BACK 2 START DISCOVERY

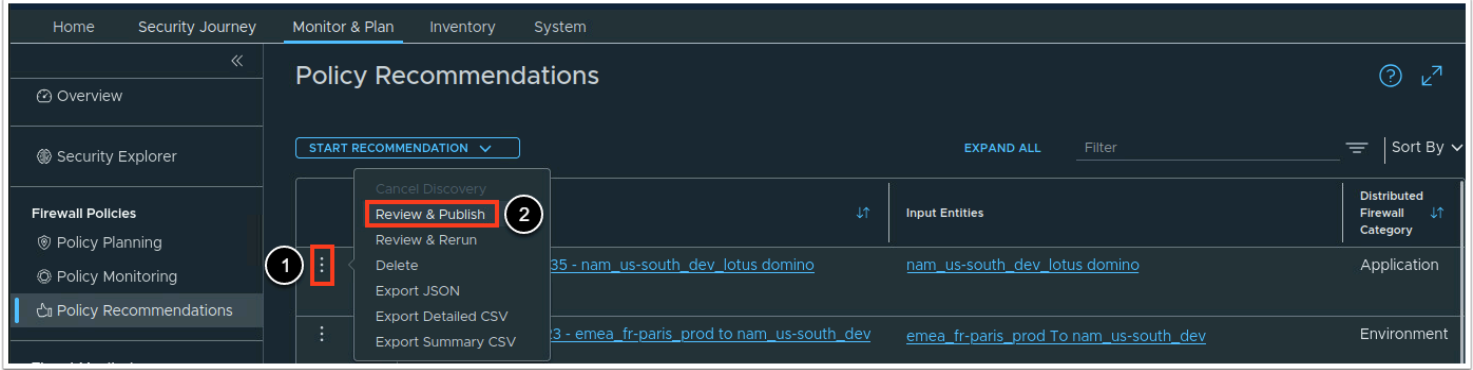
1. Recommendations can be configured to generate L4 Services Rules or L7 Context Profile Rules. Confirm **L4 Services** is selected
2. Click **START DISCOVERY**

# View in Recommendations Tab



- 1. Click **VIEW IN POLICY RECOMMENDATIONS**

# Open Recommended Rules Page



- 1. When the status is "Ready to Publish", click the **3 dots** to the left of the recommendation name
- 2. Click **Review & Publish**

# Review Recommended Rules

< REC 260824 06:34:35 - nam\_us-south\_dev\_lotus domino

Review Recommendations > Sequence & Publish

Showing recommended Policy Rules for the selected input entities ([nam\\_us-south\\_dev\\_lotus domino](#)) as well as the other entities they communicate with. [LEARN MORE](#)

Show Graph

Recommended Policies **Rules (7)** Groups (1) Services (0) Category: Application

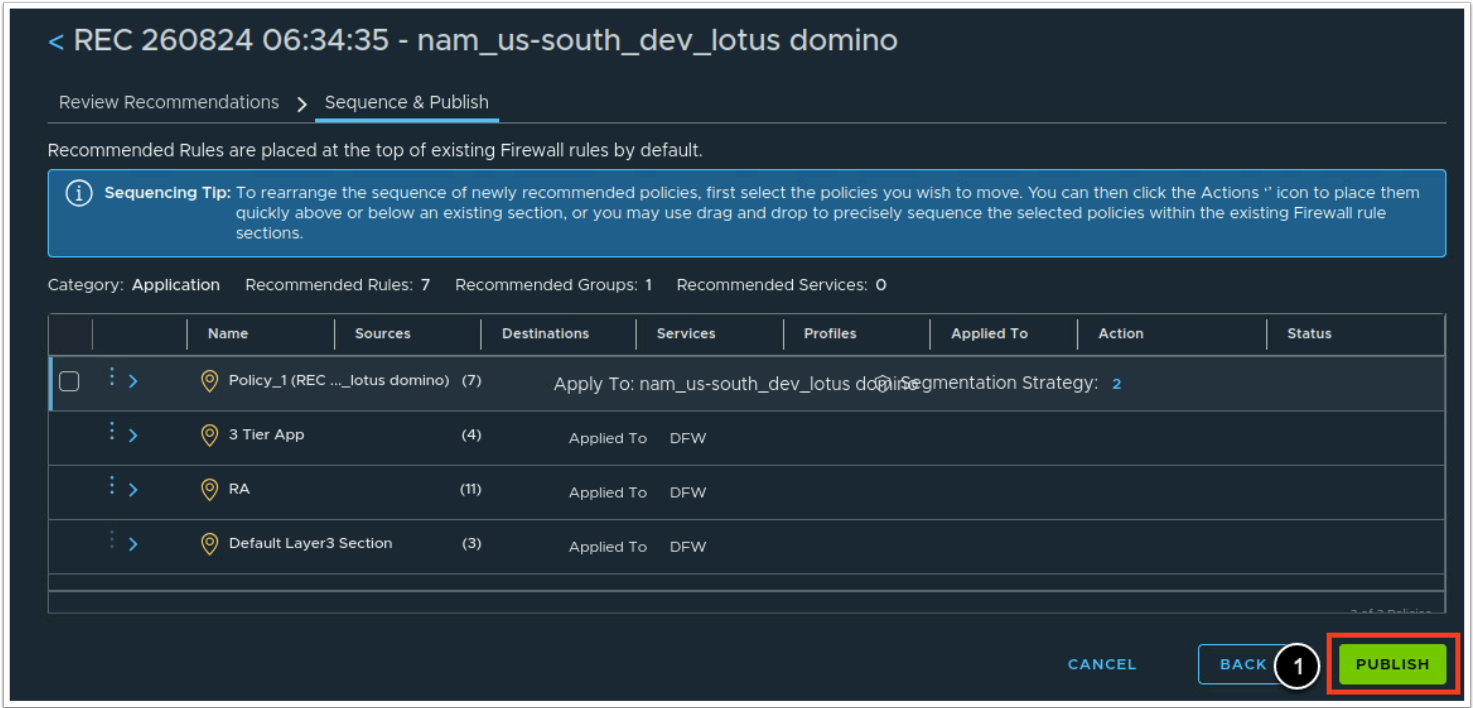
REPLACE DELETE Filter

|                          | Name                               | Sources                                                          | Destinations | Services | Profiles | Applied To | Action | Status |
|--------------------------|------------------------------------|------------------------------------------------------------------|--------------|----------|----------|------------|--------|--------|
|                          | Policy_1 (REC ...lotus domino) (7) | Apply To: nam_us-south_dev_lotus domino Segmentation Strategy: 2 |              |          |          |            |        |        |
| <input type="checkbox"/> | New Rule_i...omino                 | Grou...ino                                                       | nam...ginx   | HTTPS    | None     | nam...ino  | Allow  | On     |
| <input type="checkbox"/> | New Rule_i...omino                 | eme...gic                                                        | nam...ysql   | MySQL    | None     | nam...ino  | Allow  | On     |
| <input type="checkbox"/> | New Rule_i...omino                 | nam...oss                                                        | nam...ysql   | MySQL    | None     | nam...ino  | Allow  | On     |

CANCEL SAVE **PROCEED**

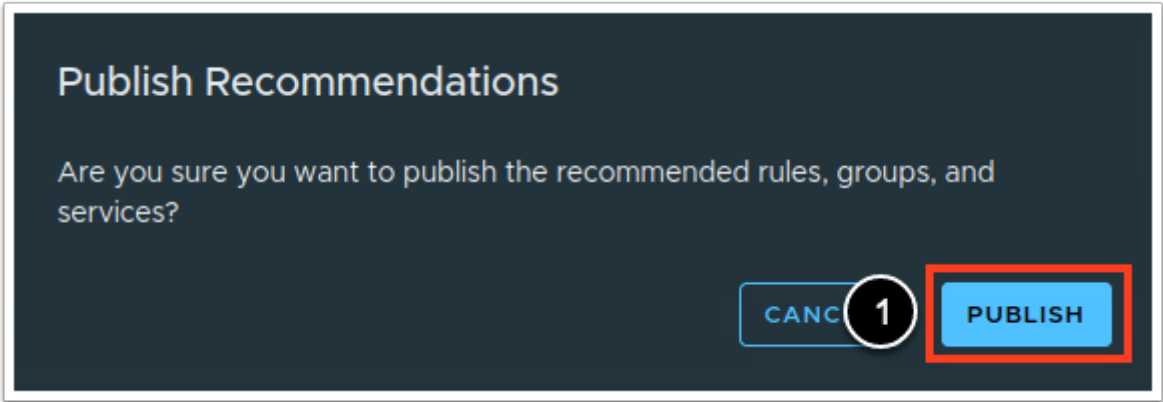
1. Review the recommended rules
  - Please note that:
    - The infra-application traffic is allowed thanks to the Application Connectivity Strategy, not with an explicit rule
    - Source and Destination are using Application Tiers when feasible.
    - When an application tier can't be selected, an existing group might be selected, or a new group might be recommended
2. Use the **scroll bar** to view the rest of the rules if needed
3. Click **PROCEED**

# Publish the Recommended Rules



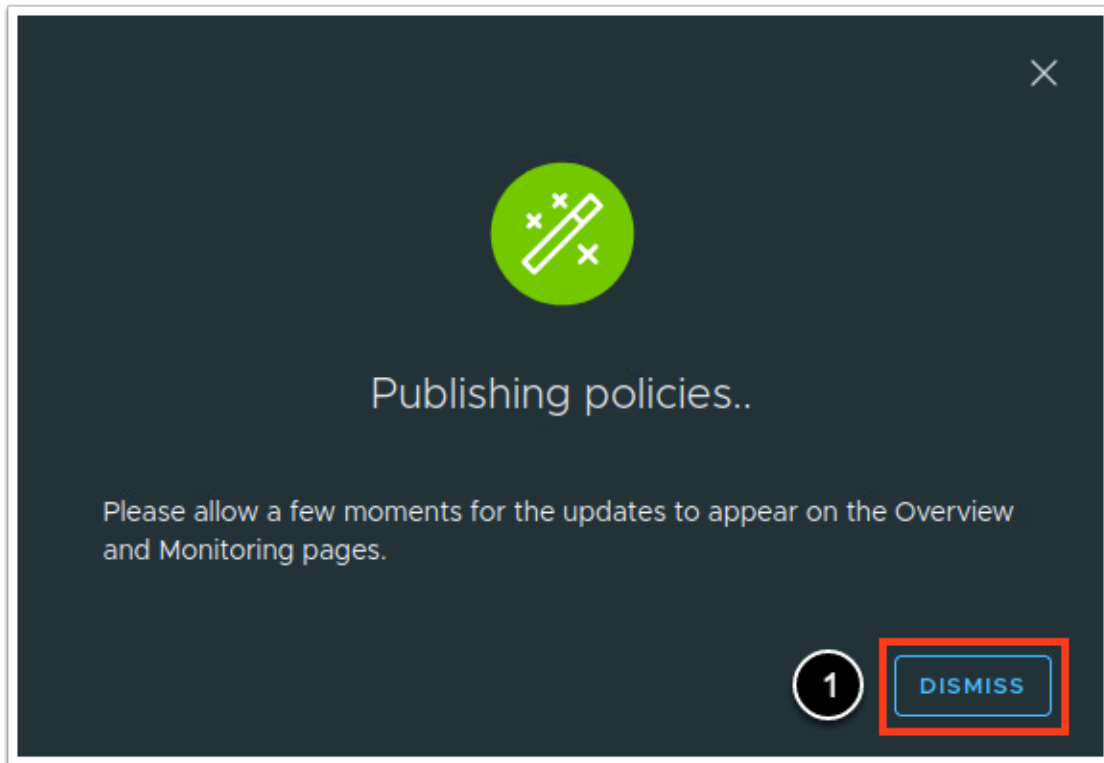
- 1. Click **PUBLISH** to publish the recommended rules to the Distributed Firewall

## Confirm Rule Publish



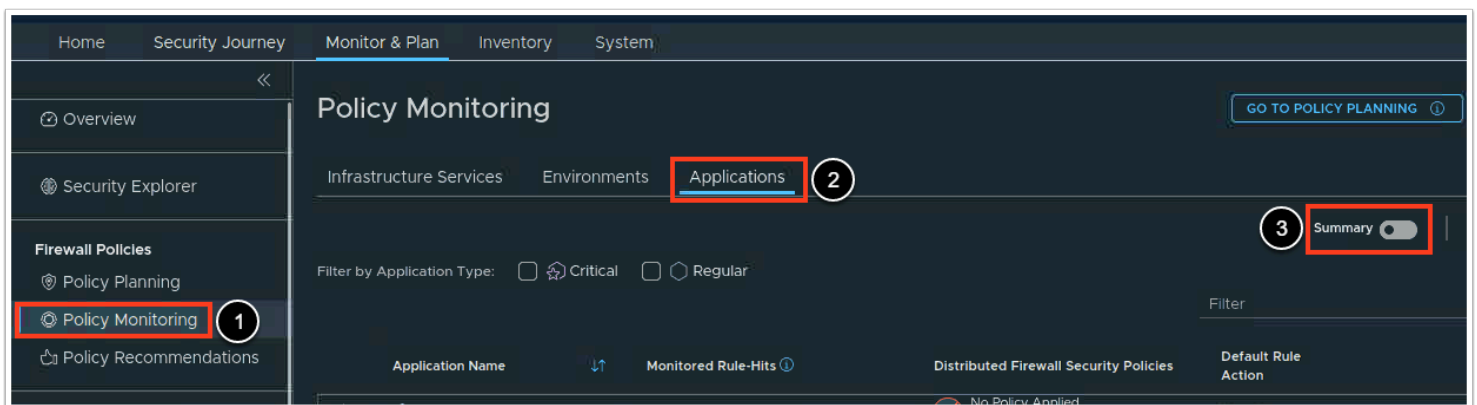
- 1. Click **PUBLISH** to confirm rule publish

## Close Rule Publish



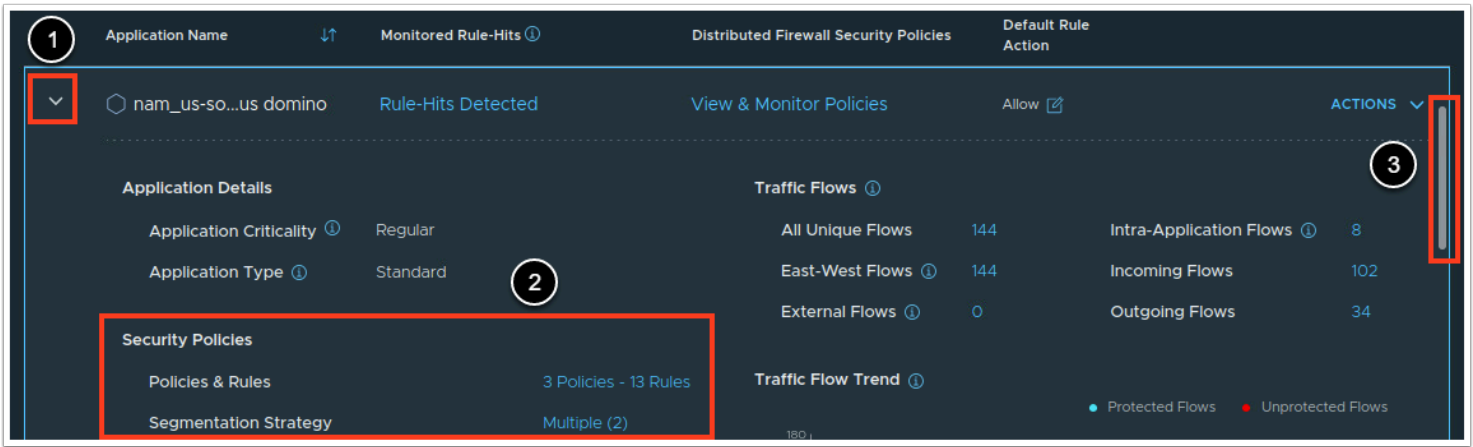
1. Click **DISMISS** to close the rule publish window

## Return to Segmentation Monitoring - Applications



1. Click **Policy Monitoring**
2. Click **Applications**
3. Click the **Summary slider** to disable and gain more screen space

# View Updated Application Monitoring



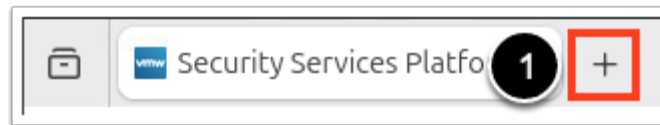
1. Locate the "nam\_us-south\_dev\_lotus domino" application and click the **arrow** to expand the Application details
2. Note the Segmentation Strategy and Policies & Rules are now populated
3. As the application is still being monitored, review any flow that hits the "Monitored Rule" (the default rule of this application in this case). You can monitor the application for multiple days or weeks depending on the application
4. Use the scroll bar to continue viewing the Application details

# View Traffic Flow Trend Changes



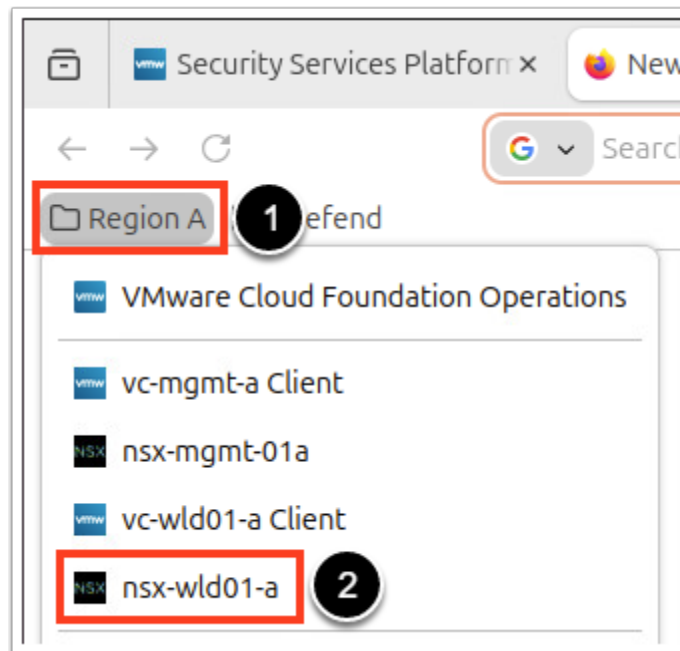
1. Note the increase in the blue line on the Traffic Flow Trend. This reflects the change in protected flows due to the publishing of recommended rules.

# Login to NSX



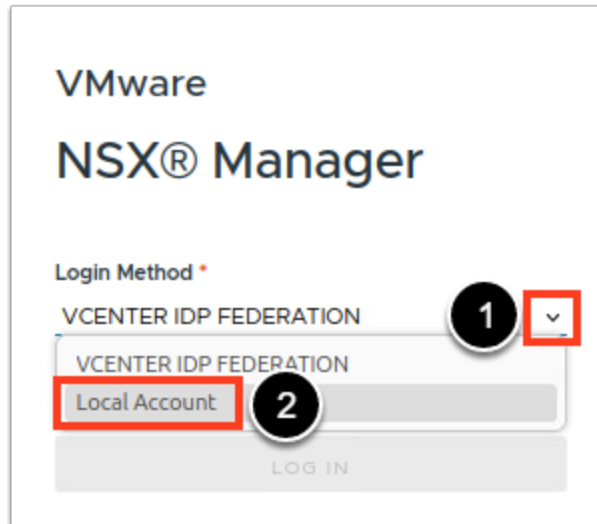
1. Click the + at the top of the browser window to open a new tab

## Open NSX Manager



1. Click on the **Region A** bookmark bar folder
2. Click **nsx-wld01-a** to launch NSX Manager

## Select Login Method



VMware  
NSX® Manager

Login Method \*

VCENTER IDP FEDERATION

VCENTER IDP FEDERATION

Local Account

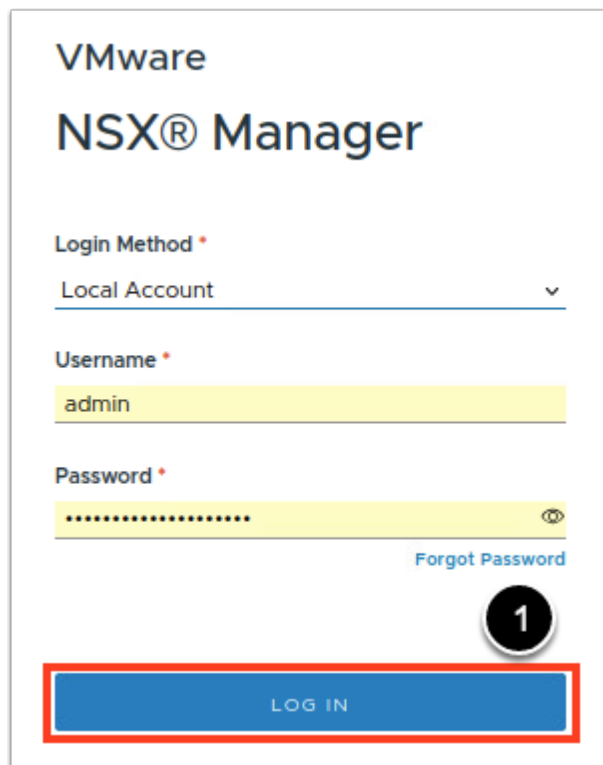
LOG IN

1

2

1. Click the **arrow** to expand the drop-down menu
2. Click **Local Account**

## Login to NSX Manager



VMware  
NSX® Manager

Login Method \*

Local Account

Username \*

admin

Password \*

.....

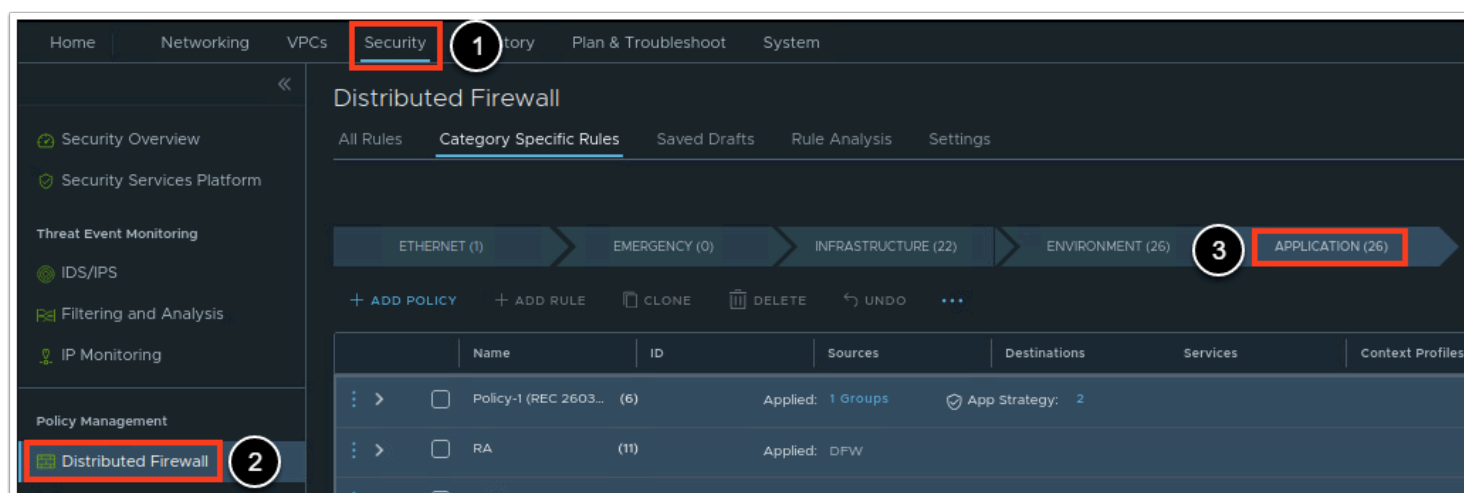
Forgot Password

LOG IN

1

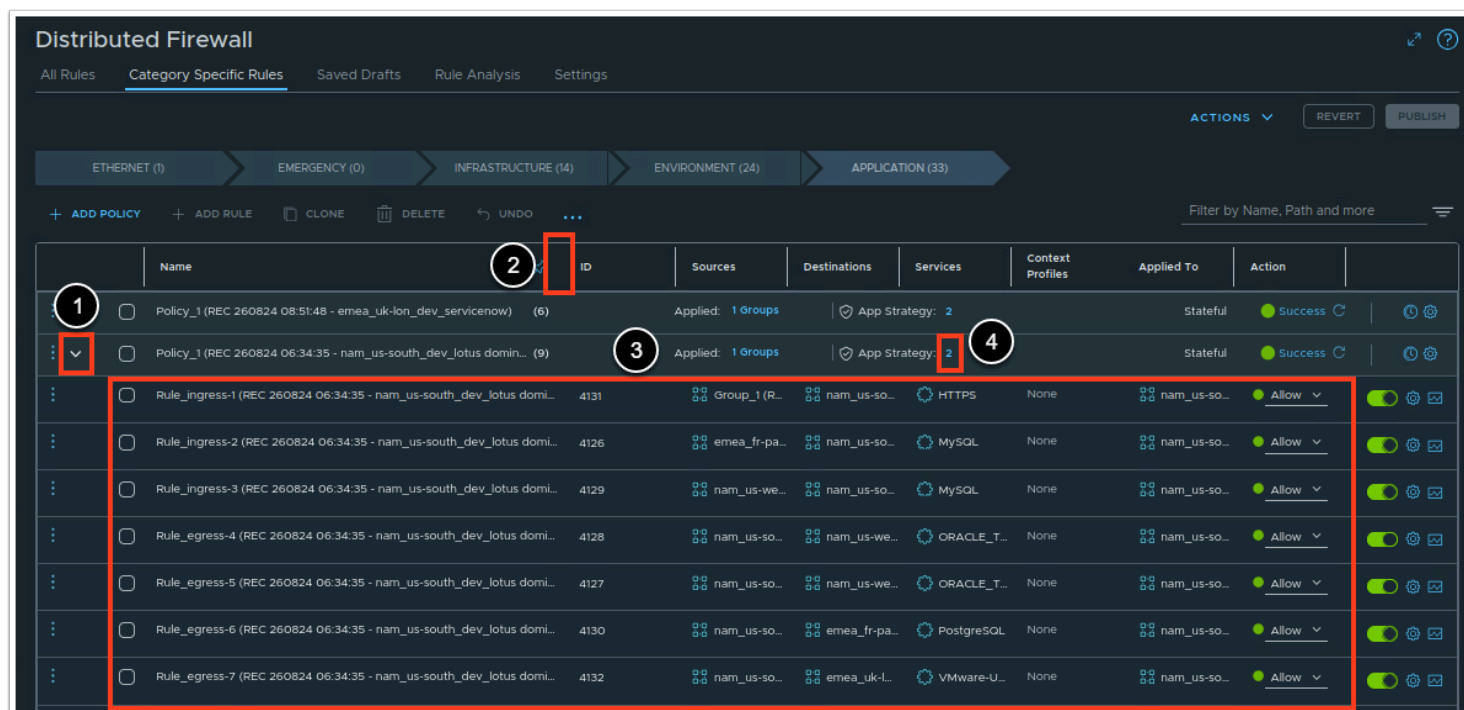
- 
1. Using the pre-filled username **admin** and pre-filled password **VMware123!VMware123!** select Login

# Validate Application Rule Publish in NSX



1. Click **Security**
2. Click **Distributed Firewall**
3. Click **Application** to look at the Application section of the DFW

## View Published Rules



1. Click the **arrow** to the left of the policy name to expand its ruleset to view the rules
2. If needed, **drag** the column header separator to expand the Name column

3. View the new rules created for the application
4. Click the "2" to the right of App Strategy to view Application Strategy details

## View Application Connectivity Strategy Details

**Application Connectivity Strategy**

Policy > Policy\_1 (REC 260824 06:34:35 - nam\_us-south\_dev\_lotus domino)

To edit below content, click [EDIT](#)

| Connectivity Strategy     | Rule ID ⓘ | Action  | Logging |
|---------------------------|-----------|---------|---------|
| Intra-application Traffic | 4125      | ● Allow | ● Off   |
| Default Rule              | 4124      | ● Allow | ● Off   |

ⓘ Please note connectivity strategy rules are executed after all other rules in the firewall policy.

[CLOSE](#)

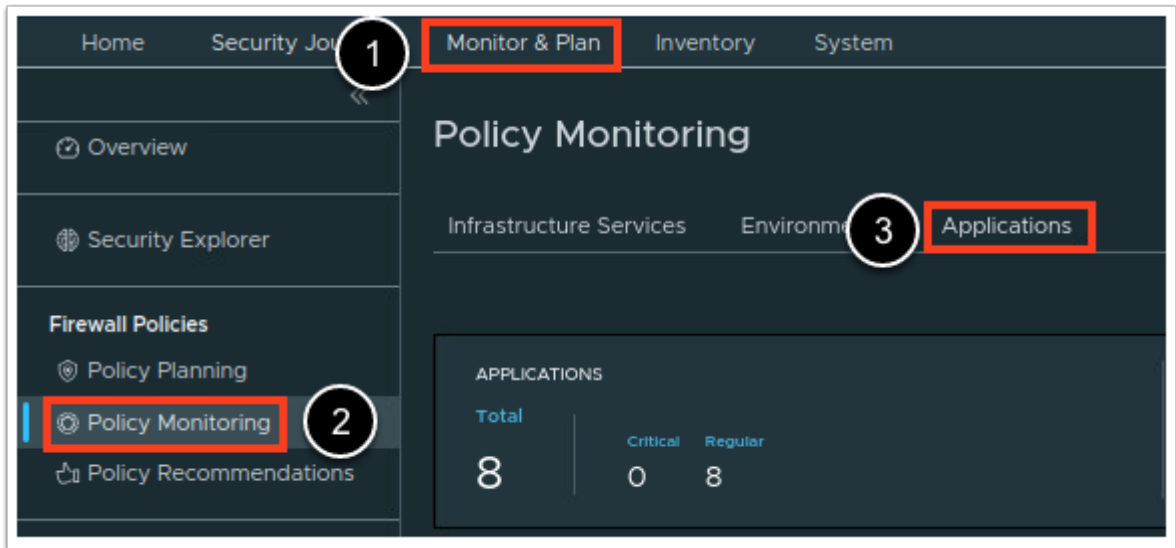
1. View the Application Connectivity Strategy details for this policy
2. Click **CLOSE**

# Lockdown Applications



1. Click on the **Security Services Platform** tab to return to SSP

## Return to Segmentation Monitoring - Applications



1. Click **Monitor & Plan**
2. Click **Policy Monitoring**
3. Click **Applications**

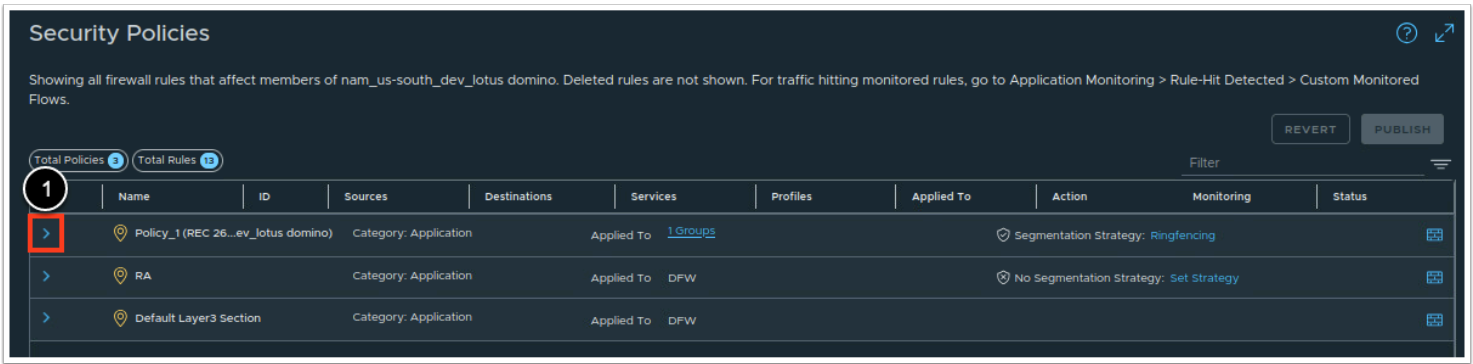
# View & Monitor Security Policies for Application



1. Locate the "nam\_us-south\_dev\_lotus domino" application
2. Click the **arrow** next to the "nam\_us-south\_dev\_lotus domino" application
3. Click **ACTIONS**
4. Click **View & Monitor Security Policies**

When no legitimate traffic hits the Monitored Rules anymore for a while, you can consider locking down the application.

## Expand the Application Policy



1. Click the **arrow** to the left of the Application Policy to expand the ruleset

# Change the Default Application Rule to Deny

Security Policies

Showing all firewall rules that affect members of nam\_us-south\_dev\_lotus domino. Deleted rules are not shown. For traffic hitting monitored rules, go to Application Monitoring > Rule-Hit Detected > Custom Monitored Flows.

Total Policies: 3 Total Rules: 12

| Name                                                                | ID   | Sources       | Destinations    | Services      | Profiles | Applied To    | Action      | Monitoring | Status |
|---------------------------------------------------------------------|------|---------------|-----------------|---------------|----------|---------------|-------------|------------|--------|
| Rule_egress-4 (REC 260824 06:34:35 - nam_us-south_dev_lotus domino) | 4126 | nam_us_nodejs | nam_us_oracle   | ORACLE_TNS    | None     | nam_us_domino | Allow       | Off        | On     |
| Rule_egress-5 (REC 260824 06:34:35 - nam_us-south_dev_lotus domino) | 4127 | nam_us_nodejs | nam_us_oracle   | ORACLE_TNS    | None     | nam_us_domino | Allow       | Off        | On     |
| Rule_egress-6 (REC 260824 06:34:35 - nam_us-south_dev_lotus domino) | 4130 | nam_us_nodejs | emea_fr_ostores | PostgreSQL    | None     | nam_us_domino | Allow       | Off        | On     |
| Rule_egress-7 (REC 260824 06:34:35 - nam_us-south_dev_lotus domino) | 4132 | nam_us_nodejs | emea_uk_ssandra | VMware_ateMgr | None     | nam_us_domino | Allow       | Off        | On     |
| allow_intra_rule                                                    | 4125 | nam_us_domino | nam_us_domino   | Any           | None     | nam_us_domino | Allow       | Off        | On     |
| <b>default_allow_rule</b>                                           | 4124 | Any           | Any             | Any           | None     | nam_us        | <b>Drop</b> | Off        | On     |

REFRESH

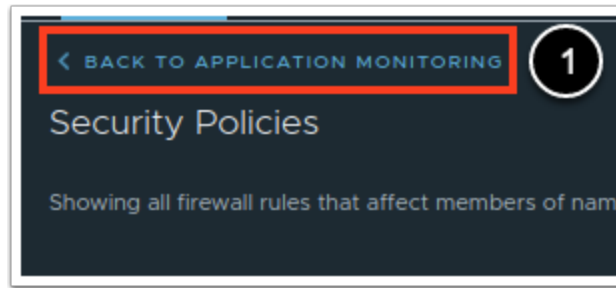
3 of 3 Policies

1. Use the **scroll bar** to scroll down in the ruleset
2. Locate the **default\_allow\_rule**
3. Click the Action drop-down menu and select **Drop**
4. An application Deny Rule is not monitored by default. If you want to monitor traffic hitting the default deny rule, you can enable monitoring on this rule by toggling the "Monitoring" switch to "On". In this lab, we will leave this in its default "Off" position
5. Click **PUBLISH**



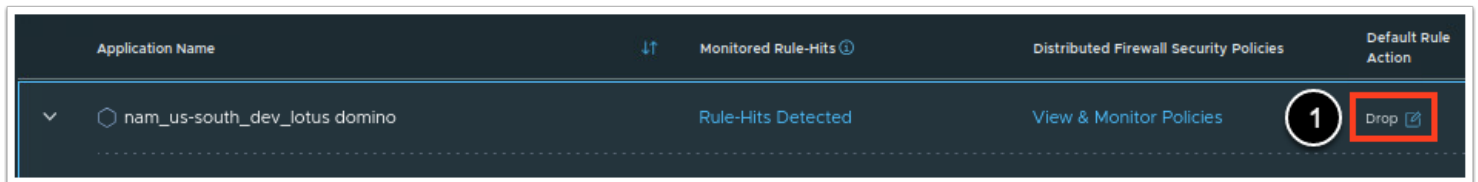
Note: It can take a few minutes for the Default Deny change to appear in the Security Policies list on this page.

## Return to Application Monitoring



1. Click **BACK TO APPLICATION MONITORING**

## Validate Application Lockdown



1. In the monitoring for the application, the Default Rule Action now shows "Drop"

 You may need to refresh the Application Monitoring list to reflect this change

## OPTIONAL: Protect and Lockdown Additional Applications

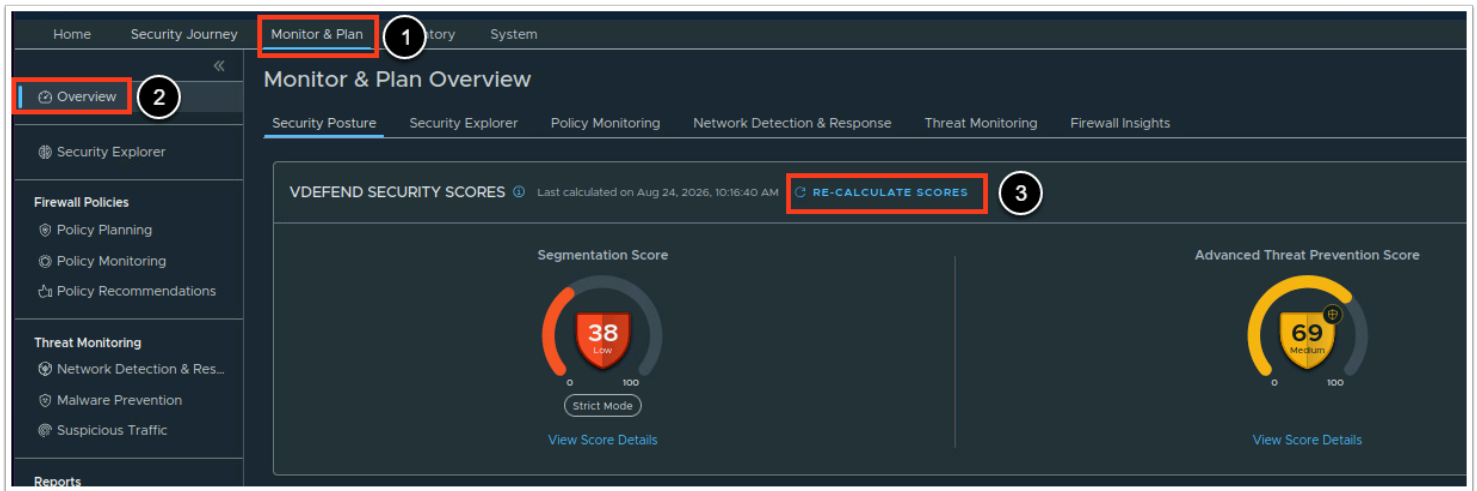
Repeat the process for the remaining Applications:

1. Run & Publish Recommendations for each Application
  - To speed up this process, you can select "Drop" as the Default Rule action for the Ringfencing Segmentation Strategy portion of the Recommendation Parameters. Because this is a lab with all "known" traffic, it's faster to run the recommendation and publish rules with a lockdown already in place. This is **NOT RECOMMENDED** in a real/production environment.
  - Remember to select "Last 1 Hour" for "Time Range for Traffic".
2. If you want your lab to look exactly the same as this lab manual:
  - Run & Publish Recommendations for all applications with a Default "Drop" action



You can create any combination of publishing/non publishing application recommendations and use different segmentation strategies in this lab. Doing so will yield different Security Segmentation Scores and Rule Breakdowns.

# Re-calculate Security Segmentation Score

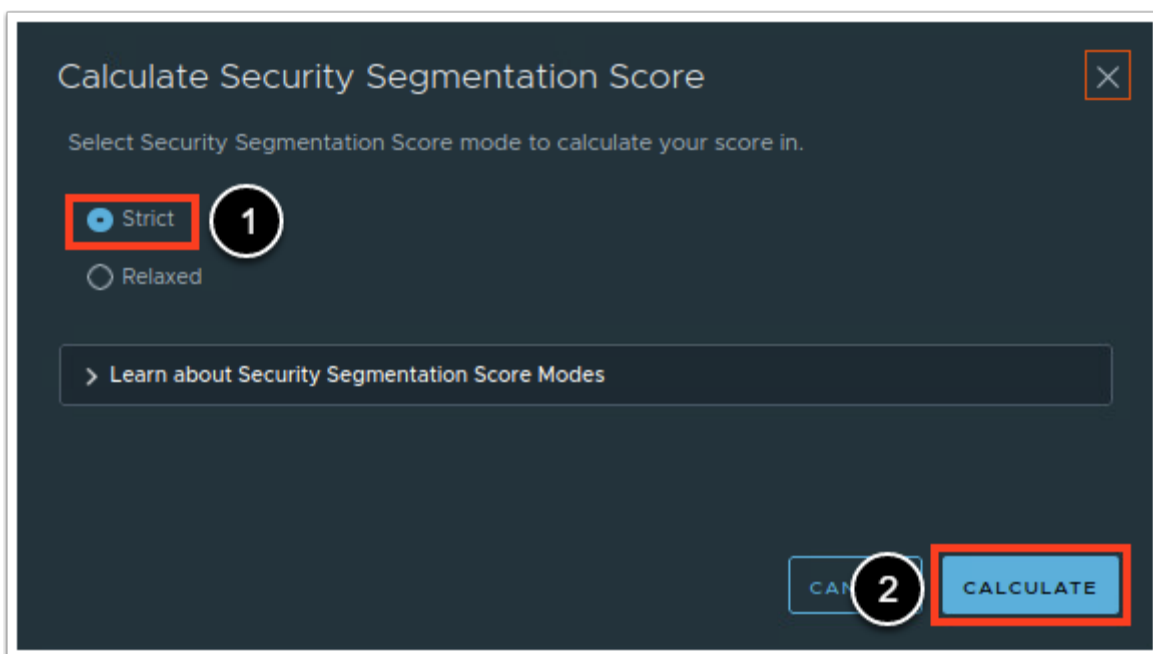


1. Click **Monitor & Plan**
2. Click **Overview**
3. Click **RE-CALCULATE SCORES**



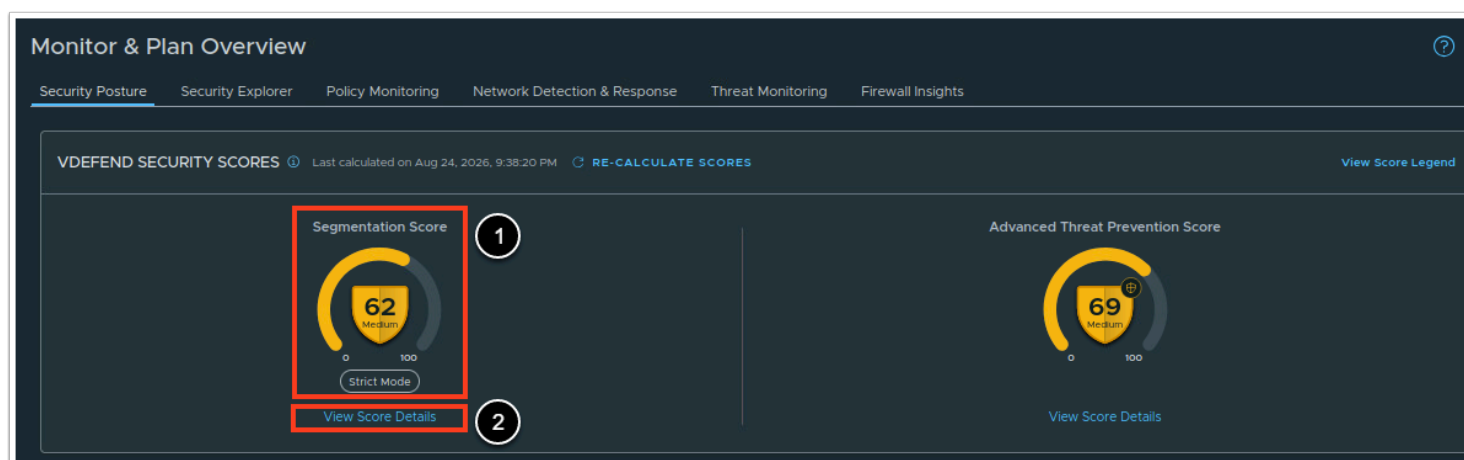
Your Security Segmentation Scores may differ based on whether you published the optional additional recommended rules earlier in this lab.

## Choose Security Segmentation Score Calculation Mode and Calculate



1. Confirm **Strict** mode is selected
2. Click **CALCULATE**

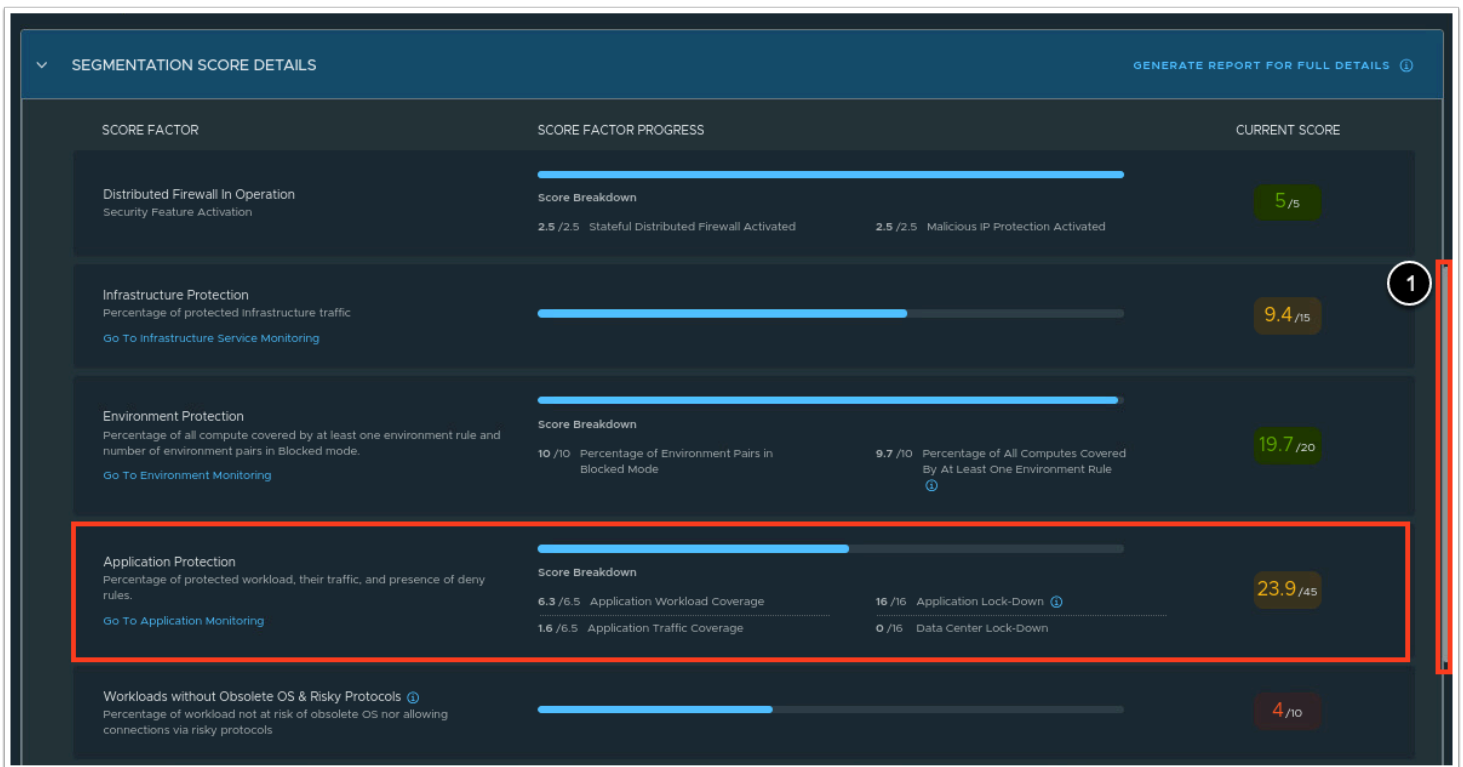
## View Updated vDefend Security Scores



1. The Segmentation Scores has increased
2. Click **View Score Details** to expand Segmentation Score Details

 Your vDefend Security Scores may differ based on whether you published the optional additional recommended rules earlier in this lab.

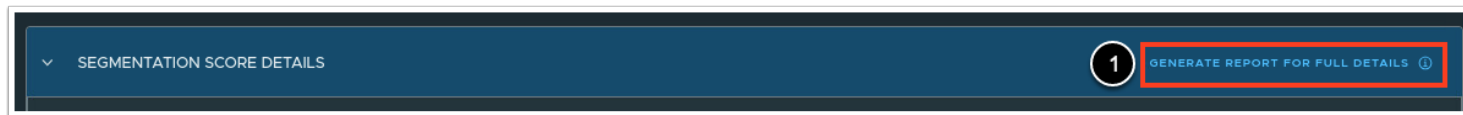
## View Updated Application Protection Score



1. Use the **scroll bar** to scroll down and view the Application Protection Score Factor.
2. The Application Protection Score has increased.

 **Note:** the score does not account for the latest 1-hour of traffic, so expect some delay in seeing the “Application Traffic Coverage” reflect your changes.

## OPTIONAL: Generate a New Security Assessment Report



1. If you would like to see what the Security Assessment Report looks like now that we have completed the Distributed Firewall and Advanced Threat Prevention with vDefend 1-2-3-4 process, click **GENERATE REPORT FOR FULL DETAILS** to walk through the report generation process again.

# Conclusion

Congratulations on completing **Module 7 Stage 4: Application Micro-segmentation!**

## Key Takeaways

- **Defined Critical Applications:** Identified and categorized critical applications within the Security Services Platform (SSP).
- **Monitored Traffic Flows:** Analyzed external, east-west, intra-application, and unprotected traffic flows using Application Monitoring.
- **Applied Ring Fencing:** Explored ring fencing concepts as an intermediate stepping stone toward full micro-segmentation.
- **Configured Connectivity Strategies:** Defined custom Connectivity Strategies to shape rule recommendation outcomes for each application.
- **Published & Validated Policies:** Deployed generated rule recommendations directly to the Distributed Firewall (DFW) and verified enforcement in NSX.
- **Enforced Application Lockdown:** Locked down application boundaries and established ongoing monitoring workflows.
- **Calculated Final Segmentation Score:** Generated a final Security Segmentation Score and Report to measure the security posture improvements achieved through the DFW 1-2-3-4 process.

## Next Steps

- Click **Show Table of Contents** in the lab interface to review previous modules or explore additional topics.
- If you have completed your session, you may now safely end your lab environment.